ANABELIAN_149_PUSH
Checked results and unresolved obligations on cotes 149 and 142

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Abstract

Status: INCOMPLETE. This is a research audit and mathematical return. No priority is claimed. The report and coverage ledger retain every requested item; an unresolved obligation is never a theorem. We distinguish checked mathematical implications from assertions inherited from earlier AI-produced returns.

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Scope and proof conventions

We use the full arithmetic quotient by the maximal geometric prime-to-p kernel, retaining the whole absolute Galois group of the base. Conjugacy of sections always means conjugacy by the geometric kernel. Every restriction to a smaller quotient proves only the corresponding necessary condition unless a converse is supplied. The attached prompt fixes the item labels; our statement identifiers have the prefix of the relevant part.

1 Part A. Point sections in characteristic p

We prove A1 for every curve with an abelian prime-to-p Galois cover of a genus-zero curve, in every characteristic. This includes every geometrically hyperelliptic curve in odd characteristic. We also prove it for every abelian prime-to-p Galois cover of a genus-one curve in odd characteristic, and in characteristic two when that quotient is supersingular. Other results include a nonabelian S_3 class and explicit ordinary characteristic-two families. The universal question for arbitrary curves remains unresolved. Both covering conditions in A2 are equivalent to equality of full prime-to-p section classes. Their final implication $P = Q$ therefore has exactly the same remaining scope as A1. The finite joint norm kernel in the first route of the prompt is determined exactly and is larger than the proposed subgroup. All statements concern geometric conjugacy of sections and retain the full arithmetic Galois group.

1.1 The covering condition is exactly section equality

A.1 (Proved; verification of R2). *Let K be finitely generated over $\mathbb{F}_p$, and let G/K be a semiabelian variety. For the continuous prime-to-p Kummer map*

$$\kappa_G : G(K) \longrightarrow H^1_{\mathrm{cont}}\left(G_K, \prod_{\ell \neq p} T_\ell G\right), \qquad \ker \kappa_G = G(K)[p^\infty].$$

In particular, equality of point sections of a smooth curve after geometric prime-to-p abelianization implies that their difference in its generalized Jacobian is p-primary torsion. The modulus is the reduced boundary of the smooth compactification.

Proof. We check the arithmetic input in the previous return. The Mordell–Weil–Lang–Néron theorem, in the form of Milne (2008, I, Theorem 16.7), says that $A(K)$ is finitely generated for an abelian variety A over such a field K. If L/K is a finite splitting field for a torus T, a normal proper model of L over its finite constant field gives

$$1 \longrightarrow k_0^\times \longrightarrow L^\times \xrightarrow{\mathrm{div}} \mathrm{Div}(X).$$

The image in the free divisor group is free, so $L^\times$ is finite torsion plus a free abelian group. Thus the subgroup $T(K) \subset (L^\times)^{\dim T}$ has this same property. For $1 \to T \to G \to A \to 1$, the image $B \subset A(K)$ is finitely generated. Choose lifts of its generators and let $H \subset G(K)$ be the group they generate. There is a finitely generated direct summand T_0 of $T(K)$ containing $H \cap T(K)$ and all its torsion. Write $T(K) = T_0 \oplus T_1$. Then

$$G(K) = (H + T_0) \oplus T_1.$$

The first summand is finitely generated and the second is free. It follows, coordinate by coordinate, that $\bigcap_{(n,p)=1} nG(K) = G(K)[p^\infty]$.

Multiplication by each n prime to p is a finite étale isogeny; the Kummer exact sequence identifies the kernel of $G(K) \to H^1(G_K, G[n])$ with $nG(K)$. These maps are continuous cohomological maps for the natural Galois action. A point in their common kernel has compatible rational division points: any finite collection of compatibility requirements is realized at a common multiple of the indices, and compactness of the product of their finite fibres supplies the inverse limit. Hence the same kernel statement holds for the full prime-to-p Tate module.

For the final assertion use the Abel map $U \to J_D$, $R \mapsto [R - P]$, defined over K by a chosen $P \in U(K)$. Pullback of the $[n]$-torsors gives a continuous morphism from the arithmetic fundamental extension of U to that of J_D. Equality of point sections therefore kills every Kummer class of $[Q - P]$. Only this direction is used below; no nonabelian conclusion is inferred from a single abelianization. □

A.2 (Proved; uniqueness of a torsion-compatible lift). *Let Ω be an algebraically closed field of characteristic $p > 0$. Let V be a smooth connected curve over Ω, with smooth proper compactification $\overline{V}$ and reduced boundary D. Let a cyclic group generated by an automorphism a of order $m > 1$, with $(m, p) = 1$, act freely on V. For $R \in V(\Omega)$ and an integer n coprime to m, the class $[aR - R] \in J_D(\Omega)$ is not killed by n. Consequently, if $f : V \to U$ is a connected étale Galois cover over Ω with group of order prime to p, and $P' \in V(\Omega)$ is fixed, then above any given $Q \in U(\Omega)$ there is at most one Q' satisfying $[Q' - P'] \in J_D(\Omega)[p^\infty]$.*

Proof. We use the induced automorphism σ on the function field and divisors, choosing its orientation so that $\sigma R = aR$; replacing a by a^{-1} has no effect on the assertion. Suppose $n[aR - R] = 0$. The divisor description of the generalized Jacobian gives $f \in \Omega(\overline{V})^\times$ such that

$$\operatorname{div}(f) = n(\sigma R - R), \qquad f(d) = 1 \quad (d \in D).$$

This description is the one in Milne (2008, III, §8): principal divisors are divided out only when the function is congruent to 1 along the modulus. The second requirement is absent if D is empty.

The divisor of $\prod_{i=0}^{m-1} \sigma^i(f)$ is zero, so this product is a nonzero constant. If $D \neq \varnothing$, evaluation at any boundary point shows that it is 1. If $D = \varnothing$, multiply f by a constant whose m-th power is its inverse. This leaves its divisor unchanged and makes its norm equal to 1. Multiplicative Hilbert 90 for $\Omega(\overline{V})/\Omega(\overline{V})^{\langle\sigma\rangle}$ then supplies g with $f = \sigma(g)/g$. For clarity, this elementary form follows by applying the operator $h \mapsto \sum_{i=0}^{m-1}(f\sigma(f)\cdots\sigma^{i-1}(f))\sigma^i(h)$ to an h for which it is nonzero; independence of distinct field automorphisms gives such an h, and shifting the sum gives the desired ratio, possibly after taking an inverse. It follows that

$$E := \operatorname{div}(g) - nR \quad \text{satisfies} \quad \sigma E = E, \qquad \deg E = -n.$$

Every interior orbit has length m, because the action is free on V. Its contribution to the degree of the invariant divisor E is therefore divisible by m. Consider a boundary orbit, let d be one of its points, and let its stabilizer have order e. Its length is m/e, and a generator of its stabilizer is $\sigma^{m/e}$. Since $(e, p) = 1$, the action on a uniformizer z at d has leading term ζz, with ζ of exact order e. Indeed the finite tame stabilizer of a point on a smooth curve acts faithfully on its tangent line; the kernel is a p-group. One can also obtain a parameter with this linear action by averaging a parameter against the tangent character.

Write $g = z^v u$, with u a local unit and $v = \operatorname{ord}_d(g)$. The value at d of $\sigma^{m/e}(g)/g$ is ζ^v. On the other hand, telescoping $f = \sigma(g)/g$ expresses this ratio as a product of m/e translates of f, all having value 1 at d. Thus $e \mid v$. The coefficient of E at d is v, since R lies in V. The whole orbit contributes $(m/e)v$, again divisible by m. Every orbit contribution is divisible by m, whence $m \mid n$. This contradicts $(m, n) = 1$ and $m > 1$.

For the last assertion, two putative lifts differ by some nonidentity deck transformation a. Their difference in J_D is p-primary, and the order of a is prime to p. The first part, with n a sufficiently large power of p, rules this out. This argument uses the entire cyclic subgroup generated by a; it does not assume that the full deck group is abelian. □

A.3 (Proved; both covering conditions in A2 are equivalent to section equality). *Let $K/\mathbb{F}_p$ be finitely generated, let Y/K be a hyperbolic curve, and let $P, Q \in Y(K)$. The following conditions are equivalent.*

(i) The continuous sections $s_P, s_Q : G_K \to E_Y^{(p')}$ are conjugate by $\Pi_Y^{(p')}$.

(ii) $[Q - P]$ is p-primary in $J_Y(K)$, and the covering condition of A2 holds for every finite extension L/K, every finite étale Galois cover $Y' \to Y_L$ of degree prime to p, and every L-rational lift P' of P.

(iii) The same condition holds for all finite étale geometrically connected covers whose geometric Galois closure has degree prime to p, without requiring the cover to be Galois over L.

In (ii), "Galois" may be interpreted in the stricter arithmetic sense: all deck transformations are defined over L. In the reverse implication it is enough to quantify over finite separable L/K. This answers the comparison requested at cote 149, manuscript p. 17, scans 35–36.

Proof. Suppose (i). Restriction along $G_L \to G_K$ preserves equality of section classes. For a cover in (iii), the group attached to it is an open subgroup $H \subset E_{Y_L}^{(p')}$, and its geometric group is $H \cap \Pi_Y^{(p')}$. The latter really is the full geometric prime-to-p group of Y': taking the core of a prime-to-p cover under the finitely many conjugates in a prime-to-p Galois closure produces a group which is an extension of a subgroup of a product of prime-to-p groups by a prime-to-p group. Its order is still prime to p. This proves cofinality of the induced topology.

The chosen P' makes a corresponding representative of $s_P(G_L)$ lie in H. Choose the representative of s_Q which equals it. In the finite geometric fibre of Y' over Q this gives a G_L-fixed point, hence $Q' \in Y'(L)$. In the arithmetic extension for Y', the two lifted section classes are equal. A.1 now gives $[Q' - P'] \in J_{Y'}(L)[p^\infty]$. Applying A.1 also to Y proves the initial difference condition. This proves (i)⇒(iii), and (iii)⇒(ii) is immediate.

For (ii)⇒(i), put $\Pi = \Pi_Y^{(p')}$ and use the actual point P as geometric base point. The extension is $E_Y^{(p')} = \Pi \rtimes s_P(G_K)$. Choose a cofinal system of open characteristic subgroups $N \subset \Pi$. Such a system exists since the geometric prime-to-p fundamental group of a curve is topologically finitely generated: SGA 1, XIII, Corollary 2.12 gives its surface presentation with $2g + \nu$ generators and the usual single relation (Grothendieck and Raynaud, 2003). The intersections of all subgroups of index at most d give one choice. Write $A_N = \Pi/N$ and let

$$H_N = N\, s_P(G_K) \subset E_Y^{(p')}, \qquad f_N : Y_N \longrightarrow Y$$

be the associated connected finite étale cover. The subgroup is open; its projection onto G_K is surjective; and $[E_Y^{(p')} : H_N] = [\Pi : N]$ is prime to p. The cover is geometrically connected and geometrically Galois with group A_N. The coset H_N supplies a distinguished $P_N \in Y_N(K)$ over P. All these data, including the maps for $N' \subset N$, are defined over K.

The conjugation action of $s_P(G_K)$ on the finite group A_N is continuous. Choose a finite Galois extension L/K whose Galois subgroup kills it. Over L, the subgroup $N\, s_P(G_L)$ is normal in $\Pi \rtimes s_P(G_L)$, with quotient A_N. Thus $(Y_N)_L \to Y_L$ is an arithmetic Galois cover of degree $|A_N|$, and (ii) supplies $Q_N \in Y_N(L)$ with

$$f_N(Q_N) = Q, \qquad [Q_N - P_N] \in J_{Y_N}(L)[p^\infty].$$

By A.2, this is the unique geometric point above Q having the displayed torsion property. Every element of G_K preserves the fibre, fixes P_N, and preserves p-primary torsion. Therefore it fixes Q_N. Since Q_N is defined over a finite separable extension, this says exactly that $Q_N \in Y_N(K)$.

If $N' \subset N$, the finite map $Y_{N'} \to Y_N$ induces the norm homomorphism of reduced-modulus generalized Jacobians. It sends $[Q_{N'} - P_{N'}]$ to the difference of the images; the condition on the reduced modulus is preserved since the norm of a function taking the value 1 at every boundary point again takes value 1 at the boundary. Hence the image of $Q_{N'}$ has p-primary difference from P_N, and uniqueness in A.2 makes it equal to Q_N.

We have obtained a compatible K-rational point over Q in every P-pointed characteristic neighbourhood. The inverse limit of these fibres is the prime-to-p étale path torsor from P to Q. The compatible points form a G_K-fixed path. Equivalently, after choosing an arbitrary reference path, its coordinate $\gamma \in \Pi$ satisfies $s_Q(g) = \gamma s_P(g)\gamma^{-1}$ for every $g \in G_K$. This is equality of the full section classes, not just of their finite images. Notice that the initial condition on $[Q - P]$ was not needed in this direction: it follows from (i). □

A.4 (Proved; the neighbourhood form of A1). *For the same Y, K, P, Q, equality of the two section classes is equivalent to the following condition: every finite étale geometrically connected cover with geometrically prime-to-p Galois closure which has a K-point over P has a K-point over Q. Consequently nonconjugacy is witnessed by one such cover having a rational point above P and none above Q.*

Proof. One direction is the fixed-fibre argument in A.3. For the reverse direction use its neighbourhoods Y_N. The finite nonempty sets $f_N^{-1}(Q)(K)$ need not have surjective transition maps, but they have a nonempty inverse limit: a finite family of compatibility conditions is realized by a point at a common refining level. Compactness of their product therefore gives a compatible family. The path-torsor argument at the end of A.3 supplies the geometric conjugator. Negating the equivalence gives the last assertion. □

1.2 The finite joint norm kernel

A.5 (Proved; exact finite norm kernels, refuting the proposed identification). *Let C be a smooth proper genus-two curve over an algebraically closed field Ω of characteristic different from 2. Choose $P \in C(\Omega)$, let $j(R) = [R - P]$, and put*

$$Z = C \times_{j, J_C, [2]} J_C, \qquad \pi : Z \longrightarrow C, \qquad G = J_C[2](\Omega) \simeq \mathbb{F}_2^4.$$

For $r = 1, 2, 3$, let $\mathcal{H}_r$ be all subgroups $H \subset G$ of codimension r, put $r_H : Z \to Z/H$, and define the finite group scheme

$$K^{[r]} = \bigcap_{H \in \mathcal{H}_r} \ker\big((r_H)_* : J_Z \longrightarrow J_{Z/H}\big).$$

As abstract groups of geometric points, noncanonically,

$$K^{[1]} \simeq (\mathbb{Z}/2)^{16} \oplus (\mathbb{Z}/4)^9,$$
$$K^{[2]} \simeq (\mathbb{Z}/2)^9,$$
$$K^{[3]} = 0.$$

In particular K_2 of R4(e) is $K^{[1]}$, of order 2^{34} and exponent 4. The suggestion $K_2 = \pi^* J_C[8]$ is false: the latter has order 2^8 and is isomorphic to $(\mathbb{Z}/4)^4$. These are geometric group identifications; no triviality of their Galois actions is asserted over a nonclosed field.

Proof. The Abel map induces an isomorphism on abelian geometric fundamental groups (Milne, 2008, III, Proposition 9.1). Thus Z is the connected mod-2 homology cover of degree 16; Riemann–Hurwitz gives $g(Z) = 17$. If H is a hyperplane, write $N_H = \sum_{h \in H} h_*$ on J_Z. Counting the hyperplanes through a vector of $\mathbb{F}_2^4$ gives

$$\sum_H N_H = 8\,\mathrm{id} + 7N_G.$$

Since $N_H = r_H^*(r_H)_*$ and $N_G = \pi^*\pi_*$, and since each π_* factors through $(r_H)_*$, the identity shows $K^{[1]} \subset J_Z[8]$. Every $H \in \mathcal{H}_2$ or $\mathcal{H}_3$ is contained in hyperplanes, so its norm factors the appropriate hyperplane norms. Hence $K^{[3]} \subset K^{[2]} \subset K^{[1]}$. All three groups are finite étale 2-primary groups. It therefore suffices to determine them on the 2-adic lattices.

Here is the exact lattice whose calculation is supplied, including the identification with the algebraic kernel. The homomorphism

$$\Phi_r : J_Z \longrightarrow \prod_{H \in \mathcal{H}_r} J_{Z/H}$$

has kernel $K^{[r]}$. Its dual, under the canonical principal polarizations, is the sum of the pullback homomorphisms r_H^*. Factor Φ_r as an isogeny onto its image followed by inclusion of an abelian subvariety. Dualizing the latter inclusion has connected kernel and is surjective on 2-adic Tate modules. Dualizing the isogeny gives the Cartier dual of its finite kernel (Milne, 2008, I, Theorem 9.1). Consequently the character group of $K^{[r]}$ has the same invariant factors as

$$\frac{T_2 J_Z}{\sum_{H\in\mathcal{H}_r} r_H^* T_2 J_{Z/H}}. \tag{A.1}$$

This use of duality retains the finite integral cokernel; replacing Tate modules by rational vector spaces would lose exactly the group being computed.

In the complex surface model $T_2 J$ is the 2-adic completion of H_1 of the curve. The map induced by r_H^* is the transfer $r_H^! : H_1(Z/H,\mathbb{Z}) \to H_1(Z,\mathbb{Z})$. For example this follows from its defining intersection identity $\langle r_H^! x, y\rangle_Z = \langle x, (r_H)_* y\rangle_{Z/H}$; it is the adjoint of pushforward, just as pullback and norm of Jacobians are adjoint. In every characteristic different from 2, SGA 1, XIII, Corollary 2.12 identifies the prime-to-characteristic geometric group with the corresponding completed surface presentation. The mod-2 homology subgroup and its intermediate subgroups are intrinsic in this identification. The group transfer on their abelianizations is defined by summing lifts of a loop and commutes with completion. Thus (A.1) is the 2-adic completion of the same integral transfer quotient in every such characteristic. This proves the required comparison, without identifying a characteristic-p full fundamental group with a characteristic-zero one.

We specify the integral calculation. Take a genus-two surface with one vertex, oriented edges a, b, c, d, and one face attached along $aba^{-1}b^{-1}cdc^{-1}d^{-1}$. In its mod-2 homology cover, vertices are $v \in \mathbb{F}_2^4$, an oriented edge (v, i) goes from v to $v + e_i$, and there is one lifted face at each v. There are 16 vertices, 64 edges and 16 faces. A spanning tree has 15 edges; the remaining 49 edges give a basis of the integral cycle lattice. In this basis the lifted face boundaries have rank 15 and primitive image, so $H_1(Z,\mathbb{Z})$ has rank 34.

For a quotient $G \to G/H$, the chain transfer sends the edge $(\bar{v}, i)$ to $\sum_{v\mapsto\bar{v}}(v, i)$. Its boundary is the transfer of the boundary, and hence it sends cycles to cycles. A spanning tree in each quotient graph supplies all its graph cycles; quotient face boundaries already land in the face-boundary lattice of Z. Accordingly it is enough to adjoin these transferred graph cycles to the 16 lifted face columns. Let M_r be the resulting integer matrix with 49 rows. Its cokernel is precisely

$$H_1(Z,\mathbb{Z}) / \sum_{H\in\mathcal{H}_r} r_H^! H_1(Z/H,\mathbb{Z}).$$

The matrix dimensions and lattice invariants are

r	$\lvert\mathcal{H}_r\rvert$	#columns	$\lvert\operatorname{coker} M_r\rvert$	$\operatorname{coker} M_r$
1	15	121	2^{34}	$(\mathbb{Z}/2)^{16} \oplus (\mathbb{Z}/4)^9$
2	35	471	2^9	$(\mathbb{Z}/2)^9$
3	15	391	1	0

(A.2)

The supplied certificates verify these finite integer assertions as follows. For each M_r they contain integer matrices X_r, Y_r, H_r with H_r square of order 49 and

$$M_r X_r = H_r, \qquad H_r Y_r = M_r. \tag{A.3}$$

These two identities prove equality of column lattices, independently of any normal-form label. The certificates give $\lvert\det H_1\rvert = 2^{34}$, $4H_1^{-1}$ integral, and $\operatorname{rank}_{\mathbb{F}_2}(M_1) = 24$. Thus its cokernel has exponent dividing 4, order 2^{34}, and $49 - 24 = 25$ generators modulo 2. If it is $(\mathbb{Z}/2)^u \oplus (\mathbb{Z}/4)^v$, these give $u + v = 25$ and $u + 2v = 34$, whence $(u, v) = (16, 9)$. Similarly

$|\det H_2| = 2^9$ and $2H_2^{-1}$ is integral, yielding $(\mathbb{Z}/2)^9$; and $|\det H_3| = 1$ yields zero. All arithmetic is exact. The first matrix is also independently reconstructed using a different spanning tree by the intermediate-quotient program, and gives the same Smith factors. The files `code/A_norm_kernel_certificate.py` and `code/A_intermediate_certificate.py` produce and verify (A.3); their JSON matrices are included. The chain maps and cycles are generated by the explicitly stated cell structure, not inferred from the expected output.

Finally $\ker(\pi^* : J_C \to J_Z) = J_C[2]$. A line bundle trivialized on the connected proper Galois cover descends from a character of G, because its descent multipliers are global units on Z, hence constants. This identifies the kernel with the 16 characters of G, all of order two, and hence with $J_C[2]$. Therefore $\pi^* J_C[8] \simeq J_C[8]/J_C[2] \simeq (\mathbb{Z}/4)^4$. For each hyperplane H, norm followed by pullback gives $(r_H)_*\pi^* a = 8 f_H^* a = 0$ for $a \in J_C[8]$. This verifies the previously known inclusion and proves that it is strict by (A.2). □

The explicit curve $y^2 = x^5 - x$ over $\mathbb{F}_3$, with $P = \infty$, already supplies a counterexample to the proposed equality of kernels: the polynomial and its derivative are coprime modulo 3, so this is a smooth genus-two curve, and the orders in A.5 are 2^{34} and 2^8. This finite check is included in `code/A_finite_route_checks.py`.

A.6 (Proved; what the finite norm tests establish). *In the situation of A.5 in odd characteristic p, assume $c = [Q - P] \in J_C(\Omega)$ is p-primary. There is a unique p-primary b with $2b = c$; put $\widetilde{P} = (P, 0)$, $\widetilde{Q} = (Q, b)$, and $\delta = [\widetilde{Q} - \widetilde{P}] \in J_Z(\Omega)$. For every intermediate quotient Z/H, a lift of Q having p-primary difference from the image of $\widetilde{P}$ is necessarily the image of $\widetilde{Q}$. If all degree-two quotient tests succeed, the prime-to-p part of δ lies in $K^{[1]}$; all degree-four quotient tests put it in $K^{[2]}$; and all degree-eight quotient tests force δ itself to be p-primary. None of these conclusions asserts that this last p-primary class is zero.*

Proof. Factor $[2] : J_C \to J_C$ through the quotient abelian variety J_C/H. The isogeny $J_C/H \to J_C$ has degree a power of two, so it has a unique p-primary preimage of c. A successful point lift in $Z/H = C \times_{J_C} (J_C/H)$ maps to that preimage under the second projection, since this projection sends the chosen point above P to zero and induces a homomorphism of Jacobians. The pair of its two coordinates therefore determines that point as the image of $\widetilde{Q}$. All quotient tests consequently apply to the same class δ.

In each of the three cases, choose a power p^a killing the finitely many norm images of δ. Then $p^a\delta \in K^{[r]}$. In particular δ is torsion. Its prime-to-p component belongs to $K^{[r]}$, since multiplication by p^a is an automorphism of that 2-primary kernel. Apply A.5. The surviving p-primary component maps to p-primary components under every norm automatically; no norm identity alone forces that component to vanish. □

A.7 (Proved; general incidence identity and the two-prime comparison). *Let $V \to C$ be a connected finite étale Galois cover of smooth proper curves over an algebraically closed field of characteristic p, with group $G = \mathbb{F}_\ell^d$, where $\ell \neq p$ and $d \geq 2$. For the norm maps to all hyperplane quotients, their common kernel on J_V is killed by ℓ^{d-1}. If a cover has deck group $\mathbb{F}_\ell^d \times \mathbb{F}_m^e$ for distinct primes $\ell, m \neq p$, the intersection of the two common norm kernels, using the hyperplanes in each factor over the corresponding complementary base, is zero. Thus if every norm of a point-divisor difference in both these families is p-primary, that difference is p-primary. This does not imply that the difference vanishes.*

Proof. For $N_H = \sum_{h \in H} h$ in $\mathbb{Z}[G]$, a nonzero vector is in $c = (\ell^{d-1} - 1)/(\ell - 1)$ hyperplanes, and zero is in $c + \ell^{d-1}$ hyperplanes. Hence

$$\sum_H N_H = \ell^{d-1}\, 1 + c N_G.$$

As in A.5, applying this identity to a common norm-kernel point gives $\ell^{d-1}x = 0$. On a product-group cover, apply the first assertion to the cover over $V/\mathbb{F}_\ell^d$ and separately over $V/\mathbb{F}_m^e$. A point in both kernels is killed by two relatively prime integers and is zero. If all norm images of δ are p-primary, choose one power p^a killing all of them. Then $p^a\delta$ lies in the zero intersection, as required. These are comparisons on one common cover. Tests on the separate covers of the original curve alone do not supply the intermediate norm tests on their common refinement. □

A.8 (Proved; exact reduction and some decided cases). *The universal assertion A1 over finite fields is equivalent to injectivity of the point-section map for every hyperbolic curve over every field finitely generated over $\mathbb{F}_p$. It is also equivalent to the universal final implication of A2. Independently of A1, injectivity holds for curves of genus zero and for affine curves of genus one, and for a proper curve C/k over a finite field whenever $J_C(k)[p^\infty] = 0$. In particular it holds over every finite constant extension when C has p-rank zero.*

Proof. In genus zero the generalized Jacobian is a torus; it has no nontrivial geometric p-primary torsion, and its Abel map is injective for a hyperbolic curve. Explicitly, after splitting the punctures, a cross-ratio coordinate gives an injective coordinate in that map. A.1 proves injectivity.

For genus one, use P as the origin of the compactification E. If a collision exists, A.1 gives a nonzero class $Q - P$ of order p^a. Choose m prime to p with $m^2 > p^a$ and use $E \setminus [m]^{-1}D \to E \setminus D$. After a finite extension its deck group and all boundary points are rational. Equality of sections lifts to a pair $0, Q'$ as in A.3. The generalized-Jacobian difference has p-power order; its projection to E has order exactly p^a, since multiplication by m is invertible on p-primary torsion. The toric kernel has no such torsion, so the generalized difference also has order p^a. There is consequently a function with divisor $p^a(Q' - 0)$ which is 1 at every point of $[m]^{-1}D$. Its degree is p^a, but the function minus 1 has at least $m^2 > p^a$ distinct zeros and no poles there. This is impossible. This verifies the step of R3 used in the reduction, for every p, including 2 and 3.

For genus at least two, the open immersion into the smooth compactification induces a continuous quotient map of arithmetic prime-to-p extensions over the identity of G_K. A geometric conjugator maps to a geometric conjugator. Thus an affine collision would be a proper collision.

We spell out the remaining specialization direction. Spread a smooth proper pair and the distinct rational sections P, Q over a smooth connected finite-type scheme $S/\mathbb{F}_q$ with function field K. Shrink S so that the boundary is finite étale and the sections are disjoint. For a closed point s put $R = \mathcal{O}^{\mathrm{h}}_{S,s}$, $L = \mathrm{Frac}(R)$, and $k = \kappa(s)$. The field k is finite and $G_L \twoheadrightarrow \pi_1(R) = G_k$ is surjective: a connected finite étale cover of the normal integral henselian local scheme remains connected over its fraction field. SGA 1, XIII, Proposition 4.3 supplies the geometric prime-to-p exact sequence for the family; Examples 4.4 verify its hypotheses for our smooth proper compactification, relative normal-crossing boundary and section. Its 4.5.3 identifies the relative geometric group after base change. In particular, with base points chosen along P, there is a continuous specialization homomorphism

$$\mathrm{sp} : E^{(p')}_{Y_L} \longrightarrow E^{(p')}_{Y_s}$$

whose restriction to the geometric kernels is an isomorphism, whose arithmetic quotient is $G_L \twoheadrightarrow G_k$, and which satisfies

$$\mathrm{sp}\, s_{P_L} = s_{P_s}\, \mathrm{pr}, \qquad \mathrm{sp}\, s_{Q_L} = s_{Q_s}\, \mathrm{pr}\,.$$

A geometric conjugator over L therefore specializes to a geometric conjugator over k, since $G_L \to G_k$ is surjective. The points remain distinct by the chosen disjointness. A collision over K thus produces a collision over a finite field of the same type. The converse universal implication holds because finite fields are finitely generated over their prime fields. Combining this with the genus-zero and genus-one cases gives the first equivalence, and A.3 gives the second.

Finally A.1 puts a collision difference in $J_C(k)[p^\infty]$. If that group is zero, the Abel embedding forces $P = Q$. If the p-rank is zero, the geometric p-primary torsion point group is zero, so the same argument works over every finite extension. This statement concerns geometric points of the torsion scheme, not its possibly nonreduced local part. □

1.3 An explicit characteristic-two separation

A.9 (Proved; a degree-27 cover in characteristic two). *Let $C/\mathbb{F}_2$ be the smooth proper curve with affine equation*

$$Y^2 + (x^2 + x)Y = x^5 + x^3 + x^2 + x.$$

It has genus two and exactly three $\mathbb{F}_2$-points $P_0 = (0, 0)$, $P_1 = (1, 0)$ and P_∞. Their pairwise differences in J_C have exact order two. Nevertheless their section classes in the geometrically prime-to-2 arithmetic fundamental group remain pairwise distinct after every finite constant extension.

More explicitly, over $k = \mathbb{F}_{16}$ there is a connected finite étale cover of degree 27, with geometric Galois closure of odd degree, having a k-point above P_∞ and none above P_0. Thus this example satisfies the requested whole-group separation; it is not evidence of conjugacy inferred from finite quotients.

Proof. Away from $x = 0, 1, \infty$ set $y = Y/(x^2 + x)$. The equation becomes

$$y^2 + y = x + \frac{1}{x} + \frac{1}{x + 1}.$$

This Artin–Schreier double cover of the projective line has three simple poles. At each pole the different exponent is two, giving $2g(C) - 2 = -4 + 6 = 2$ and hence genus two. Each pole has one rational point above it; there are no other $\mathbb{F}_2$-values of x. The divisors of x, $x + 1$ and $x/(x + 1)$ give twice the three pairwise differences. None is principal, since a function with one simple pole would make C rational. This also verifies the complete failure of odd abelianized sections to distinguish these three points.

Choose $\alpha \in \mathbb{F}_4$ with $\alpha^2 + \alpha + 1 = 0$, and $\beta \in k$ with $\beta^2 + \beta = \alpha^2$. Define

$$\begin{aligned} h &= x^2 + x, \qquad f = x^5 + x^3 + x^2 + x, \\ u &= x^2 + \alpha, \\ A &= x^3 + (\beta + 1)x^2 + \beta x + 1. \end{aligned}$$

A direct expansion in $k[x]$ gives

$$A^2 + hA + f = u^3. \tag{A.4}$$

Put $F = A + Y \in k(C)^\times$. The hyperelliptic involution sends Y to $Y + h$, so $F\,\iota(F) = u^3$. The function F has a pole of order six at P_∞ and no other pole. The only root of u is $r = \alpha^2$, and $h(r) = 1$. Thus the two points above r are unramified for x; exactly one, call it R, has $F(R) = 0$. Equation (A.4) gives

$$\operatorname{div}(F) = 6R - 6P_\infty.$$

In particular the normalization D of C in $k(C)(z)$, $z^3 = F$, is étale over C: at every point the valuation of F is divisible by three, and three is invertible. It is geometrically connected. Otherwise F would be a cube over $\overline{k}(C)$, whose cube root had its only pole of order two at P_∞. Such a function is in the span of $1, x$, by the hyperelliptic degree-two pencil, whereas its cube cannot have the nonzero Y coefficient occurring in F. Since k contains the cube roots of unity, $D \to C$ is Galois of degree three. Riemann–Hurwitz now gives $g(D) = 4$.

The involution $(x, z) \mapsto (x, u/z)$ lifts the hyperelliptic involution. Its invariant function $w = z + u/z$ satisfies

$$w^3 + uw = h.$$

Consequently D has a degree-two map r_D to the plane cubic

$$E: \quad W^3 + (X^2 + \alpha Z^2)W = X^2 Z + X Z^2. \tag{A.5}$$

This cubic is geometrically smooth: the derivatives in X and Z are Z^2 and X^2, so a singular projective point would have $X = Z = 0$, and the derivative in W then forces $W = 0$. With $O = [1 : 0 : 0]$ it is an elliptic curve. The map has degree two because z satisfies $z^2 + wz + u = 0$ over $k(E)$, and the displayed involution is nontrivial.

At P_∞ the ratio F/x^3 has residue 1. The three points of D above it are rational over k; choose P' with residue $z/x = 1$. Since u/x^2 has residue 1, the ratio w/x has residue zero at this point, so $r_D(P') = O$. At P_0 one has $F = 1$, hence the three lifts have $z = 1, \alpha, \alpha^2$. Their images on E are

$$[0 : \alpha^2 : 1], \quad [0 : \alpha^2 : 1], \quad [0 : 0 : 1]. \tag{A.6}$$

They are nonzero points of exact order three in $E(k)$.

For completeness the last finite assertion is checked by an exhaustive exact program, including the group operation. In the field model $k = \mathbb{F}_2[t]/(t^4 + t + 1)$ the program takes $\alpha = t^2 + t$ and $\beta = t^2$. It finds exactly nine points on (A.5) over k. For a smooth plane cubic and an arbitrary rational origin O, the sum of two points is obtained by taking the third intersection of their chord or tangent and then the third intersection of the line joining that point to O. This rule is valid without assuming O is a flex: subtracting the two line divisors gives $P + Q \sim O + (P \oplus Q)$. The program computes all sums, verifies associativity on all 9^3 triples, and obtains one point of order one and eight points of order three. It also checks (A.4), all three values in (A.6), smoothness, and the base-curve counts $|C(\mathbb{F}_2)| = 3$, $|C(\mathbb{F}_4)| = 3$, $|C(\mathbb{F}_{16})| = 31$. The program and its output are `code/A_char2_cubic_test.py` and its JSON output.

Now take the fibre product

$$Z' = D \times_{r_D, E, [3]} E \longrightarrow D \longrightarrow C.$$

It has degree $9 \cdot 3 = 27$ and is finite étale. It is geometrically connected: norm and pullback for the degree-two map $D \to E$ compose to [2], so $T_3 J_D \to T_3 E$ is surjective; its quotient modulo three is the monodromy of this pullback of [3]. The action is therefore transitive on its nine geometric sheets. Its geometric Galois closure has odd degree: take the compositum of the three conjugates of its degree-nine Galois cover over the cyclic degree-three cover D/C. The resulting group is an extension of a subgroup of $E[3]^3$ by $\mathbb{Z}/3$. The pair (P', O) is a k-point above P_∞. Since $E(k)$ has exponent three, $[3]E(k) = \{O\}$; (A.6) shows that no point above P_0 lifts to $Z'(k)$. This proves the asserted finite separating cover.

To see persistence after arbitrary finite constant extension, suppose the two section classes became equal. Extend constants further to contain k. The chosen P' on D would then have a matching lift Q' over P_0 whose generalized-Jacobian difference is 2-primary, by A.3. Applying r_{D*} would make $[r_D(Q') - O]$ 2-primary in E, contradicting the exact order three of every point in (A.6). Thus no such extension exists. Finally $x \mapsto x + 1$ and $x \mapsto 1/x$ each change

the Artin–Schreier right side by 1 and lift over $\mathbb{F}_4$ by a translation $y \mapsto y + c$, $c^2 + c = 1$. They permute the three branch points transitively on unordered pairs. The same nonconjugacy conclusion therefore holds for each pair among P_0, P_1, P_∞. □

A.10 (Proved; a point-separating quotient criterion). *Let C/k be a smooth proper geometrically connected curve over a finite field of characteristic p. Suppose that, after a finite constant extension, there is a geometrically connected finite étale cover $D \to C$ with geometrically prime-to-p Galois closure, together with morphisms*

$$r_i : D \longrightarrow B_i \qquad (1 \le i \le t)$$

to abelian varieties satisfying $B_i(\overline{k})[p^\infty] = 0$. Suppose that the product of the r_i is injective on geometric points. Then the point-section map for C is injective after every finite constant extension.

This criterion applies to the ordinary genus-two curve in A.9. Thus its point-section map is injective over every finite extension of $\mathbb{F}_2$, for all its rational points, not only for the three points rational over $\mathbb{F}_2$.

Proof. Suppose that $P, Q \in C(L)$ give equal section classes for some finite constant extension L/k. Extend constants further so that the cover and the maps are defined and some chosen point P' of D above P is rational. By A.3 there is a point Q' above Q such that $[Q' - P'] \in J_D[p^\infty]$. Applying the homomorphism of Albanese varieties induced by each r_i gives $r_i(Q') - r_i(P') \in B_i(\overline{k})[p^\infty] = 0$. The joint injectivity of these maps implies $Q' = P'$ and hence $Q = P$. This proves the criterion without asserting that a single fixed finite cover distinguishes all pairs over all constant extensions.

We verify the criterion in A.9, retaining its notation. The elliptic curve E in (A.5) is defined over $\mathbb{F}_4$. The following complete affine point count uses $\alpha^2 + \alpha + 1 = 0$:

x	$x^2 + \alpha$	$x^2 + x$	$\{w \in \mathbb{F}_4 : w^3 + (x^2 + \alpha)w = x^2 + x\}$
0	α	0	$\{0, \alpha^2\}$
1	α^2	0	$\{0, \alpha\}$
α	1	1	$\varnothing$
α^2	0	1	$\{1, \alpha, \alpha^2\}$

There are also the two points $O = [1 : 0 : 0]$ and $T = [1 : 1 : 0]$ at infinity. Hence $|E(\mathbb{F}_4)| = 9$, and the characteristic polynomial of its $\mathbb{F}_4$-Frobenius π is $(X + 2)^2$, by the degree and point-count formula for Frobenius (Milne, 2008, II, Theorem 1.1). The action on a prime-to-two Tate module is faithful, so the characteristic-polynomial identity gives $(\pi + [2])^2 = 0$ in $\mathrm{End}(E)$ (Milne, 2008, I, Theorem 10.15). A nonzero endomorphism of an elliptic curve is an isogeny and cannot have zero square. Therefore $\pi = [-2]$. Frobenius is purely inseparable, so [2] has no nonzero geometric kernel point. Iterating this observation proves

$$E(\overline{\mathbb{F}}_2)[2^\infty] = 0. \tag{A.7}$$

Put $v = u/z$ on D, and let $d : z \mapsto \alpha z$ generate the cubic deck group. Define two morphisms to this same elliptic curve by the rational formulas

$$r_0 = (x, w_0), \quad w_0 = z + v, \qquad r_1 = (x, w_1), \quad w_1 = \alpha z + \alpha^{-1} v.$$

They extend to morphisms on the smooth proper curve D; they are r_D and $r_D \circ d$. We check that they jointly separate all geometric points, including points where these formulas have zeros or poles.

At a finite value of x with $u \neq 0$, both z and v are regular units. Equality of both image points implies equality of x, w_0, w_1. The coefficient matrix

$$\begin{pmatrix} 1 & 1 \\ \alpha & \alpha^{-1} \end{pmatrix}$$

has determinant $\alpha + \alpha^{-1} = 1$. It therefore recovers both z and v. Since $Y = z^3 - A$, these values recover the point on the normalization D at such a point.

At $x = r = \alpha^2$, one has $u = 0$ and $h = 1$. On the branch R where $F = 0$, the valuations of u, F, z, v are respectively $2, 6, 2, 0$. Thus $z = 0$, $v \in \mu_3$, and

$$(w_0, w_1) = (v, \alpha^{-1}v).$$

On the other branch, $F = 1$, so $z \in \mu_3$, $v = 0$, and

$$(w_0, w_1) = (z, \alpha z).$$

Each formula distinguishes its three points, and the two sets of pairs are disjoint because $\alpha \neq \alpha^{-1}$.

Above $x = \infty$ the three points have residues $\xi = z/x \in \{1, \alpha, \alpha^2\}$. The residue of v/x is ξ^{-1}. The image pairs (r_0, r_1) are, in this order,

$$(O, T), \qquad (T, T), \qquad (T, O),$$

which are distinct. Images of finite points have $Z \neq 0$ and cannot coincide with these. This proves the required geometric point injectivity. Combining it with (A.7) and the already proved cubic étaleness proves the second assertion. □

1.4 A limitation of the purely inseparable route

A.11 (Proved; minimal torsion functions are separable). *Let C be a smooth proper curve over an algebraically closed field Ω of characteristic p, and let $P \neq Q$ have difference of exact order p^a, with $a \geq 1$, in $J_C(\Omega)$. Every function f with divisor $p^a(Q - P)$ defines a separable morphism $C \to \mathbf{P}^1$ of degree p^a. If C has genus two and positive p-rank, such differences occur with arbitrarily large a. Thus the existence of infinitely many p-primary point differences, by itself, does not force these functions to be purely inseparable.*

Proof. If $f = g^p$, then $\operatorname{div}(g) = p^{a-1}(Q - P)$, contradicting the exact order. Thus $f \notin \Omega(C)^p$. Over the perfect field Ω, the kernel of $d : \Omega(C) \to \Omega^1_{\Omega(C)/\Omega}$ is $\Omega(C)^p$, so $df \neq 0$. The differential criterion for a function on a smooth curve says that $\Omega(C)/\Omega(f)$ is separable. Its degree is the pole degree p^a.

For genus two choose a geometric Weierstrass point B, so $K_C \sim 2B$, and let ι be the hyperelliptic involution. Riemann–Roch shows that each degree-two divisor class is effective. Hence every $c \in J_C(\Omega)$ has a presentation

$$c = [P + R - 2B] = [P - \iota R].$$

In particular the difference map $C \times C \to J_C$ is surjective on geometric points. Positive p-rank means that $J_C(\Omega)[p^\infty] \simeq (\mathbf{Q}_p/\mathbf{Z}_p)^f$ with $f > 0$; it contains points of exact order p^a for every a. Each such point has the displayed difference presentation. Over $\overline{\mathbb{F}}_p$ each pair is defined over some finite constant extension. This conclusion concerns individual Jacobian differences and does not supply compatible pairs in the prime-to-p tower. □

A.12 (Proved; necessary growth of lifted torsion orders). *Let $C/\mathbb{F}_q$ be smooth proper geometrically connected of genus $g \geq 2$, and suppose that distinct $P, Q \in C(\mathbb{F}_q)$ have equal point-section classes in the geometrically prime-to-p group. Put $J = J_C$ and $|J(\mathbb{F}_q)[p^\infty]| = p^v$. There is an integer $N \geq 1$ such that for every positive $m \equiv 1 \pmod{N}$ there is a geometrically connected finite étale cover $X_m \to C_{\mathbb{F}_{q^m}}$, with abelian geometric Galois group of order prime to p and degree*

$$d_m = \frac{|J(\mathbb{F}_{q^m})|}{p^v},$$

with the following property. For a chosen rational lift P_m, the unique lift Q_m whose difference from P_m is p-primary has exact order p^{a_m} satisfying

$$p^{a_m} \;\geq\; \frac{2d_m}{q^m+1} \;\geq\; \frac{2(q^{m/2}-1)^{2g}}{p^v(q^m+1)}. \tag{A.8}$$

In particular these orders are unbounded. A uniform bound on the orders of compatible lifted differences, if established on these covers, would contradict a collision.

Proof. Let F be the q-power Frobenius on J. Its action on the finite group $B = J(\overline{\mathbb{F}}_q)[p^{v+1}]$ has finite order; take that order for N. If $m \equiv 1 \pmod{N}$, F^m and F have the same fixed points in B. This implies

$$A_m := J(\mathbb{F}_{q^m})[p^\infty] = A := J(\mathbb{F}_q)[p^\infty].$$

Indeed $A \subset A_m$ and $|A| = p^v$. If $|A_m| > p^v$, then either A_m is killed by p^{v+1}, or an element of larger order supplies a cyclic subgroup of order p^{v+1} in $A_m[p^{v+1}]$. In either case $|A_m[p^{v+1}]| > p^v$, contrary to equality of fixed groups in B.

Over $k_m = \mathbb{F}_{q^m}$, the Lang isogeny $F^m - 1 : J \to J$ is finite étale with constant kernel $J(k_m)$: its derivative is -1, and its geometric kernel is the finite group of Frobenius-fixed points. Quotient its source by the constant subgroup A to obtain

$$\lambda_m : J/A \longrightarrow J, \qquad \ker \lambda_m = J(k_m)/A.$$

This is finite étale of degree d_m, prime to p, and

$$\lambda_m((J/A)(k_m)) = A. \tag{A.9}$$

For if a class $\bar{b}$ is rational, then $F^m b - b \in A$. Conversely, if $c \in A$, choose a geometric b with $F^m b - b = c$; the class of b modulo A is rational and maps to c. This proves (A.9).

Pull back λ_m by $j : C \to J$, $R \mapsto [R - P]$, to obtain X_m. The Abel map surjects on the prime-to-p geometric abelian fundamental group (Milne, 2008, III, Proposition 9.1 and §9); therefore this pullback of the connected isogeny is geometrically connected. Its deck group is the constant group $J(k_m)/A$. The fibre above P has a rational point given by zero and hence consists of d_m rational points. By A.1, $[Q - P] \in A$; (A.9) gives a rational point in the fibre above Q, so it too has d_m rational points. Thus

$$|X_m(k_m)| \geq 2d_m.$$

Equality of the original section classes restricts to G_{k_m} and lifts with the chosen P_m, by A.3. Hence a matching Q_m exists and is unique by A.2. Its difference is nonzero, since the Abel map of the positive-genus curve X_m is injective and P_m, Q_m lie over distinct points.

The exact order p^{a_m} supplies a function with divisor $p^{a_m}(Q_m - P_m)$. The function can be defined over k_m: the one-dimensional space of functions having that divisor descends by multiplicative Hilbert 90. It is a nonconstant map $X_m \to \mathbf{P}^1$ of degree p^{a_m}. Each rational fibre has at most p^{a_m} rational points, so $|X_m(k_m)| \leq p^{a_m}(q^m + 1)$. This proves the first inequality in (A.8). The second follows from the Frobenius point-count formula and the absolute value $q^{1/2}$ of its $2g$ eigenvalues on J (Milne, 2008, II, Theorem 1.1). The lower bound tends to infinity because $g \geq 2$. □

1.5 Hyperelliptic curves in characteristics congruent to two modulo three

A.13 (Proved; a controlled Hesse correspondence). *Let Ω be algebraically closed of characteristic different from $2, 3$. Let $g : D \to E$ be a ramified degree-two morphism of smooth proper connected curves, where E has genus one, and choose a branch point $O \in E(\Omega)$ as its origin. There is a connected finite étale cover $a : W \to D$ of degree 9 or 18, whose geometric Galois closure has order divisible only by $2, 3$, and a morphism*

$$r : W \longrightarrow E_*, \qquad E_* : v^2 = t^3 + 1,$$

such that, for any $S, T \in W(\Omega)$,

$$r(S) = r(T) \quad \Longrightarrow \quad g(a(T)) = g(a(S)) \text{ or } g(a(T)) = 2O - g(a(S)). \tag{A.10}$$

All these data descend to a finite constant extension when the original data are defined over $\overline{\mathbb{F}}_p$.

Proof. We use the geometry underlying Bogomolov and Tschinkel (2002, §4, especially Lemma 4.3), keeping track of the degrees and the implication (A.10). The Hesse normal-form theorem (Artebani and Dolgachev, 2009, Lemma 2.1, with the characteristic hypothesis in §2) puts the elliptic curve, embedded by $|3O|$, in the form

$$E : \quad X^3 + Y^3 + Z^3 + \lambda XYZ = 0, \qquad O = (1 : 0 : -1).$$

The coordinate permutations and diagonal cube-root scalings permute its nine flexes transitively, so the prescribed origin can be put at the displayed flex. The flexes are precisely $E[3]$; explicitly, for $\zeta^3 = 1$, $\zeta \neq 1$, they are the points with one zero coordinate and the ratio of the other two equal to $-1, -\zeta, -\zeta^2$.

Projection from O gives a degree-two morphism

$$\pi : E \longrightarrow \mathbf{P}^1, \qquad t = (X + Z)/Y.$$

The apparent base point at O is removed by the linear system $|3O - O| = |2O|$. More explicitly, put $X = 1$ and $Z = -1 + s$ near O. The equation has linear term $3s - \lambda Y$, so $\pi(O) = \lambda/3$. Smoothness is equivalent to $\lambda^3 \neq -27$; hence this value is distinct from the four values in (A.11). This also checks the exceptional fibre at the base flex directly. It is the quotient by $(X : Y : Z) \mapsto (Z : Y : X)$. This involution fixes O and hence is inversion for the group law with origin O. Indeed a nonidentity involution fixing the origin is a group endomorphism u with $(u - 1)(u + 1) = 0$; nonzero endomorphisms of an elliptic curve are isogenies, so $u = -1$. The eight nonzero three-torsion points have exactly the four images

$$B_* := \pi(E[3] \setminus \{O\}) = \{\infty, -1, -\zeta, -\zeta^2\}. \tag{A.11}$$

Each fibre in (A.11) consists of an opposite pair of nonzero three-torsion points. It is unramified for π, since a nonzero three-torsion point is not two-torsion.

Form $D_3 = D \times_{g,E,[3]} E$ and denote its projections by $a_3 : D_3 \to D$ and $b_3 : D_3 \to E$. It is smooth and connected: the degree-two extension $\Omega(D)/\Omega(E)$ and the degree-nine Galois extension supplied by [3] have trivial intersection. Thus a_3 is a connected étale $E[3]$-torsor of degree nine. Since g is branched at O, b_3 has ramification index two above each point of $E[3]$. Consequently the morphism $\pi b_3 : D_3 \to \mathbf{P}^1$ has even local ramification index at every point above B_*.

The smooth double cover $\pi_* : E_* \to \mathbf{P}^1$ given by $v^2 = t^3 + 1$ is branched exactly at B_*. Normalize the fibre product $D_3 \times_{\mathbf{P}^1} E_*$ and choose a connected component W. Its map to D_3 is étale: the pullback Kummer function has even valuation at every point, and two is invertible.

Its degree is one or two. The resulting $a : W \to D$ has degree nine or eighteen. The Galois closure over D is obtained by adjoining the at most nine deck conjugates of this quadratic extension over D_3; its group is an extension of a subgroup of $(\mathbb{Z}/2)^9$ by $(\mathbb{Z}/3)^2$. This proves the claimed prime control, including the case of a split quadratic pullback.

Let $r : W \to E_*$ be the other projection. If $r(S) = r(T)$, then $\pi(b_3(S)) = \pi(b_3(T))$, where the intermediate projection from W is understood. The quotient description of π gives $b_3(T) = b_3(S)$ or $-b_3(S)$. Applying [3] gives (A.10). Every map, component and chosen point used here is defined by finitely many elements of $\overline{\mathbb{F}}_p$, so a finite constant extension suffices for all descent assertions needed below. □

A.14 (Proved; collisions have equal images in a ramified elliptic quotient). *Let k be finite of characteristic $p \geq 5$ with $p \equiv 2 \pmod 3$, and let $g : D \to E$ be a ramified degree-two morphism of smooth proper geometrically connected curves over k, where E has genus one. If two point sections of D are equal in the geometrically prime-to-p group, then their images under g are equal. This holds after every finite constant extension.*

Proof. First, $E_* : v^2 = t^3 + 1$ has $p + 1$ points over $\mathbb{F}_p$. The map $t \mapsto t^3$ permutes $\mathbb{F}_p$, since $(3, p - 1) = 1$; therefore the quadratic-character sum of $t^3 + 1$ is zero. The Frobenius characteristic polynomial is $X^2 + p$, giving $F^2 = [-p]$. Thus $[p]$ is purely inseparable and

$$E_*(\overline{k})[p^\infty] = 0. \tag{A.12}$$

Here the characteristic-polynomial identity and faithfulness on a prime-to-p Tate module are the same standard facts used in A.10. This argument verifies the supersingularity needed in the proof directly.

Suppose that $P, Q \in D(k)$ have equal section classes. Finite constant extensions preserve that equality. Choose an origin on E after such an extension and form the connected étale degree-nine cover

$$D_3 = D \times_{g,E,[3]} E, \qquad g_3 : D_3 \to E.$$

Connectedness follows from the coprime degrees two and nine, as in A.13. Choose a geometric lift P_3 of P and extend constants to make it rational. The section-lifting argument of A.3 gives a lift Q_3 of Q whose full prime-to-p section class equals that of P_3.

Choose a branch point b of g. Every point of $[3]^{-1}b$ is a branch point of g_3. In particular we can choose O_0, O_1 there with $O_1 - O_0$ of exact order three, and make these points rational by a further finite extension. For each $j = 0, 1$, apply A.13 to g_3 with origin O_j. Its cover has geometrically prime-to-p Galois closure because $p \neq 2, 3$. Lift the equal section classes with a chosen lift of P_3 to that cover. By A.1 the difference of the two images on E_* is p-primary, and (A.12) makes the images equal. Formula (A.10) therefore says, writing $b_P = g_3(P_3)$ and $b_Q = g_3(Q_3)$,

$$b_Q = b_P \quad \text{or} \quad b_Q = 2O_j - b_P \qquad (j = 0, 1).$$

If $b_Q \neq b_P$, these two equations give $2(O_1 - O_0) = 0$, contrary to its exact order three. Hence $b_Q = b_P$, and applying [3] gives $g(Q) = g(P)$. All comparisons took place after finite extensions along actual Galois subgroups; thus the argument applies over every finite constant extension of the original field. □

A.15 (Proved; A1 for all hyperelliptic curves when $p \equiv 2 \pmod 3$, $p \geq 5$). *Let k be finite of characteristic $p \geq 5$, $p \equiv 2 \pmod 3$. For every smooth proper geometrically connected hyperelliptic curve C/k of genus at least two, the map from rational points to geometric conjugacy classes of sections of $E_C^{(p')} \to G_k$ is injective, after every finite constant extension. In particular this proves A1 for every genus-two curve in these characteristics.*

By the specialization comparison in A.8, the same conclusion holds for geometrically hyperelliptic proper curves and their hyperbolic open subcurves over fields finitely generated over $\mathbb{F}_p$ in these characteristics.

Proof. We first establish a consequence that applies in every hyperelliptic genus in the stated characteristics. If two points of a hyperelliptic curve H have equal sections, then they have the same image under its hyperelliptic map $x_H : H \to \mathbf{P}^1$. To prove this, extend the finite field so that the branch points and a coordinate on the quotient are defined. Choose four of the $2g(H) + 2$ branch points, and let $E \to \mathbf{P}^1$ be the double cover branched at those four points. The normalization D of $H \times_{\mathbf{P}^1} E$ is a connected étale double cover of H. On function fields the two quadratic characters have distinct branch sets, which proves connectedness, and all pullback valuations are even, which proves étaleness. Its map $D \to E$ has degree two and is ramified above the remaining branch points of H; this set is nonempty because $g(H) \geq 2$. Lift the two equal sections to D with a chosen rational lift of the first point, using A.3 and a finite extension for that lift. A.14 says that the resulting points have equal images on E. Their images under x_H are therefore equal, proving the consequence.

We now prove injectivity by induction on $g = g(C) \geq 2$. After a finite extension split the branch set of the hyperelliptic map into disjoint subsets

$$S = S_2 \sqcup S_{2g}, \qquad |S_2| = 2, \quad |S_{2g}| = 2g.$$

Let $B \to \mathbf{P}^1$ and $H \to \mathbf{P}^1$ be the double covers with branch sets S_2 and S_{2g}, respectively. Their genera are zero and $g - 1$. Since their branch sets are disjoint, the fibre product

$$V = B \times_{\mathbf{P}^1} H$$

is smooth and connected: at a branch point of either factor, the other factor is étale; the quadratic function-field extensions are independent. The diagonal involution of this fibre product acts freely, and its quotient is the double cover branched at S, hence is C after the chosen constant extension. Thus $V \to C$ is a connected étale double cover.

The map $V \to B \simeq \mathbf{P}^1$ is a hyperelliptic double cover with $4g$ branch points, so $g(V) = 2g - 1$. If distinct points of C had equal sections, choose their lifted equal sections at points P_V, Q_V of V, as above. The consequence proved in the first paragraph, applied to the hyperelliptic curve V, gives equal images of these points on B. If $g = 2$, the curve H is elliptic and $V \to H$ is a ramified double cover, with ramification above the four points lying over S_2; A.14 gives equal images on H. If $g > 2$, functoriality gives equal point-section classes on the hyperelliptic curve H of genus $g - 1$, and the induction hypothesis gives equal images on H. The two projections $V \to B, H$ jointly distinguish every geometric point, since V is their smooth fibre product. Therefore $P_V = Q_V$, contradicting their distinct images on C. This proves the induction.

For the final assertion, spread the hyperelliptic double cover and its distinct branch divisor along with the two sections. After shrinking the base, the same smooth hyperelliptic type persists on the closed finite-field fibre. The specialization map of A.8 carries a putative collision to a collision on that fibre, which we have just excluded. An open-curve collision maps to one on its proper compactification. These are the required directions of the comparison. □

A.16 (Proved; a degree-16 or degree-32 correspondence). *In A.13 one may replace the target by $E_\dagger : v^2 = t^3 - t$, the characteristic assumption by* $\operatorname{char}\Omega \neq 2$*, and the cover degree by* 16 *or* 32*. The geometric Galois closure then has* 2*-power order. Consequently A.14 and A.15 also hold in every characteristic* $p \equiv 3 \pmod 4$*, including* $p = 3$*.*

Proof. Put the elliptic curve with prescribed branch origin in Legendre form $E : y^2 = x(x-1)(x-\lambda)$, where $\lambda \neq 0, 1$, and write $\pi = x$. Choose $a, b \in \Omega$ with $a^2 = \lambda$, $b^2 = 1 - \lambda$. The duplication identities are

$$x([2]T) = \frac{(x(T)^2 - \lambda)^2}{4x(T)(x(T)-1)(x(T)-\lambda)},$$
$$x([2]T) - 1 = \frac{(x(T)^2 - 2x(T) + \lambda)^2}{4x(T)(x(T)-1)(x(T)-\lambda)}.$$

They follow by substituting the tangent slope in the Weierstrass addition law. Thus the four values

$$B_\dagger = \{a, -a, 1+b, 1-b\}$$

are the images of four opposite pairs of exact four-torsion points: halves of $(0,0)$ and $(1,0)$. These values are distinct. Indeed $a, b \neq 0$, and an equality $\pm a = 1 \pm b$ would imply $a = 0$ or $a = \pm 1$ after squaring, contrary to $\lambda \neq 0, 1$. They are unramified values of π. Their ordered cross-ratio is

$$\frac{(a-1-b)(-a-1+b)}{(a-1+b)(-a-1-b)} = \frac{2ab}{-2ab} = -1,$$

using $a^2 + b^2 = 1$. A projective transformation therefore sends these branch values to $\{\infty, 0, 1, -1\}$. The double cover branched at $B_\dagger$ is geometrically isomorphic to $E_\dagger$.

Pull $g : D \to E$ back by $[4]$. The resulting D_4 is connected: the ramified quadratic extension $\Omega(D)/\Omega(E)$ cannot be contained in the unramified extension defined by $[4]$; since it is quadratic, their intersection is trivial. Its map to D is an étale $E[4]$-torsor of degree sixteen, and its map to E is ramified above every point of $E[4]$. Pulling back the double cover branched at $B_\dagger$ and normalizing therefore gives an étale cover of D_4 of degree one or two. Its Galois closure over D is an extension of a subgroup of $(\mathbb{Z}/2)^{16}$ by $E[4]$, hence a 2-group. The argument using the quotient by inversion gives exactly the reflection implication (A.10).

If $p \equiv 3 \pmod 4$, the quadratic-character terms for t and $-t$ in the count of $E_\dagger(\mathbb{F}_p)$ cancel. Hence the count is $p+1$, its Frobenius satisfies $F^2 = [-p]$, and $E_\dagger(\overline{\mathbb{F}}_p)[p^\infty] = 0$. To repeat A.14, first pull back the ramified double cover by $[4]$, choose two branch origins whose difference has exact order four, and apply the correspondence with both origins. The two reflection alternatives would imply twice this difference is zero, which is false. This proves the elliptic-image conclusion. The induction of A.15 only uses characteristic different from two, so it then proves the hyperelliptic conclusion in the asserted additional characteristics. The exact duplication and cross-ratio identities are independently checked in `code/A_hesse_correspondence_checks.py`. □

1.6 Removing the congruence restriction in odd characteristic

A.17 (Proved; a prime-to-p torsion cross-ratio configuration). *Let* $\Omega = \overline{\mathbb{F}}_p$ *with* $p \neq 2$*, let* E/Ω *be an elliptic curve with origin* O*, and let* $\pi : E \to \mathbf{P}^1$ *be its quotient by inversion. For every* $c \in \Omega \setminus \{0, 1\}$ *there exist points* $T_1, T_2, T_3, T_4 \in E(\Omega)$ *of order prime to* p *such that none is two-torsion, their images under* π *are distinct, and the ordered cross-ratio of those images is* c*.*

Proof. We state the torsion-density theorem being used. If A is an abelian variety over $\overline{\mathbb{F}}_p$ and $Z \subset A$ is an irreducible reduced closed subscheme not contained in a translate of a proper abelian subvariety, then the points of Z of order prime to p are Zariski dense in Z. This is the contrapositive of the Anderson–Indik consequence stated in Yang (2016, Proposition 2.3, p. 5), with its reference to Tamagawa (2004, p. 704). We have opened the former text; the latter is an indirect reference whose original PDF could not be retrieved. The underlying surjectivity of the primary projection for generating curves is also stated and proved in Bogomolov and Tschinkel (2005, Theorem 1.1). We now verify all geometric hypotheses of the density statement in this application.

On $(\mathbf{P}^1)^4$ use homogeneous coordinates $(X_i : Y_i)$ and put $\Delta_{ij} = X_iY_j - X_jY_i$. Let

$$H_c : \quad \Delta_{13}\Delta_{24} - c\Delta_{14}\Delta_{23} = 0.$$

On the locus of four distinct coordinates this is the orbit of $(\infty, 0, 1, c)$ under PGL_2; the unique transformation taking the first three entries to a given ordered triple determines the fourth. Consequently this open locus is integral of dimension three. Its complement in H_c is a union of triple-diagonals of dimension at most two: if one pair of coordinates agrees, the equation and $c \neq 0, 1$ force a third to agree. Every component of the displayed nonzero Cartier divisor has dimension three. It follows that H_c is integral, with line bundle $\mathcal{O}(1, 1, 1, 1)$; generic reducedness on the open locus and the absence of embedded components of a hypersurface justify the scheme-theoretic assertion.

Let $Z = (\pi^4)^{-1}(H_c) \subset E^4$. We verify irreducibility, including the boundary, rather than assuming that a finite pullback preserves it. On the open copy of PGL_2, the four double covers are defined by square classes $f(M(t_i))$, where E is written $y^2 = f(x)$, $M \in \mathrm{PGL}_2$ and $(t_1, t_2, t_3, t_4) = (\infty, 0, 1, c)$. Fix any finite branch value b of π. For each i, the divisor

$$D_{i,b} = \{M : M(t_i) = b\}$$

is irreducible: it is a coset of a Borel subgroup. Evaluation at t_i is smooth, so $f(M(t_i))$ has odd valuation there. At its generic point the other three functions have valuation zero, since their arguments avoid the finite branch set. These four valuations show that the square classes are independent. Their compositum is therefore a field of degree 16 over $\Omega(H_c)$.

The morphism π^4 is finite flat of degree sixteen. Its pullback to the integral H_c is finite flat and hence torsion-free over every affine domain in H_c. Each such coordinate algebra embeds in its generic fibre, the field just identified. Thus the entire pullback Z is integral; there are no extra components supported on the diagonals. It is an ample Cartier divisor in E^4, because its line bundle is the pullback of the ample $\mathcal{O}(1, 1, 1, 1)$ under a finite morphism.

An ample integral divisor in this four-dimensional abelian variety cannot lie in a translate $u + B$ of a proper abelian subvariety. Such a B would have dimension three, and $Z = u + B$. Its associated line bundle would be pulled back from the elliptic quotient E^4/B, hence would be trivial on the positive-dimensional fibres; it would not be ample. The torsion-density theorem applies to Z.

The locus where the four images are distinct and every T_i avoids $E[2]$ is a nonempty open subset of Z. The finite-flat description, or a generic projective transformation, proves its nonemptiness. A prime-to-p torsion point of E^4 in this open has exactly the four coordinates asserted. In particular they have a common finite exponent n with $(n, p) = 1$. □

A.18 (Proved; supersingular correspondences in every odd characteristic). *Let $\Omega = \overline{\mathbb{F}}_p$ with $p \neq 2$, and let $g : D \to E$ be a ramified degree-two morphism of smooth proper connected curves with $g(E) = 1$. For each chosen branch origin $O \in E(\Omega)$ there are a connected finite étale cover $a : W \to D$ whose geometric Galois closure has order prime to p, a supersingular elliptic curve E_0/Ω, and a morphism $r : W \to E_0$ satisfying the reflection implication (A.10). All data descend to a finite constant extension of a field of definition of g.*

Over every finite field of odd characteristic, equal geometrically prime-to-p point-section classes on D therefore have equal images under g.

Proof. A supersingular elliptic curve exists in every odd characteristic. One explicit way to choose one is to take c a root of the nonconstant polynomial

$$H_p(T) = \sum_{i=0}^{m} \binom{m}{i}^2 T^i, \qquad m = (p-1)/2,$$

and put $E_0 : v^2 = t(t-1)(t-c)$. The Legendre Hasse-invariant criterion is the one stated in Finotti (2009, §1, equations (1.2)–(1.3)). Its use here is valid for every $p > 2$, including three. The root cannot be zero because $H_p(0) = 1$, nor one because Vandermonde's identity gives $H_p(1) = \binom{p-1}{m} \equiv (-1)^m \pmod{p}$. Thus the displayed curve is smooth, and $E_0(\Omega)[p^\infty] = 0$.

Apply A.17 to (E, O) and this c. Let n be a common prime-to-p exponent of the four resulting torsion points. The set $B = \{\pi(T_i) : 1 \leq i \leq 4\}$ consists of four unramified values of π. A projective transformation identifies its ordered configuration with $(\infty, 0, 1, c)$, so the double cover of $\mathbf{P}^1$ branched at B is isomorphic over Ω to E_0.

Form $D_n = D \times_{g,E,[n]} E$. It is connected even if n is even: a ramified quadratic extension cannot be a subextension of the étale Galois cover $[n]$. Thus the quadratic extension and the $[n]$ extension are linearly disjoint. The map $D_n \to D$ is an étale $E[n]$-torsor of degree n^2, while $D_n \to E$ is ramified with index two above every point of $E[n]$. Every point of $\pi^{-1}(B)$ belongs to $E[n]$, since these fibres are the pairs $\{T_i, -T_i\}$. The normalized pullback of the double cover branched at B is consequently étale over D_n. Choose a connected component W. Its degree over D_n is one or two, and its Galois closure over D has group an extension of a subgroup of $(\mathbb{Z}/2)^{n^2}$ by $E[n]$. Its order is prime to p.

The projection $r : W \to E_0$ has the same reflection property as in A.13: equal images imply that the two intermediate points of E are equal or negatives; applying $[n]$ yields (A.10). All equations, torsion points and chosen components are defined over a finite subfield of Ω.

For the final assertion, start with a hypothetical equality of the section classes of $P, Q \in D(k)$. Choose a prime $\ell \neq 2, p$ and first pull back g by $[\ell]$. By A.3, after a finite extension, there are lifts P_ℓ, Q_ℓ with equal full prime-to-p section classes. A fibre over a branch point of g supplies two branch origins of this new double cover whose difference has exact order ℓ. Apply the first assertion with each origin separately. Their lifts have p-primary divisor difference by A.1; their images on the corresponding supersingular elliptic curves coincide. The two reflection alternatives therefore force equal intermediate images on E, as in A.14, since twice an element of exact order ℓ is nonzero. Applying $[\ell]$ gives $g(P) = g(Q)$. Restrictions to finite constant extensions and lifts of actual equal section classes are precisely the comparison maps proved in A.3. □

A.19 (Proved; A1 for every hyperelliptic curve in odd characteristic). *Let $p \neq 2$ and let K be a field finitely generated over $\mathbb{F}_p$. Let C/K be smooth proper geometrically connected, geometrically hyperelliptic, and of genus at least two. The point-section map*

$$C(K) \longrightarrow \{\textit{sections of } E_C^{(p')} \to G_K\}/\Pi_C^{(p')}$$

is injective. The same holds for every hyperbolic open subcurve of C. In particular A1 is proved for every genus-two curve in every odd characteristic, with no restriction on the finite field or on its size.

Proof. Over finite fields, repeat the proof of A.15 using the elliptic-image assertion of A.18. All covers in that induction are étale double covers or the covers of A.18, so their geometric Galois closures remain prime to p. The branch sets can be split and the chosen lifts made rational after finite constant extensions, which preserve a hypothetical section equality. The disjoint-branch fibre product is smooth, and its two projections separate all geometric points, exactly as proved there. This settles every hyperelliptic genus by the same induction. The specialization argument in the last paragraph of A.15, using A.8, gives the assertion over K; functoriality to the proper compactification gives it for open subcurves. For these curves over finite fields the exceptional set of Saïdi and Tamagawa (2009, Proposition 1.8) is therefore empty: commensurable decomposition groups become equal after restriction to an open subgroup of G_k, which would give equal point-section classes over a finite constant extension. □

A.20 (Proved; elementary abelian two-group covers of a genus-zero curve). *Let K be finitely generated over $\mathbb{F}_p$, $p \neq 2$, and let C/K be smooth proper geometrically connected of genus at least two. Suppose that $C_{\overline{K}}$ admits a faithful action of $(\mathbb{Z}/2)^r$, $r \geq 1$, with genus-zero quotient. Then the geometrically prime-to-p point-section map is injective for C and for each of its hyperbolic open subcurves. This includes nonhyperelliptic curves, for example the smooth plane quartic $X^4 + Y^4 + Z^4 = 0$ in every odd characteristic: changing coordinate signs gives a Klein-four group, with genus-zero quotient $U^2 + V^2 + W^2 = 0$.*

Proof. We first work over a finite field, extending it to define the action and all quotients. We induct on r. For $r = 1$ the curve is hyperelliptic and A.19 applies. Suppose $r \geq 2$, and that P, Q have equal point-section classes. For each nonidentity involution s, consider the degree-two quotient

$$q_s : C \longrightarrow C/\langle s \rangle.$$

If any quotient has genus zero, C is hyperelliptic and we are done. If it has genus one, the map is ramified by Riemann–Hurwitz, since $g(C) \geq 2$, and A.18 gives $q_s(P) = q_s(Q)$. If its genus is at least two, the quotient has the faithful action of $(\mathbb{Z}/2)^r/\langle s \rangle$ with the same genus-zero final quotient. Functoriality of section classes and the induction hypothesis again give $q_s(P) = q_s(Q)$.

These equalities force $P = Q$ on every fibre, including all ramified fibres. Indeed choose independent involutions s, t. If $P \neq Q$, equality in the three quotients for s, t, st would give

$$Q = sP = tP = (st)P.$$

But $sP = tP$ implies $(st)P = P$, contradicting the third equality. This proves the finite-field assertion over all constant extensions.

For the asserted finitely generated field, first take a finite extension defining the action. Spread the smooth proper curve, the faithful action and its quotient maps over a finite-type base over $\mathbb{F}_p$. After shrinking, their genera are constant and the action remains faithful; its order is invertible and the quotients are smooth curves. A hypothetical collision specializes by A.8 to one on a closed finite-field fibre in the same class, which has just been excluded. Functoriality to the compactification proves the open case. □

A.21 (Proved; a general elliptic-image test). *Let k be finite of characteristic p, with no restriction on p, and let $g : D \to E$ be a finite separable morphism of smooth proper geometrically connected curves, with $g(E) = 1$. Suppose that there is a geometric point $b \in E$ and an integer $m > 1$ prime to p such that every local ramification index over b is divisible by m. Equal geometrically prime-to-p point-section classes on D then have equal images under g, over every finite constant extension.*

Proof. Choose a prime $e \mid m$. We adapt the explicit construction of Poonen (2005, §3.1), while verifying prime-to-p Galois closure and keeping the point-image information. Take b as elliptic origin and choose a prime $\ell \geq 5$, $\ell \neq p$. A connected component D_ℓ of $D \times_{g,E,[\ell]} E$ is an étale cover of D whose geometric Galois closure has order dividing ℓ^2: the whole pullback is an $E[\ell]$-torsor and a connected component is a torsor under its stabilizer. The second projection $D_\ell \to E$ is surjective and has all local ramification indices divisible by e above every point of $E[\ell]$.

Write $x : E \to \mathbf{P}^1$ for the inversion quotient. Every nonzero ℓ-torsion point has trivial stabilizer under inversion, since ℓ is odd. Its image is therefore an unramified fibre even in characteristic two. Choose distinct values a_i arising from these pairs. When $p \neq 2$ and $e \geq 3$, use four such values and let

$$H : \quad z^e = \frac{(x - a_1)(x - a_2)}{(x - a_3)(x - a_4)}.$$

It has genus $e - 1 \geq 2$, and its map to the z-line has degree two; hence it is hyperelliptic. For $e = 2$, which necessarily has $p \neq 2$, use six values and the genus-two curve

$$H : \quad z^2 = \frac{(x - a_1)(x - a_2)(x - a_3)}{(x - a_4)(x - a_5)(x - a_6)}.$$

When $p = 2$, the prime e is odd. Use three chosen values, move them to $0, 1, \infty$, and take instead

$$H : \quad z^e = x(x - 1), \qquad \text{equivalently } x^2 + x = z^e.$$

We verify the required point injectivity for this target directly, including the elliptic case $e = 3$. Its genus is $g = (e-1)/2$ by tame Riemann–Hurwitz for the degree-e map to the x-line. At the unique point at infinity, $v_\infty(z) = -2$ and $v_\infty(x) = -e$. Since e is odd, the leading Laurent term gives $v_\infty(dx) = -e-1$. Differentiation gives $dx = z^{e-1}dz$, hence $v_\infty(dz) = e-3 = 2g-2$. The affine curve is étale over the z-line, so $z^i\,dz$, $0 \leq i \leq g-1$, form a basis of regular differentials. Cartier kills the terms with even i and sends those with odd i to $z^{(i-1)/2}\,dz$, with square-root semilinearity on constants. The operator is therefore nilpotent.

There is consequently no nonzero geometric two-torsion in J_H. Indeed such a class represented by a divisor D would give $\operatorname{div}(f) = 2D$ with f not a square; then df/f would be a nonzero regular Cartier-fixed differential. Regularity follows locally by writing $f = t^{2a}u$ with u a unit. To check the fixed identity directly in the separating variable z, write $f = a^2 + zb^2$. Then

$$\frac{df}{f} = \frac{b^2}{f}\,dz = \left(\left(\frac{ab}{f}\right)^2 + z\left(\frac{b^2}{f}\right)^2\right)dz,$$

whose Cartier image is itself. Thus $J_H(\overline{k})[2^\infty] = 0$, and A.1 and the Abel map prove point-section injectivity on H, for every finite constant extension. This also treats $e = 3$, when H is an elliptic curve.

In every case, each local valuation of the defining Kummer function on D_ℓ is divisible by e. A connected component of the normalized pullback is therefore étale over D_ℓ, of degree one or

e. The geometric Galois closure of its composite cover of D has order divisible only by ℓ, e, by adjoining its deck conjugates. Both primes are different from p.

Lift equal section classes to this cover using A.3. Their images on H are equal by A.19 when $p \neq 2$ and by the Cartier calculation just given when $p = 2$. Consequently their intermediate elliptic images have the same x-coordinate, and so differ at most by inversion. Applying $[\ell]$ gives the same reflection alternative (A.10) with origin b. No connectedness of the entire isogeny pullback was assumed in this argument.

To remove reflection, first pull back the original g by $[q]$, for a prime $q \neq 2, p$, and fix a connected component. Lift the equal sections there. Every point of $[q]^{-1}b$ is a fibre all of whose ramification indices remain divisible by m. Choose two such points whose difference has exact order q. Apply the construction with each as origin to the same pair of lifted points. The two reflection alternatives would force this difference to be killed by two. Hence the intermediate elliptic images are equal; applying $[q]$ proves the assertion for g. All finite data and the chosen lifts are made rational by finite constant extensions, which preserve section equality. □

A.22 (Proved; cyclic prime-degree covers of a genus-zero curve). *Let K be finitely generated over $\mathbb{F}_p$, with no restriction on the characteristic p. Let C/K be smooth proper geometrically connected of genus at least two. Suppose that, over $\overline{K}$, it admits a cyclic cover of prime degree $\ell \neq p$ of a genus-zero curve. Then its geometrically prime-to-p point-section map is injective. The conclusion also holds for every hyperbolic open subcurve of C.*

Proof. We first work over a finite field and allow finite constant extensions to define all covers and chosen points. The case $\ell = 2$ necessarily has $p \neq 2$ and is A.19. In the rest of the proof ℓ is odd. Every branch fibre of a cyclic ℓ-cover is totally ramified, because ℓ is prime. If there are r branch points, tame Riemann–Hurwitz gives

$$g(C) = \frac{(\ell - 1)(r - 2)}{2}. \tag{A.13}$$

For $p = 2$, the curve $H : z^\ell = x(x - 1)$ has $J_H(\overline{k})[2^\infty] = 0$, as proved by the Cartier calculation in A.21. Hence it has point-section injectivity for every odd ℓ, including the elliptic case $\ell = 3$.

We first prove a fibre statement, independently of the induction below: a collision on such a cover has equal images on its genus-zero base. If $\ell \geq 5$, select three branch points and choose a coordinate sending them to $0, 1, \infty$. The curve

$$H : \quad z^\ell = x(x - 1)$$

is a cyclic ℓ-cover branched at exactly these three points. It is hyperelliptic: the function z gives a degree-two map, and its genus is $(\ell - 1)/2 \geq 2$. If C and H define the same extension of the rational function field, injectivity follows from A.19 in odd characteristic and from the preceding Cartier calculation in characteristic two. Otherwise their extensions are linearly disjoint, since both have prime degree and are Galois. The normalization V of their fibre product is a connected cyclic étale ℓ-cover of C. Tameness and the matching index ℓ at every branch point of H prove étaleness, including over infinity. Equal sections lift to V by A.3 and have equal images on H by the just-established injectivity for that target. Their original x-coordinates are therefore equal.

For $\ell = 3$, use the same three-branch curve H, now of genus one. Formula (A.13) and $g(C) \geq 2$ imply $r \geq 4$. Thus C and H are distinct covers, and the normalized fibre product V is again a cyclic étale cubic cover of C. In characteristic two the target is the two-rank-zero elliptic curve just treated, so the lifted images on H are equal. In odd characteristic its cyclic cubic map to H is ramified over a branch point of C outside the three selected ones; all its

ramification indices there are three. A.21 gives equality of the H-images of the lifted collision, hence equality of the original genus-zero coordinates. This proves the fibre statement for every ℓ under consideration.

We now induct on the number r of branch points. The induction only asserts injectivity for source curves of genus at least two; lower-genus quotients will be treated separately. Write the given Kummer extension as

$$\overline{k}(C) = \overline{k}(x)(y), \qquad y^{\ell} = c(x).$$

Choose two branch points a_1, a_2. Let $u \not\equiv 0$ modulo ℓ be the valuation of c at a_1. Choose a rational function $b(x)$ with divisor $u(a_1 - a_2)$, choosing an integer representative for u, and define the cyclic cover

$$B: \quad v^{\ell} = b(x).$$

It has exactly two branch points and genus zero. Its function field differs from that of C, because C has at least three branch points. The normalized fibre product V of C and B is consequently connected and is a cyclic étale ℓ-cover of C.

Put $h(x) = c(x)/b(x)$ and let H_2 be the smooth curve with Kummer equation $w^{\ell} = h(x)$. The branch at a_1 cancels. The branch at a_2 either cancels or remains, and every other branch remains. Thus the number of its branch points is $r - 1$ or $r - 2$, and it is nontrivial, since at least one branch different from a_1, a_2 remains. In particular $g(H_2) < g(C)$. On function fields

$$\overline{k}(V) = \overline{k}(x)(y, v) = \overline{k}(x)(v, w), \qquad w = y/v. \tag{A.14}$$

The two Kummer characters b, h are independent; otherwise $c = bh$ would also be a power of b modulo ℓ-th powers and could have no branch outside $\{a_1, a_2\}$. Thus $V \to B$ and $V \to H_2$ are cyclic covers of degree ℓ.

Suppose that distinct $P, Q \in C(k)$ form a collision. The fibre statement gives a common base value $x(P) = x(Q)$. This value is not a branch point of C, since a branch fibre contains only one geometric point. Lift the collision to $P_V, Q_V \in V$ by A.3 after a finite constant extension. Applying the already proved fibre statement to the cyclic cover $V \to B$, whose source has genus $\ell(g(C) - 1) + 1 \geq 2$, gives equal B-images.

If $g(H_2) \geq 2$, induction gives equal H_2-images. If $g(H_2) = 1$ and p is odd, the map $V \to H_2$ is fully ramified above every point over a_1: the cover H_2 is unramified there, while B has index ℓ. A.21 therefore gives equal H_2-images. If $g(H_2) = 1$ and $p = 2$, formula (A.13) forces $\ell = 3$ and exactly three branch points. Their three nonzero Kummer exponents sum to zero modulo three, so they are all one or all two. A projective coordinate change and, if needed, inversion of the Kummer generator identify this curve with $z^3 = x(x - 1)$. The preceding Cartier calculation gives its two-rank-zero point injectivity. If $g(H_2) = 0$, apply the fibre statement to $V \to H_2$. In every case both images are equal.

It remains to check point separation by (A.14), rather than infer it from a birational function-field identity at a possibly singular point. Off the branch set of C, both B and H_2 are étale over the common genus-zero base. Their fibre product is therefore smooth and normal there, and agrees with its normalization V. Both P_V and Q_V lie in this locus. Equal images in B and H_2 imply $P_V = Q_V$, contradicting their distinct images on C. This closes the induction, including its lowest-branch cases, for which H_2 has genus zero or one.

Finally spread the cyclic action, the quotient and its distinct tame branch points over a finite-type model of K, after a finite extension defining them. On a sufficiently small smooth locus this remains a cyclic prime-degree cover with the same ramification data. Specialization of a collision by A.8 contradicts the finite-field result. Proper compactification gives the assertion for open subcurves. □

A.23 (Proved; all cyclic prime-to-p covers of a genus-zero curve). *Let K be finitely generated over $\mathbb{F}_p$, and let C/K be smooth proper geometrically connected of genus at least two. Suppose that $C_{\overline{K}}$ admits a cyclic Galois map of degree $n > 1$, with $p \nmid n$, to a genus-zero curve. Then its geometrically prime-to-p point-section map is injective, as is the point-section map of each hyperbolic open subcurve. There is no restriction on p or on the number of prime factors of n.*

Proof. We work first over a finite field and make finite constant extensions whenever needed to define the covers, roots of unity, branch points and chosen lifts. We induct on n; the prime cases have also been proved separately in A.22. Only curves of genus at least two enter the induction assertion. A genus-one quotient is treated by A.21, and a genus-zero quotient supplies a new rational base.

Suppose first that $n = ab$ with $a, b > 1$ coprime. Let C_a, C_b be the intermediate quotient curves of degrees a, b over the given genus-zero base. Their function fields have compositum $\overline{k}(C)$. Let P, Q be a collision. If either quotient has genus zero, C itself is a cyclic cover of that quotient of degree $b < n$ or $a < n$, so the outer induction already gives $P = Q$. Otherwise a quotient of genus at least two has equal images by the induction hypothesis. A quotient of genus one also has equal images: the Galois map from C is ramified by Riemann–Hurwitz, and every index in a branch fibre is the same nontrivial divisor of its prime-to-p degree. Thus A.21 applies.

The maps to C_a, C_b jointly distinguish all geometric points, including points over their common branch values. To verify this last assertion, take a point of the fibre product and let e_a, e_b be the two local ramification indices. They divide the coprime integers a, b. Over the algebraically closed residue field, the completed tame local product has equation $u^{e_a} = v^{e_b}$, after absorbing units into the parameters. Its normalization has exactly $\gcd(e_a, e_b) = 1$ point above the selected point. Since the function fields have the asserted compositum, C is this normalization and the joint map is injective on geometric points. Hence $P = Q$.

It remains to treat $n = \ell^a$. Let C_0 be the unique degree-ℓ quotient over the genus-zero base. In a Kummer presentation

$$\overline{k}(C) = \overline{k}(x)(y), \qquad y^n = c(x),$$

the branch points of C_0 are exactly the points where the valuation of c is not divisible by ℓ. These are exactly the points at which C has full ramification index n. Write r_{full} for their number. If $g(C_0) = 0$, the cyclic map $C \to C_0$ has degree $n/\ell < n$, and the outer induction settles C. Otherwise $r_{\mathrm{full}} \geq 3$ when ℓ is odd and $r_{\mathrm{full}} \geq 4$ when $\ell = 2$.

We next prove a fibre statement for degree n, using only the outer induction on smaller degrees: every collision on C has equal images on the genus-zero base. The case $g(C_0) = 0$ is already settled. If $\ell \geq 5$, choose three full-index branch points and take the cyclic degree-ℓ cover

$$H: \quad z^\ell = x(x-1)$$

after moving those three points to $0, 1, \infty$. Its genus is $(\ell - 1)/2$, and it has point-section injectivity by A.19 for odd p and by the Cartier calculation in A.21 for $p = 2$. If $\ell = 3$, take the same three-branch cover, which now has genus one. If $\ell = 2$, take the elliptic double cover branched at four chosen full-index points; here p is necessarily odd.

Choose a connected component V of the normalized fibre product of C and H. It is an étale cyclic cover of C of degree one or ℓ. Indeed H is branched only at the selected points, and its tame ramification index ℓ divides the full index n on C; tame base change cancels the ramification. If H has genus at least two, a lifted collision has equal H-images by its just-cited injectivity. If H has genus one, the map $V \to H$ is Galois cyclic, of degree n or n/ℓ. Its source has genus at least two since $V \to C$ is étale. Thus the map is ramified, and a branch fibre has

a common nontrivial prime-to-p ramification index. A.21 again gives equal H-images. The original base coordinates are equal in either case. This proves the fibre statement without using the inner induction that follows.

For this fixed $n = \ell^a$, we now induct on r_{full}, with the cases $g(C_0) = 0$ provided by the outer induction. Choose two full-index branch points a_1, a_2. Put $u = v_{a_1}(c)$, which is a unit modulo ℓ, and choose $b(x)$ with

$$\mathrm{div}(b) = u(a_1 - a_2).$$

The curve $B : v^n = b(x)$ has genus zero and full ramification at exactly these two points. Its function field is linearly disjoint from that of C. In fact any nontrivial intersection of the cyclic ℓ-power extensions would contain their degree-ℓ subfields; the latter cannot agree, since C_0 has at least three branch points while the corresponding quotient of B has exactly two.

Consequently the normalization V of $C \times_{\mathbf{P}^1} B$ is connected and is a cyclic étale cover of C of degree n. Matching full tame indices cancel ramification over a_1, a_2, and B is unramified elsewhere. Define H_2 by

$$w^n = h(x), \qquad h(x) = c(x)/b(x).$$

At a_1 the valuation is zero. At a_2 full ramification may remain, may become partial, or may disappear; no new branch point occurs. Every other full-index branch point remains. There is at least one such point, so h has exact Kummer order n and H_2 is a connected cyclic degree-n cover. Its number of full-index branch points is strictly smaller. The invertible change of Kummer generators $w = y/v$ gives

$$\overline{k}(V) = \overline{k}(x)(y, v) = \overline{k}(x)(v, w). \tag{A.15}$$

The two latter characters are independent: the change $(c, b) \mapsto (b, c/b)$ is an invertible change of basis of $(\mathbb{Z}/n)^2$. Thus both maps $V \to B$ and $V \to H_2$ are cyclic of degree n.

Suppose $P \neq Q$ is a collision on C. The fibre statement gives a common base value. It is different from a_1, a_2, since each of those full-index fibres consists of one geometric point. After a finite constant extension, lift the collision to P_V, Q_V on V by A.3. Apply the degree-n fibre statement to $V \to B$: its source has genus $n(g(C) - 1) + 1 \geq 2$, so the B-images agree. If $g(H_2) \geq 2$, the inner induction, or its outer-induction base case, gives equal H_2-images. If $g(H_2) = 1$, the map $V \to H_2$ is fully ramified of index n over every point above a_1, because H_2 is unramified there and B has index n. A.21 gives the required equality. If $g(H_2) = 0$, apply the already proved degree-n fibre statement to $V \to H_2$.

These two equalities imply equality of the lifted points. Indeed outside $\{a_1, a_2\}$ the map $B \to \mathbf{P}^1$ is étale, so $B \times_{\mathbf{P}^1} H_2$ is étale over the smooth curve H_2 and is already normal. By (A.15) it agrees there with V. Both lifted points lie in this locus. Equal B- and H_2-images therefore give $P_V = Q_V$, contrary to their distinct images on C. Notice that no claim of total ramification at the other branch points was used. This completes both inductions.

The same spreading and specialization as in A.22 applies to the finite tame cyclic action of order n; after shrinking the model, the quotient and its ramification indices persist. A.8 therefore gives the assertion over K. Functoriality to the proper compactification gives the assertion for open subcurves. □

A.24 (Proved; all abelian prime-to-p Galois covers of a genus-zero curve). *Let K be finitely generated over $\mathbb{F}_p$. Let C/K be smooth proper geometrically connected of genus at least two. Suppose that $C_{\overline{K}}$ has a faithful action of a finite abelian group G of order prime to p, with genus-zero quotient. Then its geometrically prime-to-p point-section map is injective. The same holds for every hyperbolic open subcurve. This theorem includes A.19, A.20 and A.23, but also allows noncyclic abelian groups with arbitrary prime-to-p exponents in every characteristic.*

Proof. Over a finite field, after defining the action and its quotients by a finite constant extension, induct on $|G|$. The cyclic case is A.23. For a nontrivial character $\chi : G \to \mathbb{Z}/n$, understood to have image of order n, put $C_\chi = C/\ker\chi$. It is a cyclic degree-n cover of the common genus-zero quotient. If any C_χ has genus zero, then $C \to C_\chi$ has group $\ker\chi$ of smaller order. The induction hypothesis settles C at once.

In the remaining case let P, Q be a collision on C. For each χ with $g(C_\chi) \geq 2$, A.23 gives equal C_χ-images by functoriality. If $g(C_\chi) = 1$, the map $C \to C_\chi$ is ramified by Riemann–Hurwitz. It is Galois of prime-to-p order, so any branch fibre has equal nontrivial ramification indices. A.21 gives equal C_χ-images also in this case. This argument uses the all-characteristic form of A.21; no classification of elliptic automorphisms in characteristic two is needed.

It remains to check that these quotient maps jointly separate geometric points at every ramified fibre. Their equalities first give the same point of C/G, so $Q = gP$ for some $g \in G$. Let $I \subseteq G$ be the stabilizer of P. Equality of the two images in $C/\ker\chi$ is exactly the assertion

$$g \in (\ker\chi)I.$$

Every character of the finite abelian group G/I is a character of G trivial on I, and such characters separate its elements. The displayed containment for all χ therefore forces $g \in I$. Hence $Q = P$.

Over a finitely generated field, extend finitely to define the action and spread the curve, faithful action and quotient over a finite-type model. Since $|G|$ is invertible, after shrinking the quotient is again smooth of genus zero and the action remains faithful. A.8 specializes any collision to the finite-field case just excluded. Passing to the compactification proves the open assertion. □

A.25 (Proved; ramified double covers of elliptic curves). *Let K be finitely generated over $\mathbb{F}_p$, with $p \neq 2$. Let C/K be smooth proper geometrically connected of genus at least two. If $C_{\overline{K}}$ admits a degree-two map to a genus-one curve, then its geometrically prime-to-p point-section map is injective. More generally the same holds if $C_{\overline{K}}$ has a faithful elementary abelian two-group action with genus-one quotient. The conclusions include every hyperbolic open subcurve.*

Proof. First suppose that $C \to E$ is a double cover over a finite field, allowing finite constant extensions to define all points and covers. It is ramified because $g(C) \geq 2$. Let S be its reduced branch set, of even cardinality r. Riemann–Hurwitz gives $g(C) = 1 + r/2$. We induct on r. When $r = 2$, the curve has genus two and A.19 applies.

For $r \geq 4$, choose distinct $a_1, a_2 \in S$. There is a double cover $B \to E$ branched exactly at these two points: over the algebraic closure, multiplication by two on $\mathrm{Pic}^0(E)$ is surjective, so $\mathcal{O}_E(a_1 + a_2)$ has a square root of degree one. The section with zero divisor $a_1 + a_2$ defines the required smooth double cover. It is connected because it is ramified, and it has genus two. These finite data descend to a finite constant extension.

The quadratic extensions defining C and B differ, since their branch sets differ. Their normalized fibre product V is connected and is a double étale cover of C: quadratic ramification cancels at a_1, a_2, and B is unramified elsewhere. Let $H_2 \to E$ be the third quadratic quotient, whose Kummer character is the product of the two original characters. Its branch set is exactly $S \setminus \{a_1, a_2\}$, so it is connected and has genus $g(C) - 1 \geq 2$. The induction hypothesis gives point-section injectivity for H_2, while A.19 gives it for B.

A collision on C lifts to a collision on V by A.3. Both quotient images agree by these two injectivity statements. Their equality forces equality of the points on V: the branch sets of B and H_2 over E are disjoint, so their fibre product is smooth everywhere and equals its normalization V. Thus the two projections jointly separate every geometric point.

This completes the branch-count induction without needing to infer point separation from a function-field identity at a singular point.

For an elementary abelian group $(\mathbb{Z}/2)^r$, induct on r, using the double-cover result for $r = 1$. For each nontrivial involution s, the quotient $C/\langle s\rangle$ still maps to the genus-one final quotient. It cannot have genus zero. If its genus is at least two, the smaller-rank induction gives equal quotient images of any collision; if its genus is one, A.21 gives that equality for the ramified double map from C. For $r \geq 2$, the three quotients attached to independent involutions s, t and st jointly separate every point, by exactly the orbit argument in A.20. Therefore the collision is trivial.

Spread the tame map or tame finite group action and its genus-one quotient over a finite-type model of K, after a finite extension defining them. A.8 specializes a hypothetical collision to the finite-field case just proved. Functoriality gives the open assertion. □

1.7 A family of ordinary genus-two curves in characteristic two

A.C2.1 (Proved). *Let K be finitely generated over $\mathbb{F}_2$, let $a, b, c \in K^\times$, and let $C_{a,b,c}$ be the smooth proper curve with function field*

$$y^2 + y = ax + \frac{b}{x} + \frac{c}{x+1}.$$

It is an ordinary genus-two curve. If

$$(a + b + c)^3 + abc = 0, \tag{C2.1}$$

then its geometrically prime-to-two point-section map is injective. The same conclusion holds for every hyperbolic open subcurve and after any finite extension of K. It also holds for any curve that becomes isomorphic to such a curve after a finite extension.

Over an algebraic closure, condition (C2.1) is equivalent to

$$b = av^3, \qquad c = a(v + 1)^3 \qquad (v \neq 0, 1). \tag{C2.2}$$

For each such v there is a connected cyclic étale cover $D \to C$ of degree three and two maps $D \to E$ to one supersingular elliptic curve whose product is injective on geometric points.

Proof. Work initially over an algebraic closure Ω of K. The three simple Artin–Schreier poles are $0, 1, \infty$. Their different exponents are two, so Riemann–Hurwitz gives genus two. Write

$$h = x^2 + x, \qquad Y = hy, \qquad f = ax^5 + (a + b + c)x^3 + cx^2 + bx.$$

The equation is $Y^2 + hY = f$. There is one point P_∞ above infinity, with pole orders two and five for x and Y, respectively. The regular differentials dx/h and $x\,dx/h$ form a basis. Cartier fixes each of them: write $1/h = h/h^2$ and use $\mathcal{C}((x^2 + x)dx) = dx$ and $\mathcal{C}((x^3 + x^2)dx) = x\,dx$ in characteristic two. Thus Cartier is bijective and the curve is ordinary.

For the parameterization put $B = b/a$, $C = c/a$. Equation (C2.1) becomes $(1 + B + C)^3 + BC = 0$. If $B + C \neq 0$, set $v = (C + 1)/(B + C)$. Direct substitution gives $v^2 + v = B + C + 1$ and $v^3 = B$, whence $(v + 1)^3 = C$. If $B = C$, the equation implies $B = C = 1$; either root of $v^2 + v + 1 = 0$ works. Conversely (C2.2) gives (C2.1) by expansion. Nonvanishing of b, c excludes $v = 0, 1$.

Choose $d \in \Omega$ satisfying $d^2 + d = a^2v^2 + a$, and set

$$u = x^2 + v^2, \qquad A = ax^3 + dx^2 + (d + a + b + c)x + b.$$

A coefficient calculation gives

$$A^2 + hA + f = a^2u^3. \tag{C2.3}$$

Let $F = (A+Y)/a$. The hyperelliptic involution ι sends Y to $Y+h$, so $F\iota(F) = u^3$. At $x = v$, the map $C \to \mathbf{P}^1$ is unramified, since $h(v) = v(v+1) \neq 0$. Of its two points, precisely one, denoted R, has $F(R) = 0$. The conjugate value is $h(v)/a \neq 0$. Because u has order two there, (C2.3) gives

$$\operatorname{div}(F) = 6R - 6P_\infty.$$

The normalization in $z^3 = F$ is therefore étale. It is connected: if F were a cube, its cube root would have its only pole at P_∞ of order two and would lie in $L(2P_\infty) = \langle 1, x\rangle$. Its cube could not contain the nonzero Y coefficient of F. Denote this cyclic cover by D; it has genus four.

Put $w = z + u/z$. The involution $z \mapsto u/z$ gives a map of degree two from D to

$$E: \quad a\big(W^3 + (X^2 + v^2Z^2)W\big) = X^2Z + XZ^2, \qquad O = [1:0:0]. \tag{C2.4}$$

The derivatives with respect to X and Z are Z^2 and X^2. A singular projective point would therefore have $X = Z = 0$, and the derivative with respect to W would then force $W = 0$. Thus this is a smooth plane cubic.

We verify supersingularity without a characteristic restriction beyond two. For a smooth plane cubic $G(X, W, Z) = 0$ in characteristic two, the scalar of Frobenius on $H^1(E, \mathcal{O}_E)$ is the coefficient of XWZ in G, in the usual one-dimensional Cech description. Indeed the exact sequence $0 \to \mathcal{O}_{\mathbf{P}^2}(-3) \xrightarrow{G} \mathcal{O}_{\mathbf{P}^2} \to \mathcal{O}_E \to 0$ identifies this group with $H^2(\mathcal{O}_{\mathbf{P}^2}(-3))$, whose generator is represented by $1/(XWZ)$. Frobenius takes it to $G/(X^2W^2Z^2)$; every cubic monomial except XWZ gives a zero class in this Cech group. In (C2.4) that coefficient is zero. Hence the elliptic curve has 2-rank zero, and $E(\Omega)[2^\infty] = 0$.

Let $\zeta \neq 1$ satisfy $\zeta^3 = 1$ and put $q = u/z$. The two maps

$$r_0 = (x, z+q), \qquad r_1 = (x, \zeta z + \zeta^{-1}q) : D \to E$$

extend to the smooth proper curves. We check their joint injectivity. For finite $x \neq v$, the two image coordinates recover z, q, since

$$\det \begin{pmatrix} 1 & 1 \\ \zeta & \zeta^{-1} \end{pmatrix} = 1.$$

Together with $Y = az^3 - A$ they recover the point. At $x = v$, let $\mathcal{R} = \{s : s^3 = h(v)/a\}$; it consists of three distinct nonzero elements. The three points above R have $z = 0$, $q \in \mathcal{R}$, and their two w-coordinates are $(q, \zeta^{-1}q)$. The other three have $q = 0$, $z \in \mathcal{R}$, and coordinates $(z, \zeta z)$. These six pairs are distinct. At infinity the three residues of z/x are $1, \zeta, \zeta^2$, while the residues of q/x are their inverses. Writing $T = [1:1:0] \in E$, their image pairs are

$$(O, T), \quad (T, T), \quad (T, O).$$

Every finite point maps to the affine chart $Z = 1$, so no other collision with these pairs occurs. The two maps are jointly injective on all geometric points.

All choices are defined after a finite extension of K. A hypothetical collision of point sections persists over that extension and over a further extension making a chosen lift to D rational. The fixed-fibre argument of A.3 supplies a matching lift whose Jacobian difference is 2-primary by A.1. Applying r_0 and r_1 kills this difference, because $E(\Omega)[2^\infty] = 0$. Joint injectivity forces the two lifts, and then the original points, to coincide. This is the argument of A.10, with the same Kummer input valid over finitely generated fields. A collision on an open subcurve maps to one on its proper compactification. These observations also prove the assertions about finite extensions and forms. □

A.C2.2 (Proved). *For the ordinary genus-two curve $C_{a,b,c}$ over an algebraically closed field of characteristic two, there exists a connected cyclic étale cubic cover with a supersingular elliptic quotient by a lift of the hyperelliptic involution exactly when (C2.1) holds. If (C2.1) fails, every connected cyclic étale cubic cover has ordinary Jacobian and admits only constant maps to abelian varieties of 2-rank zero.*

For the 2-rank-one genus-two normal form $y^2 + y = ax^3 + bx + c/x$, with $ac \neq 0$, every elliptic quotient of a cyclic cubic étale cover by a lifted hyperelliptic involution is ordinary as well.

In particular take $a = b = 1$ and $c = \alpha \in \mathbb{F}_4$ with $\alpha^2 + \alpha + 1 = 0$. Then $(a+b+c)^3 + abc = 1 + \alpha \neq 0$. Thus the construction using a single cyclic cubic cover and 2-rank-zero abelian targets cannot settle this curve. This is an obstruction to that specified construction; covers of other odd degrees remain available.

Proof. We check that the normal form used above exhausts cyclic cubic étale covers. Such covers over an algebraically closed field are classified, up to choice of generator, by nonzero $\eta \in J_C[3]$. Riemann–Roch supplies an effective divisor D_0 of degree two with $\eta = [D_0 - 2P_\infty]$. It is unique: a degree-two divisor with two independent sections belongs to the hyperelliptic pencil and would give $\eta = 0$. It cannot contain P_∞. Otherwise $\eta = [S - P_\infty]$ and $3\eta = 0$ would give a function with its only pole of order three at the Weierstrass point P_∞, where three is a gap. In the displayed model this also follows from $L(3P_\infty) = \langle 1, x \rangle$.

Choose a function with divisor $3D_0 - 6P_\infty$. The space $L(6P_\infty)$ has basis $1, x, x^2, x^3, Y$. Its Y coefficient cannot vanish: otherwise the divisor is invariant under the hyperelliptic involution. An invariant effective divisor of degree two is either a hyperelliptic fibre or a sum of Weierstrass points, whose class relative to $2P_\infty$ is killed by two. Neither can represent the nonzero three-torsion class η. After scaling, the function is therefore $A + Y$, with

$$A = Lx^3 + dx^2 + ex + k, \qquad L \neq 0.$$

Let $u = x^2 + u_1 x + u_0$ be the monic polynomial whose zero divisor on C is $D_0 + \iota D_0$. Taking norms and comparing the leading coefficient gives

$$A^2 + hA + f = L^2 u^3. \tag{C2.5}$$

The required cubic cover is $z^3 = (A + Y)/L$, after a harmless constant rescaling over the algebraically closed field. Thus no three-torsion class of smaller reduced degree has been omitted.

The lift $\tau : z \mapsto u/z$ of the hyperelliptic involution has elliptic quotient. Here is a ramification verification in characteristic two. The cubic cover has genus four. Over each of the three branch points of $C \to \mathbf{P}^1$, the composite S_3-cover has three points with distinct order-two inertia groups. Exactly one is fixed by the specified τ, and its different exponent is two, as in the original Artin–Schreier cover. There are no other fixed points. Riemann–Hurwitz for the quotient by τ therefore reads $6 = 2(2g(E) - 2) + 6$, so $g(E) = 1$. Its invariant functions $x, w = z + u/z$ give the integral plane cubic

$$L\big(W^3 + (X^2 + u_1 XZ + u_0 Z^2)W\big) = X^2 Z + XZ^2. \tag{C2.6}$$

It has the genus-one function field just computed, so its arithmetic genus one forces it to be smooth. The Cech calculation in A.C2.1 makes its Hasse scalar Lu_1. It is supersingular exactly when $u_1 = 0$.

Assume $u_1 = 0$. The coefficients of x^5, x, x^3 in (C2.5) give, respectively,

$$L = a, \qquad k = b, \qquad e = d + a + b + c.$$

The coefficients of $x^4, x^2, 1$ then imply

$$d^2 + d + a = a^2 u_0, \qquad u_0^2 + u_0 + 1 = ((b+c)/a)^2, \qquad u_0^3 = (b/a)^2. \tag{C2.7}$$

Choose v with $v^2 = u_0$. Since squaring is injective, (C2.7) gives $v^3 = b/a$ and $v^2 + v + 1 = (b+c)/a$. Consequently $(v+1)^3 = c/a$, which implies (C2.1). Conversely A.C2.1 constructs a connected cubic cover with supersingular quotient whenever (C2.1) holds.

For the stronger assertion let $D \to C$ be any connected cyclic cubic cover, let δ generate its deck group, and use the lift τ above. Its Prym variety P has dimension two, and

$$J_D \sim J_C \times E^2. \tag{C2.8}$$

We justify the isogeny also in characteristic two. Choose an auxiliary prime different from two and three and work on rational Tate modules. The invariant part for δ is VJ_C; its complement is VP. The hyperelliptic involution acts as -1 on VJ_C. Thus the image of VE under pullback lies in VP and equals $(VP)^\tau$. The two subspaces $(VP)^\tau$ and $\delta(VP)^\tau$ have zero intersection: an element fixed by the two corresponding reflections is fixed by δ, whereas VP has no δ-invariants. Each has dimension two, and together they span VP. The two pullback maps $E \to P$ consequently give an isogeny $E^2 \to P$. Pullback and norm for $D \to C$, whose composition is [3], give (C2.8).

If (C2.1) fails, every elliptic quotient (C2.6) is ordinary. The base curve is ordinary, so (C2.8) makes J_D ordinary. An abelian quotient of an ordinary abelian variety is ordinary, by Poincare reducibility and additivity of 2-rank under products and isogenies. A positive-dimensional abelian subvariety of a 2-rank-zero variety has 2-rank zero. Hence a homomorphism from J_D to such a variety has zero image. The Albanese property then makes every map from D to such a variety constant. Substituting $(1, 1, \alpha)$ gives the asserted explicit example.

For the 2-rank-one normal form put $Y = xy$, so $h = x$ and $f = ax^5 + bx^3 + cx$. The same reduced-divisor argument and ramification calculation give (C2.5)–(C2.6), with $h = x$. The right side of the quotient cubic is then XZ^2. The x^5 coefficient of the norm on the left is a, whereas the right coefficient is $L^2 u_1$. Thus $u_1 \neq 0$. The coefficient of XWZ in the quotient cubic remains Lu_1, so it is ordinary. The base curve has 2-rank one: Cartier fixes dx/x and kills dx, its regular differential basis. □

A.C2.3 (Proved). *Let K be finitely generated over $\mathbb{F}_2$ and let $s \in K^\times$. For the smooth proper genus-two curve*

$$C_s: \quad y^2 + y = sx^3 + x + \frac{s^5}{x}, \tag{C2.9}$$

which has 2-rank one, the geometrically prime-to-two point-section map is injective. The assertion persists after finite extensions, for finite-extension forms, and for hyperbolic open subcurves.

Equivalently this covers the 2-rank-one models $y^2 + y = ax^3 + bx + c/x$ with $abc \neq 0$ and $a^5 = b^{16}c$. There is a connected cyclic étale cover $D \to C_s$ of degree five with two jointly point-injective maps to a genus-two curve of 2-rank zero.

Proof. The two Artin–Schreier poles have orders three and one, so the genus is two. With $Y = xy$ the equation is

$$Y^2 + xY = f, \qquad f = sx^5 + x^3 + s^5 x.$$

The regular differentials $dx/x, dx$ have Cartier images $dx/x, 0$; hence the 2-rank is one. The point P_∞ above infinity again has pole orders two and five for x, Y.

Over a finite extension choose d with $d^2 + d = s^8$, and put

$$u = x^2 + s^2, \qquad A = x^5 + sx^4 + x^3 + x^2 + dx + s^5.$$

The direct identity

$$A^2 + xA + f = u^5 \tag{C2.10}$$

gives $F\iota(F) = u^5$ for $F = A + Y$. At $x = s$, the hyperelliptic map is unramified and the two values of F differ by $s \neq 0$. Exactly one point R has $F = 0$. Since u has order two there and F has pole order ten at infinity, its divisor is $10R - 10P_\infty$. Consequently $z^5 = F$ defines an étale cyclic cover D of degree five. It is connected: a fifth root of F would have its only pole of order two at P_∞, hence belong to $\langle 1, x\rangle$, which cannot produce the nonzero Y coefficient. The genus of D is six.

The involution $\tau : z \mapsto u/z$ lifts the hyperelliptic involution. Put $w = z + u/z$ and $X = x + s$, so $u = X^2$. The fifth-power identity in characteristic two gives

$$w^5 + X^2w^3 + X^4w = x.$$

Set $t = X^2 + Xw + w^2$. Its left side is wt^2, so $x = wt^2$. Substituting this into the definition of t gives

$$(t^4 + t^2 + 1)w^2 + sw + (s^2 + t) = 0.$$

Writing $H = t^4 + t^2 + 1$ and $V = Hw/s$ exhibits the quotient function field as that of

$$B_s : \quad V^2 + V = \frac{(t^4 + t^2 + 1)(t + s^2)}{s^2}. \tag{C2.11}$$

Indeed $\Omega(D)^\tau = \Omega(x, w)$, since z satisfies $z^2 + wz + u = 0$, and the displayed rational transformations identify $\Omega(x, w)$ with $\Omega(t, V)$. The right side of (C2.11) has one pole of order five. Thus B_s is smooth proper of genus two, and its regular differentials are $dt, t\,dt$. Cartier sends them to $0, dt$, respectively. It is nilpotent, so B_s has 2-rank zero and $J_{B_s}(\overline{K})[2^\infty] = 0$.

Let $r_0 : D \to B_s$ be the quotient map, and let δ generate the cyclic deck group of D/C_s. Define $r_1 = r_0 \circ \delta$. These two maps jointly separate all geometric points by a group action argument which also handles the ramification points. A fibre of a quotient by $\langle\tau\rangle$ is a geometric orbit. If distinct P, Q had both images equal, then

$$Q = \tau P = \delta^{-1}\tau\delta P.$$

The dihedral identity $\tau\delta\tau = \delta^{-1}$ would force $\delta^2 P = P$. This is impossible because a nonidentity deck transformation of the étale cyclic cover D/C_s fixes no point.

Compose r_0, r_1 with an Abel embedding of B_s into its Jacobian, choosing a base point after a finite extension if needed. The resulting two maps have targets with no geometric 2-primary torsion points and are still jointly injective. The section-lifting and Kummer argument of A.C2.1, or A.10, therefore proves the asserted point-section injectivity. Finally, $x' = bx$ transforms $y^2 + y = ax^3 + bx + c/x$ into $y^2 + y = (a/b^3)x'^3 + x' + bc/x'$. The relation $a^5 = b^{16}c$ makes $bc = (a/b^3)^5$, proving the equivalent parameter statement. The assertions about open subcurves and finite extensions follow as in A.C2.1. □

1.8 A thirteen-fold cover of a second characteristic-two curve

A.C2.4 (Proved). *Let K be finitely generated over $\mathbb{F}_2$. The geometrically prime-to-two point-section map is injective for the curve*

$$C_0 : \quad y^2 + y = x^3 + \frac{1}{x}, \tag{C2.12}$$

and for its finite-extension forms and hyperbolic open subcurves. There is a connected cyclic étale cover of degree thirteen with two jointly point-injective quotient maps to a genus-six curve of 2-rank zero.

Proof. The change $u = 1/x$, $V = u^2 y$ gives

$$V^2 + u^2 V = u^5 + u. \tag{C2.13}$$

Write P_∞ for the unique point above $u = \infty$ and P_0 for that above $u = 0$. In this model the Artin–Schreier pole orders are one at infinity and three at zero. In particular, C_0 has genus two and 2-rank one. One can see the latter from its differential basis $du/u^2, du/u$: Cartier kills the first and fixes the second.

We first exhibit a thirteen-torsion point on a specified translate of the curve in its Jacobian. Choose

$$\alpha^2 + \alpha + 1 = 0, \qquad \beta^2 + \alpha^2 \beta = 1,$$

so that $\alpha, \beta \in \mathbb{F}_{16}$, and set $R = (\alpha, \beta)$ on (C2.13). Define

$$\begin{aligned} B(u) &= u^4 + \alpha u^3 + u + 1, \\ A(u) &= \alpha(1+\beta)u^6 + \alpha^2 \beta u^5 + \alpha u^4 + \alpha(1+\beta)u^3 \\ &\qquad + (1 + \alpha\beta)u^2 + \alpha u + \alpha^2. \end{aligned}$$

Direct expansion in this field gives

$$A^2 + u^2 AB + (u^5 + u)B^2 = (u + \alpha)^{13}. \tag{C2.14}$$

For $G = A + BV$, one has $G(R) = 0$ and $G(\iota R) = \alpha \neq 0$. Its pole order at P_∞ is thirteen: A has pole order twelve, while BV has pole order thirteen. Taking the norm in (C2.14), with the conjugate factor a unit at R, therefore gives

$$\operatorname{div}(G) = 13R - 13P_\infty.$$

The class $L = \mathcal{O}(R - P_\infty)$ is nontrivial, since a function with one simple pole would make the curve rational. It has exact order thirteen. The cover D defined by $z^{13} = G$ is étale and connected; otherwise G would be a thirteenth power, again giving a function with divisor $R - P_\infty$. Its genus is fourteen.

Here is the complete Frobenius calculation which controls the 2-rank of this cover. On a smooth curve in characteristic two, let $\mathcal{B}$ be the image of $d : F_*\mathcal{O}_C \to F_*\Omega^1_C$, with the module structure induced by absolute Frobenius. The sequence

$$0 \longrightarrow \mathcal{O}_C \longrightarrow F_*\mathcal{O}_C \longrightarrow \mathcal{B} \longrightarrow 0 \tag{C2.15}$$

is exact. Locally in a parameter t, the image is the free rank-one module of differentials $h(t)^2 dt$, so $\mathcal{B}$ is a line bundle. A rational locally exact differential with ordinary divisor $2D_1$ is a rational section of $\mathcal{B}$ with divisor D_1; this follows from the same square-coefficient local description.

For (C2.13), the different formula gives

$$\operatorname{div}(du) = 4P_0 - 2P_\infty, \qquad \operatorname{div}(u) = 2P_0 - 2P_\infty.$$

The locally exact differential $d(1/u) = du/u^2$ therefore has divisor $2P_\infty$. Consequently

$$\mathcal{B} \simeq \mathcal{O}(P_\infty).$$

Tensor (C2.15) by L and use the projection formula. The map on first cohomology is the semilinear Frobenius map

$$H^1(C, L) \longrightarrow H^1(C, L^2).$$

Both spaces have dimension one, by Riemann–Roch and nontriviality of these degree-zero line bundles. Its kernel contains, and equals the image of, $H^0(C, \mathcal{B} \otimes L) = H^0(C, \mathcal{O}(R))$, which has dimension one. Indeed the preceding $H^0(C, L^2)$ is zero. Thus this Frobenius map is zero.

Over an algebraic closure the eigenspace decomposition for the cyclic cover is

$$H^1(D, \mathcal{O}_D) = \bigoplus_{i \in \mathbb{Z}/13} H^1(C, L^i).$$

Each nonzero-index summand is one-dimensional. Frobenius sends the summand indexed by i to that indexed by $2i$. The powers

$$1, 2, 4, 8, 3, 6, 12, 11, 9, 5, 10, 7$$

are all the nonzero residues modulo thirteen. Thus every nonzero-index summand lies on a single Frobenius cycle, containing the zero map just found. The twelfth Frobenius power is zero on the sum of these summands. The trivial summand is $H^1(C, \mathcal{O}_C)$ and has stable Frobenius rank one. The Artin–Schreier cohomology sequence identifies 2-rank with the dimension of Frobenius-fixed cohomology (Yang, 2016, Definition 1.1 and the following discussion), so D has 2-rank one as well.

The norm identity lifts the hyperelliptic involution to

$$\tau : z \longmapsto (u + \alpha)/z.$$

If δ generates the cyclic deck group, then $\tau\delta\tau = \delta^{-1}$. The quotient $B = D/\langle\tau\rangle$ has genus six. Indeed, above each of the two branch points of $C \to \mathbf{P}^1$, exactly one of the thirteen points is fixed by τ. Their different exponents are respectively four and two, unchanged by the étale local base change $D \to C$. Riemann–Hurwitz gives $26 = 2(2g(B) - 2) + 6$, hence $g(B) = 6$.

The same dihedral representation argument as in A.C2.2 gives

$$J_D \sim J_C \times J_B^2.$$

For clarity, the complement of the cyclic invariants in a rational Tate module has dimension twenty-four. Its fixed subspace under τ, and its translate by δ, each have dimension twelve. Their intersection is zero, since an element in it is fixed by δ^2 and therefore by δ, whose order is odd. These are precisely the two pulled-back copies of VJ_B. The two maps from J_B thus give the claimed isogeny on the Prym. Additivity of 2-rank and the calculation $f(D) = f(C) = 1$ imply $f(B) = 0$.

Two deck-conjugate quotient maps $D \to B$ jointly separate points: a nontrivial collision would again give a fixed point of δ^2, as in A.C2.3. After composing with an Abel embedding into J_B, their targets have no nonzero geometric 2-primary torsion points. The section-lifting argument of A.10 proves the claimed injectivity. All equations, the deck transformation and chosen base points are defined after finite extensions; this proves the stated field and form variants. □

1.9 The exact theta-orbit criterion and its remaining arithmetic condition

A.C2.5 (Proved). *Let C be a smooth proper genus-two curve over an algebraically closed field Ω of characteristic two. Let $\mathcal{B}$ be its line bundle of locally exact differentials, with the absolute-Frobenius module structure, and put*

$$\Theta = \{L \in \mathrm{Pic}^0(C) : H^0(C, \mathcal{B} \otimes L) \neq 0\}.$$

For an odd integer $n > 1$, let L have exact order n, and let $D \to C$ be the associated connected cyclic étale cover. Write $\mathcal{O}_n$ for the orbits of multiplication by two on $(\mathbf{Z}/n) \setminus \{0\}$. Then

$$f(D) = f(C) + \sum_{\substack{O \in \mathcal{O}_n \\ L^i \notin \Theta \text{ for all } i \in O}} |O|. \tag{C2.16}$$

The set Θ is symmetric under $L \mapsto L^{-1}$. If every orbit meets $\{i : L^i \in \Theta\}$, a lift of the hyperelliptic involution has a quotient B of genus $(n-1)/2$ and 2-rank zero. Two deck-conjugate quotient maps $D \to B$ jointly separate geometric points. In particular, for an odd prime ℓ, the sufficient condition is satisfied whenever

$$L \in \Theta[\ell] \setminus \{0\}, \qquad \langle 2, -1 \rangle = (\mathbf{Z}/\ell)^\times. \tag{C2.17}$$

Over a field finitely generated over $\mathbb{F}_2$, a curve satisfying this geometric condition has an injective geometrically prime-to-two point-section map, as do its finite-extension forms and hyperbolic open subcurves.

Proof. The exact sequence (C2.15) is valid for every smooth curve in characteristic two. Its local description also proves $\mathcal{B}^{\otimes 2} \simeq \Omega^1_C$: if t is a local parameter, a locally exact differential is $h(t)^2 dt$, so the transition functions for $\mathcal{B}$ square to those for Ω^1_C. Thus $\deg \mathcal{B} = 1$. Riemann–Roch gives

$$h^0(\mathcal{B} \otimes M) = h^0(\Omega^1_C \otimes \mathcal{B}^{-1} \otimes M^{-1}) = h^0(\mathcal{B} \otimes M^{-1})$$

for every degree-zero M, proving the symmetry.

For $0 \neq i \in \mathbf{Z}/n$, both L^i and L^{2i} are nontrivial of degree zero. They have no global sections, and their first cohomology spaces have dimension one. Tensoring (C2.15) by L^i therefore identifies the kernel of

$$F_i : H^1(C, L^i) \longrightarrow H^1(C, L^{2i})$$

with $H^0(C, \mathcal{B} \otimes L^i)$. Hence F_i is zero precisely when $L^i \in \Theta$, and otherwise it is a bijective semilinear map between one-dimensional spaces. The cyclic-cover decomposition

$$H^1(D, \mathcal{O}_D) = \bigoplus_{i \in \mathbf{Z}/n} H^1(C, L^i)$$

is respected by Frobenius, with indices multiplied by two. On an orbit meeting Θ, the $|O|$th Frobenius power is zero. On an orbit disjoint from Θ, Frobenius is bijective and its stable rank is $|O|$. The index-zero summand has stable rank $f(C)$. The Artin–Schreier description of 2-rank now proves (C2.16). This proof treats all nonzero characters, including characters of proper conductor when n is composite.

The hyperelliptic involution ι acts by -1 on $\mathrm{Pic}^0(C)$, so it lifts to an automorphism τ of the cyclic cover which conjugates a deck generator δ to δ^{-1}. Such a lift is an involution: $\tau^2 \in \langle \delta \rangle$ commutes with τ, and inversion fixes no nonidentity element of an odd cyclic group. At a branch point of $C \to \mathbf{P}^1$, the action of τ on the n points above it is an affine inversion of the cyclic torsor, so exactly one of those points is fixed. The completed local map $D \to C$ at that point is an isomorphism, and the different exponent for $D \to D/\langle \tau \rangle$ equals the exponent

for $C \to \mathbf{P}^1$. The latter different has total degree six, since $g(C) = 2$. Since $g(D) = n + 1$, Riemann–Hurwitz gives

$$2n = 2(2g(B) - 2) + 6, \qquad g(B) = (n-1)/2.$$

The dihedral argument of A.C2.2 applies to every odd n:

$$J_D \sim J_C \times J_B^2. \tag{C2.18}$$

Here is the precise linear-algebra verification. In an auxiliary rational Tate module at a prime different from two and from the divisors of n, let W be the complement of the cyclic invariants. Its nontrivial cyclic characters occur in pairs λ, λ^{-1}, and τ exchanges the members of each pair. Consequently W^τ has half the dimension of W. The subspaces W^τ and δW^τ intersect trivially: an element in the intersection is fixed by both corresponding reflections, hence by δ^2, hence by δ, whereas W has no cyclic invariants. Both subspaces are pulled-back Tate modules of the quotient Jacobian. They therefore span W and give (C2.18). The cyclic-invariant part is VJ_C, on which τ acts by -1.

If every orbit meets Θ, formula (C2.16) and additivity of 2-rank under isogeny give $f(B) = 0$. Two quotient maps associated with the reflections τ and $\delta^{-1}\tau\delta$ jointly separate points: a nontrivial collision would be a fixed point of δ^2, impossible because the odd cyclic deck action is free. Composing with Abel embeddings into J_B and applying A.10 proves the point-section assertion. The covers, quotient maps and chosen base points descend after a finite extension, as required there.

Finally suppose $n = \ell$ is prime and (C2.17) holds. If $L \in \Theta$, symmetry also gives $L^{-1} \in \Theta$. Every nonzero residue modulo ℓ then belongs to the doubling orbit of 1 or of -1, proving the required orbit condition. In particular, the condition is weaker than requiring two itself to be a primitive root: for $\ell = 7$ there are two doubling orbits, exchanged by sign, and both are met simultaneously. □

A.C2.6 (Unresolved). *For an arbitrary genus-two curve over $\overline{\mathbb{F}}_2$, the existence of an odd-order torsion point satisfying the all-orbits condition of A.C2.5 is not established here. Prime-to-two torsion density on Θ does not supply the missing order and orbit restrictions.*

Exact remaining obligation. The theta curve is a translate of the Abel image of C and generates its Jacobian. Yang's Proposition 2.3 therefore gives density of odd-order torsion on it over $\overline{\mathbb{F}}_2$ (Yang, 2016). That statement does not prescribe a prime order ℓ, the subgroup $\langle 2, -1\rangle$ modulo ℓ, or which other multiples of a chosen point remain on Θ. Formula (C2.16) identifies the exact missing condition: every nonzero doubling orbit must contain a theta point. For a composite order, the orbits of proper-order characters must also be treated. A single point of exact order n only guarantees a zero map on its own orbit and, by symmetry, on the opposite orbit.

This distinction is substantive. In the 2-rank-one genus-two family of A.C2.2 every cyclic cubic étale cover has an ordinary elliptic involution quotient. Formula (C2.16) therefore implies $\Theta \cap J_C[3] = \{0\}$ in this family. Thus even the smallest prescribed odd prime need not occur on the nonzero theta locus. Any cyclic cover whose degree is divisible by three retains an ordinary contribution from these order-three characters, regardless of theta intersections among its primitive characters. The ordinary family off the hypersurface of A.C2.1 has the same nonzero-three-torsion obstruction.

Nor does a single theta intersection give a quantitative small-2-rank tower. Formula (C2.16) gives

$$\frac{f(D)}{n} = \frac{f(C)}{n} + \frac{1}{n} \sum_{O \text{ disjoint from } \Theta} |O|.$$

To make this ratio small, theta intersections must eliminate almost all characters. Density alone supplies no such proportion. No assertion that arbitrary curves admit these covers, or that normalized 2-rank tends to zero in suitable prime-to-two towers, is used in the positive conclusions. □

1.10 A weaker criterion from the deck action on two-primary torsion

A.C2.7 (Proved). *Let C be a smooth proper geometrically connected curve of genus at least two over a field K finitely generated over $\mathbb{F}_2$. Suppose that, over $\overline{K}$, it admits a connected nontrivial finite étale Galois cover $D \to C$ of odd degree, with group G, for which*

$$G \longrightarrow \mathrm{Aut}_{\mathbb{F}_2}(J_D[2](\overline{K})) \tag{C2.19}$$

is not faithful. Then the geometrically prime-to-two point-section map of C is injective. The conclusion also holds for its finite-extension forms and hyperbolic open subcurves.

More precisely, a pair of distinct points giving equal section classes would force (C2.19) to be faithful for every such cover. If $m_2(G)$ denotes the least dimension of a faithful representation of G over $\mathbb{F}_2$, the same hypothetical pair would force

$$f(D) - f(C) \geq m_2(G). \tag{C2.20}$$

In particular, a single such cover with $f(D) = f(C)$ suffices for point-section injectivity.

Proof. Work first over an algebraic closure. The curve D is not hyperelliptic. Otherwise its hyperelliptic involution is unique, so every deck transformation descends to its genus-zero quotient. Choose an element $\sigma \in G$ of odd prime order. Its image on $\mathbf{P}^1$ has the same order, since the kernel of descent is the order-two hyperelliptic group. A finite odd-order automorphism of $\mathbf{P}^1$ in characteristic two has a fixed geometric point. Its fibre on D has at most two geometric points, so σ fixes a point in that fibre. This contradicts the free deck action of an étale Galois cover.

Let $P', Q' \in D(\overline{K})$ lie over distinct points of C. The class $T = [Q' - P']$ has trivial stabilizer in G. Indeed, $\delta T = T$ for $1 \neq \delta \in G$ would give

$$\delta Q' + P' \sim Q' + \delta P'. \tag{C2.21}$$

The two effective degree-two divisors in (C2.21) are distinct: their terms lying over each of the two base points would otherwise force both $\delta P' = P'$ and $\delta Q' = Q'$. A linear equivalence between distinct effective degree-two divisors on a curve of genus at least two gives a degree-two map to $\mathbf{P}^1$, after removing any common base point. A degree-one map is excluded by the genus. Thus (C2.21) would make D hyperelliptic, a contradiction.

The étale two-adic Tate module of J_D is free of rank $f(D)$, and its geometric two-primary torsion is the quotient of its rational Tate module by this lattice. An odd-order automorphism acting trivially on $J_D[2](\overline{K})$ acts trivially on the whole two-adic module. This can be checked without any torsion-free congruence-group citation: if a matrix A of odd order m satisfies $A \equiv 1 \pmod{2^r}$, then

$$A^m \equiv 1 + m(A-1) \pmod{2^{r+1}}$$

implies $A \equiv 1 \pmod{2^{r+1}}$. Starting at $r = 1$ and iterating proves $A = 1$. Thus any element in the kernel of (C2.19) fixes every two-primary torsion point of J_D.

Now suppose that distinct $P, Q \in C(K)$ give equal section classes. Descend the cover and its deck transformations after a finite extension of K, and extend again to make a chosen lift P' rational. By A.3 there is a lift Q' with $T = [Q' - P']$ two-primary. A nonidentity element in the kernel of (C2.19) would fix T, contrary to its trivial stabilizer. This proves the criterion

and the necessity of faithfulness. The same argument after restriction proves the stated form and open-subcurve variants.

For the quantitative assertion, the averaging idempotent $|G|^{-1}\sum_{g\in G} g$ acts integrally on the two-adic Tate module, since $|G|$ is odd. Its invariant summand is the pullback of the two-adic Tate module of J_C: pullback followed by norm is multiplication by $|G|$, and norm followed by pullback is the sum of deck transformations. This summand has rank $f(C)$. The complementary summand has rank $f(D) - f(C)$, and its reduction modulo two must be a faithful G-representation because the invariant summand is trivial. This proves (C2.20). □

A.C2.8 (Proved). *In the genus-two cyclic situation of A.C2.5, let S be the union of the nonzero doubling orbits that miss Θ. The kernel of the deck action on geometric two-primary torsion has order*

$$d = \gcd\big(n, \{i : i \in S\}\big), \tag{C2.22}$$

where $d = n$ if S is empty. Consequently $d > 1$ already implies point-section injectivity by A.C2.7.

In particular, if $n = \ell^a$ is an odd prime power, it suffices that every doubling orbit on $(\mathbf{Z}/n)^\times$ meets $\{i : L^i \in \Theta\}$. No condition is needed on characters of smaller order. The simple sufficient condition is

$$L \in \Theta \text{ of exact order } \ell^a, \qquad \langle 2, -1\rangle = (\mathbf{Z}/\ell^a)^\times. \tag{C2.23}$$

For a cyclic group of odd prime-power order ℓ^a one has $m_2(G) = \mathrm{ord}_{\ell^a}(2)$.

Proof. The Artin–Schreier description of étale H^1 identifies its scalar extension with the stable Frobenius part of $H^1(D, \mathcal{O}_D)$. The calculation of A.C2.5 therefore shows that its nontrivial cyclic characters are exactly those indexed by S, with multiplicity one. Dualizing to geometric two-torsion may invert all characters; it does not change their common kernel. A power δ^k acts trivially exactly when $n \mid ik$ for every $i \in S$. The number of such powers is the greatest common divisor in (C2.22).

If $n = \ell^a$, this kernel is nontrivial exactly when no primitive character survives, since a primitive index is coprime to n, whereas all nonprimitive indices are divisible by ℓ. Symmetry of Θ gives (C2.23) as in A.C2.5. For example at $n = 9$, a theta point of exact order nine kills the single orbit of all six primitive characters. The order-three block may remain ordinary, but the deck action then has a kernel of order three, which suffices here. This does not assert existence of such a nine-torsion theta point on every curve.

Finally, an irreducible $\mathbb{F}_2$-representation of a cyclic group corresponds to a doubling orbit of its geometric characters. A faithful character of order ℓ^a has orbit length $\mathrm{ord}_{\ell^a}(2)$. Any faithful representation of this cyclic prime-power group must contain such a character: otherwise all character orders divide ℓ^{a-1} and a subgroup of order ℓ acts trivially. One faithful orbit already gives a faithful representation, proving the formula for $m_2(G)$. □

The full-quotient obstruction in A.C2.6 and the weaker deck criterion are compatible. Surviving order-three characters prevent the entire Prym from having 2-rank zero, but at degree 3^a, $a > 1$, they need not make the deck action faithful. Torsion density still does not guarantee the prime-power order or simultaneous primitive-orbit intersections required by (C2.23). A degree-sized lower bound in place of (C2.20), or a construction forcing a sufficiently small normalized 2-rank in general towers, has not been established here.

A.26 (Proved; a nonabelian symmetric-group quotient class). *Let K be finitely generated over $\mathbb{F}_p$, with $p \nmid 6$. Let C/K be smooth proper geometrically connected of genus at least two. Suppose that $C_{\overline{K}}$ has a faithful S_3-action with genus-zero quotient, and that the quotient by a subgroup generated by a transposition has genus at most two. Then A1 holds for C and every hyperbolic open subcurve. The three transposition quotients are isomorphic, so the genus condition is independent of the selected subgroup.*

Proof. We first work over a finite field after a finite constant extension defining the action. Put $N = A_3$ and $B = C/N$. It is a double cover of the genus-zero final quotient. If $g(B) = 0$, then C is a cyclic cubic cover of a genus-zero curve and A.23 applies. If $g(B) \geq 2$, it is hyperelliptic and A.19 shows that any collision on C has equal B-images. If $g(B) = 1$, the cyclic cubic map $C \to B$ is ramified because $g(C) \geq 2$, so A.21 gives the same equality.

Let $H_s = C/\langle s \rangle$ for each transposition s. If their common genus is zero, C is hyperelliptic and A.19 settles the result. If it is one, either A.25 settles C directly or A.21 gives equal H_s-images. If it is two, A.19 gives equal H_s-images. Thus in the remaining cases a collision has equal images on B and all three curves H_s.

These maps jointly distinguish every geometric point. The two points have the same final base value, so write $Q = gP$ for $g \in S_3$, and let I be the stabilizer of P. Tameness makes I cyclic; it is trivial, a transposition subgroup, or N. If $I = 1$, equal images in B and one H_s force $g \in N \cap \langle s \rangle = 1$. If $I = \langle s \rangle$, its own quotient has fibre $\langle s \rangle P = \{P\}$ above the image of P, so equality in H_s gives $Q = P$. If $I = N$, the same argument applies to the fibre of $C \to B$. Hence no nontrivial collision exists. Spreading the tame action and specializing by A.8 gives the assertion over K; functoriality gives the open assertion.

For a concrete branch pattern in this class, let a be the number of transposition branch points and b the number of three-cycle branch points. Tame Riemann–Hurwitz in the regular and degree-three permutation actions gives

$$g(C) = -5 + \frac{3a}{2} + 2b, \qquad g(B) = \frac{a-2}{2}, \qquad g(H_s) = -2 + \frac{a}{2} + b.$$

The pattern $(a, b) = (6, 1)$ therefore has genus six, with $g(B) = g(H_s) = 2$. It exists over a finite extension of every $\mathbb{F}_p$ with $p \nmid 6$: the branch-cycle tuple

$$(12), (12), (12), (12), (12), (23), (132)$$

has product one and generates S_3. The tame prime-to-p presentation in SGA 1, XIII, Corollary 2.12 realizes it on a seven-punctured projective line, after choosing distinct branch points over $\overline{\mathbb{F}}_p$. The resulting finite cover and its smooth compactification descend to a finite constant field. No claim about the full automorphism group of these examples is needed. □

A.27 (Proved; abelian covers of genus one with exponent dividing 120). *Let K be finitely generated over $\mathbb{F}_p$, with $p \neq 2$. Let C/K be smooth proper geometrically connected of genus at least two. Suppose that $C_{\overline{K}}$ has a faithful action of a finite abelian group G of order prime to p, with genus-one quotient, and that the exponent of G divides 120. Then A1 holds for C and every hyperbolic open subcurve. In particular this treats every cyclic cover of a genus-one curve of degree $3, 4, 5$, or 8 prime to p, with arbitrary allowed partial ramification in degrees four and eight. For $p = 3$ or 5 the prime-to-p hypothesis excludes the corresponding primary factors of G.*

The independent branch-cancellation proof below is retained as an earlier route. Its conclusion is subsumed by A.29, which removes the exponent bound by a different divisor construction.

Proof. We work first over finite fields, making finite constant extensions to define all data. There are six steps.

Two fully ramified branch points. Let $B \to E$ be a cyclic cover of degree $n \in \{3, 4, 5\}$, with $p \nmid n$, branched at exactly two points A, B_0, with full index n at both. The local Kummer exponents are opposite units modulo n. Choose an elliptic inversion ι exchanging A and B_0; its origin is a half of their sum. Write the Kummer equation and its divisor as

$$z^n = c, \qquad \operatorname{div}(c) = nD + u(A - B_0), \qquad \deg D = 0, \quad \gcd(u, n) = 1.$$

Since ι acts as minus one on $\operatorname{Pic}^0(E)$, the divisor $D + \iota^* D$ is principal. After adjusting a constant choose h with

$$\operatorname{div}(h) = D + \iota^* D, \qquad c\,\iota^* c = h^n.$$

Thus ι lifts to $\tau(z) = h/z$. If n is odd, the constant $\iota^* h/h$ has both order dividing n and order dividing two, so it is one and $\tau^2 = 1$. For $n = 4$ this constant is a priori 1 or -1. At any of the four fixed points T of ι, the branch pair is absent, so $v_T(c)$ is divisible by four and

$$v_T(h) = 2v_T(c)/4$$

is even. A local parameter for the tame involution has $t \mapsto -t$; an anti-invariant rational function has odd leading valuation. Therefore $\iota^* h = -h$ is impossible, and again $\tau^2 = 1$. If δ generates the cyclic deck group, then $\tau\delta\tau = \delta^{-1}$.

Riemann–Hurwitz gives $g(B) = n$. For odd n, each of the four unramified cyclic fibres above a fixed point of ι has exactly one τ-fixed point: on a cyclic torsor of odd order an inversion has one fixed point. Hence the quotient $H = B/\langle\tau\rangle$ has genus $(n-1)/2$. For $n = 4$ such a fibre has zero or two fixed points. The total fixed count r_τ is between zero and eight, and tame Riemann–Hurwitz reads

$$6 = 2(2g(H) - 2) + r_\tau.$$

Thus r_τ is two or six and $g(H)$ is two or one. In all cases the reflection quotient has genus one or two. When its genus is two, A.19 gives equal H-images of a collision on B. When its genus is one, the degree-two map $B \to H$ is ramified, so A.21 gives the same conclusion. The same applies to the quotient map composed with δ.

A.21 applied to $B \to E$ also gives equal E-images. A distinct pair with equal E-images lies away from A, B_0, where the cyclic action is free. Equal images under the two reflection quotients would give

$$Q = \tau P = \delta^{-1}\tau\delta P = \delta^{-2}\tau P,$$

so $\delta^2 Q = Q$. This is impossible on that locus, since δ^2 is nontrivial for $n = 3, 4, 5$. The two full branch fibres themselves have one geometric point each. This proves A1 for every such two-branch cover.

Arbitrary cyclic degrees three and five. Let $C \to E$ have prime degree $\ell = 3$ or 5. Its nonempty branch set has cardinality $r \geq 2$: the sum of its nonzero Kummer residues is zero modulo ℓ. The case $r = 2$ is the preceding step. For $r \geq 3$, choose two branch points a_1, a_2 and a cyclic degree-ℓ cover $B \to E$ branched exactly there, with exponents $u, -u$, where u matches that of C at a_1. Such a cover exists by divisibility of $\operatorname{Pic}^0(E)$ by ℓ; its character can be represented by a rational function with divisor $u(a_1 - a_2) + \ell D$ for a degree-zero divisor D. It has the two-branch injectivity just proved.

The two prime-degree fields are distinct since their branch sets differ. Their normalized fibre product V is a connected cyclic étale ℓ-cover of C. The ratio Kummer character defines a third quotient $H_2 \to E$, whose branch set loses a_1 and possibly a_2, with every other branch unchanged. It remains a connected degree-ℓ cover and has fewer branch points, so induction

gives its injectivity. A collision on C has equal E-images by A.21. If the points are distinct, this common image avoids the entire branch set, since every branch fibre of C is total. Lift the collision to V. Its images on B and H_2 agree. Their fibre product is étale over E at the common base value and is therefore already normal there. It agrees with V at both points, so those points coincide. This proves the branch-count induction. A putative one-point branch set on H_2 is excluded by the same sum-of-residues relation; the inductive endpoint is again a two-branch cover.

Arbitrary cyclic degree four. Let $C \to E$ have degree four, and let $C_0 \to E$ be its quadratic quotient. The branch points of C_0 are exactly the full-index-four branch points of C. If there are none, C_0 is a connected unramified quadratic cover of E, hence has genus one. The double cover $C \to C_0$ is covered by A.25.

Otherwise the number of full-index points is positive and even. Choose two of them, a_1, a_2, and a cyclic degree-four cover $B \to E$ with exactly these branch points, matching the exponent of C at a_1 and using the opposite exponent at a_2. We can and do choose its quadratic quotient different from C_0. To justify this choice even when C_0 has only these two branch points, twist the Kummer character of B by an unramified cyclic character of order dividing four. The reduction of these characters modulo two exhausts the four unramified quadratic characters. Explicitly, functions with divisor $4(T - O)$, for $[T - O] \in E[4]$, give the characters, and their quadratic reductions give $2[T-O] \in E[2]$. Multiplication by two from $E[4]$ to $E[2]$ is surjective. At most one of these four quadratic twists equals C_0. The twist changes none of the branch indices or residues of B.

Distinct quadratic subfields make the cyclic degree-four fields of C and B linearly disjoint. Their normalization V is consequently a connected cyclic étale degree-four cover of C. The ratio character $h = c/b$ has exact order four: the change $(c, b) \mapsto (b, c/b)$ is an invertible change of basis in the character group $(\mathbb{Z}/4)^2$. Let H_2 be its degree-four quotient over E. The branch at a_1 disappears. At a_2 the two odd exponents add to an even exponent, so it is no longer a full-index point; it may have index two or be unramified. Every other full-index point is unchanged. Thus their number drops by two, allowing induction on this number, with the unramified-quadratic case already supplying its base.

More precisely, a collision on C has equal E-images by A.21; a distinct pair avoids a_1, a_2, whose fibres are total. Lift to V. Its B-images agree by the two-branch result. If $g(H_2) \geq 2$, induction gives equal H_2-images. If $g(H_2) = 1$, the map $V \to H_2$ is fully ramified of index four over every point above a_1, since H_2 is unramified there and B has index four. A.21 gives the same equality. Genus zero cannot occur because H_2 maps separably to E. Away from a_1, a_2, $B \to E$ is étale, so $B \times_E H_2$ is already smooth and normal and equals V. Its two projections therefore separate the lifted points. This proves the degree-four assertion, including all partial ramification.

Cyclic degree eight. First let $B_8 \to E$ have exactly two full-index branch points. The Kummer/Picard argument in the first step again lifts inversion exchanging them to an involution τ: in the notation there, $v_T(h) = 2v_T(c)/8$ is even at each inversion-fixed point, ruling out the anti-invariant root. Now $g(B_8) = 8$, so the fixed count of any reflection is two or six. The two reflection classes have fixed counts summing to eight. Indeed on each of the four fixed elliptic fibres the reflections act on $\mathbb{Z}/8$ as $j \mapsto a - j$ and $j \mapsto a + 1 - j$; exactly one of these has two fixed elements. Choose τ with fixed count six. Its quotient H_8 has genus three.

The central involution δ^4 commutes with τ and descends to a nontrivial involution on H_8. Its quotient is the reflection quotient H_4 of the cyclic degree-four cover $B_4 = B_8/\langle\delta^4\rangle$. At an inversion-fixed elliptic point, the affine reflection has fixed elements modulo eight exactly when a is even, and the same is true modulo four. Each fixed count is two in that case. Thus the induced reflection on B_4 also has total fixed count six, and H_4 has genus one. Consequently $H_8 \to H_4$ is a double cover of a genus-one curve, and A.25 proves point-section injectivity for H_8. Use this target for two deck-conjugate reflection maps from B_8. Together with the equal

elliptic images from A.21, the same δ^2 argument as in the first step proves injectivity for B_8.

For an arbitrary cyclic degree-eight $C \to E$, induct on its number of full-index-eight branch points. If there are none, its quadratic quotient $C_0 \to E$ is unramified and has genus one; the degree-four map $C \to C_0$ is covered by the preceding step. Otherwise choose two full-index branch points and a two-branch cyclic degree-eight auxiliary cover, matching the first residue and using the opposite residue at the second. Twisting by unramified degree-eight characters makes its quadratic quotient different from C_0, since $[4] : E[8] \to E[2]$ is surjective. Distinct quadratic subfields imply disjointness of the cyclic degree-eight fields. The normalized fibre product is a connected cyclic étale degree-eight cover of C, and its ratio quotient $H_2 \to E$ still has degree eight. The number of full-index points drops by two, although the second selected point may retain index two or four.

The rest of the degree-four proof applies with eight in place of four. A.21 gives equal elliptic images; a distinct pair avoids the two selected full-index fibres. The auxiliary images agree by the two-branch result. The ratio images agree by induction when its genus is at least two, or by A.21 when its genus is one, because the map from the common cover is fully ramified of index eight above the canceled first point. Away from the two selected points the auxiliary cover is étale, so the final fibre product is already normal and its projections separate the lifted pair.

Cyclic degrees dividing 120. Degrees two, three, four, five and eight are now settled by A.25 and the preceding steps. If $n \mid 120$ has at least two distinct prime factors, split it as $n = ab$ with $a, b > 1$ coprime and use its two corresponding quotient curves over E. Each has genus at least one. A quotient of genus at least two is settled by induction on the number of distinct prime factors; a genus-one quotient has equal images by A.21, since the Galois map from C is ramified. The two quotient maps jointly separate geometric points because their local tame indices are coprime, by the normalization argument in A.23.

The abelian group and field extensions. For a finite abelian G as in the statement, each nontrivial character quotient is a cyclic cover of E of degree dividing 120. If its genus is at least two, the preceding step gives equal images of a collision; if it is one, A.21 gives the equality. There is no genus-zero quotient above E. With $Q = gP$ and $I = \mathrm{Stab}(P)$, these equalities give $g \in (\ker \chi)I$ for every character χ. Characters of the abelian group G/I separate points, so $g \in I$ and $P = Q$, including at every ramified fibre. Finally spread the tame action and its genus-one quotient and apply A.8 to obtain the result over K. Functoriality to the proper curve gives the open case. □

1.11 A finite obstruction to the configuration-space exactness route

A.28 (Refuted). *The proposed universal left exactness of the geometrically prime-to-p configuration-space homotopy sequence is false. More precisely, let $C/\mathbb{F}_5$ be the smooth proper genus-two curve with affine equation*

$$y^2 = x^5 - x + 1,$$

let $P = \infty$, and put $U = (C \times C) \setminus \mathrm{diag}(C)$. Then the natural map

$$\pi_1((C \setminus \{P\})_{\overline{\mathbb{F}}_5})^{(5')} \longrightarrow \pi_1(U_{\overline{\mathbb{F}}_5})^{(5')}$$

is not injective. This refutes the sufficient topological condition considered in Saïdi and Tamagawa (2009, Remark 3.13); it does not refute point-section injectivity, which holds for this C by A.19.

Proof. We first work with the topological two-point configuration space of an oriented closed genus-two surface. Write B for its pure braid group and $F \triangleleft B$ for the kernel of the projection

that forgets the second point. The group F is free on four generators a, b, d, e. Throughout this proof set

$$[u, v] = u^{-1}v^{-1}uv, \qquad r = ([a^{-1}, b][d^{-1}, e])^{-1}.$$

Thus r is the peripheral element. We use the corrected presentation of Bardakov and Bellingeri (2009, Proposition 4, preprint p. 5). Its three point-pushing conjugations A_1, B_2, A_2 restrict to the following automorphisms α, β, γ of F:

	a	b	d	e
α	$r^{-1}ar$	$r^{-1}b[a, r]$	$[r, a]d[r, a]^{-1}$	$[r, a]e[r, a]^{-1}$
β	a	b	dr	$r^{-1}er$
γ	a	b	$r^{-1}dr$	$r^{-1}e[d, r]$.

The source defines its commutator and its left-conjugation notation explicitly. It also corrects the earlier exceptional-relation misprints in Bellingeri's presentation. The certificate below checks these formulas by free reduction, including explicit inverse automorphisms and the base surface relation.

Represent a homomorphism $F \to S_4$ by the tuple of its values on (a, b, d, e), and compose permutations right to left. Let Φ be the operation on such tuples obtained by successive substitution using α, then β, then γ. This is the action induced by the element $w = A_1B_2A_2$ of B. The following small table is an exact certificate. In its last column h_i satisfies $h_i\Phi(T_i)h_i^{-1} = T_{i+1}$, with subscripts modulo five:

i	a	b	d	e	h_i
0	(34)	(234)	(134)	(143)	(24)
1	(34)	(234)	(123)	(234)	(13)
2	(34)	(234)	(123)	(143)	(234)
3	(34)	(123)	(234)	(123)	(243)
4	(34)	(234)	(134)	(124)	(13).

Each equality is a multiplication of permutations of four letters. The first tuple generates S_4: its first two entries generate the stabilizer of 1, and its third entry moves 1. Its peripheral image is (234).

The five tuples belong to different simultaneous-conjugacy classes. For rows $0, 1, 2, 4$ the first pair is the same and has trivial common centralizer in S_4, while the last pair differs. The first pair of row 3 generates S_4, whereas each of the other four first pairs generates a subgroup of order six, so row 3 cannot be conjugate to one of them. Consequently the action of w on $\mathrm{Hom}(F, S_4)/S_4$ has a cycle of length exactly five.

Suppose, for a contradiction, that $\widehat{F}^{(5')} \to \widehat{B}^{(5')}$ were injective. An injective continuous map from a compact group to a Hausdorff group identifies its source with a closed subgroup. Therefore the topology on $\widehat{F}^{(5')}$ is the induced topology. Since $|S_4| = 24$, the homomorphism $\rho : F \to S_4$ specified by T_0 extends continuously to $\widehat{F}^{(5')}$. There is an open normal subgroup $N \triangleleft \widehat{B}^{(5')}$ whose inverse image in $\widehat{F}^{(5')}$ is contained in $\ker \rho$.

Let m be the order of the image of w in the finite group $\widehat{B}^{(5')}/N$. It is prime to five. For every $x \in F$, the commutator of w^m with x lies in $N \cap \widehat{F}^{(5')}$, so

$$\rho(w^m x w^{-m}) = \rho(x).$$

The w-orbit of the simultaneous-conjugacy class of ρ therefore has length dividing m. Its length is five by the displayed table, a contradiction. This proves the topological noninjectivity.

To transfer the calculation, take the smooth proper model of $y^2 = x^5 - x + 1$ over $\mathbb{Z}_5$, together with its infinity section. Its special fibre is smooth because the derivative of the right-hand side is -1 in characteristic five. At infinity the coordinates $u = 1/x$, $v = y/x^3$ give

$v^2 = u - u^5 + u^6$, whose derivative in u is -1 at $(u, v) = (0, 0)$. The product of this model with itself is smooth and proper, and its diagonal is a smooth relative Cartier divisor. Thus the configuration space and the punctured fibre are complements of relative normal-crossing divisors in smooth proper schemes. Prime-to-five specialization is an isomorphism for both, compatibly with the fibre-inclusion map, by Grothendieck and Raynaud (2003, Exposé XIII, Corollary 2.9 and §2.10). The characteristic-zero comparison with topological fundamental groups is Grothendieck and Raynaud (2003, Exposé XII, Corollary 5.2). Apply these comparisons to the commutative square formed by the fibre and total space. The noninjectivity just proved gives the asserted noninjectivity over $\overline{\mathbb{F}}_5$. □

The complete small certificate is `code/A_configuration_obstruction_certificate.py`, with its exact output in the adjacent JSON file. It uses only the Python standard library and verifies 34 free-word and finite-permutation identities. The proof above supplies the group-topological and specialization arguments; no inferred permutation-group order is needed. The larger exploratory A_5 calculation is retained only as an auxiliary calculation and is not a proof input.

This also explains the limitation of a pro-ℓ argument based on the trivial action on punctured first homology. Such an argument does not control finite quotients involving several primes. The relative homotopy theorem of Hoshi et al. (2024, Theorem C, Corollary 1.14 and Lemma 3.5) retains the full tame group of the base and treats a relative prime-to-characteristic quotient of the fibre. Its pro-ℓ comparison does not assert the left exactness refuted here.

Why retaining the full base does not complete the point argument. There is a precise obstruction to descending the relative extension. Write $\Pi = \pi_1(C_{\overline{\mathbb{F}}_5})$ and $H = \ker(\Pi \to \Pi^{(5')})$. The outer point-pushing action on the prime-to-five punctured fibre induces an action on the finite set $\mathrm{Hom}(F, S_4)/S_4$. To transport this action, apply SGA1 XIII, Corollary 2.9 to the punctured-fibre family over the smooth lifted base curve: since $|S_4| = 24$ is invertible, the sheaf $R^1 f_* S_4$ is finite locally constant and commutes with specialization. Thus full-base specialization carries the finite torsor action compatibly; no identification of the full characteristic-zero and characteristic-five base groups is being assumed. Let $\bar{w} \in \Pi$ be the specialization of the word used above. In the procyclic closure of $\bar{w}$, take its 5-primary component: equivalently use the exponent $\varepsilon \in \widehat{\mathbb{Z}}$ whose 5-adic component is 1 and whose other prime-adic components are zero. Then $\bar{w}^\varepsilon \in H$, but it still permutes the displayed five classes in a five-cycle. Thus H acts nontrivially through *outer* automorphisms of the prime-to-five fibre.

If the relative extension with this full prime-to-five fibre were the pullback of an extension over $\Pi^{(5')}$, its outer action would factor through $\Pi^{(5')}$, and H would act trivially on the same five classes. Hence such descent is impossible. In the arithmetic setting, the relative extension retaining the full base can be pulled back along a full point section. Equality of the prime-to-p images of two sections does not identify their full lifts, so it does not identify these relative pullbacks by the proposed argument.

This is a loss in the section comparison, not a claim that the abstract curve or relative group is undetermined by its arithmetic prime-to-p group: the finite-field anabelian theorem already reconstructs the curve. To finish this route one would need either a means of identifying the relevant full-base lift of an individual prime-to-p section without assuming point separation, or an anabelian reconstruction theorem for the strictly smaller fibre quotient surviving in the full prime-to-p configuration group. Neither input is established here. The ordinary punctured-curve theorem in Saïdi and Tamagawa (2009, Corollary 3.10) does not supply the latter input.

1.12 All tame abelian covers of elliptic curves in the stated characteristics

A.29 (Proved; abelian prime-to-p covers of genus one without an exponent bound). *Let K be finitely generated over $\mathbb{F}_p$, and let C/K be smooth proper geometrically connected of genus at least two. Suppose that $C_{\overline{K}}$ has a faithful action of a finite abelian group G of order prime to p, with genus-one quotient E. If p is odd, or if $p = 2$ and E is supersingular, then the geometrically prime-to-p point-section map of C is injective. The same holds for every hyperbolic open subcurve. There is no restriction on the exponent of G. In particular, the odd-characteristic conclusion subsumes A.25 and A.27.*

Proof. We first work over a finite field, allowing finite constant extensions to define all the finite geometric data used below. It suffices initially to treat a cyclic cover $C \to E$ of degree $n > 1$, with $(n, p) = 1$. It is ramified since $g(C) \geq 2$. Choose an origin on E and a Kummer equation

$$z^n = c \in \overline{k}(E)^*.$$

Set $N = 2n$ if p is odd and $N = n$ if $p = 2$. In the latter case n is odd. Pull back by the étale isogeny $[N] : E' \to E$, where E' denotes another copy of E, and choose a connected component C' of the normalization. Then $C' \to C$ is a geometrically prime-to-p Galois étale cover: the full pullback is an $E[N]$-torsor, and a component is a torsor under its stabilizer. It has genus at least two. The map $C' \to E'$ is cyclic of degree dividing n, and the chosen root z still satisfies

$$z^n = c' := [N]^* c.$$

No connectedness of the entire pullback is assumed.

Let a be the nonzero residue modulo n of the valuation of c at one of its branch points. The inverse image of this point is a coset of $E'[N] \simeq (\mathbb{Z}/N)^2$, and the valuation of c' has the same residue a at every point of that coset. At each such point the local ramification index of every component is

$$e = \frac{n}{\gcd(n, a)}. \tag{A.16}$$

This formula is valid even when the chosen component has degree smaller than n.

An explicit second-difference identity. For integers i put $q_i = i(i-1)/2$. This sequence is periodic modulo n with period N, because

$$q_{i+N} - q_i = Ni + \frac{N(N-1)}{2} \equiv 0 \pmod{n}.$$

Here $N = 2n$ always works, while $N = n$ works for odd n. Moreover $q_{i-1} + q_{i+1} - 2q_i = 1$. Therefore, on a cyclic row of N points with generator d,

$$\sum_{i \bmod N} \delta_{x+id} = \sum_{i \bmod N} q_i\big(\delta_{x+(i+1)d} + \delta_{x+(i-1)d} - 2\delta_{x+id}\big) \pmod{n}. \tag{A.17}$$

Choose a basis d, d_2 of $E'[N]$ and sum (A.17) over the N rows of each branch coset. Multiplying by its coefficient a expresses the full branch divisor of c' modulo n as a sum of symmetric triples, each coefficient being an integer multiple of the original coefficient a of its coset.

For a centre $x \in E'$ let $u_x : E' \to \mathbf{P}^1$ be the elliptic inversion quotient with origin x. The generator d has order $N > 2$, so the two points $x + d, x - d$ are distinct and are unramified for this double map. After scaling its coordinate, the rational function

$$h_x = u_x - u_x(x + d)$$

has exactly the divisor

$$\operatorname{div}(h_x) = (x + d) + (x - d) - 2(x). \tag{A.18}$$

In characteristic two the inversion map is still separable of degree two, and nonzero odd-torsion points are unramified, so the same divisor formula holds.

List the triples with their coefficients as (h_j, c_j), where c_j is a multiple of the coefficient a on its branch coset. Formula (A.17) gives

$$\operatorname{div}\left(c' \Big/ \prod_j h_j^{c_j}\right) \equiv 0 \pmod{n}. \tag{A.19}$$

Terms with $c_j \equiv 0 \pmod{n}$ may be omitted. For each remaining term, choose a root $w_j^n = h_j^{c_j}$ and let B_j/E' be its connected smooth cover. Put $r_j = \gcd(n, c_j)$ and $m_j = n/r_j > 1$. Since h_j has simple zeros, this cover has degree m_j. After rescaling constants and using Bézout's identity it has the equivalent equation

$$s_j^{m_j} = h_j. \tag{A.20}$$

Indeed, writing $c_j = r_j b_j$ with $\gcd(b_j, m_j) = 1$, a component has $w_j^{m_j} = \zeta h_j^{b_j}$; remove ζ by a constant rescaling and choose $ub_j + vm_j = 1$ to set $s_j = w_j^u h_j^v$.

The auxiliary curves separate their point sections. Suppose first that p is odd. In coordinates with origin the centre of the triple, write E' as $v^2 = R(u)$, where R is a separable cubic. If $h_j = u - b$, then b is not a branch value, because the zeros in (A.18) are unramified. Thus (A.20) gives

$$B_j: \quad v^2 = R(s_j^{m_j} + b).$$

The polynomial on the right is squarefree of degree $3m_j$: its three values before extracting m_j-th roots are nonzero, and $(m_j, p) = 1$. Hence B_j is hyperelliptic with

$$g(B_j) = \frac{3m_j - \gcd(m_j, 2)}{2} \geq 2.$$

A.19 supplies point-section injectivity for every B_j. In particular, no genus-one auxiliary exception is needed.

If $p = 2$ and E' is supersingular, it has a Weierstrass equation

$$v^2 + v = u^3 + Au + B$$

over the algebraic closure. For completeness, in a general characteristic-two Weierstrass equation, a nonzero coefficient a_1 gives a nonzero two-torsion point at $a_1 u + a_3 = 0$, using the formula for inversion. Supersingularity therefore forces $a_1 = 0$; smoothness forces $a_3 \neq 0$. Scaling and translating the u-coordinate gives the displayed form. After (A.20) the auxiliary curve is

$$v^2 + v = (s_j^{m_j} + b)^3 + A(s_j^{m_j} + b) + B.$$

It is an Artin–Schreier curve with one pole, of odd order $3m_j$. The differential calculation in A.21 applies verbatim to any polynomial of odd degree: if its genus is g, then $s_j^i\, ds_j$, $0 \leq i \leq g - 1$, form a basis of regular differentials, and Cartier kills even indices and sends an odd index i to $(i - 1)/2$. It is nilpotent. As in A.21, a nonzero geometric two-torsion class would produce a nonzero regular Cartier-fixed logarithmic differential. Thus $J_{B_j}(\bar{k})[2^\infty] = 0$. A.1 and the Abel map give point-section injectivity for B_j over every finite constant extension.

An étale common cover and an unramified residual character. Let V be the connected normalization in the compositum of the function fields of C' and all the B_j. It is an abelian Galois cover of E' of exponent dividing n. At any point in a branch coset with coefficient a, each auxiliary local exponent is either c_j, $-2c_j$, or zero, with $a \mid c_j$ as integers. Its ramification index therefore divides $n/\gcd(n, a)$. No auxiliary branches outside the branch cosets. Tame

cyclic extensions of the completed local field over an algebraically closed residue field have compositum ramification index the least common multiple of their indices. Consequently $V \to C'$ is étale, and the inertia order of V/E' at any point is exactly that of C'/E' there.

The cover $V \to C'$ is Galois of prime-to-p order. The composite $V \to C$ has prime-to-p geometric Galois closure: take all conjugates over the Galois cover $C' \to C$. Their compositum has group a subgroup of a finite product of the prime-to-p groups of V/C', extended by the prime-to-p group of C'/C. Equivalently it embeds in the corresponding finite wreath product. This checks the exact lifting hypothesis of A.3.

Inside $\overline{k}(V)$ put

$$w_H = z \Big/ \prod_j w_j.$$

By (A.19), the connected cover H/E' defined by w_H is unramified; its degree divides n, so H has genus one. This also includes the degree-one case. Since C'/E' has a nonempty branch set, choose a branch point t. The cover H/E' is unramified there, while V/E' has the nontrivial prime-to-p index e of C'/E'. Thus all indices of $V \to H$ above any point of H over t equal e. A.21 shows that every collision on V has equal images on H.

Inertia retains exactly the point information required. Suppose P, Q is a collision on C. By A.3 and the checked Galois-closure condition, after a finite constant extension it has compatible lifts P_V, Q_V forming a collision on V. Their images on every B_j agree by the injectivity proved above, and their images on H agree by A.21. In particular they have the same image t_0 on E'. Write $Q_V = gP_V$ with $g \in \Gamma := \mathrm{Gal}(V/E')$, and let $I \subset \Gamma$ be the inertia group at P_V.

Use the chosen Kummer roots to define characters $\chi_0, \chi_j, \chi_H : \Gamma \to \mu_n$ for z, w_j, w_H. They satisfy

$$\chi_0 = \chi_H \prod_j \chi_j. \tag{A.21}$$

Let e_0 be the ramification index of C'/E' at t_0, with $e_0 = 1$ allowed. Its inertia image is $\chi_0(I) = \mu_{e_0}$. Equal B_j-images mean $\chi_j(g) \in \chi_j(I)$; by the ramification divisibility,

$$\chi_j(I) \subseteq \mu_{e_0}.$$

The residual cover is unramified, so $\chi_H(I) = 1$. Equal H-images therefore give $\chi_H(g) = 1$. Equation (A.21) now yields $\chi_0(g) \in \mu_{e_0} = \chi_0(I)$. Equivalently $g \in (\ker \chi_0)I$, so the two points have the same image on C', and hence $P = Q$ on C. This argument includes every partially ramified fibre. It does not require the projections to the auxiliary curves to distinguish all points of the normalization V.

For a general finite abelian group G, take all its nontrivial cyclic character quotients C_χ. They map separably to E, so none has genus zero. If $g(C_\chi) \geq 2$, the cyclic theorem gives equal C_χ-images of a collision on C. If its genus is one, the Galois map $C \to C_\chi$ is ramified by Riemann–Hurwitz, and A.21 gives the same equality. These equalities imply the same E-image. Write $Q = gP$ and $I = \mathrm{Stab}_G(P)$. For every character, $g \in (\ker \chi)I$. Characters of G/I separate its elements, so $g \in I$ and $P = Q$.

Finally define the finite tame action and genus-one quotient over a finite extension of K, spread them over a smooth finite-type model, and apply A.8. In characteristic two the supersingular condition persists: the elliptic quotient has identically zero j-invariant. The resulting finite-field fibres therefore remain in the proved class. Specialization excludes a collision over K, and functoriality to the proper compactification proves the assertion for open subcurves. □

The exact second-difference, ramification-divisibility, genus, and inertia-subgroup calculations are independently reproducible with `code/A_elliptic_divisor_certificate.py`; its adjacent JSON output includes even and composite degrees and all partial branch

coefficients in the tested degrees. The displayed identities above prove these facts for all n; the finite computation is a check, not a replacement for them.

A.29a (Proved; an obstruction over an ordinary elliptic quotient in characteristic two). *Let $c \in \mathbb{F}_4$ satisfy $c^2 + c + 1 = 0$, let*

$$E: \quad y^2 + y = x + c/x,$$

and let B be the smooth proper curve with function field

$$B: \quad y^2 + y = s^3 + c + \frac{c}{s^3 + c}. \tag{A.22}$$

The elliptic curve E is ordinary, and $B \to E$, given by $x = s^3 + c$, is exactly a degree-three auxiliary of the construction in A.29, with its symmetric triple centred at the origin and its two zeros at nonzero three-torsion points. Nevertheless, over $\overline{\mathbb{F}}_2$ there is no nonconstant morphism from B to any abelian variety of 2-rank zero. Thus replacing each ordinary-elliptic auxiliary by a direct map to such an abelian variety cannot finish the missing characteristic-two case. This is not a counterexample to A1.

Proof. The Artin–Schreier map $E \to \mathbf{P}^1_x$ has two simple poles, at zero and infinity, so $g(E) = 1$. Its two ramification points differ by a nonzero two-torsion class, as $\operatorname{div}(x) = 2(P_0 - P_\infty)$. Thus E is ordinary. Set $Y = xy + c$. Then

$$Y^2 + xY = x^3 + c^2.$$

Choose d with $x(d) = c$, so $y(d)^2 + y(d) = c^2$. The tangent line at d is

$$Y = (y(d) + c^2)x + c^2.$$

Since $(y(d)+c^2)^2+(y(d)+c^2) = c$, substitution into the Weierstrass equation gives $(x+c)^3 = 0$. Hence d is a nonzero flex, and has exact order three. Equivalently, the three-division polynomial is $x^4 + x^3 + c^2$, which vanishes at $x = c$. Thus $s^3 = x + c$ is the auxiliary associated to $(d) + (-d) - 2(P_\infty)$.

In (A.22) there are three simple finite poles, at the distinct roots a_i of $s^3 + c$, and one pole of order three at infinity. Their different exponents are respectively $2, 2, 2, 4$. Riemann–Hurwitz gives $g(B) = 4$. More explicitly, ds has order two at each finite ramification point and order zero at infinity. The four regular differentials

$$ds, \quad \frac{ds}{s - a_1}, \quad \frac{ds}{s - a_2}, \quad \frac{ds}{s - a_3}$$

are independent, hence form a basis. Cartier kills ds and fixes the three logarithmic forms. Its stable rank is three, so B has 2-rank three. One can also read the lower bound from the three independent classes $[P_{a_i} - P_\infty]$ of order two: a principal combination with coefficients zero or one would have an invariant defining function under the Artin–Schreier involution, hence a function of s, whose valuation at every ramification point is even. The logarithmic differential injection used in A.21 gives the matching upper bound from the displayed Cartier-fixed space.

The following exact counts include the unique normalized point over each rational pole and the point at infinity:

r	1	2	3	4
$\lvert B(\mathbb{F}_{4^r})\rvert$	7	33	82	225.

The certificate `code/A_ordinary_elliptic_obstruction.py` verifies them in two ways: exhaustive enumeration of y for each nonpole s, and the Artin–Schreier trace criterion. It uses

explicit irreducible binary polynomials of degrees $2, 4, 6, 8$, whose irreducibility is itself checked by exhaustive trial division. Thus the counts are reproducible finite arithmetic. Newton's identities and the genus-four functional equation give the Frobenius characteristic polynomial (Milne, 2008, II, Theorem 1.1)

$$\begin{aligned} P(X) &= (X^2 - X + 4)\left(X^6 + 3X^5 + 9X^4 + 20X^3 + 36X^2 + 48X + 64\right) \\ &= X^8 + 2X^7 + 10X^6 + 23X^5 + 52X^4 + 92X^3 + 160X^2 + 128X + 256. \end{aligned} \tag{A.23}$$

The four power sums used here are $-2, -16, -17, 32$.

There is no root of (A.23) of the form 2ζ with ζ a root of unity. If such a root has order m, then $\varphi(m) \le 8$. Every prime divisor ℓ of m must satisfy $\ell - 1 \le 8$, so it is $2, 3, 5$, or 7; their exponents are at most $4, 2, 1, 1$, respectively. Filtering these possibilities by the totient condition gives exactly

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 15, 16, 18, 20, 24, 30.$$

For every member of this finite list the certificate performs the rational Euclidean algorithm and verifies

$$\gcd\big(P(2X), \Phi_m(X)\big) = 1.$$

It constructs each cyclotomic polynomial recursively from $X^m - 1$, so no numerical approximation or root recognition is involved.

If J_B had a nonzero quotient of 2-rank zero and dimension r, Poincaré reducibility and additivity of the 2-rank under isogeny would give $3 \le 4 - r$. Thus $r = 1$, and this quotient would be a supersingular elliptic curve. Over a finite extension of $\mathbb{F}_4$ it would be isomorphic to the supersingular curve in A.10, whose $\mathbb{F}_4$-Frobenius is $[-2]$. The corresponding quotient of the Tate module would force some root α of P to satisfy $\alpha^a = (-2)^a$ for an integer $a > 0$. This says that $\alpha/2$ is a root of unity, just excluded. Finally a nonconstant map from B to an abelian variety of 2-rank zero induces a positive-dimensional such quotient of J_B, after translating one point to the origin. No such map exists. □

A.29b (Proved; the obstruction survives the translated cubic refinement). *Use E, B, d from A.29a and put $h = x + c$, with $\operatorname{div}(h) = (d) + (2d) - 2(O)$. Let $\tau(t) = t + d$, and write $h_i = \tau^{i*}h$ for $i = 0, 1, 2$. Let V be the connected normalization obtained by adjoining cube roots of h_0 and h_1 to $\overline{k}(E)$. Then $V \to B$ is étale of degree three and $g(V) = 10$. It has a Heisenberg group of order 27 over $E/\langle d\rangle$, with central inertia of order three. Nevertheless, V also admits no nonconstant map to any abelian variety of 2-rank zero, even over $\overline{\mathbb{F}}_2$. Thus this particular first cubic refinement does not remove the obstruction in A.29a.*

Proof. In the order $(O, d, 2d)$ the three divisor vectors are

$$(-2, 1, 1), \qquad (1, 1, -2), \qquad (1, -2, 1). \tag{A.24}$$

Their sum is zero, so $h_0h_1h_2 = \kappa$ is a nonzero constant. Moreover

$$\operatorname{div}(h_1/h_0) = 3(O) - 3(2d).$$

The corresponding unramified cubic character is nontrivial: if h_1/h_0 were a cube, its cube root would have divisor $(O) - (2d)$, impossible since this is a nonzero order-three class on the elliptic curve. The classes of h_0, h_1 in $\overline{k}(E)^*/\overline{k}(E)^{*3}$ are independent. Indeed their branch residues force the sum of the coefficients of any relation to be zero modulo three, leaving a power of this nontrivial unramified character. Hence V/E has group

$$A = \{(\epsilon_0, \epsilon_1, \epsilon_2) \in \mu_3^3 : \epsilon_0\epsilon_1\epsilon_2 = 1\} \simeq (\mathbb{Z}/3)^2.$$

Choose $u_i^3 = h_i$ with $u_0u_1u_2 = \kappa^{1/3}$. At each of the three branch points, all residues in (A.24) equal one modulo three. Consequently the inertia subgroup is

$$Z = \{(\zeta, \zeta, \zeta) : \zeta \in \mu_3\} \subset A,$$

of order three, and there is no other ramification. The indices on V/E and B/E therefore agree at every point, proving that V/B is étale. Its degree is three, and $g(V) - 1 = 3(g(B) - 1) = 9$.

Translation τ lifts by cycling u_0, u_1, u_2. This lift has order three, since it restores each generator after three iterations and preserves their prescribed product. It cyclically permutes the three coordinates of A. Thus $V \to E/\langle d\rangle$ has group $A \rtimes \langle\tau\rangle$, of order 27 and exponent three. Its centre and commutator subgroup are both Z; these assertions follow directly from the permutation action on triples with coordinate sum zero modulo three. This is the Heisenberg group of order 27. Its inertia remains the central subgroup Z.

Let B_i be the quotient retaining u_i and the coordinate on E. Each B_i is isomorphic to B by translation. Its degree-three kernel $H_i \subset A$ is the subgroup with i-th coordinate one. Choose a generator z of Z and a generator b of one of the H_i. The three kernels are, in some order,

$$H_i = \langle bz^i\rangle, \qquad i = 0, 1, 2.$$

The quotient $E_2 = V/Z$ is an unramified degree-three cover of E, hence has genus one. The action of b on E_2 is a deck translation of this elliptic isogeny, and so acts trivially on J_{E_2}.

Write $N_H = \sum_{a\in H} a$ for the corresponding endomorphism of J_V. The equality of integral group-ring elements

$$\sum_{i=0}^{2} N_{H_i} = 3\,\mathrm{Id} + (b + b^2)N_Z$$

follows by expanding the three order-three subgroup sums. The image of N_Z lies in the pullback of J_{E_2}, because N_Z is pullback after the norm for $V \to E_2$. The translation observation therefore gives

$$\sum_{i=0}^{2} N_{H_i} = 3\,\mathrm{Id} + 2N_Z, \qquad N_Z^2 = 3N_Z$$

as endomorphisms of J_V. In particular,

$$\left(\sum_{i=0}^{2} N_{H_i}\right)(9\,\mathrm{Id} - 2N_Z) = 27\,\mathrm{Id}\,. \tag{A.25}$$

For $q_i : V \to B_i$ we have $q_i^* q_{i,*} = N_{H_i}$. Thus the image of the homomorphism

$$\Phi : \prod_{i=0}^{2} J_{B_i} \longrightarrow J_V, \qquad (a_i) \longmapsto \sum_i q_i^* a_i$$

contains the image of the left-hand side of (A.25). Multiplication by 27 is surjective on J_V, so Φ is surjective. This is the needed norm-image assertion; it is not inferred by counting dimensions.

If a nonconstant map from V to an abelian variety T of 2-rank zero existed, translating one value to the origin would give a nonzero homomorphism $J_V \to T$. Its composite with each q_i^* is zero by A.29a, since $B_i \simeq B$. Surjectivity of Φ would make the homomorphism zero, a contradiction. The finite group and integral group-ring calculations are verified in `code/A_cubic_refinement_certificate.py` and its adjacent JSON. The code separates the group-ring identity from the geometric translation relation $(b - 1)N_Z = 0$, which is justified above. No conclusion is drawn about other odd étale covers of B or V, or about A1 itself. □

A.29c (Proved; separation despite the direct-target obstruction). *For the ordinary-elliptic auxiliary*

$$B: \quad y^2 + y = s^3 + c + \frac{c}{s^3 + c}, \qquad c^2 + c + 1 = 0,$$

of A.29a, A1 holds over every finite extension of $\mathbb{F}_4$. Thus the absence of a nonconstant map from either B or its first translated refinement V to an abelian variety of 2-rank zero does not prevent point-section injectivity.

Proof. Put $\delta(s, y) = (cs, y)$ and $\iota(s, y) = (s, y + 1)$. These automorphisms commute. Write O_B for infinity and W_0, W_1, W_2 for the finite Weierstrass points, lying over $s^3 + c = 0$, and number them cyclically under δ. The δ-fixed points are O_B and the two points with $s = 0$. Under $B \to E$, all three W_i map to the nonzero two-torsion point T of E above $x = 0$.

Let

$$A = \big(\ker(\mathrm{Nm}_{B/E} : J_B \to E)\big)^0, \qquad \gamma(P) = [P - \delta P].$$

The image of γ lies in A: it is connected, has zero norm, and contains zero. The degree-three norm and pullback give an isogeny decomposition, so $\dim A = 3$ and the 2-rank of A is two. By A.29a its $\mathbb{F}_4$-Frobenius polynomial is

$$R(X) = X^6 + 3X^5 + 9X^4 + 20X^3 + 36X^2 + 48X + 64. \tag{A.26}$$

Two divisor facts. First, every nonzero value of γ has a unique geometric preimage. Indeed, $\gamma(P) = \gamma(Q)$ implies

$$P + \delta Q \sim Q + \delta P.$$

If these effective divisors are equal, either $P = Q$ or both points are δ-fixed. Otherwise they are two fibres of the hyperelliptic pencil. Consequently $\delta Q = \iota P$ and $\delta P = \iota Q$, which imply $\delta^2 P = P$, hence $\delta P = P$. Then $Q = \iota P$ is fixed as well. The zero fibre consists exactly of the δ-fixed points, by injectivity of the Abel map.

Second, for non-Weierstrass points P, Q,

$$\gamma(P) - \gamma(Q) \notin A[2](\overline{\mathbb{F}}_2) \setminus \{0\}. \tag{A.27}$$

The nonzero points of this two-dimensional $\mathbb{F}_2$-space are precisely $[W_i - W_j]$ for $i \neq j$. They lie in A because adjacent differences are $\gamma(W_i)$; the independent classes $[W_i - O_B]$ from A.29a show that their span has rank two. If (A.27) failed, we would have

$$P + \delta Q + W_j \sim Q + \delta P + W_i, \qquad i \neq j. \tag{A.28}$$

Each side has exactly one Weierstrass summand. If neither side contains a hyperelliptic pair, uniqueness of reduced divisors of degree three makes the divisors equal, which their different Weierstrass summands exclude. If exactly one contains such a pair, reduction gives degrees one and three, again impossible. If both contain pairs, they are $(P, \delta Q)$ and $(Q, \delta P)$; removing them would give $[W_j - O_B] = [W_i - O_B]$, also impossible.

Only the elementary degree-two and degree-three cases of hyperelliptic reduction are needed here. They can also be proved without an algorithm: a function of degree $d \leq 3$ outside the hyperelliptic rational subfield gives a birational curve of bidegree $(2, d)$ in $\mathbf{P}^1 \times \mathbf{P}^1$. Its arithmetic genus is $d - 1 \leq 2$, contradicting $g(B) = 4$. Thus every moving system of those degrees is the hyperelliptic pencil plus its fixed divisor. The quadratic hyperelliptic extension is separable in characteristic two, so this argument applies without changing characteristic.

The Frobenius bound. The norm relation gives $1 + \delta + \delta^2 = 0$ on A. The geometric étale Tate module $T_2 A$ has rank two over $\mathbb{Z}_2$ and is a torsion-free rank-one module over

$$\mathcal{O} = \mathbb{Z}_2[\zeta_3],$$

the unramified quadratic discrete valuation ring. It is therefore free of rank one. Since δ is defined over $\mathbb{F}_4$, Frobenius Φ commutes with this action and acts on that module by a unit $\lambda \in \mathcal{O}^*$. This module argument is essential; point counts alone would not establish the asserted module structure.

Exact determinants of powers of the integral companion matrix of (A.26) give

$$\begin{aligned} |A(\mathbb{F}_{4^3})| &= 291772 = 2^2 \cdot 72943, \\ |A(\mathbb{F}_{4^6})| &= 70459436736 = 2^6 \cdot 1100928699. \end{aligned} \tag{A.29}$$

The geometric 2-primary point group is $\mathrm{Frac}(\mathcal{O})/\mathcal{O}$. The kernel of $\Phi^r - 1$ on it has order $2^{2v_{\mathcal{O}}(\lambda^r-1)}$. Consequently

$$v_{\mathcal{O}}(\lambda^3 - 1) = 1, \qquad v_{\mathcal{O}}(\lambda^6 - 1) = 3. \tag{A.30}$$

Suppose $\gamma(P)$ has exact order 2^a, $a \geq 4$. For $\sigma = \Phi^{6\cdot 2^{a-4}}$ the identity $(1+v)^2 - 1 = 2v + v^2$ shows

$$v_{\mathcal{O}}\big(\lambda^{6\cdot 2^{a-4}} - 1\big) = a - 1.$$

Here one starts with $\lambda^6 = 1 + 8u$, u a unit; every subsequent squaring increases the valuation by exactly one. Hence $\sigma\gamma(P) - \gamma(P)$ is a nonzero point of $A[2]$. Since Weierstrass points have γ-value of order at most two, both P and σP are non-Weierstrass. This contradicts (A.27). Every 2-primary value of γ therefore has order at most eight. By (A.30), Φ^6 fixes $A[8]$; the unique-preimage assertion proves

$$0 \neq \gamma(P) \in A[2^\infty] \quad \Longrightarrow \quad P \in B(\mathbb{F}_{4096}). \tag{A.31}$$

No bound uniform in curves or covers is being assumed.

The complete finite certificate. Set $H = s^3 + c$ and $Y = Hy$. Then

$$Y^2 + H(s)Y = F(s), \qquad F = H^3 + cH = s^9 + cs^6 + s^3 + c. \tag{A.32}$$

Both independent programs use $\mathbb{F}_2[t]/(t^{12} + t^3 + 1)$ and verify that $t + 1$ has a cycle of all 4095 nonzero residues. This verifies that the quotient is a field; in binary coefficient encoding $c = 72$. Exhaustive enumeration gives 4206 points, including infinity. Because $\gamma(\delta P) = \delta\gamma(P)$ and $\gamma(\iota P) = -\gamma(P)$, one representative from each orbit under δ and ι suffices. There are 702 affine orbits, of which two are exceptional: the two fixed points with $s = 0$, and the three finite Weierstrass points. Their γ-orders are respectively one and two. Infinity is fixed as well.

Here is a direct linear-algebra certificate excluding $8\gamma(P) = 0$ on every other orbit. For $Q = \delta P$, the principal divisor of $(s - s(Q))^8$ is $8Q + 8\iota Q - 16O_B$. Thus $8(P - Q)$ is principal if and only if there is a nonzero element of $L(16O_B)$ vanishing to order at least eight at both P and ιQ. The pole orders of s and Y are two and nine. Riemann–Roch gives $\dim L(16O_B) = 13$, and its following 13 linearly independent elements are a basis:

$$1, s, s^2, \ldots, s^8, \quad Y, sY, s^2Y, s^3Y. \tag{A.33}$$

Independence also follows by separating the rational subfield from its Y-multiple. At either non-Weierstrass point, $s - s(P)$ is a uniformizer and $H(P) \neq 0$. Expanding (A.32) uniquely determines the first eight coefficients of Y recursively, since the coefficient of each new unknown is $H(P)$. The sixteen vanishing conditions on the basis (A.33) form a 16 by 13 matrix over $\mathbb{F}_{4096}$.

The self-contained program `code/root_A29c_linear_certificate.py` verifies the series equation coefficient by coefficient and performs exact Gaussian elimination. Every one of the 700 nonexceptional orbit matrices has rank 13. Its adjacent JSON records the ranks, point

count, and a hash of all matrix entries. A nonzero numerator with the specified zeros would have a pole of order exactly 16 at infinity: its required zero divisor already has degree 16. Thus the matrix criterion is an equivalence, including the reverse implication, and proves the desired exclusion.

An independent certificate, `code/A_genus4_prym_experiment.py`, computes the same packet by exact Mumford divisor arithmetic. Its composition and reduction formulas, with their ideal and principal-divisor proofs, are given in `audit/A_genus4_prym_certificate.md`. It verifies every division and curve congruence, compares 2103 Cantor doublings with independent Hensel doublings, and checks 702 inverse additions and 702 deck-cycle sums. It also computes (A.29) by exact integer determinants. The two methods agree. Together with (A.31), their conclusion is

$$0 \neq \gamma(P) \in A[2^\infty] \quad \Longrightarrow \quad P \in \{W_0, W_1, W_2\}. \tag{A.34}$$

The translated cover eliminates the packet. Suppose distinct P, Q have equal prime-to-two section classes over a finite extension. The ramified cyclic map $B \to E$ and A.21 force their E-images to agree. Thus $Q = \delta^j P$, $j = 1$ or 2, and A.1 makes $[P - Q]$ 2-primary. If $j = 2$, use $[P - \delta^2 P] = -\delta^2\gamma(P)$; in either case (A.34) puts P, Q in the finite Weierstrass triple. Their common image on E is T.

Lift this collision through the étale cubic cover $V \to B$ of A.29b, using A.3 and a finite constant extension. Its central inertia subgroup is Z and $E_2 = V/Z$ is elliptic. The cyclic cubic map $V \to E_2$ is ramified, with nine branch points, so A.21 forces the two lifts to have equal E_2-images. They differ by a nonidentity element $z^j \in Z$, since their images on B are distinct.

For $i = 0, 1, 2$, the quotient B_i retaining the root of $h_i = \tau^{i*}h$ is identified with B by the translation $t \mapsto t + id$ on E. This identification sends z to the same deck transformation δ. The two lifts therefore give a nontrivial δ-pair on each B_i with 2-primary divisor difference. The hyperelliptic involution on B_i covers $t \mapsto -t - 2id$; its Weierstrass points lie precisely over $-id$ and $T - id$. For $i = 1, 2$, neither base point equals T, because id is nonzero of order three and T has order two. Hence the images over the actual base point T are non-Weierstrass. This contradicts (A.34) under the indicated identification $B_i \simeq B$. The collision is impossible. □

1.13 What remains of A1 and A2

A.30 (Unresolved; exact remaining assertion). *For every finite field k of characteristic p, every smooth proper geometrically connected C/k of genus at least two, and distinct $P, Q \in C(k)$, the two classes of sections in $E_C^{(p')}$ are distinct. Equivalently, the conditions of A.3 force $P = Q$ in this range. The remaining cases include arbitrary curves outside the abelian prime-to-p genus-zero quotient class of A.24, the genus-one quotient class of A.29, the S_3 class of A.26, and the other cases proved above; in characteristic two, this includes hyperelliptic curves outside the cases proved above. No general proof or counterexample is supplied here. Thus A1 and the final implication in A2 are not finished in their full scope.*

The inspected result of Saïdi and Tamagawa (2009, Proposition 1.8(vi) and Corollary 1.11(iii)) is finiteness of an exceptional set, not its emptiness. Its proof uses a finiteness theorem for prime-supported torsion on the difference surface in the Jacobian; it supplies no sublinear bound as the curve varies in an étale tower. The refinement (Saïdi and Tamagawa, 2017) still retains exceptional sets. Our searches are recorded separately; we make no priority claim for A.2–A.4.

The tame-Belyi theorem of Kedlaya et al. (2023, Theorems 1.2 and 1.5) does not by itself reduce the remaining assertion to A.24. A tame map need not have prime-to-p Galois-closure order, and the map from its Galois closure to the original curve need not be étale. Even for a nonabelian prime-to-p Galois cover, cyclic character quotients only determine a point modulo

the commutator subgroup and its stabilizer. The argument of A.24 does not remove that remaining fibre ambiguity.

On scan 36 the phrase "de X" in the covering sentence has to be interpreted mathematically as a cover of Y, the curve containing P, Q: otherwise its fibres above these points are not defined. The visible wording was inspected in the supplied image; this is an editorial correction supported by the types of the maps, not an assertion that the ink reads Y. Likewise "p-torsion" has to mean p-primary torsion, with no uniform exponent asserted.

2 Part B. Limit actions and reduction

Unless a different hypothesis is stated, R is a henselian discrete valuation ring of residue characteristic zero, K is its fraction field, and the residue field k is finitely generated over $\mathbb{Q}$. We distinguish $E_{U_K} = \pi_1(U_K)$ from the fundamental group of a smooth model over R. This distinction is essential in the contact calculation below. The manuscript anchor for B0–B7 and B9–B11 is cote 149, manuscript pp. 49–51, scans 99–104, especially equations (30)–(31) and conditions a)–f). We use the author's page numbers at the top of recto pages; the sequential numbers at the lower left are a different pagination.

2.1 B0: comparison and the realization obstruction

B.1 (Proved). *For a torus S and an abelian variety A over a field k finitely generated over $\mathbb{Q}$, $\mathrm{Hom}_{G_k}(TS, TA) = 0$. Also $\mathrm{Hom}_{G_k}(V_\ell A, \mathbb{Q}_\ell) = 0$ for every prime ℓ.*

Proof. We may replace k by a finite extension splitting S: a map nonzero before restriction is still nonzero afterwards. A nonzero continuous map of full Tate modules has a nonzero ℓ-component for some ℓ; components with different residue primes have no nonzero continuous homomorphisms. Thus it would give a nonzero G_k-equivariant map $\mathbb{Z}_\ell(1) \to T_\ell A$. Choose a smooth integral model M of k over its algebraic constant number field, and shrink M until A extends to an abelian scheme. Both local systems factor through $\pi_1(M)$, and $G_k \twoheadrightarrow \pi_1(M)$ is surjective because a connected finite etale cover of normal M has connected generic fibre. Equivariance therefore holds for $\pi_1(M)$. At a closed point z of M, the pullback of the proposed map, after an etale path identifies geometric fibres, is still nonzero and is $G_{\kappa(z)}$-equivariant. The field $\kappa(z)$ is a number field. This contradicts Stix's Lemma 19, which states the torus–abelian variety vanishing over the number fields in question (Stix, 2008a, Lemma 19). The result invoked there is Ribet's Theorem 1: for an abelian variety over a number field, its torsion over the extension obtained by adjoining all roots of unity is finite (Katz and Lang, 1981, Appendix, Theorem 1). Indeed a nonzero image of $\mathbb{Z}_\ell(1)$ would give infinitely many torsion points over that extension. This verifies the precise implication from the opened theorem rather than treating an earlier AI return as authority.

For the second assertion the Weil pairing identifies $(V_\ell A)^*$ with $V_\ell A^\vee(-1)$. A map to $\mathbb{Q}_\ell$ is consequently a map $\mathbb{Q}_\ell(1) \to V_\ell A^\vee$; multiplication by an appropriate power of ℓ makes it integral. The first assertion excludes it. □

B.2 (Proved). *Let $1 \to \Pi \to E \to G \to 1$ be a continuous extension of profinite groups with Π topologically finitely generated and centre-free. Its outer action $\rho : G \to \mathrm{Out}(\Pi)$ determines the extension: the map*

$$E \longrightarrow \mathrm{Aut}(\Pi) \times_{\mathrm{Out}(\Pi)} G, \qquad e \longmapsto (\mathrm{conj}_e, \bar{e})$$

is an isomorphism of profinite extensions. This is the part of F149-C4 used in R6 and R11; it requires no 2026 theorem about relative curves.

Proof. The topology on $\mathrm{Aut}(\Pi)$ is the inverse-limit topology on its actions on finite characteristic quotients. Finite generation makes those quotients cofinal among finite quotients, so $\mathrm{Aut}(\Pi)$ is profinite. The displayed map is continuous. Its kernel is $Z(\Pi) = 1$. Given (a, g) on the right, lift g to $e \in E$. Equality of outer classes gives $a = \mathrm{int}(h)\,\mathrm{conj}_e$ for some $h \in \Pi$; hence (a, g) is the image of he. A continuous bijection between compact Hausdorff groups is a topological isomorphism. In the applications the geometric group is a hyperbolic surface group or a nonabelian free profinite group. Centre-freeness in both cases, including every nonempty prime set, is the explicit statement of Mochizuki (2007, Remark 1.1.3). □

B.3 (Known). *For a stable marked model in the stated characteristic-zero setting, its log geometric group is $\Pi_{\mathcal{G}} \simeq \Pi_U$. A geometric node of thickness m gives an injected edge subgroup $\Pi_e \simeq \widehat{\mathbb{Z}}(1)$. After an unramified extension defining the node and its branches, the node log group and its map to the base log group are*

$$T_P \rtimes G_k \longrightarrow T \rtimes G_k, \qquad (a, b, c, g) \longmapsto (c, g), \qquad T_P = \{(a, b, c) : a + b = mc\}.$$

Its geometric kernel is the antidiagonal $\{(a, -a, 0)\}$, not the whole rank-two node inertia group. Every residue splitting makes this antidiagonal essentially cyclotomically normalized.

Proof. Here are the comparison inputs, with their hypotheses. Illusie's Theorem 7.6 identifies finite Kummer etale covers of a log regular fs scheme with finite etale covers of its trivial-log open tamely ramified at the boundary's codimension-one points (Illusie, 2002, Theorem 7.6). The published proof is Mochizuki's Log Purity Theorem 3.3: normalization in such a cover is log regular and the map extends uniquely log etale (Mochizuki, 1999, Theorem 3.3). Hoshi explicitly checks that the constructed chart $P \subset P_Y \subset n^{-1}P$ makes this map Kummer (Hoshi, 2009, Remark 1.10). Full faithfulness follows because maps between normalizations are determined on their dense interiors and integral closure extends every map of their function fields. Hoshi's Corollary 3.9, in the numbering of RIMS-1558 (July 2006), identifies Kummer etale cover categories of a proper log regular fs scheme over a complete local ring and its strict closed log fibre. These statements apply to the curve with the divisorial log structure of its special fibre and marks. In residue characteristic zero all the relevant ramification is tame.

Completion does not alter the groups being compared. On absolute Galois groups the natural restriction $G_{\widehat{K}} \to G_K$ is an isomorphism: it is the identity on G_k, and on inertia it identifies the compatible Kummer roots of a uniformizer, giving the identity on T. The Kummer classification follows from SGA 1, XIII, Propositions 5.1–5.2: adjoining prime-to-residue-characteristic roots of normal-crossing parameters removes tame ramification. On geometric groups the base-extension map is an isomorphism in characteristic zero. The exact comparison used here is SGA 1, XIII, Proposition 4.3 (homotopy exactness for the stated locally 1-aspherical morphisms with a section), Examples 4.4 (smooth proper pairs with relative normal-crossing boundary), and 4.5.3 (base-change compatibility of the relative pro-Σ group) (Grothendieck, 2003). A geometric section is available after passing to the algebraically closed base; the arithmetic map then has isomorphic kernel and quotient. These maps transport the specified splitting, not merely its existence. The decisive wording in XIII, 4.5.3 is "canoniquement isomorphe" to the pulled-back relative group; in XIII, 5.2 the extension is "de manière unique à isomorphisme unique près". The regularity and normal-crossing hypotheses of 5.1–5.2 are applied to the trait parameter; the possibly nonregular node of thickness m uses log purity and the saturated chart below, not an illicit application of regular-scheme purity to that node.

The model is a stable log curve because its classifying morphism to the stack of stable marked curves pulls back the universal log curve; at a node the smoothing parameter maps to

a unit times ϖ^m. Locally its chart is

$$P = \langle X, Y, T_0 \mid X + Y = mT_0 \rangle.$$

This monoid is saturated: in $P^{\mathrm{gp}} = \mathbb{Z}^2$, with basis (X, T_0), its cone consists of the integral pairs (r, s) with $s \geq 0$ and $s + mr \geq 0$; if $r \geq 0$ use X, T_0, and if $r < 0$ use $Y = (-1, m), T_0$. Kummer characters are therefore $\mathrm{Hom}(P^{\mathrm{gp}}, T) = T_P$. Compatible roots satisfy $x^{1/n} y^{1/n} = (\varpi^{1/n})^m$; this proves the displayed homomorphism and its cyclotomic action coordinate by coordinate.

Mochizuki's Example 2.5 identifies the geometric admissible group with the semi-graph group (Mochizuki, 2007, Example 2.5). The admissible-cover description in Mochizuki (1996, §2, Definition 2.1) uses covers which become admissible after a tame base extension. In characteristic zero the same local equations are the Kummer charts above: on a normalized component a branch has the equation $z = w^n$, while the paired branch has opposite character. Matching those branches and the common root of the smoothing parameter is exactly both gluing conditions. Normalization and the full-faithfulness already proved identify the resulting categories, including their morphisms.

For the minimal deformation log point the node-group injection is Hoshi–Mochizuki's Lemma 2.11(ii); part (iii) gives $(a, b) \mapsto a + b$ (Hoshi and Mochizuki, 2011, Lemma 2.11). Pull back the extension along $T \to \bigoplus_e T$, $c \mapsto (m_e c)_e$, dual to $\bigoplus_e \mathbb{N} \to \mathbb{N}$, $(n_e) \mapsto \sum m_e n_e$. An injection remains an injection after a fibre product. Its kernel is the displayed antidiagonal and maps onto the edge-like subgroup. This proves the assertion for arbitrary thickness, without identifying the edge with the rank-two inertia.

Finally, the inverse image of $s(G_k)$ in $T_P \rtimes G_k$ is a closed subgroup which maps onto $s(G_k)$: lift c by $(mc, 0, c)$. Conjugation on its kernel is multiplication by $\chi(g)$, since T_P is abelian. Taking its image in the arithmetic curve group proves essential cyclotomic normalization for every s. No splitting of that inverse image was assumed. □

B.4 (Proved). *R10 and R11 hold with the node notation Π_e: a stable nodal fibre prevents E_s from being the extension of a smooth hyperbolic curve after any finite residue extension; for arbitrary U, realization of ρ_s over k, realization after a finite extension of k, and potentially good marked reduction are equivalent. Consequently B9 holds.*

Proof. Mochizuki's Proposition 1.2 makes distinct edge-like subgroups distinct up to geometric conjugation and commensurably terminal. In particular a procyclic overgroup normalizes an edge group and must equal it. Nodes and cusps thus give $\nu + n$ distinct maximal procyclic classes when there are n nodes. By B.1, Stix's Theorem 20 applies to a smooth curve over k: nontrivial essentially cyclotomically normalized procyclic subgroups are precisely subgroups of cusp inertia. His Proposition 11 and Corollary 13 give their normalizers and distinctness, and Corollary 21 gives precisely one maximal class per geometric cusp (Stix, 2008a).

We must exclude a smooth realizing curve of a different type. Let $t = b_1(\mathcal{G})$. The graphic quotient of $H_1(\Pi, \mathbb{Q}_\ell)$ has dimension t and becomes trivial after a finite residue extension, for every splitting: inertia acts trivially on the graph. This is also Proposition 2.4(i) of Mochizuki (2007). The homology of a smooth curve has an exact sequence

$$0 \to \mathbb{Q}_\ell(1)^{\max(\nu-1,0)} \to H_1 \to V_\ell J \to 0$$

after the cusps are split. B.1 and the infinite cyclotomic image show it has no trivial quotient. Hence $t > 0$ is impossible for a smooth realization. For a tree, $H^2(\Pi, \mathbb{Z}/\ell)$ distinguishes proper and affine: it is $\mathbb{Z}/\ell$ for the profinite genus-g surface group, and zero for the free profinite group of a punctured hyperbolic surface. Here is the required comparison. In the tower of all finite etale covers of a smooth proper hyperbolic curve over an algebraically closed characteristic-zero field, every $H^1(-, \mathbb{Z}/\ell)$ class dies on the cover representing its torsor. Every H^2 class dies

after a further cover of degree ℓ, since pullback multiplies the orientation class by the degree. Such a cover exists at every stage because the genus is at least two. Higher cohomology vanishes. The direct limit in positive degree is therefore zero, so the Cartan–Leray comparison identifies continuous group cohomology with the curve's etale cohomology. Poincare duality gives $H^2 = \mathbb{Z}/\ell$; this is also the surface duality of Mochizuki (2007, Proposition 1.3). For an affine hyperbolic curve SGA 1, XIII, Corollary 2.12 gives a free profinite group. It is projective: lift its free generators through any finite epimorphism. Hence its continuous H^2 with finite coefficients vanishes. These are intrinsic group invariants.

In the proper tree case a single node already contradicts Stix's theorem. In the affine tree case, eliminate leaf vertices in the boundary homology relations: each node class becomes the sum of the cusp classes on one side, leaving exactly the single relation among the ν cusp classes. Therefore

$$0 \to \mathbb{Q}_\ell(1)^{\nu-1} \to H_1(\Pi, \mathbb{Q}_\ell) \to \bigoplus_v V_\ell \operatorname{Jac}(C_v) \to 0.$$

By B.1, the dimension of $\operatorname{Hom}_{G_k}(\mathbb{Q}_\ell(1), H_1)$ is $\nu - 1$. A smooth realizing affine curve has the same dimension, hence the same number ν of cusps, contradicting the $\nu + n$ classes. This also proves the counting criterion in L5, including $\nu = 0$. A genuine point or cusp action has the outer action of its smooth curve; forgetting the lift proves B9 from this obstruction.

For completeness we supply the finite-extension comparison used in the potential statement. For a finite extension K_1/K inside the chosen separable closure put

$$H = s^{-1}(G_{K_1}), \qquad G_L = G_{K_1} \cap p^{-1}(H), \qquad p : G_K \to G_k.$$

Then H is open, $s(H) \subset G_L$, $p(G_L) = H$, and L/K is finite and contains K_1. Its residue field is the field fixed by H. Field base change gives continuous extension isomorphisms

$$E_{U_L} = E_U \times_{G_K} G_L, \qquad E_{U_L} \times_{G_L, s|_H} H = E_s \times_{G_k} H.$$

Thus a nodal stable model after finite extension obstructs a proposed smooth realization after restricting to this particular open subgroup. Stable models exist after finite extension by properness of the stable pointed moduli stack, and a node persists under every further field extension of the stable special fibre (Knudsen, 1983, Theorem 2.7).

Conversely take a finite Galois extension K_1/K giving good marked reduction, with group G and residue field k_1. Uniqueness of stable models extends every generic semilinear G-isomorphism to the model. Write λ_σ for the reduced semilinear isomorphism of its special pair (C_1, D_1). The map $\alpha_s : G_k \to G_K \to G$ is continuous and has finite image. Over the finite Galois residue extension fixed by $\ker \alpha_s$, the maps $\lambda_{\alpha_s(g)}$ give an actual descent datum. It is effective: a sufficiently high power of $\omega_{C_1}(D_1)$ embeds the pair, its section space is a finite-dimensional semilinear vector space, and ordinary Galois descent descends both its homogeneous ideal and the marked divisor. Smoothness and the etale divisor property descend faithfully flatly. Call the descended complement V_s.

The comparison is equivariant for $Q = G \times_{\mathrm{Gal}(k_1/k)} G_k$. The natural surjection $G_K \to Q$ has kernel I_{K_1}, which acts trivially by good reduction. Restriction along s is $g \mapsto (\alpha_s(g), g)$, exactly the descended action. Notice that the second coordinate retains the entire arithmetic action on geometric fundamental groups; this action is not asserted to factor through a finite group. B.2 now identifies the extensions. This proves all three equivalences for the specified s. □

2.2 B1–B2: twists, homology and weights

B.5 (Known). *For a split stable marked model, positive Kummer inertia acts by the profinite multi-twist $\prod_e \tau_e^{m_e}$, and its map from T is injective if a node exists. On homology it acts by $x \mapsto x + \sum_e m_e \langle x, \delta_e \rangle \delta_e$. Exchanging branches inverts the oriented edge generator; it does not insert an additional minus sign in the unoriented Dehn twist.*

Proof. The input is Hoshi and Mochizuki (2012, Theorem 4.8(iv)–(v), Lemma 5.4(ii)–(iii), Corollary 5.9): for a semi-graph of pro-Σ pointed-stable-curve type its Dehn group is the product of one cyclotome per node; the log deformation map has coordinates the lengths m_e; conjugation permutes those coordinates and multiplies them by the cyclotomic character. These statements hold for every nonempty set of primes excluding the residue characteristic. Here that is all primes. The chart $xy = \varpi^m$ of B.3 fixes the positive Kummer normalization; pulling a loop in the smoothing parameter back under $t_e = \varpi^{m_e}$ multiplies its exponent by m_e. The topological sign agrees with the plumbing convention $y = \varpi^m / x$. Across a separating edge it conjugates the second vertex group by $\gamma_e^{m_e}$; across an HNN edge it sends the stable letter u to $u\gamma_e^{m_e}$. These are identities on finite coverings and therefore pass to the profinite groups. The corresponding topological formula is also explicitly Theorem (3.2) of Asada et al. (1992).

Changing paths conjugates the construction by the geometric group and therefore does not change its outer class. Reversing an edge replaces its generator by the inverse and also reverses the crossing direction; the two changes cancel in the twist. If a node is present, the coordinate map $T \to T$, $a \mapsto m_e a$, is injective, proving inertia injectivity. On a crossing cycle the twist adds m_e copies of the vanishing circle; signed crossing number is $\langle x, \delta_e \rangle$. The circles are disjoint, so their pairwise intersections vanish and the contributions add. This derives the abelian formula from the actual group automorphism. In the nonsplit case Galois permutes nodes and multiplies an oriented edge by χ times the sign of its branch permutation. In the homology formula both occurrences of δ_e change sign, while in the Dehn-coordinate formula only χ and the node permutation remain. □

B.6 (Proved). *The filtration and dimensions in B2/L2 hold. In the nonsplit version the weight-zero piece is $H_1(\mathcal{G}, \mathbb{Q}_\ell)$ with its graph action, the weight-minus-one piece is $\bigoplus_v V_\ell \operatorname{Jac}(C_v)$ with component permutations and descent action, and the weight-minus-two piece is the module of boundary cycles of the cut surface, with branch signs and Tate twist. There is a natural exact sequence*

$$0 \longrightarrow (\mathbb{Q}_\ell[D(\bar{k})]/\mathbb{Q}_\ell)(1) \longrightarrow W_{-2} \longrightarrow H^1(\mathcal{G}, \mathbb{Q}_\ell)(1) \longrightarrow 0$$

when $\nu > 0$; for $\nu = 0$ the first term is zero. The phrase "node part" means the image of monodromy, a complement over $\mathbb{Q}_\ell$, rather than an unspecified canonical direct summand of the integral boundary lattice.

Proof. Cut the smooth surface at its vanishing circles and retain the actual punctures. Boundary homology of the resulting components, modulo opposite branch identifications and the sum-of-boundaries relation at every vertex, is W_{-2}. Filling those boundaries gives the homology of the normalized components; killing all vertex homology gives graph homology. The cellular chain complex of this decomposition proves the three successive quotients. Counting relations in the connected graph gives respectively $t + \max(\nu - 1, 0)$, $2\sum_v g_v = 2(g - t)$, and t. Every construction uses boundary or vertex maps of the semi-graph, hence is Galois-equivariant in the log comparison B.3.

Log inertia acts trivially on these quotients: edge twists change a crossing by a boundary class and act trivially on the normalized components. Consequently their G_k actions are independent of the residue splitting. For an oriented cycle $a = (a_e)$ in the graph the monodromy pairing is $q(a, b) = \sum_e m_e a_e b_e$. Its matrix on a cycle basis has positive determinant: over $\mathbb{R}$ it

is positive definite, since all $m_e > 0$. Its determinant is a nonzero integer, hence nonzero in every $\mathbb{Q}_\ell$. The Picard–Lefschetz formula of B.5 therefore identifies the graph quotient with the t-dimensional image of $N = \log \tau = \tau - 1$ in W_{-2}. The quotient of boundary cycles by the cusp subspace is the dual graph homology with Tate twist, giving the displayed exact sequence.

There is a genuine open-curve caution here. If "node part" were read as the span of all individual vanishing-cycle classes, that strengthened reading would be false. For $\mathbf{P}^1 \setminus \{0, t, 1, \infty\}$ the stable graph is a tree, so $t = 0$ and $N = 0$, but the separating node has nonzero homology class $a_0 + a_t$ in the cusp augmentation module. It is in the radical of the intersection form. Thus the isomorphism asserted for N is to its image, which projects isomorphically onto the dual graph quotient, and not to the span of every boundary circle called a node.

Finally choose a finite residue extension splitting the finite graph and marks and models of all normalized component Jacobians. Specialize those models at a common good finite-field point. Frobenius on the three graded pieces has complex absolute values 1, $q^{1/2}$ and q, respectively, with the convention of arithmetic Frobenius on homology. Thus their characteristic polynomials have disjoint roots. A shared constituent would give a common Frobenius eigenvalue on the corresponding subquotient, a contradiction. The same argument works after any prescribed finite extension of k. The statement is about constituents, not about splitting the possibly nontrivial extensions between the pieces. □

2.3 B3–B4: the exact abelian criterion

The margin of cote 149, manuscript p. 49, scan 100, says "je l'ai démontré, dans le contexte des VA, il y a belle lurette". The statement that the opened SGA 7 text supplies is the abelian-variety criterion below. The computations distinguish that criterion from good reduction of the curve; the handwriting alone does not certify a stronger historical claim.

B.7 (Proved). *For any henselian trait of residue characteristic zero, the following are equivalent: inertia is trivial on $H_1(U_{\bar{K}}, \mathbb{Z}_\ell)$ for one prime ℓ; it is trivial on Π^{ab}; the Jacobian of $\bar{U}$ has good reduction and inertia fixes every geometric point of D. Equivalently the generalized Jacobian extends as an extension of an abelian scheme by a torus. The same statement about one prime $\ell \neq p$ and all prime-to-p homology holds in residue characteristic p.*

Proof. The generalized Jacobian J_D fits into $0 \to S_D \to J_D \to J \to 0$, where $S_D = (\mathrm{Res}_{D/K} \mathbb{G}_m)/\mathbb{G}_m$ if $D \neq \varnothing$ and $S_D = 0$ otherwise. Its Tate module is $H_1(U_{\bar{K}}, \mathbb{Z}_\ell)$; the torus character lattice is the degree-zero lattice on $D(\bar{K})$. The resulting exact sequence is

$$0 \to (\mathbb{Z}_\ell[D(\bar{K})]/\mathbb{Z}_\ell)(1) \to T_\ell J_D \to T_\ell J \to 0.$$

For ℓ invertible in R, inertia acts trivially on $\mathbb{Z}_\ell(1)$. Its permutation action on the displayed quotient is faithful when $\nu \geq 2$; for $\nu = 1$ the one point is automatically fixed, and for $\nu = 0$ there is no condition. Trivial action on $T_\ell J_D$ therefore implies unramified marks and trivial action on $T_\ell J$.

SGA 7 I, IX, Corollary 5.10 says that an abelian variety has good reduction if and only if its ℓ-divisible group extends over the trait, assuming $\mathrm{char}\, K = 0$ when $\ell = p$. For $\ell \neq p$ an extension is equivalent to an unramified Tate module, since finite etale covers of a henselian trait are precisely unramified Galois modules. This gives the claimed good reduction of J (Grothendieck, 1972, IX, Corollary 5.10 and Proposition 5.13(a)).

Conversely, unramified marks extend S_D to a torus over R. After a finite unramified splitting extension, an extension of J by S_D is specified by the homomorphism from the character lattice of S_D to $J^\vee(K)$. Every value extends uniquely to an R-section of the proper abelian scheme $J^\vee$. The Poincare biextension then constructs the extended group extension;

uniqueness and the original Galois action descend it over R. Its ℓ^n-torsion is finite etale for ℓ invertible in R, proving unramifiedness of every such Tate module. In residue characteristic zero this includes every prime, so the product is Π^{ab}. This also shows why collisions of distinct marked sections in a chosen curve model do not obstruct this abelian criterion: their divisor classes still extend to the proper dual abelian scheme. □

B.8 (Proved). *For residue field finitely generated over $\mathbb{Q}$, the complete implication table for the five conditions of scans 100–103 is as follows. A row is the antecedent; a column is the consequent.*

	a	b	c	d	e
a	–	yes	yes	yes	yes
b	yes	–	yes	yes	yes
c	no	no	–	yes	no
d	no	no	yes	–	no
e	no	no	no	no	–

Proof. Tamagawa's Theorem (5.3), for a DVR and a hyperbolic smooth marked generic curve, equates good marked reduction with trivial prime-to-residue-characteristic outer inertia, and with trivial outer pro-ℓ inertia for one ℓ (Tamagawa, 1997, Theorem (5.3)). In residue characteristic zero this is $a \Leftrightarrow b$. Inner automorphisms act trivially on abelianization, so the characteristic quotient maps give $b \Rightarrow c \Rightarrow d$; B.7 gives $d \Rightarrow c$. Good reduction gives the special-fibre action, hence $a \Rightarrow e$.

For all negative implications from c or d, use $\mathbf{P}^1 \setminus \{0, t, 1, \infty\}$ over $\mathbb{Q}((t))$. Blowing up the meeting sections $0, t$ gives two rational components, each with three special points, joined at a node of thickness one. The nontrivial Dehn twist of B.5 is a separating twist; its vanishing class is a sum of cusp classes and is in the radical of intersection pairing, so homological inertia is trivial. B.4 excludes e. The model is stable nodal, so a, b fail.

For negative implications from e, use $y^2 = x^3 - t^2 x$ minus the origin over $\mathbb{Q}((t))$. With $r^2 = t$, put $x = r^2 X$, $y = r^3 Y$; the equation becomes $Y^2 = X^3 - X$. The nontrivial inertia element acts as $[-1]$ on the elliptic curve and as -1 on its Tate module. Thus b, c, d fail and a fails by Tamagawa. Good reduction after adjoining r and B.4 give e for every splitting. The coordinate substitutions, discriminants and homology matrices are checked in `code/BC_checks.py`. □

B.8u (Proved). *With the full geometric group in b, c, e and $\ell \neq p$ in d, the table of universal implications over henselian traits of positive residue characteristic is*

	a	b	c	d	e
a	–	no	no	yes	no
b	yes	–	yes	yes	yes
c	no	no	–	yes	no
d	no	no	no	–	no
e	no	no	no	no	–

The same table gives the universal implications when restricted to equal characteristic $p > 0$. This is a statement allowing p to vary; the counterexamples below are given at $p = 3$ and $p = 7$, not asserted to occur by the same formulas at every prime. The separate mixed-characteristic table, where b, c, e have impossible antecedents, is C.3. The prime-to-p table is C.5.

Proof. The implications $b \Rightarrow a$ and $a \Rightarrow d$ follow from Tamagawa's criterion and smooth prime-to-p specialization, respectively; $b \Rightarrow c \Rightarrow d$ is functorial abelianization and projection to the ℓ-component. B.20 proves $b \Rightarrow e$ in equal characteristic, by a full specialization argument for proper curves and by the impossibility of b for affine curves. In mixed characteristic the implication is vacuous by C.3. For all negative cells B.21 gives a, d true and b, c, e false; B.22 gives c, d true and a, b, e false; and B.23 gives e true and a, b, c, d false. These exhaust the twenty ordered pairs. Constant proper hyperbolic curves show that b is possible in equal characteristic, so its positive row is not merely vacuous there. □

2.4 B5–B6: what varies with the splitting

B.9 (Refuted). *The splittings themselves do not form a torsor under $H^1(k, T)$ as L4 says. After a reference splitting s_0 is fixed, they form a torsor under $Z^1_{\mathrm{cts}}(G_k, T)$; their T-conjugacy classes form a torsor under $H^1(k, T)$. If the outer inertia image has finite order n, good marked reduction holds over $K(\varpi^{1/n})$, and the realizing curve V_s is the twist of V_{s_0} by the image of this cohomology class under $H^1(k, T) \to H^1(k, \mu_n) \to H^1(k, \mathrm{Aut}(C', D'))$.*

Proof. Write $G_K = T \rtimes G_k$ using s_0, and write $s(g) = (a(g), g)$. Multiplication gives $a(gh) = a(g) + \chi(g)a(h)$, exactly the continuous cocycle condition. Conjugation by $u \in T$ replaces $a(g)$ by $a(g) + u - \chi(g)u$. These are distinct splittings whenever the coboundary is nonzero; for example complex conjugation over $\mathbb{Q}$ gives $u - \chi(g)u = 2u \neq 0$ for $u \neq 0$. Thus the distinction is substantive. Quotienting by these coboundaries gives precisely the asserted H^1 torsor.

A finite quotient of $T = \widehat{\mathbb{Z}}(1)$ of order n has kernel nT. The extension obtained by adjoining an nth root of a uniformizer is totally ramified of degree n and has inertia nT in the chosen separable closure. Tamagawa's criterion therefore gives good marked reduction over it. Conversely good reduction after finite extension makes an open subgroup of T act trivially, so the inertia image is finite. This proves the finiteness criterion without replacing "good" by "potentially good".

Take s_0 to fix compatible roots of ϖ. The special pair over $K(\varpi^{1/n})$ has a descent action of $T/nT = \mu_n$; conjugation by $s_0(G_k)$ is the cyclotomic action, making $\mu_n \to \mathrm{Aut}(C', D')$ a G_k-equivariant homomorphism. If the root extension is not Galois, these statements are made after adjoining μ_n and descending the resulting finite etale group scheme; no constant cyclic Galois group over k is silently assumed. The finite descent datum in B.4 changes by exactly the image of $a(g)$ in this automorphism group. The standard cocycle construction of a twist gives the displayed formula. Changing a by a coboundary gives an isomorphic descended curve, as required. □

B.10 (Proved). *No isomorphism-invariant predicate on the residue action alone can be equivalent to good marked reduction. A predicate equivalent to e is that there are precisely ν geometric conjugacy classes of maximal essentially cyclotomically normalized procyclic subgroups. The number ν can itself be recovered from the action, so this is an intrinsic predicate on the stated class of actions. Adding triviality of the original inertia action in* $\mathrm{Out}(\Pi)$ *makes it equivalent to a.*

Proof. For $y^2 = x^3 - t^2x$ minus the origin and the reference splitting in B.9, the residue action is that of the constant once-punctured elliptic curve $Y^2 = X^3 - X$ over $\mathbb{Q}$. The first family has bad marked reduction and the constant family has good reduction. Thus the residue action cannot distinguish them. For genuine point actions choose $(0, 0)$ on both affine elliptic curves: the coordinate isomorphism and its $[-1]$ descent datum fix this point, so their residue point actions agree as well. The count criterion follows from B.4: after the explicit restriction

comparison there, a stable nodal model contributes at least $\nu + n$ such classes, while a smooth model has exactly ν. Essential normalization is unchanged by passing to an open subgroup: intersect a normalizing group with its inverse image for one direction, and use the same normalizing group for the other. This proves the reduction to a stable model and covers the proper case $\nu = 0$.

To make the predicate intrinsic, put $M = H_1(\Pi, \mathbb{Q}_\ell)$. If $H^2(\Pi, \mathbb{Z}/\ell) \neq 0$, set $\nu(\rho) = 0$. Otherwise let

$$a = \max_{k'/k \ mfinite} \dim \mathrm{Hom}_{G_{k'}}(M, \mathbb{Q}_\ell), \qquad b = \max_{k'/k \ mfinite} \dim \mathrm{Hom}_{G_{k'}}(\mathbb{Q}_\ell(1), M).$$

These maxima exist since the dimensions are bounded integers and both maxima can be attained after one common finite extension. Use B.4 to reach a stable model compatibly with the given splitting and then split its graph and marks by a further residue extension. B.1 and B.6 give $a = t$ and $b = t + \nu - 1$: every map to the trivial module kills W_{-1}, and every map from $\mathbb{Q}_\ell(1)$ lands in W_{-2}. The same vanishing holds after every further finite extension. Hence $\nu(\rho) = b - a + 1$ in the affine case. The predicate is that the above intrinsic number of cyclotomic maximal classes equals $\nu(\rho)$. Finally the added inertia condition is exactly Tamagawa's criterion, and good reduction already implies e. □

2.5 B7: the residue-prime warning

B.11 (Known). *Over a henselian mixed-characteristic trait an abelian variety has good reduction if and only if its p-divisible group extends over the trait. For a smooth proper genus-at-least-two curve with a semistable model over a complete characteristic-zero DVR with perfect residue field of characteristic p, and a base point extending to a section of that model, good reduction is equivalent to crystallinity of its unipotent p-adic etale fundamental group, equivalently of every $E^{\mathrm{et}}_{n,b}$. These are distinct criteria; the first does not detect good reduction of the curve.*

Proof. The first assertion is exactly SGA 7 I, IX, Corollary 5.10 and Proposition 5.13(a), including their explicit $\mathrm{char}\, K = 0$ hypothesis for $\ell = p$ (Grothendieck, 1972). The second is Andreatta et al. (2015, Theorems 1.6 and 1.9); their introduction fixes completeness, perfect residue field, properness, genus at least two, semistability and an integral base point before stating these theorems. We retain all of those hypotheses. The Galois exact sequence is now $1 \to I_K \to G_K \to G_k \to 1$ with $1 \to P_K \to I_K \to \prod_{\ell \neq p} \mathbb{Z}_\ell(1) \to 1$. Replacing I_K by the full cyclotomic T is false.

The example $y^2 = (x^3 - 5^6)(x^3 - 1)$ over $\mathbb{Q}_5$ has stable special fibre two smooth elliptic curves meeting in a separating node of thickness one. Indeed separate the branch clusters by $xz = 5^2$ and set $A = (1 - z^3)(x^3 - 1)$; at the join normalization has $V^2 = xA$, $W^2 = z/A$, $VW = 5$. The unit $A(0, 0) = -1$ and the implicit function criterion give the regular nodal chart. The two components have equations $Y^2 = 1 - X^3$ and $V^2 = x(x^3 - 1)$, smooth in characteristic five. Thus the Separating-node Picard–Lefschetz gives trivial prime-to-five homological inertia, so the abelian criterion B.7 proves that the Jacobian has good reduction. Equivalently the relative degree-zero Picard scheme of a compact-type stable curve is an abelian scheme, with special fibre the product of the component Jacobians (Dupuy and Rabinoff, 2017, Introduction, compact-type models). Its 5-divisible group therefore extends. The nodal fibre is stable, so the curve has bad reduction. At either rational point at infinity the proper model supplies an integral base point. AIK therefore implies that at least one $E^{\mathrm{et}}_{n,b}$ is not crystalline. The polynomial and local chart identities are checked in `BC_checks.py`. □

B.11a (Proved). *Let K be a complete discretely valued field of characteristic zero with perfect residue field of characteristic p, and let C/K be a smooth proper geometrically connected curve of genus at least two with $b \in C(K)$. Then C has good reduction if and only if every representation $E^{\mathrm{et}}_{n,b}$ of its pointed unipotent p-adic fundamental group is crystalline. No semistability assumption is necessary in this proper statement.*

Proof. The universal unipotent objects and their fibres are defined from $\pi_1(C_{\bar{K}}, b)$ independently of a model. The first nonconstant graded piece, at $n = 2$, is dual to $H^1_{\mathrm{et}}(C_{\bar{K}}, \mathbb{Q}_p)$; this is the identification appearing in Andreatta et al. (2015, Theorem 1.7(ii)). Crystalline representations are closed under subquotients and duals, so crystallinity of every $E^{\mathrm{et}}_{n,b}$ implies crystallinity of $V_p \operatorname{Jac}(C)$. Their Theorem 1.2 then gives good reduction of the Jacobian.

Deligne–Mumford's Theorem 2.4 says that a proper genus-at-least-two curve has stable reduction if and only if its Jacobian has semiabelian reduction (Deligne and Mumford, 1969, Theorem 2.4, printed p. 89). Its Section 2 assumes the residue field algebraically closed. To apply it here, pass to the strict henselization of R; the perfect-residue-field hypothesis makes its residue field algebraically closed. Good reduction of the Jacobian is preserved by this base change. The unique stable model supplied there descends: it and its isomorphisms are of finite presentation, hence defined over a finite unramified extension, and uniqueness of stable models makes the generic Galois descent datum extend and satisfy its cocycle identity. Effective descent for the canonically polarized model gives a stable model over R. This is precisely the descent mechanism in the proof of the same theorem on printed p. 92. Its resolution has regular semistable total space, by Proposition 2.3 of that paper. Properness extends b to an R-point of this model. All hypotheses of AIK's Theorem 1.9 now hold, so C has good reduction. Conversely, good reduction supplies such a proper smooth model and integral base point, and AIK's theorem makes every $E^{\mathrm{et}}_{n,b}$ crystalline. □

B.11b (Proved). *For a pair $(E, \{O\})$ of type $(1, 1)$ over the complete mixed-characteristic field with perfect residue field of B.11a, good marked reduction is equivalent to crystallinity of $H_1((E \setminus \{O\})_{\bar{K}}, \mathbb{Q}_p)$, with no semistability assumption.*

Proof. With one puncture the boundary augmentation module is zero, so this representation is V_pE. AIK's Theorem 1.2 equates its crystallinity with good reduction of E. The identity section extends over the abelian scheme and is a relative etale divisor, hence gives good marked reduction. Conversely a good marked model gives good reduction of E. In the local field case the same relative criterion is explicitly recorded in Porowski (2023, Theorem 3.1 and Introduction); the additional $p \geq 5$ and torsion/discrete-section assumptions of his absolute reconstruction theorems are not needed for this relative assertion. □

B.11c (Known). *For a stable marked log curve over a standard log point over a characteristic-zero field, its unipotent log de Rham fundamental group's outer monodromy is trivial if and only if the underlying stable curve is smooth. A section specializing to the smooth unmarked locus gives trivial genuine monodromy in the smooth case. This is a marked log de Rham theorem, not by itself a mixed-characteristic etale crystalline comparison.*

Proof. These are exactly Chiarellotto et al. (2020, Theorem 6.5 and Proposition 6.8); Definition 6.4 includes marked stable curves, and Definition 6.6 defines the good-section condition as lying in the relative trivial-log locus. Their Theorem 6.13 proves the nontrivial outer monodromy in the minimal semistable case. The mixed-characteristic application in their Theorem 6.29 explicitly assumes a proper curve and invokes AIK's comparison. It does not state the additional marked etale comparison required for all of B7. □

B.11d (Proved). *Let K be complete discretely valued of characteristic zero, with perfect residue field of characteristic $p > 0$, and let $U = \bar{U} \setminus D$ be any hyperbolic curve. Good marked reduction is equivalent to the following criterion in p-adic data of finite covers of Π_U:*

There exist a finite unramified *extension L/K, a geometrically connected finite Galois cover $V \to U_L$ of degree prime to p, and $v \in V(L)$ such that its smooth proper compactification C has genus at least two, the finite map $C \to \bar{U}_L$ is ramified at every geometric cusp, and the proper unipotent p-adic fundamental group of C based at v is crystalline. Equivalently, every universal unipotent representation $E^{\mathrm{et}}_{n,v}$ of G_L for C is crystalline.*

In the forward direction the cover can have degree a power of any fixed prime $\ell \geq 5$, $\ell \neq p$. For $D = \varnothing$ the identity cover is allowed. No semistability assumption on U or C is needed.

Proof. We first identify the data attached to the original group. Such a cover is given by a continuous quotient $q : E_{U_L} \twoheadrightarrow Q$ whose restriction to Π_U is surjective. Put $H = \ker(q|_{\Pi_U})$. The proper geometric group of its compactification is

$$\Pi_C = H \Big/ \overline{\langle H \cap \gamma I_c \gamma^{-1} : c,\ \gamma \in \Pi_U \rangle}^{\mathrm{normal}}, \tag{B.11d.1}$$

where I_c ranges over the geometric cusp groups. Indeed a finite etale cover of V extends etale to C exactly when its inertia at each missing point vanishes, by normalization and the local parameter criterion. Quotienting the arithmetic subgroup $\ker q$ by the same normal subgroup gives E_C. The point v defines its actual Galois action; its $\mathbb{Q}_p$-unipotent completion is the data in the criterion. We use these proper quotients of finite-index subgroups of Π_U, with inherited arithmetic action and peripheral data, rather than claiming direct crystallinity of Π_U at every base point.

Suppose the pair has good reduction. A finite unramified extension splits its etale marked divisor and supplies an integral point b of the smooth open model: a nonempty smooth special fibre has a point over a finite separable residue extension, and henselian lifting supplies b. We choose a finite ℓ-group quotient in which every cusp survives. If $\nu \geq 2$, abelianization modulo ℓ suffices, since the boundary classes have their sole relation that their sum is zero and each class is nonzero. If $\nu = 1$, then $g \geq 1$, and the geometric pro-ℓ group is free on $a_1, b_1, \ldots, a_g, b_g$, with cusp $(\prod_i [a_i, b_i])^{-1}$. Map a_1, b_1 to the two elementary generators of the Heisenberg group over $\mathbb{F}_\ell$, all other generators to one; their commutator is a nonzero central element. These presentations are SGA 1, XIII, Corollary 2.12 (Grothendieck, 2003).

Intersecting all automorphic images of the kernel makes it characteristic. Only finitely many occur: open subgroups of a fixed index in a finitely generated profinite group are determined by homomorphisms to a fixed finite symmetric group and a point stabilizer. The new quotient Q embeds in a finite product of the old finite ℓ-group, so remains an ℓ-group and no cusp becomes trivial. Every ramification index e_c is at least ℓ. The genus of the compactification satisfies

$$2g(C) - 2 = |Q| \left(2g - 2 + \sum_{c \in D} (1 - e_c^{-1}) \right). \tag{B.11d.2}$$

For $g = 0$, $\nu \geq 3$ and $\ell \geq 5$ bound the parentheses below by $-2 + 3(1 - 1/5) > 0$; for $g = 1$, the nonempty ramification set makes them positive; for $g \geq 2$ positivity is automatic. Thus $g(C) \geq 2$. When D is empty we instead use the identity cover, with $g \geq 2$.

The integral section b splits the arithmetic geometrically pro-ℓ extension. Smooth-pair comparison identifies its *actual* action with the action of its special-fibre point, so inertia is trivial. The exact inputs are SGA 1, XIII, Proposition 4.3, Examples 4.4 and 4.5.3, with ℓ invertible on the trait (Grothendieck, 2003). The finite action on the characteristic quotient Q therefore factors through G_k. A further finite unramified extension kills it. Projection to Q is then a homomorphism of the semidirect product, trivial on the section of b, and defines the desired constant-deck Galois cover with an L-point v above b.

The same quotient of the smooth open model's arithmetic group extends the cover etale over that model. Normalize the proper model in its function field. Ramification occurs only along the horizontal marks, with indices powers of ℓ. SGA 1, XIII, Propositions 5.1–5.2 give, after an etale change, the regular Kummer chart $z = w^{e_c}$. Thus the normalization is smooth over the trait at the missing points as well. The proper curve C has good reduction, and v is integral; AIK supplies the crystalline representations (Andreatta et al., 2015, Theorems 1.6 and 1.9).

Conversely suppose the criterion holds. B.11a gives good reduction of the proper curve C, without semistability in advance. Its deck automorphisms extend to its smooth proper model $\mathcal{C}/R_L$, and their special action remains faithful by the finite unramified stable-curve Isom scheme (Deligne and Mumford, 1969, Theorem 1.11). The projective quotient $\mathcal{X} = \mathcal{C}/Q$ exists by the invariant graded ring of the canonical polarization. We prove locally that it is smooth and that its reduced branch divisor is finite etale.

After strict henselization and completion, take a geometric special point with stabilizer $H \subset Q$. Its tangent character is faithful. Indeed a nontrivial finite-order automorphism of $\bar{k}[[z]]$ whose order is prime to p cannot have derivative one: its first nonzero higher coefficient is multiplied by that order under iteration, and cannot vanish. An automorphism identical on the completed local ring is identical on the curve, contrary to faithfulness of the special action. Thus H embeds in $\bar{k}^*$ and is cyclic, of order e.

If $e > 1$, a generator h has special derivative $\zeta \neq 1$. The formal equation $h(z) - z = 0$ has invertible derivative, so Hensel's lemma gives a unique fixed section lifting the special point. Shift z to vanish on that section, lift ζ to the eth roots of unity in the trait, and put

$$w = e^{-1} \sum_{i=0}^{e-1} \zeta^{-i} h^i(z).$$

Its special linear coefficient is nonzero and $h(w) = \zeta w$. The completed invariant ring is consequently $R_L[[w]]^H = R_L[[w^e]]$. For $H = 1$ the ring is simply $R_L[[w]]$. To obtain the local ring of the global quotient, the completed semilocal ring of a Q-orbit is a product of these local rings; taking Q-invariants reduces it to H-invariants in one factor. This proves smoothness of $\mathcal{X}$. At a ramified point the branch locus is exactly the section $w^e = 0$. No branch point appears only in the special fibre, because every special fixed point lifted uniquely, and no two branch points merge in this chart. Properness then makes the full reduced branch divisor finite etale. Smoothness and this divisor property descend through the strictly local base changes.

The generic quotient is $\bar{U}_L$, and its reduced branch divisor is exactly D_L by hypothesis. Thus $\mathcal{X}$ with that divisor is a good marked model. Finally L/K is unramified, so $I_L = I_K$ under the usual common separable closure identification. Good marked reduction over L makes the pro-ℓ outer inertia trivial already over K; Tamagawa's criterion gives good marked reduction over R (Tamagawa, 1997, Theorem 5.3). □

This treats every marked type, including $(0, 3)$, $(0, 4)$ and $(1, 1)$, under the complete/perfect-residue hypotheses, through proper AIK on a finite cover. A direct marked comparison on the original open curve's unipotent group is not presumed. B.11f–B.12 give a separate finite-cover p-divisible-group criterion over arbitrary residue fields.

B.11e (Proved). *Let R be henselian, of mixed characteristic, with perfect residue field, and let $\widehat{K}$ be the fraction field of its completion. Good marked reduction over R is equivalent to the criterion B.11d applied to $U_{\widehat{K}}$. Its Galois representations are attached to the original arithmetic group via*

$$G_{\widehat{K}} \simeq G_K, \qquad E_{U_{\widehat{K}}} \simeq E_U.$$

In this formulation the auxiliary points in B.11d are points over finite unramified extensions of $\widehat{K}$; no descent of those points to K is asserted.

Proof. The natural continuous outer map $G_{\widehat{K}} \to G_K$ is an isomorphism by the Krasner comparison, in exactly the henselian-DVR scope of Minamide and Tsujimura (2022, Lemma 3.1). Its map on residue Galois groups is the identity, so it identifies inertia. Characteristic-zero geometric base change identifies the geometric curve groups, including the cusp subgroups, and the two arithmetic exact sequences therefore identify $E_{U_{\widehat{K}}}$ with E_U. Good marked reduction over R implies it over $\widehat{R}$, hence the finite-cover crystalline criterion by B.11d. Conversely that criterion implies good marked reduction over $\widehat{R}$. Thus the pro-ℓ outer inertia action is trivial there, for $\ell \neq p$, and the identified inertia action is trivial over K. Tamagawa's criterion over the original DVR gives good marked reduction over R. This last argument avoids any unproved algebraization from a formal model and retains the perfect residue field needed by AIK. □

B.11f (Proved). *Let R be a henselian DVR with fraction field K of characteristic zero and residue field of characteristic $p > 0$, with no perfection assumption. Let C/K be a smooth proper geometrically connected curve of genus $h \geq 2$, and fix one prime $\ell \neq p$. Define $\mathcal{P}_\ell(C/R)$ by the following finite collection of conditions:*

1. $\mathrm{Jac}(C)[p^\infty]$ *extends to a p-divisible group over R.*
2. *For each connected geometric cyclic etale cover $D \to C_{\bar{K}}$ of degree ℓ, there is a finite extension L/K and a descent $D_L \to C_L$ such that $\mathrm{Jac}(D_L)[p^\infty]$ extends to a p-divisible group over R_L.*

Here R_L is the valuation ring of the unique extension of the valuation. The extensions in the second condition may be ramified. There are exactly $(\ell^{2h}-1)/(\ell-1)$ isomorphism classes of these unlabelled geometric covers, so one common finite extension can be used for all of them. Then

$$C \text{ has good reduction over } R \quad \Longleftrightarrow \quad \mathcal{P}_\ell(C/R).$$

The conditions are conditions on integral p-adic Galois data of Π_C and its cyclic index-ℓ subgroups: their abelianized pro-p groups are the corresponding Jacobian Tate modules.

Proof. In characteristic zero a p-divisible group over a field is an etale system of finite group schemes, determined by its integral Tate module. For the Jacobian this module is $H_1(\Pi_C, \mathbb{Z}_p)$. Condition (1) therefore asks for an integral Barsotti–Tate extension of this representation. Condition (2) asks for potentially such extensions for a finite collection of open geometric subgroups, with the arithmetic actions obtained after descent of the covers. SGA 7 I, IX, Corollary 5.10 and Proposition 5.13(a)–(b) equate these conditions respectively with good and potentially good reduction of the Jacobians; at the residue prime their hypothesis is $\mathrm{char}\, K = 0$, without a perfect-residue assumption (Grothendieck, 1972). Different finite-field descents of a fixed geometric cover become isomorphic over a further finite extension. Thus the potential condition is independent of the descent chosen.

Connected cyclic degree-ℓ covers correspond to nonzero continuous characters $\Pi_C \to \mathbb{F}_\ell$, up to multiplication by $\mathbb{F}_\ell^*$. Since $H^1(C_{\bar{K}}, \mathbb{F}_\ell)$ has dimension $2h$, this gives the asserted finite count. Finite presentation descends each cover and its deck action to a finite extension of K; taking the compositum of the finitely many extensions proves the common-field assertion. Inner automorphisms act trivially on their abelianizations, so the resulting Tate-module actions do not depend on a point section.

We may prove the geometric assertion after completing R. The natural $G_{\widehat{K}} \to G_K$ is an isomorphism by Minamide and Tsujimura (2022, Lemma 3.1), and it identifies inertia and finite extensions. Characteristic-zero geometric comparison then identifies the arithmetic fundamental groups, their finite etale covers and the representations just described. Good

reduction of a Jacobian, and potential good reduction after a finite extension, are preserved in both directions by this comparison: use the same inertia action on a Tate module at an auxiliary prime different from p and the SGA 7 criterion. Good reduction of C is likewise preserved in both directions by Tamagawa's pro-r outer-inertia criterion for $r \neq p$ (Tamagawa, 1997, Theorem 5.3). Thus both sides of the claimed equivalence are unchanged. This reduction ensures that the proper lifting theorem SGA 1, X, Theorem 2.1 is used with its literal *complete local base* hypothesis.

If C is good, its Jacobian is an abelian scheme, proving (1). For (2), prime-to-p specialization for a smooth proper curve identifies each geometric degree-ℓ cover with a cover of the geometric special fibre. That cover descends over a finite separable residue extension and lifts over the corresponding finite unramified extension of the complete trait. The exact inputs are SGA 1, X, Corollary 3.9 and Theorem 2.1 (Grothendieck, 2003). Its lifted proper etale cover is smooth, so its Jacobian is good. This proves the required potential condition for every geometric cover.

Conversely assume (1) and (2). By (1) the Jacobian is good over R. Take a finite extension over which C has a stable model; stable reduction applies over an arbitrary residue field, by properness of the stable-curve stack (Knudsen, 1983, Theorem 2.7). Pass to a further finite unramified extension when splitting geometric components and nodes is necessary. Work on the geometric special fibre C_0 and its connected dual graph Γ. The toric rank of the Jacobian's Neron identity component is $b_1(\Gamma)$. Precisely, SGA 7 I, IX, (12.3.6)–(12.3.7) and (12.4.1) identify its character lattice with the kernel of the oriented edge-to-vertex incidence map (Grothendieck, 1972). If the total stable model has a singularity $xy = \varpi^m$, its regular resolution only subdivides that edge, leaving this kernel unchanged; the same Picard–Neron comparison applies. Since the Jacobian remains good under this base change, $b_1(\Gamma) = 0$. Thus Γ is a tree.

Suppose there is a node e. Removing e partitions Γ into two nonempty subtrees. Both sides contain a component of positive genus. Indeed, if one side had r rational components, its total number of incident branches, including the one attachment at e, would be $2(r-1)+1 = 2r-1$. Stability of every unmarked rational component would require at least $3r$, a contradiction. Choose one positive-genus component on each side and a nonzero $\mathbb{F}_\ell$ character of its geometric fundamental group.

There is an etale $\mathbb{F}_\ell$-torsor $D_0 \to C_0$ with these characters on the two chosen components and zero characters on the others. To verify this gluing assertion, normalization gives the exact sequence of etale constant sheaves

$$0 \longrightarrow \mathbb{F}_{\ell,C_0} \longrightarrow \nu_*\mathbb{F}_{\ell,\widetilde{C}_0} \longrightarrow \bigoplus_e \mathbb{F}_{\ell,e} \longrightarrow 0.$$

At a node the last map is the difference of the two branch values. The skyscraper terms have no positive cohomology over the algebraically closed residue field. The associated sequence therefore surjects $H^1(C_0, \mathbb{F}_\ell)$ onto $\bigoplus_v H^1(C_v, \mathbb{F}_\ell)$, with kernel $H^1(\Gamma, \mathbb{F}_\ell) = 0$. This constructs the desired class. Its restriction to each side of e is nonzero, so its cover of that connected side is connected: a torsor under a group of prime order is either connected or trivial. Above e there are exactly ℓ nodes. Contracting the two connected lifted side graphs leaves two vertices joined by ℓ edges. Consequently

$$b_1(\Gamma_{D_0}) \geq \ell - 1 > 0. \tag{B.pdiv.1}$$

All the data of this finite etale torsor descend to a finite *separable* extension of the residue field: SGA 1, IX, Theorem 4.10 makes finite etale covers invariant under a purely inseparable extension, and finite presentation then descends them from the separable closure (Grothendieck, 2003). After the corresponding unramified extension of the already chosen complete stable trait, SGA 1, X, Theorem 2.1 lifts the cover and its $\mathbb{F}_\ell$-torsor structure uniquely over the

proper stable model. Its generic fibre is geometrically connected: after any finite extension a disconnected generic cover would disconnect the normal finite etale total cover, whereas its connected geometric special fibre forbids that. The lifted model is again stable, since an etale cover preserves the nodal local equations and ampleness of the relative dualizing bundle. It has the graph in (B.pdiv.1).

The geometric generic cover just obtained is one of those in (2). But its Jacobian has positive toric rank, and this remains true after every finite field extension. Indeed the base-changed stable model has the same geometric graph; resolving thicker nodes only subdivides edges. The character-lattice formula therefore retains positive rank. It cannot become an abelian scheme. A descent of this same geometric cover satisfying (2) would become isomorphic to it over a common finite extension, contradicting this persistent positive toric rank. Thus C_0 has no nodes, and C has potentially good reduction.

It remains to descend good reduction, rather than merely assert potential goodness. Choose a finite Galois stable extension and its now smooth model. The finite stable-curve Isom scheme extends the semilinear descent automorphisms, and specialization identifies their actions on the prime-to-p Tate module of the Jacobian (Deligne and Mumford, 1969, Theorem 1.11). By (1), the original inertia acts trivially on that Tate module. Each resulting special-curve automorphism is therefore the identity. One can use Deligne and Mumford (1969, Theorem 1.13, Lemma 1.14) directly: for a nonidentity automorphism of a smooth curve of genus $h \geq 2$, the graph has nonnegative finite intersection with the diagonal, whereas trivial action on H^1 would give intersection number $2 - 2h < 0$ by their Lefschetz–Weil calculation. This argument also covers wild finite automorphisms. Smooth specialization is functorial for these model automorphisms. It follows that the entire original inertia acts trivially in the outer group of the geometric pro-r fundamental group, $r \neq p$. Tamagawa's criterion now gives good reduction over the original trait. This avoids an unsupported descent through a wild quotient. □

For the requested example $y^2 = (x^3 - 5^6)(x^3 - 1)$ over $\mathbb{Q}_5$, B.11 gives two elliptic special components and a separating node. The Jacobian's 5-divisible group extends, but a nontrivial cyclic etale double cover on each elliptic component gives, by the preceding construction, a geometric double cover of the genus-two generic curve whose stable graph has two vertices and two edges. Its genus is three and its Jacobian has toric rank one after every finite extension. Thus condition (2) of $\mathcal{P}_2$ fails, explicitly distinguishing the curve from its good Jacobian. The graph and genus calculation is included in `BC_pdiv_cover_checks.py`.

B.12 (Proved). *Let R be any henselian mixed-characteristic DVR, with residue characteristic $p > 0$, and let $U = \bar{U} \setminus D$ be hyperbolic. Fix a prime $\ell \geq 5$, $\ell \neq p$. Good marked reduction is equivalent to the following criterion in p-adic data attached to finite covers of Π_U:*

There exist a finite unramified *extension L/K and a geometrically connected finite Galois cover $V \to U_L$ of degree prime to p whose smooth proper compactification C has genus at least two, whose extension $C \to \bar{U}_L$ is ramified at every geometric cusp, and for which $\mathcal{P}_\ell(C/R_L)$ holds.*

When D is empty the identity cover is permitted. In the forward direction the cover can have degree a power of ℓ. No perfection, completeness, semistability or chosen point hypothesis is required. For perfect residue fields the crystalline finite-cover criterion of B.11d–B.11e remains an alternative. A crystalline comparison for the original open curve is not asserted.

Proof. The proper group Π_C is the quotient of the geometric subgroup of the cover by the closed normal subgroup generated by its cusp inertias, exactly as in (B.11d.1). Its cyclic index-ℓ subgroups and the integral p-adic abelianizations used in B.11f are therefore data of finite covers and quotients of Π_U, with their inherited arithmetic actions. Finite extensions and good reduction may again be compared with the completion by the same Galois comparison

and Tamagawa criterion as in B.11f. Alternatively all constructions below can be made over the original henselian trait.

Suppose good marked reduction holds. The finite-cover construction in B.11d, up to the application of AIK, uses only henselian lifting and the prime-to-p smooth-pair and Kummer comparisons, so it applies to the present residue field. Explicitly, a smooth nonempty special open has a point over a finite separable residue extension: use an etale chart to affine space over the separable closure, then descend the point. Henselian lifting and an unramified extension provide an integral point. If there are at least two cusps, the modulo-ℓ abelian boundary quotient is nonzero at every cusp. If there is only one cusp, positive genus and the Heisenberg ℓ-group quotient make its commutator nonzero. A characteristic refinement remains a finite ℓ-group quotient. The actual action based at the integral point is unramified on this quotient; an additional finite unramified extension kills its finite action and gives a constant-deck Galois cover with a point above the chosen point. The genus estimate (B.11d.2), with $\ell \geq 5$, makes its compactification have genus at least two. The regular tame charts $z = w^{e_c}$ give a good proper model of that compactification. Each of these steps uses the same SGA 1 statements and proofs specified in B.11d and does not use residue perfection. B.11f now supplies $\mathcal{P}_\ell$.

Conversely, B.11f gives good reduction of the proper covering curve C. Its constant deck group Q extends faithfully to the special fibre of its smooth stable model, by the finite unramified Isom scheme of Deligne and Mumford (1969, Theorem 1.11). Its order is invertible on the trait. We spell out the one local issue needed over an imperfect residue field in the tame quotient argument of B.11d.

At a geometric fixed point the stabilizer H embeds faithfully in the multiplicative group of the tangent line. A prime-to-p finite-order automorphism with derivative one would have its first nonzero higher coefficient multiplied by its order under iteration; it must therefore be the identity on the formal local curve, hence on the connected special curve. Faithfulness excludes this. Thus H is cyclic; if nontrivial its invariant tangent subspace is zero. The fixed locus of H is formally smooth over R_L near this point. Indeed, for a square-zero extension of test rings, smoothness first gives a lift of a fixed point. The differences of its H-translates form a cocycle in the relative tangent module; averaging, since $|H|$ is invertible, changes the lift to an H-fixed lift. This establishes the infinitesimal lifting criterion. The zero invariant tangent space makes the fixed locus etale there. In particular its residue extension is separable, so the required fixed section is available after an etale base change even when the original residue field is imperfect.

At that section the eigenparameter construction of B.11d gives $h(w) = \zeta w$ and

$$R_L[[w]]^H = R_L[[w^{|H|}]].$$

The same computation with the semilocal product over an orbit gives the local ring of the projective quotient $\mathcal{C}/Q$. Thus this quotient is smooth and its reduced branch locus consists etale-locally of disjoint sections. Every special fixed point lifts, so no branch point occurs only in the special fibre. To check that different branch sections cannot collide, translate their ramification points by the deck group so that any alleged collision occurs at the same special point of $\mathcal{C}$. Its cyclic stabilizer contains both nontrivial generic stabilizers, since their fixed equations persist under specialization. In the eigenparameter w, every nontrivial subgroup of this tame stabilizer has the same unique local fixed section $w = 0$. The two purported distinct ramification sections must therefore coincide; so do their images. Properness makes the reduced branch divisor finite etale over R_L. Its generic fibre is exactly D_L, since the cover is etale over U_L and ramified at every cusp. The quotient is therefore a good marked model of $(\bar{U}_L, D_L)$.

Finally L/K is unramified, so the identified inertia groups are equal. Tamagawa's prime-to-p criterion descends good marked reduction to R. If one made the constructions after completion, the same comparison used at the start descends the conclusion to the original henselian trait. This proves the criterion in the full stated scope. □

2.6 B8–B10: effective actions and the contact map

B.13 (Proved). *Let $(\mathcal{X}, \mathcal{D})/R$ be a smooth marked model, let $x \in U(K)$, and suppose its proper extension meets a rational special cusp c. Write $z(x) = \varpi^{\mu} u$, where z is a boundary parameter, $u \in R^*$ and $\mu > 0$. The map $\lambda : G_K \to \pi_1(\mathcal{X} \setminus \mathcal{D}) \simeq \pi_1(U_k)$ induced by x has inertia map $T \xrightarrow{[\mu]} I_c \hookrightarrow \Pi$ and image of index μ in the cusp normalizer. Its restriction along $s_a(g) = (a(g), g)$ is the cuspidal section of Kummer class $\mu[a] + \kappa_k(\bar{u}) \in H^1(k, T)$. These are statements after specialization to the total smooth model. They are not statements that the generic point section has index μ in a generic cusp decomposition group.*

Proof. The prime-to-residue-characteristic Kummer covers of the punctured local chart are dominated by $w^n = z$, by SGA 1, XIII, Propositions 5.1–5.2. Pullback gives $w^n = \varpi^{\mu} u$. Units have nth roots over the strict henselization, so inertia acts by μ times the uniformizer character. The peripheral injection follows from the punctured-surface presentation, including one cusp: its word is a product of commutators, whose nonzero degree-two class in every free pro-ℓ quotient shows infinite order. Thus $T \to \Pi$ is injective before abelianization. The full map lands in $I_c \rtimes G_k$ and is $(\tau, g) \mapsto (\mu\tau + b(g), g)$, where $[b] = \kappa_k(\bar{u})$; this follows by the same root computation on units. Its intersection with I_c is μI_c, and it surjects onto G_k, giving the index. Substituting $\tau = a(g)$ proves the residue cocycle formula. A generic point section of $E_{U_K} \to G_K$, by definition, has trivial intersection with its geometric kernel. This proves that transferring the index assertion to that extension would be a type error. In positive residue characteristic the same proof uses $T_{p'}$, kills P_K, and replaces the index by the prime-to-p part of μ. □

B.14 (Proved). *Suppose the residue outer action is realized by V_s in B.4. A section from a K-point or a rational cusp of U restricts, after an explicitly chosen finite residue extension, to a point or cusp action of V_s. For an interior point meeting the special boundary the residue Kummer class is the class in B.13; a point reducing away from the boundary gives its ordinary special-fibre point section. A cuspidal section remains cuspidal, with its Kummer class restricted along the chosen splitting.*

Proof. Choose a finite extension giving good marked reduction and then take $H = s^{-1}(G_{K_1})$ and G_L as in B.4. The section of the specified point, or the specified cuspidal section, restricts through the Cartesian arithmetic extension over G_L. Properness extends the point to the good compactification. If its reduction avoids the boundary, functoriality and smooth specialization identify its action with that of this special point. If it meets the boundary, B.13 identifies the action with a cuspidal section, including the unit and multiplicity contributions. A rational generic cusp extends to a section of the etale boundary, and its normalizer specializes to that cusp normalizer; the root calculation transports an arbitrary cuspidal Kummer twist. Over a further finite residue extension the descent datum defining V_s is trivial, so these are literally point or cusp actions on the base change of V_s. This is the exact sense of the remark on cote 149, manuscript p. 51, scans 103–104: these constructions do not produce a nongeometric genuine action above a geometrically realized outer action. □

B.15 (Proved). *For a K-point specializing to a node e of a split stable marked model, $\Pi_e \subset C_{\Pi}((s_x \circ s)(G_k^{\circ}))$ for every residue splitting. If the point specializes to a marked point, the corresponding cusp inertia is contained in this centralizer.*

Proof. At a node the section factors through the local log point group of B.3. If the point has local coordinates $x = \varpi^a u$, $y = \varpi^{m-a} v$, its inertia coordinates are $(a\tau, (m-a)\tau, \tau)$, with the unit cocycles on the arithmetic coordinates. On G_k° the cyclotomic character is one. The whole rank-two Kummer kernel is abelian, so all these elements commute with its antidiagonal Π_e, regardless of a, m or the splitting cocycle. At a cusp B.13 gives the same argument with its rank-one kernel. The upper bounds are proved separately in B.15u, using arithmetic cusp uniqueness and the profinite Bass–Serre tree. In particular this local calculation does not identify outer fixed conjugacy classes with elements fixed by the genuine action. □

B.15a (Proved). *Put $F = k(\mu_\infty)$, where $k/\mathbb{Q}$ is finitely generated. For a hyperbolic curve Y/F, an arbitrary continuous section of $\pi_1(Y) \to G_F$ fixes at most one lifted geometric cusp. No extension of this section to G_k is required.*

Proof. We give the arithmetic input and the finite-cover argument explicitly. For every finite extension F'/F and every abelian variety A/F', one has

$$T(A)^{G_{F'}} = 0, \qquad \bigcap_{n \geq 1} nA(F') = 0.$$

Here is the spreading argument extending Ribet's number-field theorem (Katz and Lang, 1981, Appendix, Theorem 1). Write $F' = k_0(\mu_\infty)$ with $k_0/\mathbb{Q}$ finitely generated and descend the finite data to k_0. Let a be its algebraic constant number field, and spread A to an abelian scheme over a smooth geometrically integral a-variety S with function field k_0. A rational torsion point over F' extends over $S_{a(\mu_\infty)}$: its finite etale division scheme has a generic section, and normality extends that section. Evaluation at a closed point of S, after passage to its number-field residue and its cyclotomic closure, is injective on torsion sections: the equalizer of two sections of a finite etale scheme is open and closed. Ribet's finiteness therefore gives the first assertion. This argument also works after every finite extension of F.

For the second assertion first consider a number-field cyclotomic closure $a(\mu_\infty)$. A point divisible by every integer has compatible factorial division points, by compactness of their finite nonempty sets. Descend the abelian variety and the point to a finite subextension a_1. The Kummer cocycle on G_{a_1} vanishes on the normal subgroup $G_{a(\mu_\infty)}$. Its values are therefore fixed by that subgroup, so the first assertion makes the cocycle zero. All division points are then defined over a_1. Reduction in a completion at a good finite place, modulo successive powers of its maximal ideal, puts the point in the intersection of the kernels of finite quotient groups. It is zero. For general F', spread a proposed nonzero infinitely divisible point as a section over S and remove its intersection with the identity section. Every division point extends over $S_{a(\mu_\infty)}$ by the same finite etale argument. Specialization gives a nonzero infinitely divisible point over a number-field cyclotomic closure, a contradiction.

Suppose now that two distinct lifted cusps are fixed by the section $\sigma : G_F \to \pi_1(Y)$. A finite geometric etale cover separates their images. Refine it by a characteristic cover whose proper compactification has genus at least two. Such a geometric cover exists: for a punctured genus-zero curve, choose three punctures and restrict a Fermat quartic cover of the resulting three-punctured line; for genus one, an etale cover first supplies two boundary points and a double cover branched there has genus two; in proper genus at least two the identity suffices. A characteristic refinement of the Galois closure preserves this genus bound by Riemann–Hurwitz. The subgroup generated by this characteristic geometric subgroup and $\sigma(G_F)$ descends the cover to F. The two fixed cusps become distinct F-rational points a, b of its proper compactification $\overline{Y}$.

Both points induce the same section of $\pi_1(\overline{Y}) \to G_F$, because cusp inertia is killed on compactification. Under the Abel map $z \mapsto [z - b]$, their difference has zero Kummer class

for every multiplication cover of $\mathrm{Jac}(\overline{Y})$, hence $[a-b] \in \bigcap_n n\,\mathrm{Jac}(\overline{Y})(F) = 0$. A function with divisor $a-b$ would give a degree-one map to $\mathbb{P}^1$, contrary to the positive genus. This proves uniqueness, including distinct lifts of the same base cusp. It is the finite-neighbourhood proof of Stix's packet theorem, with its arithmetic hypothesis verified over F (Stix, 2008a, Theorem 14 and its proof; Appendix). □

B.15u (Proved). *Let $(\mathcal{C},\mathcal{D})/R$ be a split stable marked model in the characteristic-zero setting of Part B. Let $x \in U(K)$ and let $s : G_k \to G_K$ be any residue splitting. For the genuine action specified by the point section, put $H = (s_x \circ s)(G_k^\circ)$. Then*

$$C_\Pi(H) = \begin{cases} 1, & \text{if } x \text{ specializes to a smooth unmarked point,} \\ I_c, & \text{if } x \text{ specializes to a marked point } c, \\ \Pi_e, & \text{if } x \text{ specializes to a node } e. \end{cases}$$

The inertia groups are the actual lifted subgroups selected by the local point path. The equality is independent of the positive contact multiplicity at a mark and of the two positive contact orders $a, m_e - a$ at a node. In particular no other lifted node group or cusp group is contained in this centralizer.

Proof. We specify both the arithmetic vertex groups and the profinite-tree argument; passing to an abstract tree or assuming that arbitrary fixed subgraphs are path connected would not suffice.

Remove the open edges (the marked cusps) from the geometric semi-graph of B.3, retaining the vertex groups and the closed-edge groups. This finite graph of profinite groups has injective vertex and edge maps. Its standard Bass–Serre graph $\mathcal{T}$ is a profinite tree, with

$$V(\mathcal{T}) = \coprod_v \Pi/\Pi_v, \qquad E(\mathcal{T}) = \coprod_e \Pi/\Pi_e.$$

The sums are finite, so the edge set is compact. The geometric stabilizers are the indicated conjugates. These are the construction and exact tree theorem of Zalesskii and Mel'nikov (1989, §3.7 and Proposition 3.8, pp. 419–420). The one-vertex graph with no nodes is included directly. The split arithmetic action acts continuously on this tree and preserves the two branches of each lifted edge; in particular there are no edge inversions.

For a lifted vertex v, let Y_v be its normalized component minus the nodes and marks. Its vertical logarithmic chart is the constant chart $\mathbb{N} \to \mathcal{O}_{Y_v}$, $1 \mapsto 0$, pulled back from the base log point. The Kummer covers of this chart are the roots of the base parameter. After adjoining these roots the remaining finite cover is etale on Y_v. Thus the arithmetic logarithmic vertex group D_v has the Cartesian description

$$D_v \simeq \pi_1(Y_v) \times_{G_k} G_K. \tag{B.vertex}$$

The projection to $\pi_1(Y_v)$ kills precisely the vertical inertia, which intersects the geometric vertex group trivially and centralizes it. This is also the vertex part of the normalized-component gluing in B.3: the chart is strict over the base log point away from its deleted special points, and adjoining a root of the base chart leaves an ordinary finite etale component cover. Conversely these component covers and the compatible base roots give every such local Kummer cover. The injections in B.3 identify its geometric kernel with Π_v, and its projection onto G_K is surjective. Its image is the entire arithmetic stabilizer: any other stabilizer element can be multiplied by a local element with the same image in G_K to lie in $N_\Pi(\Pi_v) = \Pi_v$, the terminality assertion of Mochizuki (2007, Proposition 1.2). Conjugating the chosen lift conjugates this entire Cartesian description.

Pulling (B.vertex) back along s identifies $D_v \cap E_s$ with $\pi_1(Y_v)$. Consequently, for *every* H-fixed vertex, even one not fixed by the whole point-section image of G_k, projection gives an actual continuous section

$$G_{k(\mu_\infty)} \longrightarrow \pi_1((Y_v)_{k(\mu_\infty)}).$$

It projects to the identity on $G_{k(\mu_\infty)}$ because H already does so. The incident edges at v are the lifted cusps of Y_v corresponding to node branches; the identification follows from their coset descriptions Π_v/Π_e. Distinct incident edges give distinct lifted cusps, including when their images are the same branch at a finite level. This identification is equivariant for D_v. Therefore two distinct incident H-fixed edges would give two fixed lifted cusps for the displayed section, contrary to B.15a. Every vertex of $\mathcal{T}^H$ has degree at most one.

Theorem 2.8 of Zalesskii and Mel'nikov (1989, pp. 413–414) applies to an arbitrary profinite group acting on a full profinite tree; it gives that $\mathcal{T}^H$ is a subtree. It is nonempty: the local point path fixes the vertex of the specialization if it is smooth, and fixes the lifted node edge if the specialization is nodal. Its edge set is closed in the compact edge set of $\mathcal{T}$, hence compact. A connected profinite graph with compact edges and more than one vertex has an edge incident to every vertex (Zalesskii and Mel'nikov, 1989, Lemma 2.14, p. 417). For clarity, this last fact follows by taking finite graph quotients: the closed subsets of edges whose endpoint agrees with a given vertex in each quotient have the finite intersection property.

It follows that $\mathcal{T}^H$ is either one vertex or one edge. Indeed, in the nontrivial case every vertex has degree exactly one. The two endpoint maps identify its vertex set with two disjoint copies of its compact edge space: they are continuous bijections onto disjoint closed sets covering all vertices, hence homeomorphisms. If the edge space had two points, a nontrivial clopen partition would give a disconnected finite quotient graph. This contradicts connectedness. This argument does not collapse an edge and does not presume compactness of the edges after such a collapse.

Write $C = C_\Pi(H)$. It preserves $\mathcal{T}^H$, since its elements commute with H. At a node the fixed tree already contains the local edge, so it is exactly that edge. The action has no inversions, hence C lies in its geometric stabilizer Π_e. B.15 gives the reverse inclusion for every splitting and for all contact orders.

If the specialization is smooth and unmarked, the action at its fixed vertex, under (B.vertex), is the ordinary point action of Y_v at that special point. R15 gives $C \cap \Pi_v = 1$. An incident fixed edge would have its nontrivial inertia centralized by H: the edge action is the cyclotomic character, which is trivial on G_k°. Thus no such edge exists, and the fixed tree is just v. Since C preserves it, $C \subset \Pi_v$ and $C = 1$.

If the specialization is the mark c, the action at the fixed vertex is the cuspidal action computed in B.13. R15 gives $C \cap \Pi_v = I_c$. An incident fixed node edge would instead give a second fixed lifted cusp of Y_v, contradicting B.15a. Again the fixed tree is just v, and consequently $C = I_c$. Distinct lifted peripheral groups are distinct maximal procyclic groups by the peripheral terminality in Mochizuki (2007, Proposition 1.2); hence the displayed equalities exclude every other node or cusp group. The multiplicities affect the local Kummer homomorphism, but B.15 and the tree upper bounds show that they do not affect this centralizer. □

2.7 B11: reconstruction from inertia and arithmetic monodromy

B.16 (Known). *Among stable marked models in this residue-characteristic-zero setting, the dual semi-graph is functorially determined by the geometric group with its G_K outer action and its specified inertia subgroup. Once the edge classes and Dehn coordinates are identified, the closed full-profinite inertia image determines every positive integral thickness. For proper curves the inertia image alone already determines the graph, the full edge classes, and the*

thicknesses. For affine curves the same conclusion holds if the geometric cusp classes are supplied as additional data.

Proof. First consider two proper stable curves and a continuous isomorphism of their full geometric groups carrying one closed outer inertia image onto the other. If those images are nontrivial, B.5 identifies each with T. The isomorphism between them preserves their unique pro-ℓ Sylow subgroups. In Example 2.5 of Mochizuki (2007) take the closed subgroup $H = I_k^{\log}$ and its image J; this is allowed without requiring H to be open. The action is ℓ-logarithmically full because its ℓ-inertia is infinite. Both graphs are noncuspidal. Consequently Corollary 2.7(iii), with Σ the set of all primes, says that the given isomorphism is graphic. This recovers the full edge classes, not merely their images in one pro-ℓ quotient. If the inertia image is trivial, Tamagawa implies that the stable model is smooth, whose one-vertex graph is determined by the geometric genus. For affine curves with cusp classes given, apply instead Hoshi and Mochizuki (2011, Corollary 4.2) to the inertia representations: they have NN-type by B.5 and the positive thicknesses, and the isomorphism is group-theoretically cuspidal by hypothesis. That corollary gives the same full graphic conclusion.

We also check the arithmetic reconstruction route specified in L8. Choose a prime ℓ and pass through the characteristic quotient $\Pi \twoheadrightarrow \Pi^{(\ell)}$. An equivariant isomorphism of full groups induces an equivariant isomorphism of these quotients and of their outer image groups. In the nodal case B.5 gives infinite ℓ-inertia, which is the logarithmic fullness required in Mochizuki (2007, Corollary 2.8). The action is arithmetically full because it is the image of the whole log Galois group. The residue field embeds in a finitely generated extension of $\mathbb{Q}_r$ for a prime $r \neq \ell$: choose a transcendence basis over a number field, embed that number field in a finite extension of $\mathbb{Q}_r$, and take a finite extension containing the remaining algebraic generators. Thus the prime-set hypothesis of that corollary is satisfied with $\Sigma = \{\ell\}$, not incorrectly with all primes. The given compatibility with the specified I_K matches ℓ-inertia. The corollary says the isomorphism is graphic and hence reconstructs the dual semi-graph. If inertia is trivial the model is smooth by Tamagawa; the single vertex genus and cusp number are recovered from homology, properness and Stix's cusp criterion as in B.4.

With a graphic identification in hand, Theorem 4.8(iv) of Hoshi and Mochizuki (2012) identifies the Dehn group with $\prod_e T$. Projection of the inertia image to the e coordinate is $m_e\widehat{\mathbb{Z}}$. This ideal determines all $v_\ell(m_e)$, hence the unique positive integer m_e. A chosen generator can change by a unit of $\widehat{\mathbb{Z}}$ without changing those ideals. If only prime-to-p coordinates were retained, the p-part of m_e would be lost. This proves the precise conditional thickness assertion. The affine extraction without cusp data is proved next by a different reconstruction theorem. □

B.16u (Proved). *Let $(\mathcal{C}, \mathcal{D})/R$ be a stable marked model in the residue-characteristic-zero setting of Part B. Put $J = \rho(I_K) \subseteq \mathrm{Out}(\Pi)$, with the full geometric profinite group Π. The pair (Π, J) determines the set*

$$\{([\Pi_e]_\Pi, m_e) : e \text{ a geometric node}\}.$$

This holds for affine curves without a supplied set of cusp classes, and is functorial under every continuous isomorphism of the geometric groups carrying one closed inertia image onto the other. No generator of J is part of the data. The thicknesses use the normalized valuation of the trait over which the stable model is given.

Proof. An unramified finite extension splitting the special fibre changes neither geometric groups, geometric node classes, thicknesses nor inertia. We can therefore split the fibre for the proof without adding data to the reconstruction. If $J = 1$, B.5 says there are no nodes.

Suppose there is a node. Then B.5 identifies J with $\widehat{\mathbb{Z}}$ and, after a complex topological comparison and a choice of orientation, identifies a topological generator with the completion of

$$\varphi = \prod_e \tau_e^{m_e} \in \mathrm{Mod}(S),$$

where S is a compact oriented surface whose boundary circles represent the cusps. The node circles form a disjoint essential multicurve, and every component S_v cut out by that multicurve has $\chi(S_v) < 0$. These facts are exactly stability: a disk or an annulus component would be an unstable rational component, and a peripheral or parallel pair of node circles would cut off such a component. A base point outside the twist annuli gives an actual automorphism of $\Gamma = \pi_1(S)$. We use this choice only to prove the reconstruction formula; it is not retained as input.

The group constructed from the inertia image. Because Π is finitely generated and centre-free, B.2 constructs, from the given pair alone, the profinite extension

$$1 \longrightarrow \Pi \longrightarrow E_J := \mathrm{Aut}(\Pi) \times_{\mathrm{Out}(\Pi)} J \xrightarrow{q} J \longrightarrow 1.$$

The chosen automorphism $\widehat{\varphi}$ generates a procyclic subgroup of $\mathrm{Aut}(\Pi)$ which maps isomorphically onto J: the composite $\widehat{\mathbb{Z}} \to \mathrm{Aut}(\Pi) \to J$ is an isomorphism by B.5. Consequently

$$E_J \simeq \Pi \rtimes_{\widehat{\varphi}} \widehat{\mathbb{Z}} \simeq \widehat{\Gamma \rtimes_{\varphi} \mathbb{Z}}.$$

For completeness, the second isomorphism is full profinite completion, not a procyclic approximation. Characteristic finite-index subgroups of the finitely generated group Γ are cofinal among its finite-index subgroups. Thus φ extends to the indicated continuous action on $\widehat{\Gamma} = \Pi$. Every homomorphism $\Gamma \rtimes \mathbb{Z} \to Q$ to a finite group extends on Γ to Π and on $\mathbb{Z}$ to $\widehat{\mathbb{Z}}$; the semidirect relation extends by continuity. Conversely their product has dense image. These facts give both universal maps and show they are inverse. The marked normal subgroup $\Pi \subset E_J$ is preserved throughout.

The mapping torus and its geometric decomposition. Let M_φ be the mapping torus of the multitwist, represented by a homeomorphism equal to the identity off the disjoint twist annuli and on ∂S. It is compact and orientable, with one boundary torus for each cusp, and its fundamental group is $\Gamma \rtimes_\varphi \mathbb{Z}$. The interior has universal cover $\widetilde{\mathrm{int}\, S} \times \mathbb{R} \simeq \mathbb{R}^3$. Irreducibility descends from this cover by Hatcher (n.d., Proposition 1.6). Each boundary torus is incompressible: in the surface-bundle extension its group is generated by a nontrivial boundary loop and a lift of the generator of $\mathbb{Z}$, and projection to $\mathbb{Z}$ first kills the second exponent in any proposed relation, after which surface-group injectivity kills the first. The same argument applies to each node torus.

Cutting at the node tori gives the product Seifert pieces $M_v = S_v \times S^1$. There are no solid-torus, annular-product or twisted-Klein-bottle pieces since $\chi(S_v) < 0$. These tori are exactly the JSJ tori. Here are the relevant minimality checks. None is boundary-parallel, and no two are parallel: either possibility would make the corresponding essential surface circles peripheral or freely homotopic, respectively. Such circles are excluded by the stable multicurve. On the torus associated to e, choose the surface circle c_e and the circle fibre t_v on one side. The circle fibre on the other side is

$$t_w = c_e^{\epsilon_e m_e} t_v, \qquad \epsilon_e \in \{1, -1\}. \tag{B.JSJ.1}$$

Indeed the two product trivialisations of the annulus mapping torus differ by exactly m_e full twists. This also follows directly from the separating-edge or HNN formula in B.5. Thus the two primitive fibre slopes have intersection number $m_e > 0$.

Deleting any of these cuts leaves an essential torus, so the joined piece is not atoroidal. It cannot be Seifert either. In an irreducible Seifert piece an essential torus is vertical or horizontal (Hatcher, n.d., Proposition 1.12). A horizontal torus gives a finite torus cover of the base orbifold and hence a virtually solvable fundamental group, whereas the joined piece contains the nonabelian free group $\pi_1(S_v)$. It must therefore be vertical. The induced Seifert fibrations on each $S_v \times S^1$ have the product fibre slope on the boundary: the only exceptions to boundary-slope uniqueness are a solid torus or an I-bundle over a torus or Klein bottle (Hatcher, n.d., Lemma 1.17), already excluded here. Their slopes would have to coincide across the omitted cut, contrary to (B.JSJ.1). This argument also applies to several omitted cuts and to a loop edge. Minimality and uniqueness in the torus-decomposition theorem identify the displayed decomposition with the JSJ decomposition (Hatcher, n.d., Theorem 1.9).

The exact profinite JSJ input. We use Xu (2025, Theorem 5.5), in the opened arXiv version 4, whose hypotheses are compact orientable irreducible 3-manifolds with empty or incompressible toral boundary, excluding closed Sol manifolds. It states that every isomorphism of their profinite fundamental groups induces a congruent isomorphism of their JSJ graphs of profinite groups. In particular it preserves vertex and edge groups up to ambient conjugacy, with the two incidence homomorphisms. No peripheral marking is a hypothesis. The same source, Propositions 5.2–5.4, supplies efficiency, the standard profinite tree and the boundary version used in that theorem. Our manifold satisfies these hypotheses; when there are nodes its nonabelian product pieces exclude the Sol exception. This is a different input from the cuspidal or noncuspidal hypotheses of the combinatorial anabelian results used in B.16.

Accordingly E_J recovers the profinite groups

$$G_v = \widehat{\pi_1(S_v)} \times \widehat{\mathbb{Z}}, \qquad A_e = \widehat{\pi_1(T_e)} \simeq \widehat{\mathbb{Z}}^2,$$

and their incidences. The geometric node group is recovered by

$$\Pi_e = A_e \cap \Pi. \tag{B.JSJ.2}$$

This is an equality with the full closed cyclic subgroup: the surface bundle map on $A_e = \overline{\langle c_e, t_v \rangle}$ sends c_e to zero and t_v to a generator of J, so its kernel is precisely $\overline{\langle c_e \rangle}$.

The JSJ theorem initially gives ambient E_J conjugacy. This yields exactly the required Π conjugacy. For any $g \in E_J$, choose $h \in A_e$ with $q(h) = q(g)$; such an h exists because $q(A_e) = J$. Then $gh^{-1} \in \Pi$, and h normalizes $A_e \cap \Pi$. Hence the E_J conjugates of this node group are already its Π conjugates. The same adjustment applies after an isomorphism between two reconstructed extensions.

An intrinsic integral thickness formula. The surface factor of G_v is a free profinite group of rank at least two, hence centre-free; thus $Z(G_v)$ is exactly its product $\widehat{\mathbb{Z}}$ fibre. For the two incidence embeddings $\iota_{e,i} : A_e \hookrightarrow G_{v_i}$, put

$$F_{e,i} = \iota_{e,i}^{-1}(Z(G_{v_i})), \qquad i = 0, 1.$$

Both are closed primitive cyclic subgroups of A_e. They remain two different subgroups when $v_0 = v_1$, because the two incidence embeddings are retained. Formula (B.JSJ.1), completed profinitely, gives

$$\boxed{m_e = [A_e : F_{e,0}F_{e,1}].} \tag{B.JSJ.3}$$

In the basis (c_e, t_v) the two columns are $(0, 1)$ and $(\epsilon_e m_e, 1)$, so their cokernel is $\widehat{\mathbb{Z}}/m_e\widehat{\mathbb{Z}}$, of order m_e. This finite index is preserved by every congruent isomorphism of the JSJ groups. It

uses no orientation or chosen generator of J. It also explains directly why retaining all primes recovers the full positive integer rather than only selected valuations.

Finally an isomorphism $\alpha : \Pi \to \Pi'$ carrying J onto J' induces $E_J \to E_{J'}$ by conjugation on automorphisms. Applying the JSJ theorem, (B.JSJ.2) and (B.JSJ.3) proves the stated functorial reconstruction. If a ramified extension of degree d is specified, its inertia image is the corresponding open subgroup and the same construction gives the base-changed thickness dm_e, as it must. No thickness independent of the normalization of the trait is asserted. □

2.8 B12–B14: the three explicit tests

B.17 (Proved). *For the Tate elliptic curve E_q over $\mathbb{Q}((q))$ with its origin removed, the stable marked special fibre is a rational nodal curve with one mark, one loop edge of thickness one and one cusp. In the presentation $\Pi = \widehat{\langle a, b\rangle}$, node inertia is $\overline{\langle a\rangle}$, cusp inertia is generated by $[a, b]^{-1}$, and positive inertia is $a \mapsto a$, $b \mapsto ba$. The graded dimensions of B2 are $(1, 0, 1)$ in weights $(0, -1, -2)$. All five conditions a–e are false.*

Proof. The Tate plumbing identifies the two branches of the normalization by $xy = q$; its special normalization is $\mathbf{P}^1$ with branch points $0, \infty$ and the origin mark 1. It is stable, has one node and the stated thickness. Cutting the torus along a gives the HNN presentation; the single crossing b gives the stated twist. Its matrix on homology in the basis (a, b) is $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Thus b, c, d fail and so does a. The weight-zero quotient is nonzero, so B.4 excludes e for every splitting. More explicitly, the Tate-module extension is $0 \to \mathbb{Q}_\ell(1) \to V_\ell E_q \to \mathbb{Q}_\ell \to 0$ with Kummer class of q. On the reference splitting fixing its roots this class is zero; on s_a it is the ℓ-component of $[a]$. This verifies explicitly both the possible change of the extension and independence of its graded pieces. The surface matrix calculation is checked in the code. □

B.18 (Proved). *For $U : y^2 = x^5 - t$ minus infinity over $\mathbb{Q}((t))$, good marked reduction is obtained over $r^{10} = t$ by $x = r^2 X$, $y = r^5 Y$. The special pair is $Y^2 = X^5 - 1$ with its unique infinity, so there are no nodes after this extension. Outer inertia has order ten and acts on homology with characteristic polynomial $\Phi_{10}(z) = z^4 - z^3 + z^2 - z + 1$. The answers to a, b, c, d, e are respectively no, no, no, no, yes. For $[s - s_0] \mapsto a \in \mathbb{Q}^*/\mathbb{Q}^{*10}$, the realizing curve is $y^2 = x^5 - a$ minus infinity, with the cocycle convention of B.9.*

Proof. The indicated substitution gives $Y^2 = X^5 - 1$ exactly; the degree-five polynomial is square-free in characteristic zero, and its smooth proper completion has genus two and a single rational point at infinity. A generator $r \mapsto \zeta_{10} r$ acts on the special coordinates by $(X, Y) \mapsto (\zeta_{10}^{-2} X, \zeta_{10}^{-5} Y)$, of order ten. On holomorphic differentials dX/Y, $X\,dX/Y$ the eigenvalues are ζ_{10}^3, ζ_{10}, and on full homology the eigenvalues are the four primitive tenth roots. Thus the order is already ten on homology, and b, c, d fail. Tamagawa excludes a; B.4 gives e. The sole graded piece after the good extension is weight minus one of dimension four.

An explicit group description retaining the peripheral element is useful: let $\Gamma = \widehat{C_5 * C_2} = \langle u, v \mid u^5 = v^2 = 1\rangle$ and map it onto C_{10} by $u \mapsto 4$, $v \mapsto 5$. Then $\Pi = \ker(\Gamma \to C_{10})$ is free profinite of rank four, and its cusp group is generated by $(uv)^{10}$, with inverse permitted by orientation. This is the orbifold covering $X^5 = z$, $Y^2 = z - 1$ of the affine line, with orbifold orders five and two at $0, 1$. The kernel is torsion-free and has Euler characteristic $10(1/5 + 1/2 - 1) = -3$, proving its rank. Conjugation by $u^4 v$, whose image is $1 \in C_{10}$, describes the positive inertia outer class, up to the chosen orientation convention. Finally if $b^{10} = a$, the invariant coordinates $x = b^2 X$, $y = b^5 Y$ for the twisted descent datum satisfy $y^2 = x^5 - a$. This proves the twist formula for every splitting class. The substitutions and cyclotomic polynomial are verified symbolically. □

B.19 (Proved). *For $\mathbf{P}^1 \setminus \{x^3 = t\}$ over $\mathbb{Q}((t))$, good marked reduction is obtained over $r^3 = t$, $x = rX$, with special pair $(\mathbf{P}^1, \{X^3 = 1\})$. There are no nodes after this extension. The group is free profinite of rank two with cusp generators c_0, c_1, c_2 and relation $c_0c_1c_2 = 1$; inertia cyclically permutes the three cusp conjugacy classes and has order three in the outer group. Its homology polynomial is $z^2 + z + 1$. Conditions a, b, c, d, e have values no, no, no, no, yes. If $[s - s_0]$ maps to $a \in \mathbb{Q}^*/\mathbb{Q}^{*3}$, then $V_s = \mathbf{P}^1 \setminus \{x^3 = a\}$.*

Proof. The substitution is exact, and the divisor $X^3 = 1$ is finite etale over $\mathbb{Q}$, though its points split only over $\mathbb{Q}(\zeta_3)$. Inertia acts by $X \mapsto \zeta_3^{-1}X$. It permutes the cusp classes cyclically; on their augmentation module the nontrivial eigenvalues are the primitive cube roots. Thus the homological action has order three. This proves failure of b, c, d, and failure of a follows either from the ramified divisor or Tamagawa. Good reduction after the cubic extension gives e. Over $\mathbb{Q}(\zeta_3)$ the B2 piece is $\mathbb{Q}_\ell(1)^2$; over $\mathbb{Q}$ it retains the Galois permutation action on the three roots, as required by the nonsplit form of B.6. Twisting X by a cube root of a gives precisely the displayed divisor. This example checks that "good reduction of the curve" cannot replace good reduction of the marked pair, even in type $(0, 3)$. □

2.9 The full group in equal characteristic: completion of B3

In this subsection R is a henselian DVR of equal characteristic $p > 0$, $K = \mathrm{Frac}(R)$, and k is its residue field. The conditions a, b, c, d, e have the meanings in B3, with the *full* geometric fundamental group in b, c, e, and $\ell \neq p$ in d. A residue splitting is assumed given whenever e is mentioned. These are modern extensions of cote 149, manuscript pp. 49–51, scans 99–104; we do not attribute the positive-characteristic conclusions below to the manuscript.

B.20 (Proved). *For a proper hyperbolic curve C/K, trivial full outer inertia implies that C has good reduction and that the full geometric specialization map*

$$\mathrm{sp} : \pi_1(C_{\bar{K}}) \longrightarrow \pi_1(\mathcal{C}_{\bar{k}})$$

is an isomorphism. Its outer equivariance identifies the action restricted along every *residue splitting with that of the smooth special fibre. In particular $b \Rightarrow e$. For an affine hyperbolic curve in equal characteristic p, full inertia acts nontrivially already on $H^1_{\mathrm{et}}(U_{\bar{K}}, \mathbb{F}_p)$. Consequently full b and c are impossible in the affine case.*

Proof. We first prove the proper assertion. Trivial full outer inertia implies triviality on every characteristic pro-ℓ quotient. Tamagawa's good-reduction criterion therefore gives a smooth proper model $\mathcal{C}/R$ (Tamagawa, 1997, Theorem 5.3). The specialization map is surjective: SGA 1, X, Corollary 2.4 applies to every proper smooth morphism over a locally noetherian base, not just to a complete trait (Grothendieck, 2003, X, Corollary 2.4). We prove its injectivity by treating *all* finite geometric Galois covers, including those of degree divisible by p.

Let $D_{\bar{K}} \to C_{\bar{K}}$ be a connected finite etale Galois cover with group Q and degree d. Finite presentation allows us, after a finite separable extension L/K, to descend the cover, all its deck automorphisms and a point of D to L. The latter gives compatible sections of the arithmetic sequences of D and C, and hence actual Galois actions on $H = \pi_1(D_{\bar{K}}) \triangleleft \Pi = \pi_1(C_{\bar{K}})$. For $\sigma \in I_L$, the action on Π is inner by hypothesis. The induced outer action on H consequently belongs to the finite image of Π/H in $\mathrm{Out}(H)$: conjugation by an element of H is inner on H. This conclusion does not require a choice of continuous conjugating elements in Π.

The continuous outer action of I_L on H thus has finite image. Its kernel is open in I_L; choose an open subgroup of G_L whose intersection with I_L lies in that kernel. After replacing L by the corresponding finite extension, the outer inertia action on H, and therefore on its

pro-ℓ quotient, is trivial. Applying the same good-reduction criterion to the proper curve D gives a smooth proper model $\mathcal{D}/R_L$. Here $g(D) = 1 + d(g(C) - 1) \geq 2$.

We now justify extension of the *map*; good models alone would not suffice. The chosen L-point of D extends by properness and gives an R_L-section of $\mathcal{D}$ and, by its image on the generic fibre, an R_L-section of $\mathcal{C}$. The relative Abel map embeds $\mathcal{C}$ as a closed subscheme of its relative Jacobian J. The Neron mapping property extends the generic map $\mathcal{D}_L \to \mathcal{C}_L \hookrightarrow J_L$ uniquely across the smooth R_L-scheme $\mathcal{D}$. For precise sources, SGA 7 I, IX, 1.1 defines that mapping property on *all* smooth schemes over the trait, and Theorem 12.1(c)–(d), especially (12.1.8)–(12.1.10), identifies the relative degree-zero Picard scheme with the identity component of the Neron model (Grothendieck, 1972, IX, 1.1 and Theorem 12.1). One may apply the mapping property to the Neron model first: the extended section maps to zero, and the connected special fibre of $\mathcal{D}$ then maps into its identity component J. These identifications can be checked over the strict henselization and descend by their uniqueness. The extended map factors through the closed subscheme $\mathcal{C}$: the pulled-back defining ideal vanishes on the dense generic fibre and $\mathcal{O}_{\mathcal{D}}$ is torsion-free over the DVR. We obtain $f : \mathcal{D} \to \mathcal{C}$.

Choose a relatively ample line bundle M on $\mathcal{C}$. The degree of f^*M on fibres of the proper flat family $\mathcal{D}/R_L$ is constant, by constancy of its Euler characteristic. Its generic degree is positive. Therefore the map on the special fibre is nonconstant. All its fibres are finite, so f is proper and quasi-finite, hence finite. At a point of $\mathcal{D}$ above a point of $\mathcal{C}$, both local rings are regular and have the same dimension. A system of parameters from the latter is a system of parameters in the former because the map is finite. Since a regular local ring is Cohen–Macaulay, this is a regular sequence, and the local flatness criterion gives flatness. Thus f is finite flat of rank d.

Every generic deck automorphism extends uniquely to $\mathcal{D}$. Here the exact input is Deligne–Mumford's Theorem 1.11: the isomorphism scheme of two stable curves is finite and unramified; its properness extends the automorphism over the DVR (Deligne and Mumford, 1969, Theorem 1.11, p. 84). Specialization of these automorphisms is injective. Indeed the equalizer of two sections of an unramified separated scheme is open and closed; if it contains the closed point of a local trait, it is the entire trait. They commute with f by uniqueness on the dense generic fibre. The special map, of total degree $d = |Q|$, consequently has d distinct deck automorphisms. Its separable degree is therefore at least d, so it is a separable Galois map of degree d and has no inseparable factor.

The genera of the smooth proper fibres are constant, and the generic cover is etale. Hence

$$2g(\mathcal{D}_k) - 2 = d\big(2g(\mathcal{C}_k) - 2\big).$$

Riemann–Hurwitz on the special fibres makes its effective different zero. Thus the special map is etale. A finite flat morphism which is etale on both fibres of a trait is etale; this follows either from the fibrewise criterion or from the vanishing of its module of relative differentials. We have proved that the given geometric cover extends etale after a finite extension of the trait.

Specialization is compatible with that extension and with this finite etale cover. Accordingly the quotient $\Pi \to Q$ factors through $\pi_1(\mathcal{C}_{\bar{k}})$. The change of algebraically closed fields does not change these *proper* geometric groups (Grothendieck, 2003, X, Corollary 1.8). Every finite quotient of Π has now been treated; their kernels have trivial intersection, proving injectivity of sp. The construction of specialization from the arithmetic fundamental-group diagrams in SGA 1, X, 2.1–2.4 also proves its outer G_K-equivariance, with G_K acting on the special group through G_k. Since sp is an isomorphism, restriction along any $s : G_k \to G_K$ is exactly the special-fibre outer action. This proves e over k itself.

For the affine assertion, passing to a finite separable extension preserves any putative triviality of inertia. We may therefore split a boundary point P and choose, by Riemann–Roch,

a function $f \in \mathcal{O}(U)$ whose only pole is at P, of order $m > 0$ prime to p. We choose m sufficiently large. If ϖ is a uniformizer, let a satisfy $a^p - a = \varpi^{-1}$. This defines a totally ramified Artin–Schreier extension of degree p: the valuation of a is $-1/p$, so the extension has ramification index p. Some $\tau \in I_K$ satisfies $\tau(a) = a + 1$. On the affine geometric curve, the Artin–Schreier sequence gives

$$H^1_{\mathrm{et}}(U_{\bar{K}}, \mathbb{F}_p) = \mathcal{O}(U_{\bar{K}})/(h^p - h).$$

Then $\tau[af] - [af] = [f] \neq 0$. To check the inequality, if $h^p - h = f$, any pole of h gives a pole of f of order divisible by p, and no pole can disappear. The unique pole of f has order m prime to p, a contradiction. Thus the inertia action on the mod-p abelian quotient is nontrivial, proving the last assertion. □

B.21 (Proved). *Let $k = \mathbb{F}_9 = \mathbb{F}_3(\alpha)$ with $\alpha^2 = -1$, $R = k[[t]]$, $K = k((t))$, and $\lambda = -1 + t$. The smooth proper genus-two curve*

$$C: \quad y^2 = (x^2 + \alpha)(x^2 + \alpha - 1)(x^2 + \alpha - \lambda)$$

has a, d true and b, c, e false. Failure of e holds for every residue splitting and after every finite residue extension.

Proof. Consider the smooth elliptic model

$$E: \quad v^2 = u(u-1)(u-\lambda) = u^3 - tu^2 + \lambda u.$$

Its discriminant $16\lambda^2(1-\lambda)^2$ is a unit. For a characteristic-three cubic $v^2 = u^3 + a_2u^2 + a_4u + a_6$, the nonzero three-torsion points are the flexes, with $a_2u^3 + a_2a_6 - a_4^2 = 0$. This is also the explicit calculation of van der Geer and Moonen (2011, Example 5.27). In our case their u coordinates satisfy $u^3 = -\lambda^2/t$.

We use this formula on the *etale Verschiebung kernel*, avoiding the imperfection pitfall in substituting K^{sep} for $\bar{K}$ in $E[3]$. Let $F : E \to E^{(3)}$ be relative Frobenius and $V : E^{(3)} \to E$ Verschiebung, so $VF = [3]$; these maps and the identity over a characteristic-p base are constructed in van der Geer and Moonen (2011, Definitions 5.18 and Propositions 5.19–5.20). The two nonzero points of $\ker V$ have coordinates

$$X = -\frac{\lambda^2}{t}, \qquad Y^2 = X(X-1)(X-\lambda^3) = -\frac{\lambda^4(\lambda-1)^4}{t^3}.$$

Indeed Frobenius is radicial and bijective on algebraically closed points, and carries the two flexes to these points; $\deg V = 3$. On putting $z = Yt^2/(\lambda^2(\lambda-1)^2)$ we obtain

$$z^2 = -t.$$

This quadratic polynomial is separable and Eisenstein. Thus the two nonzero points of the *finite etale* K-group scheme $\ker V$ are exchanged by inertia. Notice that the assertion concerns a cover $E^{(3)} \to E$ which is etale over K, not a possibly nonetale 3-torsion subgroup of E over this imperfect field; the distinction is explicitly warned about in van der Geer and Moonen (2011, Caution 5.24).

The map $C \to E$, $(x, y) \mapsto (x^2 + \alpha, y)$, has degree two. The displayed sextic expands over $\mathbb{F}_9[[t]]$ as

$$f(x) = x^6 - tx^4 + \big((\alpha+1)t - 1\big)x^2 + \alpha + (\alpha+1)t.$$

At $t = 0$ its derivative is x, while $f(0) = \alpha \neq 0$. The two points at infinity are smooth since the leading coefficient is one and 2 is invertible. Hence the proper model is smooth of genus two,

proving a and d. Pulling back the degree-three etale Verschiebung cover along $C \to E$ gives a geometrically connected cover: the two function-field degrees two and three are coprime, and the latter extension is Galois over the algebraic closure. Thus the geometric abelianized group of C has the inertia-ramified quotient $\ker V(\bar{K})$. It follows that c, and therefore b, is false.

We compute the geometric 3-ranks to exclude e. On the holomorphic basis $\eta_0 = dx/y$, $\eta_1 = x\,dx/y$, Cartier's formula gives

$$\mathcal{C}(\eta_0) = \big((\alpha+1)t - 1\big)^{1/3}\eta_0, \qquad \mathcal{C}(\eta_1) = (-t)^{1/3}\eta_1.$$

Here we compute over an algebraic closure: write $1/y = f(x)/y^3$ and use $\mathcal{C}(x^{3j+2}dx) = x^j dx$. The semilinear map is invertible generically and has stable rank one at $t = 0$. Thus $f(C_{\bar{K}}) = 2$ and $f(C_{\bar{k}}) = 1$.

Suppose the full residue action were realized by a smooth hyperbolic curve W over a finite extension k'/k. The intrinsic pro-prime-to-3 quotient then has the same outer action as $C_{k'}$: good proper specialization identifies it independently of the splitting. Cohomology at any $\ell \neq 3$ distinguishes a proper curve from an affine curve and determines its genus; hence W is proper of genus two. Centre-free extension reconstruction B.2 identifies the associated geometrically prime-to-3 arithmetic groups. Saidi–Tamagawa's exact finite-field anabelian theorem applies: their Theorem 3.9 assumes all primes other than the field characteristic in the geometric group and concludes that an isomorphism of those arithmetic groups comes from a scheme isomorphism (Saïdi and Tamagawa, 2009, Theorem 3.9). Consequently W is isomorphic to $C_{k'}$ as a scheme; a possible automorphism of its finite field of constants does not affect its 3-rank. But its full geometric group was assumed isomorphic to $\pi_1(C_{\bar{K}})$, and $\dim_{\mathbb{F}_3} H^1(\pi_1(-), \mathbb{F}_3)$ is the 3-rank. It would have to be both two and one. This proves failure of e, for all the splittings and finite residue extensions asserted. □

B.22 (Proved). *Over $\mathbb{F}_7((t))$ the proper genus-two curve*

$$D: \quad y^2 = (x^3 - t^6)(x^3 - 1)$$

has c, d true and a, b, e false for the full-group conditions.

Proof. The normalized compact-type model is the same explicit model as in B.11, with uniformizer t. We recall its checks. In the inner chart $x = t^2X$, $y = t^3Y$, the special component is $Y^2 = 1 - X^3$. The outer component normalizes to $V^2 = x(x^3 - 1)$, which becomes $Y^2 = 1 - X^3$ under $X = 1/x$, $Y = V/x^2$. Near the connecting point put $xz = t^2$ and $A = (1 - z^3)(x^3 - 1)$. The normalization has

$$V^2 = xA, \qquad W^2 = z/A, \qquad VW = t.$$

Since $A(0,0) = -1$, this is a regular separating node of thickness one. Both components are smooth elliptic curves in characteristic seven, and the node gives a stable compact-type special fibre. The model therefore remains nodal after every finite extension, so a is false and Tamagawa makes b false. C.5 excludes even prime-to-7 residue realization, hence excludes full e.

The relative Jacobian J is an abelian scheme with special fibre the product of the two component Jacobians. Each is ordinary: the Hasse coefficient of x^6 in $(1 - x^3)^3$ is $3 \neq 0$ in $\mathbb{F}_7$. For $\ell \neq 7$ good reduction gives trivial inertia on $T_\ell J$. For the residue prime we give the additional argument rather than appealing to a prime-to-p criterion. The iterated Verschiebung

$$V^n : J^{(7^n)} \longrightarrow J$$

is finite flat, of rank 7^{2n}, and its special kernel is etale because the special abelian surface is ordinary. Thus its kernel is finite etale over R: the nonetale locus of this finite flat scheme is closed, its image in the local trait is closed, and it misses the closed point. Over an algebraically closed field, relative Frobenius is a bijection on points and identifies these kernels with the etale 7^n-torsion quotients. They consequently give unramified 7-primary geometric Tate modules. More directly, the etale isogenies V^n form the cofinal 7-power covers of the ordinary abelian variety, all extended over R. Hence inertia is trivial on every primary part of $\pi_1(D_{\bar{K}})^{\mathrm{ab}} = \pi_1(J_{\bar{K}})$. This proves c and d. □

B.23 (Proved). *Over $K = \mathbb{F}_3((t))$ the proper genus-two curve*

$$C: \quad y^2 = t(x^5 - x)$$

has e true for every residue splitting, whereas a, b, c, d are false.

Proof. The constant curve $C_0 : Y^2 = x^5 - x$ over $\mathbb{F}_3$ is smooth of genus two: the polynomial and its derivative are coprime. Over $L = K(r)$, $r^2 = t$, the substitution $y = rY$ identifies C_L with $(C_0)_L$. The nontrivial inertia element acts on C_0 as the hyperelliptic involution ι. It acts as -1 on every prime-to-3 Tate module of the Jacobian: the quotient is $\mathbf{P}^1$, so $1 + \iota = 0$ on its degree-zero Picard variety. Thus d, c, b fail, and good reduction would contradict $a \Rightarrow d$.

Because the curve is *proper*, base extension from an algebraic closure of the finite field to $\bar{K}$ preserves its full geometric fundamental group. Its full outer G_K-action is the constant $G_{\mathbb{F}_3}$-action twisted by the quadratic character $\chi : G_K \to \langle \iota \rangle$. For any given splitting s, the character $\chi \circ s$ defines a quadratic twist of C_0 over $\mathbb{F}_3$; ordinary finite Galois descent makes this a smooth proper genus-two curve. Its full outer action is exactly the restriction along s. This proves e already over $\mathbb{F}_3$. Properness in this argument is essential; we do not substitute an affine constant curve and assume invariance of its full wild fundamental group under extension of algebraically closed fields. □

B.24 (Proved). *The same failure of $a \Rightarrow b, c, e$ occurs explicitly at $p = 2$. Over $R = \mathbb{F}_2[[t]]$, let C be the smooth proper normalization of*

$$y^2 + tz^3y + y = z^9.$$

Then C_K is a proper genus-four curve with a, d true and full b, c, e false. Failure of e again holds for every residue splitting and after every finite residue extension. The ramified character which detects c here occurs on a cyclic quotient of order four.

Proof. Start with the smooth elliptic scheme

$$E: \quad y^2 + txy + y = x^3.$$

Its discriminant is $1 + t^3$, a unit. Its generic fibre is ordinary since its Weierstrass coefficient $a_1 = t$ is nonzero, whereas its special fibre $y^2 + y = x^3$ is supersingular; the elementary characteristic-two criterion is van der Geer and Moonen (2011, Example 5.26). The divisor of x is $P + Q - 2O$, where $P = (0, 0)$, $Q = (0, 1)$, and O is the origin at infinity. These are three disjoint sections. Adjoining $z^3 = x$ gives the displayed curve. The normalization is smooth over R: away from the three sections the cover is etale, and at each section its tame cyclic local normalization, after an etale change extracting a cube root of the unit, is a smooth parameter with base parameter its cube. All three ramification indices are three, and Riemann–Hurwitz gives $2g(C) - 2 = 3(3 - 1) = 6$.

We compute a ramified etale cover of the generic elliptic curve. Put

$$x = t^2X + t^{-1}, \qquad y = t^3Y + t^{-3}, \qquad B = t^{-3}, \qquad A = t^{-12} + t^{-9}.$$

Its generic equation becomes

$$Y^2 + XY = X^3 + BX^2 + A. \tag{B.24.1}$$

For a point (X, Y) with $X \neq 0$, the tangent slope is $X + Y/X$. Substitution in the duplication formula gives

$$X(2P) = X(P)^2 + A/X(P)^2.$$

The unique nonzero geometric two-torsion point has $X = 0$. Consequently the points of exact order four have $X^4 = A$. Under twice iterated relative Frobenius their images are the two primitive points of the etale kernel of $V^2 : E^{(4)} \to E$. Writing their coordinates as (A, Y_4), we have

$$Y_4^2 + AY_4 = A^3 + B^4A^2 + A^4.$$

Set $w = Y_4/A + A + B^2 + B$. The equation is exactly

$$w^2 + w = B = t^{-3}. \tag{B.24.2}$$

The right side has odd pole order three, so this Artin–Schreier extension is totally ramified of degree two: any root has valuation $-3/2$. It exchanges the two primitive generators of the etale cyclic group $\ker V^2(\bar{K}) \simeq \mathbb{Z}/4$. The relative Frobenius map is used only to locate this etale kernel; no identification of all geometric torsion points with points over K^{sep} is made. Its pullback along the degree-three map $C \to E$ is geometrically connected, since three and four are coprime. Thus it gives an inertia-ramified quotient of $\pi_1(C_{\bar{K}})^{\mathrm{ab}}$. This proves failure of c and b, despite good reduction.

For completeness, the special curve $y^2 + y = z^9$ has 2-rank zero directly by Cartier. The different exponent at its unique point over $z = \infty$ is ten, so dz has divisor $6P_\infty$; a basis of regular differentials is $dz, z\,dz, z^2dz, z^3dz$. Cartier kills the even powers and sends $z\,dz$ to dz and z^3dz to $z\,dz$. Its third iterate is zero. Generically, pullback and norm for the degree-three map to the ordinary E have composite multiplication by three, so the etale two-primary Tate module of E is a direct summand of that of $\mathrm{Jac}(C)$. In particular $f(C_{\bar{K}}) \geq 1$. The special 2-rank is zero, whereas the generic one is positive. If a full residue realization existed, prime-to-two specialization, the properness/genus invariants and Saidi–Tamagawa Theorem 3.9 would identify it with the special curve, exactly as in B.21. Full mod-two homology contradicts the two different ranks. This proves the claim. □

The exact identities, Hasse coefficients, smoothness and node checks in B.21–B.24 are verified in `code/BC_equal_char_checks.py`; all 35 named checks pass. They verify the explicit examples, whereas the all-cover specialization argument is proved in B.20.

3 Part C. Finite residue fields

We now assume explicitly that $K/\mathbb{Q}_p$ is finite, with residue field $k = \mathbb{F}_q$, except where C.3 and the implication-table conclusion of C.5 state broader scopes. A superscript (p') on a geometric group leaves the arithmetic quotient G_K unchanged. These statements extend the questions on cote 149, manuscript pp. 49–51, scans 99–104, to the finite-residue-field setting requested in Part C; they are not attributed to those manuscript pages.

C.1 (Known). *Under good marked reduction, the outer action on $\Pi^{(p')}$ factors through G_k and is the outer action on the prime-to-p geometric group of the special fibre. Arithmetic Frobenius acts on the inertia of a rational cusp by multiplication by q.*

Proof. SGA 1, XIII, Examples 4.4 and 4.5.3 give the smooth-pair specialization isomorphism of geometric prime-to-p groups and its functoriality over the base (Grothendieck, 2003). Thus the representation is pulled back along $G_K \to G_k$. At a rational cusp use a local parameter z and the covers $w^n = z$, with $(n, p) = 1$. Inertia acts by $w \mapsto \zeta_n w$; conjugation by arithmetic Frobenius replaces ζ_n by ζ_n^q. Passing to inverse limits gives multiplication by q on $\prod_{\ell \neq p} \mathbb{Z}_\ell(1)$. With geometric Frobenius the inverse convention would give q^{-1}; the assertion uses arithmetic Frobenius. □

C.2 (Known). *For a split stable marked model, wild inertia acts trivially on $\Pi^{(p')}$ in the outer group, and tame inertia acts by the product of Dehn twists with node exponents m_e. Edge groups and cusp groups inject as pairwise nonconjugate maximal procyclic subgroups $\widehat{\mathbb{Z}}^{(p')}(1)$. Each has a normalizing lift of arithmetic Frobenius acting by the qth power.*

Proof. Apply the log and semi-graph comparisons of B.3 with $\Sigma = \{\ell : \ell \neq p\}$. Only Kummer indices invertible in the residue field occur. Hoshi–Mochizuki's Theorem 4.8 and Lemma 5.4 are stated for this set of primes, and give the coordinates $m_e c$ of tame inertia (Hoshi and Mochizuki, 2012). The action factors through the tame log base; its wild kernel is therefore trivial. The node map is $\{(a, b, c) \in T_{p'}^3 : a + b = m_e c\} \rtimes G_k \to T_{p'} \rtimes G_k$. The injection and commensurable terminality arguments use the same prime-to-p versions of the cited semi-graph statements. A Frobenius lift in this node group exists by its surjection to the log base; conjugation on the antidiagonal is q. The cusp calculation is C.1. No assertion of this kind for the full mixed-characteristic geometric group follows from this quotient comparison. □

C.3 (Proved). *For every hyperbolic curve over the fraction field of a henselian mixed-characteristic DVR, full-group conditions b and c of scans 101–102 are false. Its full geometric group is not isomorphic to the full geometric group of any smooth hyperbolic curve over a field of the residue characteristic, so full-group condition e is false as well. The full mixed-characteristic table is*

	a	b	c	d	e
a	–	no	no	yes	no
b	vac	–	yes	vac	vac
c	vac	vac	–	yes	vac
d	no	no	no	–	no
e	vac	vac	vac	vac	–

Here "vac" means a logically valid implication with an impossible antecedent in this explicitly stated setting.

Proof. If $g \geq 1$, $H_1(U_{\bar{K}}, \mathbb{Q}_p)$ surjects onto $V_p J$ and the latter has determinant χ_p^g by the Weil pairing. The cyclotomic character has infinite image on inertia over such a trait; the corresponding determinant statement is exactly SGA 7 I, IX, Proposition 5.14(b), whose hypothesis is a trait with fraction field of characteristic different from p, not a finite extension of $\mathbb{Q}_p$ (Grothendieck, 1972). Thus c is false. If $g = 0$, hyperbolicity gives $\nu \geq 3$. After a finite extension splitting the marks, homology is $\mathbb{Q}_p(1)^{\nu-1}$, again ramified. Since $b \Rightarrow c$ through abelianization, b is false too.

For the geometric-group assertion, degree-two cohomology at a prime $\ell \neq p$ first distinguishes proper from affine groups. In the proper case the same cohomology and

$H^1(-, \mathbb{F}_\ell)$ determine the genus g. The characteristic-zero group's maximal elementary abelian p quotient has dimension $2g$. In characteristic p the corresponding dimension is the p-rank, at most g (the etale part of the p-torsion of its Jacobian). These cannot agree for $g \geq 2$. In the affine case the characteristic-zero group is topologically finitely generated. The affine characteristic-p group is not: Artin–Schreier classes give infinitely many independent elements of $H^1_{\mathrm{et}}(V, \mathbb{F}_p)$. To see independence, choose functions with a single pole of arbitrarily large orders prime to p at one boundary point; Riemann–Roch provides them for all sufficiently large orders. Every nonzero $(h^p - h)$ with a pole has leading pole order divisible by p, so a finite nontrivial linear combination with largest chosen pole order cannot be of that form. Hence no full-group realization exists.

Good marked reduction implies d by C.1, and smooth models give examples where a, d are true but b, c, e false. The compact-type genus-two example over $\mathbb{Q}_5$ in B.11 has d true and a false. This proves every nonvacuous cell. The table is specifically mixed characteristic; it must not be exported to an unspecified equal-characteristic trait, where constant families give different full-group behavior. □

C.4 (Proved). *Let V be a smooth hyperbolic curve over $\mathbb{F}_q$ and put $\Sigma = \{\ell : \ell \neq p\}$. A nontrivial closed procyclic subgroup of $\pi_1(V_{\bar{k}})^\Sigma$ which is essentially normalized with cyclotomic action is contained in a cusp inertia subgroup. Consequently the maximal such subgroups are precisely the geometric cusp groups.*

Proof. We spell out the adaptation needed in L9 rather than cite Stix's characteristic-zero formulation as though it had the same hypotheses. The geometric group and its finite-index subgroups are the pro-Σ surface groups of SGA 1, XIII, Corollary 2.12. Their cusp injections and self-normalizers follow from Mochizuki (2007, Proposition 1.2); the stronger assertion that distinct cusp groups have trivial intersection is exactly Hoshi and Mochizuki (2011, Lemma 1.5), whose hypotheses permit every nonempty prime set Σ. These groups are torsion-free. If an abelian subgroup A meets a cusp group I nontrivially, then for $a \in A$ the intersection $I \cap aIa^{-1}$ contains that same nontrivial intersection; Lemma 1.5 gives $aIa^{-1} = I$, and self-normalization gives $a \in I$. Thus $A \subset I$. This proves explicitly the prime-to-p form of Stix's Corollary 13 needed below, including its application to a nonzero $\mathbb{Z}_\ell$ component of a procyclic subgroup.

Suppose a procyclic subgroup A is not cuspidal and is cyclotomically normalized. Choose a nonzero ℓ-component $A_\ell \simeq \mathbb{Z}_\ell$. The union W of all cusp inertia groups is compact: each is parametrized by its compact conjugacy orbit and there are finitely many geometric cusp classes. We have $A_\ell \cap W = 1$. The compact set $A_\ell \setminus \ell A_\ell$ can therefore be separated from W in a finite quotient of the geometric group. Choose an open normal subgroup N realizing this separation and stable under an open arithmetic normalizer; finite generation permits a characteristic refinement. Set $H = NA_\ell$. The finite cyclic ℓ-group H/N has the property that the images of all intersections of cusp groups with H lie in its subgroup of ℓth powers. Indeed its subgroups are linearly ordered, and an image not so contained would meet the separated generator coset. Hence the nontrivial degree-ℓ quotient H/NA_ℓ^ℓ kills every cusp inertia of the cover associated to H, and extends over its smooth proper compactification $\bar{V}'$.

For completeness, this descent is geometric. Choose a normalizer element γ projecting to Frob_q^r, where $r > 0$ is such that the arithmetic normalizer surjects onto $G_{\mathbb{F}_{q^r}}$. The universal property of $\widehat{\mathbb{Z}}$ gives a continuous homomorphism $\widehat{\mathbb{Z}} \to \overline{\langle\gamma\rangle}$, $n \mapsto \gamma^n$; its composite with $G_{\mathbb{F}_{q^r}} \simeq \widehat{\mathbb{Z}}$ is the identity. The element γ normalizes H because it normalizes A_ℓ and N is characteristic. Thus $H \rtimes \overline{\langle\gamma\rangle}$ is an open arithmetic subgroup defining a geometrically connected cover V' over $\mathbb{F}_{q^r}$. Its smooth compactification is defined over that same finite field, and conjugation by γ induces its Frobenius action on abelianized geometric homology. The composite $A_\ell \to H \to T_\ell \operatorname{Jac}(\bar{V}')$ is nonzero: the unramified degree-ℓ quotient just

constructed factors through its reduction modulo ℓ and is nonzero on A_ℓ. It is equivariant for the normalizing arithmetic group, acting by χ_ℓ on A_ℓ. Thus for some $r > 0$ it gives a nonzero map $\mathbb{Q}_\ell(1) \to V_\ell \operatorname{Jac}(\bar{V}')$ intertwining Frobenius to the rth power. On the source Frobenius has eigenvalue q^r; on the target all complex absolute values are $q^{r/2}$ by the Weil theorem for the Jacobian. This is impossible. This is exactly the finite-cover argument of Stix (2008a, Theorem 20, proof), with the unavailable characteristic-zero torus assertion replaced by the explicit finite-field weight calculation. Every cover used has geometric Galois closure of order supported on Σ. Maximality now identifies the full cusp groups. □

C.5 (Proved). *For every continuous residue splitting $s : G_k \to G_K$, the prime-to-p outer Frobenius obtained by restriction is realized by a smooth hyperbolic curve over k if and only if U has potentially good marked reduction. Allowing a power of Frobenius, or a finite residue extension, gives the same criterion. This includes extensions needed for good or stable reduction which are wildly ramified. With b, c, e interpreted using geometric prime-to-p groups, the implication table is the table of B.8. This last table of universal implications holds over all henselian traits of residue characteristic p, in equal or mixed characteristic; the realization criterion preceding it still has the finite-residue-field hypothesis.*

Proof. If good reduction holds after a finite Galois extension, use the finite descent datum of B.4 on its good special pair. Prime-to-p specialization is equivariant for every model isomorphism, including those in wild inertia. The map to $\operatorname{Gal}(K_1/K) \times_{\operatorname{Gal}(k_1/k)} G_k$ has kernel I_{K_1}, trivial on the prime-to-p geometric group. Pulling back by s therefore gives exactly the descended Frobenius action. Thus realization already holds over k.

For the converse take a finite extension with stable marked reduction, possibly wild, and use the actual restriction square $E_{U_L} \times_{G_L, s|_H} H = E_s \times_{G_k} H$ from B.4. Enlarge the residue field to split components, nodes, branches and marks. If the stable graph has a cycle, the resulting homology has a Frobenius eigenvalue one from graph homology. A smooth affine or proper curve has only eigenvalues of absolute values $q^{1/2}$ and q in its geometric homology, so cannot realize that action. If the graph is a tree, degree-two cohomology first matches properness. In the proper case a node subgroup contradicts C.4. In the affine case the multiplicity of the eigenvalue q is $\nu - 1$; the normalized component Jacobians have weight one and contribute no such eigenvalue. Thus a realizing curve has the same number of cusps. C.2 gives $\nu + n$ distinct cyclotomic maximal procyclic classes, whereas C.4 gives only ν. This excludes every nodal stable fibre. The stable fibre is therefore smooth, proving potential good reduction.

For clarity about wild inertia, once stable reduction holds over L its wild inertia acts trivially on the prime-to-p group by C.2. Hence the original wild inertia has finite image, because its intersection with G_L has finite index. The proof did not presume that image was already trivial over K. Nor did it assume that a given residue splitting restricts to G_L without the open-subgroup construction.

For the table, Tamagawa gives $a \Leftrightarrow b$, B.7 gives $c \Leftrightarrow d$, and smooth-pair prime-to-p specialization gives $a \Rightarrow e$. These implications hold for every henselian trait of residue characteristic p. Counterexamples over local fields suffice to refute every other universal implication, and can be given at every prime. The curve $\mathbf{P}^1_{\mathbb{Q}_p} \setminus \{0, p, 1, \infty\}$ has c, d true and a, b, e false: its rational marked sections give unramified abelian homology, while its stable model has one separating node of thickness one. The last assertion follows from the realization criterion just proved. This construction also works at $p = 2$ and $p = 3$.

For e true but a, b, c, d false, at an odd prime use $y^2 = x^3 - p^2 x$ minus infinity. It becomes $Y^2 = X^3 - X$ over $\mathbb{Q}_p(\sqrt{p})$, a good model because its discriminant is 64, and inertia acts by -1 on every prime-to-p Tate module. At $p = 2$ use $y^2 = x^3 + 2$ minus infinity. Over $r^2 = 2$, the substitutions $x = 2X$, $y = r(2Y + 1)$ give $Y^2 + Y = X^3$, whose discriminant is -27. The nontrivial inertia element acts as $(X, Y) \mapsto (X, -Y - 1)$, namely $[-1]$. This is a concrete

example where the good-reduction extension is wild. The substitutions and discriminants are checked in the attached program. The compact-type genus-two example in B.11 separately proves that the abelian failure is not confined to genus-zero boundary collisions. □

C.6 (Proved). *Assume split stable marked reduction over K, and let s_x be the section of $x \in U(K)$. Write $H = G_{K(\mu_\infty)}$. Up to the geometric conjugation determined by the lift of the specialization, the subgroup of $\Pi^{(p')}$ fixed by $s_x(H)$ is:*

specialization of x	$(\Pi^{(p')})^{s_x(H)}$
smooth unmarked point on C_v	Π_v
node e	Π_e
marked point c	I_c.

Here Π_v is the injected prime-to-p fundamental group of the normalized component punctured at all its branches and marks. At a node $xy = \varpi^m$ the contact integers are $a = v(x)$, $m - a = v(y)$ with $0 < a < m$; at a mark the contact integer is $\mu > 0$. The listed fixed groups do not change when these positive integers change.

Proof. All prime-to-p roots of unity generate the maximal unramified extension of a local field: the residue fields $\mathbb{F}_{q^r}$ are generated by roots of unity of order $q^r - 1$. Therefore $H \subset I_K$. Its image in tame inertia $T_{p'}$ is an open nonzero subgroup. Indeed the remaining extension by p-power roots of unity has a pro-p Galois subgroup of finite index, so its prime-to-p tame quotient is finite. Wild inertia acts trivially also on the genuine pointed action: an outer-trivial action gives, by centre-freeness, a homomorphism from the pro-p group to the pro-prime-to-p group of inner automorphisms, which must be trivial.

Work in the log extension $\Pi_I = \Pi^{(p')} \rtimes^{\mathrm{out}} T_{p'}$. For a point reducing smoothly away from marks, the image of inertia under its section is the vertex inertia I_v, the subgroup centralizing the vertex group. This follows by the local smooth chart: its generic covers restrict to the ordinary covers of the punctured normalized component. The image J of H is a nontrivial closed subgroup of I_v. Hoshi–Mochizuki's Lemma 3.6(i) states, for every such J, the exact equality $Z_{\Pi_I}(J) \cap \Pi^{(p')} = \Pi_v$ (Hoshi and Mochizuki, 2011, Lemma 3.6). Its hypotheses are the nodally nondegenerate semi-graph representation; positive thicknesses give the stricter SNN condition here by their Lemma 2.11 and the length calculation. This proves the first row, including the smooth-fibre case $\Pi_v = \Pi^{(p')}$.

At a node the local inertia image has coordinates $(a\tau, (m-a)\tau, \tau)$ in the rank-two group of B.3, and units do not alter tame inertia. It lies in the edge decomposition group D_e but in neither adjacent vertex inertia: those correspond respectively to one branch coordinate zero. Multiplication by a positive integer is injective on $T_{p'}$, so the same remains true after restriction to J. If $h \in \Pi^{(p')}$ centralizes J, then $J \subset D_e \cap D_{he}$. Proposition 3.8(i)–(ii) of Hoshi and Mochizuki (2011) says that distinct edge decomposition groups have zero intersection unless the edges share a vertex, in which case their intersection is exactly that vertex inertia. Both alternatives contradict the preceding observation. Hence $he = e$ and $h \in \Pi_e$ by edge self-normalization. The rank-two group is abelian, so every element of Π_e does centralize J. This proves equality.

At a marked point the same proof uses its local group $I_c \times T_{p'}$ and slope $(\mu\tau, \tau)$. The adjacent vertex inertia has first coordinate zero; the positive slope excludes it. Proposition 3.8 applies to cusp edges as well as nodes, giving exactly I_c. All subgroup comparisons are made in the common log extension, so the answer concerns fixed elements of the genuine pointed action, not just fixed conjugacy classes. □

C.7 (Proved). *For $U = \mathbf{P}^1_{\mathbb{Q}_5} \setminus \{0, 1, 5, \infty\}$ and $x = 2$, choose geometric cusp generators a_0, a_5, a_1 in the free pro-prime-to-five group. Then*

$$(\Pi^{(5')})^{s_2(G_{\mathbb{Q}_5(\mu_\infty)})} = \overline{\langle a_0 a_5, a_1 \rangle},$$

a free pro-prime-to-five subgroup of rank two. For good reduction the corresponding fixed prime-to-p group at an integral unmarked point is all of $\Pi^{(p')}$.

Proof. Blow up the collision of 0 and 5 on the marked projective line over $\mathbb{Z}_5$. The two resulting rational components each have three special points and meet in one node of thickness one. The point 2 lies on the component with $1, \infty$, and the node. Its node boundary word is $a_0 a_5$, so the normalized component group is the displayed free group. C.6 proves the equality. The genuine tame action can be written $a_i \mapsto d a_i d^{-1}$ for $i = 0, 5$, $a_1 \mapsto a_1$, with $d = a_0 a_5$; the code verifies the fixed boundary words and a finite quotient on which this action is nontrivial. The good-reduction assertion is the node-free case of C.6. □

C.7u (Unresolved). *The full-group fixed subgroups requested in C4, including the point 2 and the integral tangent at 0 on $\mathbf{P}^1_{\mathbb{Q}_5} \setminus \{0, 1, \infty\}$, are not determined here.*

The quotient map $\Pi \twoheadrightarrow \Pi^{(p')}$ sends fixed elements to fixed elements, but invariants need not surject through a quotient. In particular the large fixed quotient just computed supplies neither a lift nor an upper bound for the full fixed group. Wild geometric covering data and its genuine action remain to be controlled; no claim of full-group triviality or cusp equality is made from the prime-to-p calculation.

C.8 (Proved). *After choosing a place of $\bar{\mathbb{Q}}$ above p and identifying its decomposition group with $G_{\mathbb{Q}_p}$, its intersection with $G^\circ_{\mathbb{Q}}$ is $G_{\mathbb{Q}_p(\mu_\infty)}$. Thus C.6 describes the local cyclotomic-kernel fixed group at that chosen place in the stated quotient.*

Proof. The chosen place embeds the algebraic closure into an algebraic closure of its completion. Finite extensions of the henselization and completion have the same Galois category, identifying the decomposition group with $G_{\mathbb{Q}_p}$. Cyclotomic characters commute with this embedding because their actions are literally the actions on the same roots of unity. Taking kernels therefore gives the asserted intersection. Another place changes the decomposition subgroup by $G_{\mathbb{Q}}$-conjugacy, the stated ambiguity. □

4 Part D. Rational points behind sections

The results below isolate actual implications and supply abstract discontinuous actions. D.14 refutes the universal assertion D5; D1–D4 remain unresolved. In particular, we do not convert the unrefereed local claim R24 into an unconditional input. The manuscript passages are cote 149, manuscript pp. 22–31, scans 45, 48, 56, 59–63, and pp. 62v–63, scans 126–127.

4.1 The lifting assertions and their quantifiers

D.1 (Known; Bresciani's lifting criterion and its consequence). *For a smooth geometrically connected curve X/k, with k finitely generated over $\mathbb{Q}$, a continuous section is geometric or cuspidal if and only if its base change to $k(t)$ is birationally liftable. Consequently, the assertion D3, quantified over every such field and every hyperbolic curve, is equivalent to the section conjecture with cuspidal sections allowed at the rational boundary.*

Proof. The first assertion is precisely Theorem A of the opened final source Bresciani (2024). Its additional hypothesis is liftability *after the transcendental extension* $k(t)/k$; liftability only over k is not substituted for it. If D3 holds, apply it to $X_{k(t)}$ and the restriction of any given section of X/k. Theorem A makes that original section geometric or cuspidal. Conversely a geometric section at $x \in X(k)$ lifts through the decomposition group of the discrete valuation of $k(X)$ at x. In characteristic zero its exact sequence has kernel $\widehat{\mathbb{Z}}(1)$ and is split after choosing a local parameter: the coefficient action on the Puiseux extensions gives a continuous splitting. A cuspidal section already factors through that same decomposition group at the boundary, so also lifts. Composing with $G_{k(X)} \twoheadrightarrow \pi_1(X)$ gives the required original section up to geometric conjugacy; conjugating the lift adjusts the representative. These maps are continuous maps of profinite groups over G_k. This is also the derivation stated in Bresciani's Theorem B, with exactly its universal quantifier over finitely generated extensions. □

D.2 (Proved; D2 and D4 imply D1). *If D2 and D4 both hold, every continuous section of* $\pi_1(\mathbb{P}^1_{\mathbb{Q}} \setminus \{0,1,\infty\}) \to G_{\mathbb{Q}}$ *is a point section or a cuspidal section. In particular one with* $C(s) = 1$ *is a point section, using the checked cyclotomic-centralizer assertion R15.*

Proof. Lift s by D4 to $\widetilde{s} : G_{\mathbb{Q}} \to G_{\mathbb{Q}(t)}$. By D2 its image belongs to D_v for a rational place v of $\mathbb{Q}(t)$. If v belongs to the open curve, the quotient $G_{\mathbb{Q}(t)} \to \pi_1(U)$ kills its inertia. The image of D_v is the decomposition section at v, so the quotient of $\widetilde{s}$ is s_v. If v is one of the three removed points, its inertia is retained and the image belongs to the cuspidal decomposition group. Its nontrivial geometric inertia centralizes the restriction to $G^{\circ}_{\mathbb{Q}}$, because the latter has trivial cyclotomic character. Thus $C(s) = 1$ excludes this case. All choices of prolongation of v change the conclusion by conjugacy in $G_{\overline{\mathbb{Q}}(t)}$, hence by geometric conjugacy after passage to $\pi_1(U)$. □

The converse direction required to settle D3 is not the established Hom theorem for open homomorphisms: a section has no open image in the relevant geometric extension. Nor does solving separate finite embedding problems produce compatible solutions automatically. For a countable base, successive surjectivity for all pairs of open subcurves does supply a compatible sequence, as the lemma labelled `cuspbl` in the opened Bresciani source proves. Surjectivity of those transition maps is itself the missing assertion.

4.2 The adelic integrality route

D.3 (Proved; verification of the arithmetic part of R23). *Write* $\widehat{\mathbb{Q}^{\times}} = \varprojlim_n \mathbb{Q}^{\times}/(\mathbb{Q}^{\times})^n$. *Its elements are pairs*

$$(\epsilon, (a_p)_p)\,, \quad \epsilon \in \{\pm 1\}, \quad a_p \in \widehat{\mathbb{Z}}, \quad \#\{p : a_p \not\equiv 0 \pmod{n}\} < \infty \quad (n \geq 1).$$

An element belongs to $\mathbb{Q}^{\times}$ *if and only if every* a_p *is an ordinary integer and all but finitely many are zero. In particular, if all its local Kummer classes are represented by elements of* $\mathbb{Q}_p^{\times}$, *and almost all by units of* $\mathbb{Z}_p$, *it is rational. Moreover*

$$\varprojlim_n \mathbb{A}_{\mathbb{Q}}^{\times}/(\mathbb{A}_{\mathbb{Q}}^{\times})^n \simeq \widehat{\mathbb{Q}^{\times}} \times \widehat{\mathbb{Z}}^{\times}.$$

Proof. Unique factorization gives $\mathbb{Q}^{\times} = \{\pm 1\} \times \bigoplus_p \mathbb{Z}$. Reducing modulo n gives finite-support exponent vectors; taking the inverse limit gives exactly the displayed description. Conversely such compatible vectors define an element at every level. The valuation v_p extends to the continuous homomorphism into $\widehat{\mathbb{Z}}$ extracting a_p. A local representative has integral valuation;

a local unit has valuation zero. Finite integral support and the sign give a rational number with the same coordinates, and the coordinate description proves equality. This proves the stated local criterion without a global approximation assumption.

An idele has finitely many nonzero finite valuations. Multiplying by the unique rational number with those valuations and the sign of its real component leaves a finite unit idele and a positive real component. This gives the group isomorphism $\mathbb{A}_{\mathbb{Q}}^{\times} \simeq \mathbb{Q}^{\times} \times \widehat{\mathbb{Z}}^{\times} \times \mathbb{R}_{>0}$. The last factor is divisible. For the compact abelian group $B = \widehat{\mathbb{Z}}^{\times}$, the map $B \to \varprojlim B/B^n$ is surjective by compactness. Its kernel is $\bigcap_n B^n = 1$: each finite quotient of each $\mathbb{Z}_\ell^{\times}$ is killed by its order, and these quotients separate points. This proves the last isomorphism. The local-to-global assertion agrees with the opened Stix Proposition 2 (Stix, 2015); here the proof for $\mathbb{Q}$ is written independently. □

D.4 (Refuted; naive evaluation at a zero or pole). *For a birational section, the pullback of the Kummer class of $f \in \mathbb{Q}(t)^{\times}$ is not always the class of the value of f at its associated place. At a zero or pole it is a leading-coefficient class multiplied by the chosen tangential class. This corrects the unqualified formula in the proposed route to D2.*

Proof. At $t = 0$ choose the splitting of $G_{\mathbb{Q}((t))} \to G_{\mathbb{Q}}$ obtained by fixing all compatible roots of t, and twist it by the Kummer class of 2. Embed its decomposition group into $G_{\mathbb{Q}(t)}$ by choosing a prolongation of the t-adic valuation. For $f = t^m u(t)$, with $u(0) \neq 0$, its Kummer pullback is

$$\kappa\big(2^m u(0)\big).$$

To check this, choose an nth root of $u(0)$ and apply Hensel's lemma to $u(t)/u(0)$, whose constant term is one; its nth root with constant term one is fixed by the coefficient action. The root of t^m contributes the twist 2^m. This works compatibly for every n in characteristic zero. In particular the three values for $t, t-1, t+1$ are $2, -1, 1$. They cannot be values of a single field evaluation: evaluation with $t = 2$ would give $t - 1 = 1$. The place is nevertheless the rational place $t = 0$. The identities are checked symbolically in `code/root_checks.py`. □

D.5 (Known; the exact sufficient integrality condition). *Let $U = \mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$. A continuous section which is locally a point of $U(\mathbb{Q}_p)$ at every finite place and is integral on $\mathbb{P}^1_{\mathbb{Z}} \setminus \{0, 1, \infty\}$ outside one finite set of primes is the section of a point of $U(\mathbb{Q})$. This does not supply the missing uniform finite set for D1 or D2.*

Proof. For precision, Stix's Corollary 6 in the opened final arXiv source (Stix, 2015) applies to a hyperbolic curve over $\mathbb{Q}$ or an imaginary quadratic number field with a nonconstant global unit, and identifies adelic sections with rational points. Here the unit t supplies the hypothesis. We describe the proof's arithmetic step: the images of s under t and $1 - t$ are classes $a, b \in \widehat{\mathbb{Q}^{\times}}$. The local points and the uniform integrality hypothesis put them in $\mathbb{Q}^{\times}$ by D.3. Injectivity of local Kummer theory gives $x_p = a$ and $1 - x_p = b$ for every p, so $a + b = 1$ and the local support is the singleton $\{a\}$. The source's Corollary 5 converts finite support of a noncuspidal Selmer section into a point section. Its proof applies Stoll's Theorem 8.2 to the finite fibres over $\{a\}$ in all neighbourhood covers, after a cover with compactification of positive genus, and then takes their inverse limit. The hypothesis on finite descent is supplied for section-induced local points by Harari–Stix, Theorem 2.1 and Remark 2.2(1) in the attached preprint (Theorem 11 in the published numbering quoted by Stix) (Harari and Stix, 2012; Stoll, 2007). Its exceptional set of places may have Dirichlet density zero, and its condition concerns all torsors under finite group schemes. Both hypotheses have the required form for a globally induced section. We also opened the 2017 errata to Stoll: the corrections concerning complex places and connected

torsors do not change this use of Theorem 8.2 (Stoll, 2017). The rational point at the limit projects to a, and its decomposition section is s.

Stix's Theorem 2 gives, for a birationally liftable noncuspidal section, integrality at a set of primes of Dirichlet density one. That set need not be cofinite. For example any sequence of primes growing faster than $\exp(2^n)$ is infinite and has Dirichlet density zero. Thus that theorem does not supply the hypothesis used in the first paragraph. The missing passage from density one to a common finite exceptional set is retained as an unresolved step, not made by changing the meaning of "almost all". □

4.3 What the unrefereed local claim would imply

D.6 (Proved implication; conditional input R24). *If the proper local section conjecture asserted in R24 holds for all finite extensions of $\mathbb{Q}_p$, it implies the point-or-cusp section conjecture for every hyperbolic curve over each such field. No unconditional use of R24 is made in this paper.*

Proof. Fix $k/\mathbb{Q}_p$ finite and a section s of U/k. In its system of finite étale neighbourhoods, the covers whose compactification has genus at least two form a cofinal system. Indeed a geometric hyperbolic curve has such a cover: for genus zero use a sufficiently large Fermat cover after a three-puncture map; for a once-punctured genus-one curve use a finite quotient of its free fundamental group in which the boundary commutator is nontrivial, then Riemann–Hurwitz; genus at least two needs no initial cover. A characteristic refinement of the normal core descends to an s-neighbourhood over k. Further covers preserve the genus bound by Riemann–Hurwitz.

For each such neighbourhood V, the section on V induces a section on $\overline{V}$, hence a point of $\overline{V}(k)$ by the assumed proper theorem. These are nonempty compact spaces in the p-adic topology. Their continuous transition maps have a nonempty inverse limit: every finite set of compatibility conditions is realized in a common dominating neighbourhood, and compactness applies to the product. The resulting completed pro-point is fixed by $s(G_k)$. Its projection is either an interior point, whose stabilizer is its decomposition section, or a boundary point, whose stabilizer is its cusp decomposition group. This is exactly the fixed-point dictionary of the fundamental covering tower. The argument is conditional on the proper theorem; compactness cannot manufacture those nonempty sets before that input is justified. □

4.4 Abstract semilinear actions need not be continuous

D.7 (Proved; an explicit discontinuous family of actions). *There exist abstract semilinear actions of $G_{\mathbb{Q}}$ on the universal pro-cover of $U = \mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$ for which a rational function has a nonopen stabilizer. The examples constructed here normalize the original geometric deck group and have exactly one $G^{\circ}_{\mathbb{Q}}$-fixed completed pro-point. They therefore refute automatic continuity, but do not refute D5.*

Proof. Use the rational base point $x = 1/2$. Let ρ_x be its continuous semilinear action on the pointed universal pro-cover, and let $h(t) = 1 - t$. Because $h(x) = x$, pullback of pointed covers gives a unique lift $\widetilde{h}$ fixing the distinguished pro-point $\widetilde{x}$. It has order two and commutes with $\rho_x(G_{\mathbb{Q}})$: both facts follow from the corresponding identities on the pointed covering functor. It normalizes the deck group.

For distinct rational primes $p_1, p_2, \ldots$, the multiquadratic extension has Galois group $V = \prod_{n \geq 1} \mathbb{F}_2$. Independence of the prime square classes proves that the continuous restriction map $G_{\mathbb{Q}} \twoheadrightarrow V$ is surjective. Regard V as an abstract vector space. The constant-one vector

does not belong to $V_0 = \bigoplus_n \mathbb{F}_2$. Choose a linear functional $\lambda : V/V_0 \to \mathbb{F}_2$ taking its class to one, and compose with the quotient map. Such a functional exists by extending a basis. It is nonzero but vanishes on the dense subgroup V_0, so is discontinuous. Let $\epsilon : G_{\mathbb{Q}} \to \mathbb{F}_2$ be the composite. It is discontinuous: a continuous surjection of compact groups is a quotient map. It vanishes on $G_{\mathbb{Q}}^{\circ}$, since every quadratic extension of $\mathbb{Q}$ is cyclotomic.

Define

$$\rho(g) = \widetilde{h}^{\epsilon(g)} \rho_x(g).$$

Commutation and additivity of ϵ prove the action law. On constants the action is still g, so it is semilinear. The stabilizer of the function t is exactly $\ker \epsilon$, which is nonopen. Every $\rho(g)$ normalizes the original deck group. On $G_{\mathbb{Q}}^{\circ}$ the action equals ρ_x, and fixes $\widetilde{x}$.

We verify uniqueness rather than assume it from an earlier AI return. Ribet's Theorem 1 in the opened appendix to Katz–Lang (Ribet, 1981) says that the torsion of an abelian variety over a number-field cyclotomic extension is finite. It also implies that no nonzero point is divisible by every integer: choose compatible division points of such a point P and descend P and the variety to a number field k_0. Their Kummer cocycle on G_{k_0} vanishes on the normal cyclotomic subgroup. The cocycle identity makes its values invariant under that subgroup, whereas the invariant Tate module is zero by Ribet. Thus all division points are already over k_0. Embed them in $A(k_{0,v})$ at a finite place of good reduction. This is a profinite group; its finite quotients include reduction modulo every power of the maximal ideal. A point divisible by every integer maps to zero in each finite quotient, and these quotients separate points. Thus $P = 0$; reduction to the residue field alone would not suffice. This holds for every finite extension of $\mathbb{Q}^{\mathrm{ab}}$, which is $k_0(\mu_\infty)$ for some number field k_0.

Here a completed pro-point means a compatible system of geometric closed points on the proper finite levels; it does not mean the generic scheme point. If a second such pro-point were fixed, choose a finite characteristic covering level separating the two points and dominating a cover with compactification of positive genus. The subgroup generated by that characteristic geometric group and $\rho_x(G_{\mathbb{Q}^{\mathrm{ab}}})$ defines a cover over $\mathbb{Q}^{\mathrm{ab}}$. The two distinct projected points on its compactification have the same section class. Their Abel–Jacobi difference has zero Kummer class and is divisible by every integer, hence zero by the preceding argument. A principal divisor of the form $P - Q$ with $P \neq Q$ would give a degree-one map to $\mathbb{P}^1$, contradicting positive genus. This proves the uniqueness assertion. □

D.8 (Proved; a failed proposed route). *Reparameterizing the coefficient action by a nonidentity abstract automorphism of $G_{\mathbb{Q}}$ does not give an action semilinear for the specified action of $G_{\mathbb{Q}}$ on $\overline{\mathbb{Q}}$.*

Proof. If the coefficient action of g is $\alpha(g)$, semilinearity requires $\alpha(g)(a) = g(a)$ for every $a \in \overline{\mathbb{Q}}$. The action defining $G_{\mathbb{Q}}$ is faithful, so $\alpha(g) = g$ for every g. An individual field automorphism of $\overline{\mathbb{Q}}$ only conjugates the Galois action and induces a continuous inner automorphism of its Krull topological group. It does not produce the discontinuous cocycle demanded in the proposed route. □

D.9 (Proved; abelianized splitting is insufficient). *There is a finite extension of groups with perfect kernel which has no section. Its quotient obtained by abelianizing the kernel has a section. Therefore vanishing of the abelianized obstruction alone is not a valid general solution of the embedding problems in the proposed route to D3–D4.*

Proof. Put $F = \mathbb{F}_9$ and let $\sigma : x \mapsto x^3$ be its nontrivial automorphism. In $\mathrm{PGL}_2(F) \rtimes \langle \sigma \rangle$ take the subgroup G consisting of $\mathrm{PSL}_2(F)$ and the elements $[A]\sigma$ with nonsquare determinant. The

determinant square class is multiplicative and invariant under σ, so this union is a subgroup. Its quotient by $N = \mathrm{PSL}_2(F)$ is $\mathbb{Z}/2$.

If an element $[A]\sigma$ in the nontrivial coset had square one, then $AA^\sigma = \lambda I$. Applying σ and conjugating by A gives $\lambda^\sigma = \lambda$, so $\lambda \in \mathbb{F}_3^*$. Taking determinants gives

$$(\det A)^4 = \lambda^2 = 1.$$

A nonsquare in the cyclic group $\mathbb{F}_9^*$ has fourth power -1, a contradiction. Thus the nontrivial coset contains no involution and the quotient has no section.

For completeness N is perfect. The upper and lower unipotent matrices generate $\mathrm{SL}_2(F)$ by elementary row operations. Conjugation by $\mathrm{diag}(d, d^{-1})$, with $d^2 \neq 1$, multiplies the upper parameter by d^2. Its commutators with upper unipotents therefore give every upper unipotent, since $d^2 - 1 \neq 0$. The analogous lower calculation does the same for lower unipotents. Passing to the projective quotient proves perfectness. Hence $G/[N, N] = G/N$ and the abelianized-kernel extension is the identity on $\mathbb{Z}/2$.

The program `code/D_embedding_obstruction.py` enumerates all 720 projective matrices, checks the 360 nonsquare-coset elements, and verifies that none squares to the identity after the field involution. It also verifies generation of the 360-element kernel by elementary matrices. This is a counterexample to the abstract inference, not to D3 or D4 for absolute Galois groups: the latter require their particular nonabelian kernels and local conditions to be used. □

4.5 An abstract action outside every finite quotient

D.10 (Proved). *Let $\mathcal{U}$ be the universal pro-cover of $U = \mathbb{P}^1_{\overline{\mathbb{Q}}} \setminus \{0, 1, \infty\}$ and Ω its function field. There are a continuous tangential semilinear action ρ_0 of $G_\mathbb{Q}$, a completed cusp c, and an injective homomorphism*

$$T : (\mathbb{A}_f, +) \longrightarrow \mathrm{Aut}_{\overline{\mathbb{Q}}}(\mathcal{U})$$

such that every T_a fixes c,

$$\rho_0(g) T_a \rho_0(g)^{-1} = T_{\chi(g)a}, \qquad \mathrm{Stab}_{\mathbb{A}_f}(\lambda) = \widehat{\mathbb{Z}}.$$

Here λ is the coordinate on U, viewed in Ω, and $\mathbb{A}_f$ denotes the additive finite adeles.

Proof. Fix $\overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$. The classical modular uniformization identifies $\mathfrak{H}/\Gamma(2)$ with $U(\mathbb{C})$, and for $q = \exp(\pi i \tau)$ its coordinate has expansion

$$\lambda(q) = 16q - 128q^2 + 704q^3 + \cdots \in \mathbb{Q}[[q]].$$

The product formula and this uniformization are given in Calegari et al. (2025, p. 628, equation (1.0.2) and the following paragraph). Only this classical coordinate description is used, not that paper's unbounded-denominators theorem. Analytic finite-cover comparison is SGA 1, XII, Theorem 5.1 (Grothendieck and et al., 2003). Choose the compatible cusp above $\tau = i\infty$ to embed

$$\Omega \hookrightarrow \mathcal{P} := \bigcup_{n \geq 1} \overline{\mathbb{Q}}((q^{1/n})).$$

An element of Ω lifts to a meromorphic function on $\mathfrak{H}$, invariant under a finite-index subgroup of $\Gamma(2)$ and meromorphic at its cusps. Its selected Puiseux germ has algebraic coefficients. The coordinate ring of $\mathcal{U}$ consists of the elements with no poles on $\mathfrak{H}$; this will also check the affine pro-curve, not just its field.

Every finite adele is $a = r + z$ with $r \in \mathbb{Q}$ and $z \in \widehat{\mathbb{Z}}$, and $\mathbb{Q} \cap \widehat{\mathbb{Z}} = \mathbb{Z}$. Thus the quotient $\mathbb{A}_f/\widehat{\mathbb{Z}} \simeq \mathbb{Q}/\mathbb{Z}$ defines the character

$$\psi : \mathbb{A}_f \longrightarrow \mu_\infty(\mathbb{C}), \qquad \psi(r + z) = \exp(2\pi i r).$$

For a Puiseux series set

$$T_a\left(\sum_k b_k q^{k/n}\right) = \sum_k b_k \psi(ka/n) q^{k/n}.$$

Changing a denominator n to mn changes k to mk, so this formula is well-defined. It gives $T_a T_b = T_{a+b}$, fixes constants, and preserves the q-valuation.

If $z \in \widehat{\mathbb{Z}}$, then T_z fixes q and hence $\lambda(q)$. It is the usual cusp-inertia action on the Puiseux roots. Since $\Omega/\overline{\mathbb{Q}}(\lambda)$ is normal with allowed branch locus $\{0, 1, \infty\}$, this preserves Ω and the affine coordinate ring. For $r \in \mathbb{Q}$, the operator T_r is analytically $f(\tau) \mapsto f(\tau + 2r)$. The matrix $u(2r) = \left(\begin{smallmatrix} 1 & 2r \\ 0 & 1 \end{smallmatrix}\right)$ commensurates $\Gamma(2)$: clearing a denominator d shows that conjugation of $\Gamma(2d^2)$ lands in $\Gamma(2)$, and the same argument applies to the inverse. Therefore if f is invariant under Γ_f of finite index in $\Gamma(2)$, then $T_r f$ is invariant under the finite-index subgroup

$$\Gamma(2) \cap u(2r)^{-1} \Gamma_f u(2r).$$

It is meromorphic at every cusp. Consequently $T_r f$ is algebraic over $\mathbb{C}(\lambda)$ with branching only over $0, 1, \infty$.

We check descent here explicitly. The Puiseux coefficients of $T_r f$ remain algebraic. Take a polynomial relation over $\mathbb{C}(\lambda)$ and clear denominators. Its finitely many monomials in λ and $T_r f$ have Laurent coefficients in $\overline{\mathbb{Q}}$. The homogeneous linear system obtained by comparing coefficients has a nonzero complex solution, hence a nonzero solution over $\overline{\mathbb{Q}}$: its rank is determined by finite minors. Thus $T_r f$ is algebraic over $\overline{\mathbb{Q}}(\lambda)$ and belongs to Ω. Moreover $\tau \mapsto \tau + 2r$ preserves $\mathfrak{H}$ and the rational cusps, so it preserves absence of poles on $\mathfrak{H}$. The inverse is T_{-r}. Every $a = r + z$ now acts on $\mathcal{U}$ as asserted. This proof does not assume uniqueness of roots in the full automorphism group.

These automorphisms also act on the completed pro-cover. Indeed the image of any finite-level function field is contained in a further finite level. The resulting rational map between the smooth proper compactifications extends uniquely everywhere, by properness and normality for curves. The extensions are compatible under composition and passage to further levels. Applying the inverse operator proves that they define an automorphism of the completed pro-object.

The coefficient of q in $T_a\lambda$ is $16\psi(a)$. It follows that $T_a\lambda = \lambda$ if and only if $a \in \widehat{\mathbb{Z}}$: sufficiency follows because T_a then fixes q. If such an a acts trivially on all of Ω, apply it to $\lambda^{1/n} \in \Omega$ for every n. Their leading terms force $\psi(a/n) = 1$, so $a \in \bigcap_n n\widehat{\mathbb{Z}} = 0$. This proves faithfulness.

Let $\rho_0(g)$ act coefficientwise on $\mathcal{P}$, fixing every $q^{1/n}$. It restricts to the continuous tangential action on Ω supplied by $\operatorname{Spec} \mathbb{Q}((q)) \to U$, $\lambda \mapsto \lambda(q)$. The parameter q is a rational formal uniformizer; it need not belong to Ω. Directly on each Puiseux monomial,

$$g\big(\psi(a/n)\big) = \psi\big(\chi(g)a/n\big)$$

proves the displayed conjugation identity. Both types of operator preserve the selected valuation, hence its completed cusp c. □

D.11 (Proved). *There is an abstract semilinear action ρ of $G_{\mathbb{Q}}$ on $\mathcal{U}$ that normalizes no deck group of a finite hyperbolic quotient. Its $G_{\mathbb{Q}}^{\circ}$-fixed set on the completed pro-cover is precisely the single cusp c of D.10. This answers the second demand of D5. It does not settle D5's universal fixed-point assertion.*

Proof. Fix an odd prime ℓ and distinct rational primes $p_1, p_2, \ldots$ different from ℓ. Compatible radicals give Kummer cocycles and their continuous product

$$\kappa : G_{\mathbb{Q}} \longrightarrow V := \prod_{i \geq 1} \mathbb{Z}_\ell, \qquad \kappa(gh) = \kappa(g) + \chi_\ell(g)\kappa(h).$$

We have chosen a compatible basis of the Tate twist in writing these coordinates. Its restriction to $G_{\mathbb{Q}}^{\circ}$ is a continuous surjective homomorphism. Indeed, for finitely many p_i and $m \geq 1$, put

$$K = \mathbb{Q}(\mu_{\ell^m}), \qquad L = K(p_i^{1/\ell^m}).$$

The extension $K/\mathbb{Q}$ is unramified at the p_i. Valuations there show that their classes in $K^{\times}/(K^{\times})^{\ell^m}$ are independent of order ℓ^m. Kummer theory gives $T := \mathrm{Gal}(L/K) \simeq (\mathbb{Z}/\ell^m)^r$. The extension $L/\mathbb{Q}$ is Galois, and a lift of a cyclotomic automorphism with character 2 modulo ℓ acts on T by that scalar. Its commutators with T exhaust T, since $\chi_\ell - 1$ is a unit. As the quotient by T is abelian, T is exactly the commutator subgroup. Hence $L \cap \mathbb{Q}^{\mathrm{ab}} = K$, which proves the required surjection from $G_{\mathbb{Q}}^{\circ}$ at every finite level. Compactness gives the surjection onto V.

Let $e_i \in V$ be the coordinate vectors. They are linearly independent in $V \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$. Extend them to a vector-space basis and choose a linear functional taking e_i to ℓ^{-i}. Its restriction is a $\mathbb{Z}_\ell$-linear map

$$f : V \longrightarrow \mathbb{Q}_\ell, \qquad f(e_i) = \ell^{-i}.$$

This basis choice uses the axiom of choice and specifies an abstract cocycle; it is not an effective algorithm for evaluating the cocycle on arbitrary Galois elements. It is onto, because its image contains $\ell^{-i}\mathbb{Z}_\ell$ for every i. Its kernel is dense: realize any finitely many prescribed coordinates by a finitely supported v. Choose j outside those coordinates so large that $\ell^j f(v) \in \mathbb{Z}_\ell$, and use $v - \ell^j f(v) e_j$. This has the prescribed coordinates and is killed by f.

Regard $\mathbb{Q}_\ell$ as the subgroup of $\mathbb{A}_f$ supported at ℓ and set $a_g = f(\kappa(g))$. Linearity gives $a_{gh} = a_g + \chi(g) a_h$. Therefore

$$\rho(g) = T_{a_g} \rho_0(g)$$

is an action by D.10, semilinear for the specified usual action on $\overline{\mathbb{Q}}$. Moreover

$$a(G_{\mathbb{Q}}^{\circ}) = \mathbb{Q}_\ell, \qquad K_0 := \ker(a|_{G_{\mathbb{Q}}^{\circ}}) \text{ is dense in } G_{\mathbb{Q}}^{\circ}.$$

For density, a continuous surjective homomorphism of compact groups is an open quotient map; apply this to κ and the dense kernel of f.

The orbit of λ is infinite: ρ_0 fixes this rational coordinate and D.10 gives

$$\{\rho(g)\lambda : g \in G_{\mathbb{Q}}^{\circ}\} \simeq \mathbb{Q}_\ell/\mathbb{Z}_\ell.$$

We show why this rules out *every* normalized finite hyperbolic quotient, without using a congruence-quotient comparison. If a deck group H were normalized, the action would descend to its hyperbolic quotient W and the field $F = \Omega^H$. Choose a number field K defining W and finitely many function-field generators. For $g \in G_K$, compare the descended semilinear automorphism to the coefficient action of that K-model. Their quotient is in the finite group $\mathrm{Aut}_{\overline{\mathbb{Q}}}(W)$. Thus those generators have finite G_K-orbits, and then finite $G_{\mathbb{Q}}$-orbits. Constants also have finite orbits, so every element of F does. Each element of Ω is algebraic over F. Its orbit is among the roots of the finitely many conjugates of a polynomial equation over F, hence is finite. This contradicts the orbit of λ. No continuity was presumed in this argument.

It remains to determine the fixed set. Every T_a and $\rho_0(g)$ fixes c, so c is fixed by ρ. If a completed geometric closed pro-point P is fixed by $\rho(G_{\mathbb{Q}}^{\circ})$, it is fixed by $\rho_0(K_0)$. Its stabilizer under ρ_0 is closed: take cofinal characteristic finite levels, descending over $\mathbb{Q}$ under ρ_0. At each level an algebraic closed point has an open and closed Galois stabilizer, and the stabilizer of P is their intersection. Density of K_0 implies that P is fixed by $\rho_0(G_{\mathbb{Q}}^{\circ})$.

The same Kummer argument as in D.7 proves that the latter fixed set is $\{c\}$. Explicitly, if $P \neq c$, choose a characteristic covering level separating their projections and dominating the degree-nine Fermat cover $x^3 + y^3 = 1$, $\lambda = x^3$, of U. Its smooth proper compactification has

positive genus. More explicitly, if $N \subset \Pi_U$ is this characteristic subgroup, then $N\rho_0(G_{\mathbb{Q}}) \subset \pi_1(U_{\mathbb{Q}})$ defines a finite étale curve over $\mathbb{Q}$; take its smooth proper compactification over $\mathbb{Q}$. The two projected $\mathbb{Q}^{\mathrm{ab}}$-points have the same proper section class, since the common action fixes both pro-points. Their Abel–Jacobi difference has zero Kummer class and therefore vanishes by F.3, based on the precise abelian Kummer theorem of Tsujimura (2020, Theorem 3.1). Distinct points on a positive-genus smooth proper curve cannot have principal difference: that would give a degree-one map to $\mathbb{P}^1$. The contradiction proves $P = c$. □

The constructed action lies outside the finite-quotient normalization cases covered by R19, while satisfying the predicted fixed-point conclusion itself. Thus automatic finite-quotient descent cannot reduce all abstract actions to B. This construction alone neither proves nor refutes the universal assertion in B′, and does not exclude every logically different reduction to B. The separate construction in D.12–D.14 below refutes the universal assertion by an action with empty fixed set.

4.6 A discontinuous action with no cyclotomic fixed pro-point

D.12 (Proved). *For $U = \mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$, put $E = \pi_1(U)$ and $\Pi = \pi_1(U_{\overline{\mathbb{Q}}})$, and fix a decomposition embedding $G_{\mathbb{Q}_3} \hookrightarrow G_{\mathbb{Q}}$. There is an abstract, provably discontinuous section*

$$s : G_{\mathbb{Q}} \longrightarrow E$$

whose restriction to $G_{\mathbb{Q}_3}$ is geometrically conjugate to the continuous point section at the explicit transcendental point

$$a = 1 + \sum_{r=1}^{\infty} 3^{r!} \in U(\mathbb{Q}_3).$$

It induces an abstract semilinear action on the universal pro-cover and its completed proper levels, normalizing the original geometric deck group Π.

Proof. First verify transcendence, rather than assume a generic local point. Put $A_N = 1 + \sum_{r=1}^{N} 3^{r!}$. Then

$$v_3(a - A_N) = (N+1)!, \qquad 1 < A_N < 2 \cdot 3^{N!}.$$

If a nonzero $P \in \mathbb{Z}[T]$ of degree d vanished at a, then $3^{(N+1)!}$ would divide $P(A_N)$. The latter integer is nonzero for all sufficiently large N, since the A_N are distinct. On the other hand $|P(A_N)|_\infty \le C(2 \cdot 3^{N!})^d$ for a fixed C. This contradicts $(N+1)! > dN! + O(1)$. Thus a is transcendental over $\mathbb{Q}$. It is a 3-adic unit, and $v_3(a-1) = 1$, so it is an interior point of U.

Construct positive integers $x_n \ge 2$ recursively. Choose a prime $q_n \ne 3$ which divides none of $x_1, \dots, x_{n-1}$ and impose

$$\begin{aligned} x_n &\equiv a \pmod{3^n}, \\ x_n &\equiv q_n \pmod{q_n^2}, \\ x_n &\equiv 1 \pmod{q_i} \quad (1 \le i < n). \end{aligned}$$

The primes q_n are necessarily distinct, and the Chinese remainder theorem gives such an integer. One may always choose the least permitted prime and the least positive CRT representative. That representative is neither zero nor one because of its congruence modulo q_n^2. The first congruence is effective: only terms $3^{r!}$ with $r! < n$ contribute. We have

$$x_n \longrightarrow a \text{ in } \mathbb{Q}_3, \qquad v_{q_i}(x_j) = \delta_{ij} \quad (i, j \ge 1).$$

For $j < i$ the valuation assertion is the choice of q_i; for $j > i$ it is the third congruence. In particular the square classes of the x_n are independent. The exact twelve-term CRT and valuation certificate is in `code/F_D5_ultralimit_output.json`; its role is only to check this finite arithmetic input, not the ultrafilter or the fixed-point argument.

Choose any continuous representatives s_n of the global point sections at x_n. Fix a nonprincipal ultrafilter $\mathfrak{u}$ on $\mathbb{N}$ and, for every $g \in G_{\mathbb{Q}}$, define

$$s(g) = \lim_{n \to \mathfrak{u}} s_n(g) \quad \text{in } E.$$

Compact Hausdorffness gives this limit. Continuity of multiplication in E proves $s(gh) = s(g)s(h)$; continuity of $E \to G_{\mathbb{Q}}$ gives $\mathrm{pr}(s(g)) = g$. Thus s is an abstract section. This argument does not assert continuity of a limit of continuous maps with the pointwise topology.

We prove exactly the local compatibility required below. Characteristic-zero geometric base change identifies $\pi_1(U_{\mathbb{Q}_3})$ with $E \times_{G_{\mathbb{Q}}} G_{\mathbb{Q}_3}$, so the local point section s_a is viewed as a homomorphism $G_{\mathbb{Q}_3} \to E$. For a finite arithmetic quotient $\varphi : E \twoheadrightarrow F$, let $P = \varphi(\Pi)$. Its finite *étale* Galois cover $W \to U$ has constant quotient $W/P = U \times \operatorname{Spec} K_0$, where $K_0/\mathbb{Q}$ is the finite constant extension with group F/P. This is the ordinary arithmetic fundamental-group covering correspondence (Grothendieck et al., 2003, Exposé IX, §6). For b sufficiently close to a in $U(\mathbb{Q}_3)$, Hensel's lemma gives a Galois-equivariant, F-equivariant bijection

$$W_b(\overline{\mathbb{Q}}_3) \xrightarrow{\sim} W_a(\overline{\mathbb{Q}}_3)$$

inducing the identity on the indicated constant quotient. Here is the uniformity in that statement. Choose a finite Galois extension $L/\mathbb{Q}_3$ splitting the finite fibre W_a. The *étale* inverse-function form of Hensel's lemma gives, through every point of that fibre, a unique local L-analytic sheet over a sufficiently small ball about a. There are finitely many sheets, deck transformations, and elements of $\mathrm{Gal}(L/\mathbb{Q}_3)$, so one common ball works. Uniqueness of these sheets makes all their actions compatible. The constant coordinates remain their original algebraic values. This proves the asserted equivariance and the identity on W/P.

In torsor coordinates, such a bijection is multiplication by an element of P, since it is the identity modulo P. Consequently for all sufficiently large n the restrictions $\varphi s_n|_{G_{\mathbb{Q}_3}}$ and $\varphi s_a|_{G_{\mathbb{Q}_3}}$ are conjugate by P. This is geometric conjugacy, not merely conjugacy by an arbitrary arithmetic element of F. The P-conjugacy orbit of the latter homomorphism is a finite closed subset of $F^{G_{\mathbb{Q}_3}}$. Its ultralimit therefore still belongs to that orbit. For each φ there is a geometric conjugator at this finite level.

The set of elements $\gamma \in \Pi$ satisfying

$$\varphi\big(\gamma s(g)\gamma^{-1}\big) = \varphi(s_a(g)) \qquad (g \in G_{\mathbb{Q}_3})$$

is nonempty and closed in Π. Any finitely many such conditions have a common refinement given by a finite quotient of E. Compactness of Π supplies one γ satisfying all of them. Finite quotients separate elements of E, so

$$\gamma s|_{G_{\mathbb{Q}_3}} \gamma^{-1} = s_a$$

as actual homomorphisms. In particular the local restriction of the abstract section is continuous.

Its global discontinuity is independently visible in one finite quotient. The cover $z^2 = t$ gives a continuous character $\tau : E \to \mathbb{F}_2$ such that $\tau s_n = \kappa_2(x_n)$. Independence of the square classes gives a continuous surjection

$$\kappa : G_{\mathbb{Q}} \twoheadrightarrow \prod_{n \geq 1} \mathbb{F}_2, \qquad g \longmapsto (\kappa_2(x_n)(g))_n.$$

Indeed it is surjective on every finite product by Kummer theory, and compactness gives surjectivity onto the product. The ultrafilter functional

$$\epsilon : \prod_n \mathbb{F}_2 \longrightarrow \mathbb{F}_2, \qquad (v_n) \longmapsto \lim_{n\to \mathfrak{u}} v_n$$

is additive. It kills the dense direct sum but sends the constant-one vector to one, so it is discontinuous. A continuous surjection of compact Hausdorff groups is a quotient map; hence $\epsilon\kappa$ is discontinuous. By construction $\tau s = \epsilon\kappa$, proving the claimed discontinuity without any section-conjecture input.

Realize E as the arithmetic automorphism group of the universal pro-cover of U. Each $s(g)$ is a semilinear automorphism, and the homomorphism law gives the required abstract action. It normalizes Π, which is normal in E. At every finite covering field the automorphism extends uniquely to its proper normalization; compatibility gives the action on the completed pro-cover. The stabilizer of the function $\sqrt{t}$ is the nonopen subgroup $\ker(\epsilon\kappa)$, so the induced action itself is discontinuous. The explicit sequence and all finite approximations are effective; choosing a nonprincipal ultrafilter is a choice-principle step and does not give an algorithm for the abstract section.

In the crossed-cocycle form requested in R22, fix a continuous reference section ρ_0 at $1/2$ and put

$$a_g = s(g)\rho_0(g)^{-1} = \lim_{n\to \mathfrak{u}} s_n(g)\rho_0(g)^{-1} \in \Pi.$$

Then

$$a_{gh} = a_g\, \rho_0(g) a_h \rho_0(g)^{-1}, \qquad a_g \rho_0(g) = s(g).$$

Thus this explicitly specifies an abstract crossed cocycle for the reference action, and its twisted action is exactly the action constructed above. D.14 computes its fixed set as empty. □

D.13 (Proved). *Let $L/\mathbb{Q}_p$ be finite, let $\chi : G_L \to \widehat{\mathbb{Z}}^\times$ be the full cyclotomic character, and put $H = \ker \chi$. If $b \in L^*$ satisfies*

$$\kappa_p(b)|_H = 0 \quad \text{in } H^1(H, \mathbb{Z}_p(1)),$$

then b is a root of unity. In particular, if the restrictions of the point sections of $\mathbb{G}_m/L$ at $b_1, b_2 \in L^*$ are geometrically conjugate on H, then b_1/b_2 is a root of unity.

Proof. Choose compatible p-power roots and let $c : G_L \to \mathbb{Z}_p(1)$ be the resulting continuous Kummer cocycle. Since H acts trivially on the coefficient module, every H-coboundary is zero. The hypothesis therefore says $c|_H = 0$ literally. The cocycle descends to a continuous cocycle on the compact abelian quotient $A = \chi(G_L)$: its values are constant on H-cosets, and the compact quotient topology gives continuity. This proves the needed inflation assertion directly for the compact coefficient module.

There is $a_0 \in A$ with $\chi_p(a_0) \neq 1$, since a finite extension of $\mathbb{Q}_p$ does not contain all p-power roots of unity. The cocycle identity, evaluated on $a_0 a = a a_0$, gives

$$(\chi_p(a_0) - 1)c(a) = (\chi_p(a) - 1)c(a_0) \qquad (a \in A).$$

The right side is a coboundary. Thus $(\chi_p(a_0) - 1)[c] = 0$ in $H^1(A, \mathbb{Z}_p(1))$. Writing the nonzero scalar as p^m times a $\mathbb{Z}_p$-unit shows that $p^m \kappa_p(b) = 0$ after inflation to G_L. This is the elementary cocycle calculation often called Sah's lemma; no vanishing assertion for a nonabelian quotient is being used.

It follows at every finite Kummer level that $b^{p^m} \in (L^*)^{p^n}$ for every n. We record the local intersection precisely. Write

$$L^* = \varpi^{\mathbb{Z}} \times \mu_{q-1}(L) \times U^1, \qquad U^1 = 1 + \mathfrak{m}_L,$$

where q is the residue-field cardinality. Hensel's lemma gives the displayed prime-to-p roots of unity. The group U^1 is pro-p, since its quotients modulo $1+\mathfrak{m}_L^r$ have p-power order. An element divisible by every p^n in L^* has valuation zero; after removing its unique μ_{q-1} part, its image in each finite p-group quotient of U^1 is a p^n-th power for every n, and therefore identity. The filtration separates points. Hence

$$\bigcap_{n\geq 1}(L^*)^{p^n} = \mu_{q-1}(L).$$

Thus b^{p^m} is a root of unity, and so is b.

The final assertion is the Kummer description of $\pi_1(\mathbb{G}_m/L) = \widehat{\mathbb{Z}}(1)\rtimes G_L$ with the splitting at 1. Point-section classes are Kummer classes and their difference is the class of b_1/b_2 (Stix, 2008b, §2.2 and §8.1–8.3, especially Proposition 26). Restrict to H and apply the result just proved. □

D.14 (Refuted). *The universal assertion D5, that every abstract semilinear $G_{\mathbb{Q}}$-action on the universal pro-cover has exactly one $G_{\mathbb{Q}}^{\circ}$-fixed completed geometric closed pro-point, is false. The action of D.12 has* no *such fixed pro-point, even though it normalizes the original geometric deck group.*

Proof. Here $G_{\mathbb{Q}}^{\circ} = \ker\chi = G_{\mathbb{Q}(\mu_\infty)}$. The completed pro-points are exactly the compatible systems of geometric closed points on the proper finite levels, as in D.7; the generic scheme point is not part of this set. Suppose that a completed pro-point ξ were fixed by $s(G_{\mathbb{Q}}^{\circ})$, and let $x\in\mathbb{P}^1(\overline{\mathbb{Q}})$ be its image at the base compactification.

Use the fixed embedding $\overline{\mathbb{Q}}\hookrightarrow\overline{\mathbb{Q}}_3$ underlying the decomposition group in D.12, and choose a finite $L/\mathbb{Q}_3$ containing the coordinates of x. Put

$$H = G_L\cap G_{\mathbb{Q}}^{\circ} = \ker(\chi : G_L\to\widehat{\mathbb{Z}}^{\times}).$$

Conjugate s by the geometric element supplied by D.12, and conjugate ξ with it. Its base point remains x, and we now have $s|_{G_{\mathbb{Q}_3}} = s_a$. Consequently $s_a(H)$ fixes a completed pro-point over x.

Choose the following unit of the open tripod:

x	$f(t)$	$f(x)$
$x\in U(\overline{\mathbb{Q}})$	t	x
0	$1-t$	1
1	t	1
∞	$t/(t-1)$	1

In every case f is regular and nonzero at x. The morphism $f: U\to\mathbb{G}_m$ therefore sends the decomposition group of the fixed pro-point over x into the point decomposition group at $f(x)$. At a cusp its geometric inertia is killed, since the order of f there is zero. Equivalently, the covers $w^{3^n} = f(t)$ extend *étale* across x and their fibre at x consists of the 3^n-th roots of $f(x)$. Evaluating the section in these covers gives

$$\kappa_3\big(f(a)/f(x)\big)|_H = 0 \quad \text{in } H^1(H,\mathbb{Z}_3(1)).$$

This is the usual functorial evaluation of units on sections (Stix, 2008b, Proposition 26 and §8.3). Changing the selected roots changes only a coboundary; on H that coboundary is zero because the cyclotomic action is trivial. The argument applies to interior points and to each cusp, with no supposition that the original abstract global section is continuous.

Both $f(a)$ and $f(x)$ belong to L^*, and D.13 gives $f(a)/f(x) \in \mu(L)$. The number $f(x)$ is algebraic over $\mathbb{Q}$, as is every root of unity. Hence $f(a)$ would be algebraic over $\mathbb{Q}$. Each chosen f is an invertible fractional-linear function over $\mathbb{Q}$, so this would make a algebraic, contradicting D.12. Thus the fixed set is empty. It is the existence part of the asserted unique fixed point that fails in this example; no assertion of two distinct fixed points is made.

D.11 and this counterexample address the two demands of D5 separately: D.11 constructs an action normalizing no finite-quotient deck group and having exactly one fixed cusp, while the present action disproves the universal fixed-point claim. The latter uses discontinuous abstract actions and proves no counterexample to the continuous section conjecture in D1–D4. □

The former obligation D.U5 is superseded by D.12–D.14: the universal fixed-point assertion is false, with an empty fixed set already possible for an action normalizing the original deck group. D.11 separately supplies the requested action normalizing no finite-quotient deck group.

4.7 The exact unresolved obligations

D.U1 (Unresolved: D1). *For every continuous section $s : G_{\mathbb{Q}} \to \pi_1(U)$ with $C_{\Pi_U}(s(G_{\mathbb{Q}}^{\circ})) = 1$, there exists $x \in U(\mathbb{Q})$ such that s is geometrically conjugate to s_x.*

We checked the local-to-global Kummer route, finite-support descent, and the stronger birational criterion. None establishes uniform adelicity or birational liftability for an arbitrary section. Even accepting R24 would give local sections without this missing global condition. No conclusion about all Fermat twists follows from a computation for one twist with a rank-zero quotient.

D.U2 (Unresolved: D2). *Every continuous splitting $G_{\mathbb{Q}} \to G_{\mathbb{Q}(t)}$ belongs to the decomposition group of a rational place.*

D.3 settles the proposed arithmetic criterion for a given Kummer class; D.4 corrects its evaluation formula. What remains is to prove the common finiteness of the nonunit places needed for that criterion, or another condition forcing the place. Density-one integrality alone does not do so.

D.U3 (Unresolved: D3). *Every section of every hyperbolic curve over every field finitely generated over $\mathbb{Q}$ admits a birational lift.*

D.1 gives the exact equivalence with the section conjecture. No solution of the required global finite embedding problems, with compatible choices, has been constructed here. An abelianized H^2 obstruction supplies a necessary condition only; its vanishing is not a splitting of the nonabelian extension.

D.U4 (Unresolved: D4). *Every continuous section of $\pi_1(U) \to G_{\mathbb{Q}}$, for $U = \mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$, lifts through $G_{\mathbb{Q}(t)} \to \pi_1(U)$.*

D.2 proves exactly what combining this with D2 would yield. The remaining kernel is generated by the inertia of all deleted closed points; replacing it by an abelian product does not solve the original extension problem.

5 Part E. Categories of curves and the arithmetic image

The manuscript loci are cote 142, manuscript pp. 7–10, scans 15–21. We have opened the supplied images of scans 17 and 19. The phrase "programme à l'horizon" on scan 17 concerns the arithmetic action as well as the category. On scan 19 the proposed isotopy relation is explicitly tentative. Neither passage licenses identifying an arbitrary path of local homeomorphisms with a two-sided orbit. We distinguish full *étale* paths in a moduli stack, paths tame at its

compactification boundary, and its maximal prime-to-p quotient. These are three different objects.

5.1 Two obstructions to the requested positive-characteristic construction

E.1 (Proved; an obstruction with explicit hypotheses). *Let $k = \overline{\mathbb{F}}_p$ and $k' = \overline{k(t)}$. On $U = \mathbb{P}^1_k \setminus \{0, 1, \infty\}$ the connected $\mathbb{Z}/p$-torsor*

$$z^p - z = tx$$

over $U_{k'}$ does not descend to U, even after forgetting its chosen identification of the deck group with $\mathbb{Z}/p$. Consequently the base-extension map of full geometric fundamental groups is not an isomorphism. A category having the marking groupoids of the full moduli stacks as its isomorphism groupoids cannot have base extension as an equivalence in characteristic p. This remains an obstruction if only the noninvertible covering arrows are restricted to prime-to-p Galois closures.

Proof. The Artin–Schreier sequence on the affine curve gives

$$H^1(U_{k'}, \mathbb{Z}/p) = k'[x, x^{-1}, (x-1)^{-1}]/\{h^p - h\}.$$

Modulo an Artin–Schreier coboundary, a principal part has a unique representative whose pole orders are prime to p, after successively removing terms of order divisible by p. The field is perfect, so the necessary p-th roots of coefficients exist uniquely. Uniqueness follows because a nonconstant h has at every pole a leading term of $h^p - h$ of order divisible by p. Constants contribute nothing over an algebraically closed field. In the displayed class the sole pole is at infinity, of order one, with coefficient t. A class defined over k has reduced principal parts with coefficients in k. Therefore this class cannot descend. It is nonzero, which makes the torsor connected. Relabelling its deck group multiplies its class by an element of $\mathbb{F}_p^*$ and does not remove t.

Now $\mathcal{M}_{0,4}$ with ordered marks is this same curve. Its full fundamental group is therefore not invariant under this base extension. For unordered marks, the ordered moduli scheme is a finite *étale* cover of the quotient stack, with constant permutation group S_4. An isomorphism on the fundamental groups of the unordered stack, compatible with this cover, would induce an isomorphism on the corresponding open kernels, contrary to the ordered calculation. Thus either convention has the obstruction.

In the factorization category of (Z.2), an endomorphism of a fixed hyperbolic type which factors as a marking followed by a finite *étale* cover must have covering degree one. Indeed, for a degree-d cover of open curves in characteristic p, the Riemann–Hurwitz formula, including the nonnegative wild different at the boundary, gives

$$2g_X - 2 + n_X \;\geq\; d(2g_Y - 2 + n_Y).$$

Equality of the two hyperbolic types forces $d = 1$. The isomorphism groupoid is consequently precisely the specified marking groupoid, when markings are retained faithfully as in (Z.2). Its failure of base invariance already contradicts equivalence of categories.

The qualification about faithful markings matters. If an unspecified "category" is allowed to identify arbitrary paths or arbitrary covering arrows, the words "arrows include" alone do not determine its Hom sets. The Artin–Schreier example is not a proof about every such artificial quotient category. It proves the obstruction for the exact full-path construction requested by the manuscript and R26. The coefficient and pole-order calculations for $p = 2, 3, 5$ are also checked in `code/root_checks.py`; the argument above proves it for every p. □

E.2 (Refuted). *There is no type-preserving functor from the entire characteristic-zero category $\mathcal{T}$ to a category whose arrows all factor as a type-preserving marking followed by a finite* étale *map with prime-to-p Galois closure, if the functor is required to specialize every finite-cover arrow. This obstruction holds for every prime p, including 2 and 3.*

Proof. Let $C/\mathbb{C}$ be a smooth proper genus-two curve. The presentation

$$\pi_1(C^{\mathrm{an}}) = \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = 1 \rangle$$

surjects onto $\mathbb{Z}/p$ by sending a_1 to one and the other generators to zero. The connected unramified topological cover algebraizes by SGA 1 XII, Theorem 5.1 and Corollary 5.2 (Grothendieck et al., 2003). Its source D has genus $p+1$, because $2g_D - 2 = p(2g_C - 2) = 2p$.

A type-preserving functor would give an arrow from type $(p+1, 0)$ to type $(2, 0)$. A marking does not change the type. Any subsequent finite *étale* cover between these proper types in characteristic p has degree

$$\frac{2(p+1)-2}{2 \cdot 2 - 2} = p.$$

The order of its geometric Galois closure is divisible by its degree, so cannot be prime to p. No such arrow exists. This contradiction does not depend on selecting a bad model for D: it occurs in the types before any model is chosen.

Accordingly the source of a possible specialization functor must at least be restricted to arrows whose geometric Galois closures have order prime to p. Replacing all moduli markings by their maximal prime-to-p quotients is not automatically the right further repair: a prime-to-p covering group can have an automorphism group with p-torsion, and transport of such covers can detect that torsion. Tameness at the boundary is a different restriction. □

E.3 (Known, with derivation; the fixed-curve replacement). *Let $(\mathcal{C}, \mathcal{D})$ be a smooth proper curve with finite* étale *marked divisor over a strictly henselian complete discrete valuation ring of mixed characteristic $(0, p)$, with algebraically closed residue field. Specialization induces an isomorphism on the maximal prime-to-p geometric fundamental groups of the two open fibres. Hence connected covers with prime-to-p Galois closure correspond, preserving degree, deck group, and the type of source. The analogous category of covers of a fixed smooth marked curve over an algebraically closed characteristic-p field is invariant under extension of that field.*

Proof. The precise sources are SGA 1 XIII, Corollary 2.12 and Proposition 4.3, its Examples 4.4, and the base-change statement 4.5.3 (Grothendieck et al., 2003). Their hypotheses are a proper smooth relative curve, a relatively normal-crossings boundary finite *étale* over the base, and orders invertible at the specialization. A section exists by lifting a smooth residue point over the strictly henselian base. These hypotheses are satisfied here. In this setting specialization is surjective on the tame group and is an isomorphism on its maximal prime-to-residue-characteristic quotient. A connected cover with prime-to-p Galois closure is a transitive set for a finite prime-to-p quotient of that group. An isomorphism of the groups identifies these finite sets, their equivariant maps, their sizes, and their automorphism groups. Boundary inertia subgroups correspond by the same Kummer-parameter comparison; thus their ramification indices and the number of points above each mark are unchanged. Tame Riemann–Hurwitz then gives the same genus. For an extension of algebraically closed fields use the relative-group base-change statement in XIII, 4.5.3, applied over the algebraically closed base field with a chosen point; the same finite-set argument gives the asserted equivalence.

This is a theorem about covering categories over a fixed curve. It does not prove the missing assertion about paths and composition over all moduli stacks. Bertin–Romagny, Theorem 6.9,

constructs the smooth proper stable Hurwitz stack over $\mathbb{Z}[1/|G|]$; Proposition 6.14 gives its proper quasi-finite flat discriminant map, *étale* over the smooth target locus; and Theorem 6.20 identifies the non-Galois-cover stack by the quotient recording its Galois closure (Bertin and Romagny, 2007). These are the exact additional inputs available for that construction. They do not by themselves say that the *full* path groupoid of the nonproper moduli stack is invariant; E.1 proves that it is not. □

5.2 The arithmetic-image route and a theorem that it omits

E.4 (Known, with derivation). *The conjugacy class of the image of $G_{\mathbb{Q}}$ in the profinite Grothendieck–Teichmüller group has a published combinatorial characterization. It is the class of maximal closed subgroups satisfying the AA-property of Hoshi–Mochizuki–Tsujimura, Definition 5.12, with the COF, RGC and ArBC hypotheses required there. Determining this conjugacy class is therefore distinct from proving the equality $\widehat{\mathrm{GT}} = G_{\mathbb{Q}}$.*

Proof. Hoshi–Mochizuki–Tsujimura, Theorem 5.17(ii), states exactly this characterization; its input is the abstract profinite group of the second configuration space of the tripod (Hoshi et al., 2025). Definition 5.12 imposes four conditions on the reconstructed Kummer and function-field data; AA includes the second-configuration compatibility condition (iv), which cannot be dropped. Theorem 5.15 puts every such subgroup inside a conjugate of $G_{\mathbb{Q}}$, and Remark 5.12.1 verifies the conditions for $G_{\mathbb{Q}}$ itself.

Here is the maximality step. In a profinite group G, if a closed subgroup H satisfies $H \subseteq gHg^{-1}$, then $H = gHg^{-1}$. In every finite quotient the two images have equal cardinalities, so the inclusion is equality; intersecting the inverse images over all finite quotients gives equality of the closed subgroups. Thus every AA subgroup is contained in an AA subgroup conjugate to $G_{\mathbb{Q}}$, and no such conjugate is properly contained in another. This proves the stated maximal-subgroup characterization from the cited theorem, keeping its hypotheses intact.

This use does not assert that every GT element is arithmetic, or that testing only its cyclotomic component and the pentagon is sufficient. Nor does it yet construct the compatible action on every type and every puncture-filling arrow asked for in E2. □

5.3 An actual isotopy congruence and its profinite envelope

We give a precise category before asking for its algebraic comparison. Objects are oriented finite-type topological surfaces S with $c(S) = 2g(S) - 2 + n(S) > 0$. An elementary arrow is an oriented finite unramified covering or an oriented puncture filling. A puncture filling extends to an orientation-preserving homeomorphism of the compact surfaces and fills a specified subset of the finite end set. Homeomorphisms are included as degree-one arrows. A word is a finite composable string of elementary arrows. It determines a local homeomorphism on the open surfaces and a finite branched cover on their end compactifications.

We impose the following equivalence, called *diagram isotopy*. An elementary isotopy is a continuous one-parameter family of a fixed finite diagram of elementary arrows, with fixed endpoint surfaces and fixed types at each vertex. We also identify words having the same actual composite map. We take the equivalence relation generated by these identifications and by their occurrence inside a longer composable word. In particular this is explicitly a congruence, rather than a relation whose compatibility with composition is presumed. Denote the resulting category by $\mathcal{C}_{\mathrm{diag}}$.

E.5 (Proved). *Composition is well-defined on $\mathcal{C}_{\mathrm{diag}}$. The category of finite-cover isotopy classes embeds faithfully in it, and its automorphism group at S is the oriented mapping class group M_S. It contains puncture fillings and the usual forgetting functors on groupoids of surfaces with specified labelled punctures. Every Hom set has finitely many right M_S-orbits. In particular, for every arrow $u : S \to T$, its target lifting subgroup*

$$H_u = \{b \in M_T : \text{ there is } a \in M_S \text{ with } bu = ua\}$$

has finite index in M_T.

Proof. Concatenation of words is associative, and the stipulated congruence makes it independent of representatives. The degree of the compact map is a positive integer. It is constant in every elementary isotopy and multiplicative under composition; all relations preserve it. If a word of degree d joins S to T, then

$$c(S) = d\,c(T) + \Delta, \qquad \Delta \in \mathbb{Z}_{\geq 0}.$$

For a covering, $\Delta = 0$ by Euler characteristic. For a filling of $r > 0$ punctures the degree is one and $\Delta = r$. For composites the formula is $\Delta(vu) = \Delta(u) + \deg(u)\Delta(v)$, so every genuine filling contributes positively. Equivalently Δ is the sum of the local degrees at extra omitted source points whose images are interior target points; this follows by Riemann–Hurwitz on the compactifications.

If a word represents a finite covering, its endpoint type and degree force $\Delta = 0$. Every word equivalent to it also has $\Delta = 0$, so all its elementary factors are coverings. The elementary isotopies then are ordinary covering isotopies, and replacing a factorization by its composite makes no additional identifications. Thus the embedding is faithful. An invertible arrow has an inverse of positive degree, so both degrees are one. The two complexity formulas force both defects to vanish; all factors are homeomorphisms. Their isotopy classes form M_S.

Each noninvertible elementary arrow strictly decreases the positive integer c. After absorbing adjacent homeomorphisms, the length of a word from S to T is at most $c(S) - c(T)$, and its degrees are bounded by $c(S)/c(T)$. Only finitely many hyperbolic types have complexity at most $c(S)$. For a fixed target and degree there are only finitely many connected covering types: a finitely generated surface group has only finitely many homomorphisms to the finite symmetric group of that degree. Each such covering type gives one right source-mapping-class orbit.

For a filling, moving the finitely many extra points by an ambient isotopy of the target identifies any two configurations of the same cardinality. The isotopy may fix the retained punctures. The remaining difference is a source homeomorphism. Thus fillings also have finitely many right source orbits; for unlabelled extra points there is one. This assertion includes the braid motions of the extra points: their lifts act in the source stabilizer, rather than producing an omitted family of arrows.

The property of having finitely many right orbits is preserved under composition. Choose finite sets of representatives u_i and v_j for two adjacent Hom sets. Each composite has the form $v_j b u_i a$. Left multiplication by b permutes the finite set of right orbits of the first Hom set. Therefore finitely many further representatives suffice for all composites. There are only finitely many possible elementary type-and-degree patterns, proving the assertion for all words and hence for their quotient. Finally H_u is the stabilizer of the right orbit of u under the left action of M_T on this finite set. Its index is finite.

For labelled punctures, a homeomorphism preserving all labels extends over any specified subset of punctures. Isotopies extend in the same way. This defines the forgetting functor on the isomorphism groupoids; successive forgettings equal forgetting the union. The remaining type must still be hyperbolic. These are the topological functors that the moduli-stack maps in R26 are to realize. □

E.6 (Proved). *There is a Hausdorff profinite envelope of $\mathcal{C}_{\mathrm{diag}}$ with the same objects and*

$$\mathrm{Hom}_{\widehat{\mathcal{C}}_{\mathrm{diag}}}(S,T) = \coprod_{uM_S \in \mathrm{Hom}_{\mathcal{C}_{\mathrm{diag}}}(S,T)/M_S} \overline{S_u}\backslash\widehat{M}_S, \qquad S_u = \{a \in M_S : ua = u\}.$$

Here the closure is in $\widehat{M}_S$. Each Hom space is compact, composition is continuous, the original category has dense image, and $\mathrm{Aut}(S) = \widehat{M}_S$. This proves the topological completion part of E3 for the specified congruence.

Proof. We verify the construction rather than invoke F149-G6 without checking composition. Complete a right orbit $S_u\backslash M_S$ as the inverse limit of finite sets $S_uN\backslash M_S$ for finite-index normal $N \lhd M_S$. Compactness identifies that limit with the displayed closed-coset space. A target element $b \in H_u$ has a lift a with $bu = ua$. Its class in $N_{M_S}(S_u)/S_u$ is well-defined and depends homomorphically on b. Left multiplication by it acts on every finite set $S_uN\backslash M_S$, through a finite quotient of H_u. Since H_u has finite index, its full profinite completion is its closure in $\widehat{M}_T$: the normal cores in M_T of its finite-index subgroups give the required cofinality. Thus its action extends continuously. The finitely many cosets of H_u extend the action to all of $\widehat{M}_T$, permuting the finitely many right orbits in the double orbit of u.

For an original arrow $v : T \to V$, define $L_v(u\alpha) = (vu)\alpha$. It is well-defined and continuous because $S_u \subset S_{vu}$. For completed arrows set

$$(v\beta)(u\alpha) = L_v(\beta \cdot u)\alpha.$$

Replacing α by its left $\overline{S_u}$-coset makes no change by equivariance. Replacing β by its left $\overline{S_v}$-coset makes no change because $L_v(sh) = L_v(h)$ first for original s, h, then for their closures by continuity and density. Changes of right-orbit representatives are translations of the same formula. On each product of right-orbit pieces the formula is the quotient of a continuous map of compact profinite groups, so is jointly continuous. Associativity follows because the two continuous triple-composition maps agree on the dense set of original triples and the target is Hausdorff.

The orbit of the identity is $\widehat{M}_S$ and is open. If two completed arrows are inverse, approximate them by original arrows within their original orbit pieces and with both composites in the open identity orbits. Those original composites are automorphisms. Each of the approximating arrows, and hence its orbit representative, is an isomorphism. This proves that no new invertible orbit appears. It proves the asserted automorphism group and conservativity. Compactness follows from the finiteness of the right-orbit set in E.5. Any functor to a Hausdorff topological category with profinite automorphism groups extends uniquely: extend the homomorphisms on M_S to $\widehat{M}_S$, then the continuous orbit maps kill $\overline{S_u}$. Compatibility with composition again follows by density. This is the Hausdorff universal property required in R25. □

E.7 (Proved; the moving-critical-value test). *Diagram isotopy identifies the maps f_0 and f_t in R26, for $0 \le t \le 1/10$, but they are not in one two-sided orbit for $t > 0$. Thus the congruence of E.5 does not silently repeat the false two-sided-orbit assertion.*

Proof. Use the four-factor diagram

$$X_t \xrightarrow{z^3-3z+t} Y_{1,t} \hookrightarrow Y_2 \xrightarrow{w^2} Y_3 \hookrightarrow Y,$$

where

$$Y = \mathbb{P}^1 \setminus \{9, 16, \infty\}, \qquad Y_3 = Y \setminus \{0\},$$
$$Y_2 = \mathbb{P}^1 \setminus \{0, 3, -3, 4, -4, \infty\}, \qquad Y_{1,t} = Y_2 \setminus \{t-2, t+2\},$$

and X_t is the inverse image of $Y_{1,t}$ under the cubic. The cubic critical values are $t \pm 2$, always distinct from the fixed omitted values on this interval. The five relevant discriminants are $27(4-(t-c)^2)$ for $c = 0, \pm 3, \pm 4$, and never vanish. The source has twenty distinct punctures throughout. Ambient isotopies identifying these moving punctures give a family of the entire fixed diagram, so E.5 identifies its endpoints. On compactifications the finite critical values of the composite are $0, (t-2)^2, (t+2)^2$. Their number is two at zero and three at positive t, since their difference is $8t$. A two-sided homeomorphism preserves that number. The factorization and discriminants are verified exactly in `code/root_checks.py`. □

5.4 The algebraic Hausdorff presentation and its comparison

We specify the algebraic category, including its separation convention. Its objects are smooth connected hyperbolic curves over $\mathbb{C}$. Use the stacks $\mathcal{M}_{g,[n]}$ of unordered punctures; labelled or coloured variants use the corresponding finite covers. Generators are paths in their étale fundamental groupoids, actual finite étale maps, and actual puncture fillings. An actual isomorphism is identified with its canonical stack path. Besides the path-groupoid laws, impose the following relations. For every finite chain of specified covers and fillings, let $\mathcal{H}_w$ be its diagram stack. A path in $\mathcal{H}_w$ from w_0 to w_1, inducing source path α and target path β, imposes

$$\beta w_0 = w_1 \alpha.$$

Two chains of actual algebraic maps with the same composite are also identified. All these relations may occur inside longer words.

The *Hausdorff étale presentation* means the universal Hausdorff topological category with profinite automorphism groups realizing these data, continuously on each étale path groupoid. This is a universal property in terms of algebraic stacks and their étale paths alone. Equivalently, relations are imposed in a closed congruence. The following proof establishes its existence, so existence is not an extra assumption in the definition.

E.8 (Proved). *The Hausdorff étale presentation exists. After choices of surface markings, it is equivalent to $\widehat{\mathcal{C}}_{\mathrm{diag}}$ of E.6. In particular,*

$$\mathrm{Hom}(S,T) = \coprod_{uM_S \in \mathrm{Hom}_{\mathcal{C}_{\mathrm{diag}}}(S,T)/M_S} \overline{S_u} \backslash \widehat{M}_S, \qquad S_u = \{a \in M_S : ua = u\}.$$

Composition is continuous, Hom spaces are compact, and $\mathrm{Aut}(S) = \widehat{M}_S$. *The completed pure-cover category embeds faithfully, and the labelled puncture-forgetting operations agree with those induced by the algebraic moduli maps. This proves the comparison and completion required in E3 for the isotopy relation of E.5.*

Proof. First the diagram stacks used in the definition exist and are of finite type. A filling is described by a configuration of extra disjoint points, with the retained and forgotten sets distinguished. For a covering use the finitely many smooth Hurwitz stacks of the given degree and boundary ramification partitions. Non-Galois maps are included by Bertin and Romagny (2007, Theorem 6.20); its opened proof identifies the local deformation of a smooth cover with that of its marked target. Iterated fibre products over the intermediate moduli stacks give $\mathcal{H}_w$. Hyperbolicity makes the automorphism groups finite. We retain these stack automorphisms: a Hurwitz map to target moduli need not be representable when deck automorphisms occur, as the footnote to Bertin and Romagny (2007, Proposition 6.14) explicitly records.

The Betti presentation. Temporarily use ordinary paths in the analytic stacks. Contractibility of Teichmüller space and its quotient description identify the fundamental groupoid of each moduli stack with the groupoid of oriented surface markings up to isotopy

(Schneps, 2003, §2.1). An analytic diagram path, after local topological trivializations over an interval, gives a diagram isotopy with its endpoint transports. Equal algebraic composites give equal underlying maps. These observations define a functor from the Betti presentation to $\mathcal{C}_{\mathrm{diag}}$.

Conversely, choose a complex structure on each surface object. For a topological word ending in T, keep its chosen target structure and propagate that structure backwards: a covering pulls it back, and a filling restricts it to the punctured open. All factors become holomorphic, extend over their compactifications, and are algebraic. Join the resulting source structure to its chosen structure by a path of marked complex structures. Contractibility of Teichmüller space makes the path with this prescribed identity marking unique up to the relevant homotopy.

Here is the compatibility with the relations, including the family issue. A topological diagram isotopy can be represented at the level of its paths and endpoint transports by smooth finite-configuration paths and finite-cover transports. For fillings this is ordinary approximation of paths in the open configuration manifold, relative to the endpoints, followed by isotopy extension. Covering paths are uniquely transported after their starting cover is fixed. Work backwards from the target: these configuration paths and covering transports give a path in the analytic diagram stack. This asserts a continuous stack path; it does not assert that an arbitrary nonsmooth original isotopy is a holomorphic family in its real parameter. The induced source path has the prescribed identity marking, so the naturality relation gives the required equality of Betti arrows.

If two words have the same actual composite, their pulled-back source complex structures are the same. Both factorizations are simultaneously holomorphic and have equal algebraic composites. The stipulated relation therefore identifies them. For composition, transport the first diagram along the marked complex-structure path used for the second; its naturality square identifies this construction with that of the concatenated word. These arguments prove independence of the choices and give inverse functors, up to marking isomorphisms. The Betti presentation is therefore equivalent to $\mathcal{C}_{\mathrm{diag}}$. No two-sided-orbit assertion is used; E.7 is still a diagram path relation.

All étale path relations. Let $\mathcal{A}$ be any Hausdorff topological category with profinite automorphism groups, and let $F : \mathcal{C}_{\mathrm{diag}} \to \mathcal{A}$ be a functor. By the Betti comparison this is a realization of the Betti presentation. Each induced homomorphism from a moduli fundamental group to an automorphism group of $\mathcal{A}$ extends uniquely to the full profinite completion. Choosing one ordinary path to each endpoint extends this assertion to path groupoids; changing the choice changes the formula by an ordinary loop and leaves the extension unchanged.

Noohi's Theorem 20.4 and its following corollary identify these completions with the étale fundamental groups (Noohi, 2005). The finite-type complex Deligne–Mumford stacks in question have locally path connected, semilocally simply connected analytic stacks. Thus ordinary path torsors are dense in the étale path torsors, component by component. For a path γ in any one $\mathcal{H}_w$, the two sides of its naturality equation are continuous functions of γ into a Hausdorff Hom space of $\mathcal{A}$. They agree on ordinary paths, hence on their closure. The equality therefore holds for every étale path, for every diagram stack, and after every insertion into a longer word. The equal-actual-composite relations do not change. Conversely restricting a continuous étale realization to ordinary paths gives the original functor. The two constructions are inverse, also on compatible natural transformations.

Existence and separation. By E.6, $\widehat{\mathcal{C}}_{\mathrm{diag}}$ represents functors from $\mathcal{C}_{\mathrm{diag}}$ to every such $\mathcal{A}$. The preceding paragraph identifies this universal property with that of the algebraic Hausdorff étale presentation. It proves both existence and the asserted comparison.

We make the stabilizer conclusion explicit. Every Hausdorff realization kills $\overline{S_u}$, since its orbit map is continuous and kills S_u. Conversely, if $a \notin \overline{S_u}$, the category of E.6 is itself one

realization of the algebraic presentation, by the preceding construction. Its distinct cosets $\overline{S_u}$ and $\overline{S_u}a$ separate u from ua. No universal closed relation can therefore identify them. This proves the exact closure formula without a subgroup-separability hypothesis or an interchange of completion with an ordinary quotient.

The remaining structural statements follow from E.6. For a pure covering, defect zero excludes all genuine fillings at every intermediate relation by E.5. Its discrete stabilizers are the covering-only ones, and so are their closures. This proves the asserted completed embedding. Labelled forgetting maps and their composites are included in the presentation by the literal algebraic forgetting maps and their equality relations. □

E.9 (Proved). *For the degree-two cover followed by one filling*

$$f_b(z) = z^2 + b, \qquad b \in Y := \mathbb{P}^1 \setminus \{0, 1, \infty\},$$

$$X_b = \mathbb{P}^1 \setminus \{0, \infty, \sqrt{-b}, -\sqrt{-b}, \sqrt{1-b}, -\sqrt{1-b}\},$$

fix the three labelled target punctures. The diagram component has stack $Y \times B\mu_2$. Its source braid subgroup is the image of

$$\phi : \pi_1(Y) \times C_2 \longrightarrow M_{0,[6]}.$$

This is the right stabilizer of the arrow in the indicated ramification component. Its completed stabilizer is exactly $\overline{\operatorname{im}\phi}$, equal to the image of $\pi_1^{\mathrm{et}}(Y \times B\mu_2)$ in $\widehat{M}_{0,[6]}$.

Proof. The displayed map gives a finite étale double cover of $Y \setminus \{b\}$, followed by its one-puncture filling into Y. On compactifications the two branch values are b and ∞. Every double cover with this labelled target and these branch values is isomorphic to the displayed one, uniquely up to its deck involution. The family exists globally over Y, so the corresponding μ_2-gerbe is trivial: its stack is $Y \times B\mu_2$. The source family gives ϕ, with the C_2 factor supplied by the global deck involution.

The source complexity is four, the target complexity is one, and the degree is two, so the defect is two. The single omitted ramification point over the interior critical value b already contributes all of that defect. There are no further omitted interior points. Thus any factorization of this composite into coverings and fillings must first omit b on the target so that the degree-two factor is étale. Filling the source ramification point before that factor would make it ramified on the open. Apart from homeomorphisms and trivial factors, the cover-then-fill pattern is forced. Its diagram loops, with fixed labelled target, therefore give exactly the right stabilizer $\operatorname{im}\phi$.

By Noohi's comparison, the étale group is the profinite completion of $\pi_1(Y) \times C_2$. Its continuous image in $\widehat{M}_{0,[6]}$ is compact and hence closed, and the original image is dense in it. It is consequently exactly $\overline{\operatorname{im}\phi}$. No injectivity of ϕ or separability of its image is required. This concrete braid calculation is consistent with the general stabilizer proof of E.8. □

The Hausdorff clause in E.8 is deliberate. We do not assert that the naive abstract quotient of finite words in profinite paths has Hausdorff Hom spaces before this separation. E3 is answered by the stated Hausdorff presentation, with the precise closed-coset completion and its algebraic comparison.

5.5 A repaired category using boundary-tame moduli paths

The two changes in the next construction are explicit. In positive characteristic we use paths tame at the boundary of the stable moduli stack. In characteristic zero we restrict the covering arrows to those whose Galois closures have order prime to the chosen residue characteristic. Full characteristic-zero moduli markings remain allowed. The obstructions E.1 and E.2 still apply to the original unrestricted formulation.

E.10 (Proved). *Let k be algebraically closed of characteristic $p > 0$, and let $\mathcal{M} = \mathcal{M}_{g,[n],k}$ be the stack of hyperbolic curves with unordered punctures. Give its proper stable compactification $\overline{\mathcal{M}}$ the divisorial logarithmic structure of its boundary D. For $(m, p) = 1$, put*

$$\mathcal{R}_m = \sqrt[m]{(\overline{\mathcal{M}}, D)}, \qquad \mathcal{P}^{\mathrm{bt}}(\mathcal{M}) = \varprojlim_{(m,p)=1} \Pi_1^{\mathrm{et}}(\mathcal{R}_m),$$

using geometric base points in $\mathcal{M}$. This is the full boundary-tame fundamental groupoid. It is invariant under algebraically closed field extension. For a smooth marked model over a strictly henselian complete DVR of mixed characteristic $(0, p)$, it admits a surjective specialization from the full geometric generic moduli fundamental group, and the corresponding maps of path torsors are surjective. Finally, the canonical map

$$\mathrm{Aut}_k(C) \longrightarrow \mathcal{P}^{\mathrm{bt}}(\mathcal{M})(C, C)$$

is injective for every hyperbolic curve C/k of that type.

Proof. We first justify the stack comparison being used. If $\mathcal{X}$ is a proper finite-type Deligne–Mumford stack over a field or a complete noetherian local ring, its diagonal is finite. Theorem 2.7 of Edidin et al. (2001) gives a finite surjective representable map $Y \to \mathcal{X}$ from a scheme. The scheme Y and the next two terms of its Čech nerve are proper schemes; the latter are schemes because the diagonal is finite. Effective descent for finite *étale* covers along this map follows from SGA 1 IX, Theorem 4.12, after an *étale* atlas. It identifies a cover of $\mathcal{X}$ with a finite *étale* cover of Y, a descent isomorphism on $Y \times_{\mathcal{X}} Y$, and its cocycle on the triple product. Apply SGA 1 X, Corollary 1.8 to all three proper schemes for algebraically closed field extension, or X, Theorem 2.1 for passage from a complete local base to its closed fibre. Full faithfulness carries the isomorphism and the cocycle uniquely. Thus both finite-*étale* comparison equivalences hold for $\mathcal{X}$ (Grothendieck et al., 2003). This argument makes no prime-to-p assumption on the original stack inertia.

For completeness, the associated geometric specialization is surjective when $\mathcal{X}$ is smooth proper over a complete DVR R with algebraically closed residue field. A connected finite *étale* special cover extends by the equivalence just proved. The total cover is connected and regular. A connected regular noetherian DM stack is irreducible, as may be checked on *étale* charts, so its generic fibre is connected. Repeating after every finite extension of R proves geometric connectedness of that generic fibre. This is the Galois-category criterion for surjectivity.

Knudsen's Theorem 2.7 supplies proper smoothness and relative normal crossings for all stable pointed types; take the quotient by the constant permutation group for unordered punctures (Knudsen, 1983, p. 179). Locally, where the boundary parameters are $x_1, \dots, x_r$, the finite root stack is

$$\left[\operatorname{Spec} A[t_1, \dots, t_r]/(t_1^m - x_1, \dots, t_r^m - x_r)/\mu_m^r\right].$$

The logarithmic construction glues these charts even when the boundary has self-intersections or its local branches are permuted (Talpo and Vistoli, 2018, Propositions 3.4 and 3.10). The stack is proper and smooth DM when m is invertible. Its finite representable *étale* covers

restrict fully faithfully to the open moduli stack. Conversely, SGA 1 XIII, Proposition 5.2 extends a boundary-tame cover uniquely after extracting suitable prime-to-p roots of the boundary parameters. A common multiple works on finitely many charts, and uniqueness of normalization gives descent. This proves the groupoid description above. The Kummer-flat theorem about infinite root stacks is not needed.

Apply the proper-stack comparisons to every $\mathcal{R}_m$. They prove field-extension invariance. Over R, they give surjections at each root level. The inverse-limit map is still surjective: its compact image projects onto every level of a directed inverse system and therefore is dense and closed. Restriction of covers from a normal root stack to its dense open also induces a surjection on fundamental groups. Combining these maps proves the assertion for the full generic moduli group. The assertion for path torsors follows by choosing a path and using the group assertion. All ambiguities are the usual choices of base points and paths.

It remains to prove the last assertion, which will exclude a gerbe obstruction in E.11. Order the punctures; the ordering cover extends as a finite *étale* cover of the stable quotient stack, even when $p \mid n!$. In genus zero an automorphism fixing all these marks is identity. In positive genus add a full Jacobi level at an odd prime $\ell \neq p$, $\ell \geq 3$; in genus one use an ordered mark as origin. These data kill automorphisms. For genus at least two, use the faithful action on the Jacobian in Deligne–Mumford Theorem 1.13 and Lemma 1.14. In genus one use the faithful Tate-module action of origin-preserving elliptic automorphisms. A finite-order matrix in $1 + \ell M_{2g}(\mathbb{Z}_\ell)$ is identity: the first nonzero ℓ-adic term of its powers, using ℓ odd, rules out both prime-to-ℓ and ℓ-power torsion. Thus a finite-order automorphism fixing the full level acts trivially on the Tate module and then on the curve (Deligne and Mumford, 1969, Theorem 1.13, Lemma 1.14, and §5.14).

These level covers are boundary tame. On a regular chart of the stable compactification, the universal curve is already split semistable after strict localization. The prime-to-p form of the monodromy computation in B.5 factors through tame Kummer inertia, as in the proof of C.2. On homology modulo ℓ, a boundary generator acts by $1 + N_e$, with $N_e(x) = \langle x, \delta_e \rangle \delta_e$ and $N_e^2 = 0$. Consequently $(1 + N_e)^\ell = 1$. At an intersection the vanishing cycles are disjoint, so $N_e N_f = 0$ and these transvections generate an elementary abelian ℓ-group. Wild p-inertia is trivial. The level cover is therefore tame at every boundary branch; the same argument applies in the marked genus-one case. We use the prime-to-residue- characteristic monodromy theorem underlying B.5 here, not a claim about the full characteristic-p geometric group (Hoshi and Mochizuki, 2012, Theorem 4.8, Lemma 5.4). The action of $\mathrm{Aut}(C)$ on the fibre of the combined ordering and level cover is faithful. Since this finite cover occurs in the defining Galois category, its action detects every nonidentity curve automorphism in $\mathcal{P}^{\mathrm{bt}}$. □

E.11 (Proved). *There is a category $\mathcal{T}_k^{\mathrm{bt}}$ with all hyperbolic k-curves as objects, the groupoids $\mathcal{P}^{\mathrm{bt}}$ as its marking groupoids, and all actual finite* étale *maps with prime-to-p Galois closure included faithfully. Every arrow factors as a marking followed by such a cover. Covers transport along markings, compatibly with composition. Algebraically closed field extension induces an equivalence.*

Write $c(X) = 2g_X - 2 + n_X$. If $d = c(X)/c(Y)$ is not a positive integer, then $\mathrm{Hom}(X, Y)$ *is empty. Otherwise, for representatives $u : V_u \to Y$ of the finitely many isomorphism classes of covers of degree d, source type equal to that of X, and prime-to-p Galois closure, there is a natural description*

$$\mathrm{Hom}_{\mathcal{T}_k^{\mathrm{bt}}}(X, Y) = \coprod_u D_u \backslash \mathcal{P}^{\mathrm{bt}}(\mathcal{M}_{\mathrm{type}(X)})(X, V_u), \qquad D_u = \mathrm{Aut}_Y(V_u).$$

These Hom spaces are compact Hausdorff, composition is continuous, and $\mathrm{Aut}_{\mathcal{T}_k^{\mathrm{bt}}}(X) = \pi_1^{\mathrm{bt}}(\mathcal{M}_{\mathrm{type}(X)}, X)$.

Proof. We give the path-lifting argument, including its inertia issue. Let $u : V \to Y$ have Galois-closure group G of order prime to p. Use the stable Hurwitz stack with its Galois closure retained, marking every puncture of Y and every point above it. This is the stabilized compactification of Bertin–Romagny, rather than the generally nonnormal space of naive admissible covers. Proposition 5.3, Definition 6.8, Theorem 6.9, Proposition 6.14, Theorem 6.20 and Definitions 6.21–6.22 of Bertin and Romagny (2007) give its properness, its source and target maps, and the intermediate quotient constructions. All ramification points are marked, so the nodal stabilizers are cyclic and do not exchange branches. Their Theorem 5.5 gives the local discriminant parameters

$$\tau_i = t_i^{e_i}, \qquad e_i \mid |G|,$$

with the remaining deformation parameters unchanged.

Pull back a target root stack with $e_i \mid m$, and normalize. The local branches are $t_i = \zeta_i r_i^{m/e_i}$, where $\tau_i = r_i^m$. Each branch has the same deformation ring as the root-stack base. A relative automorphism fixing a branch fixes that full deformation ring and extends uniquely over its universal deformation. Its relative automorphism group is therefore the deck group of the generic smooth cover. The extra boundary automorphisms which permute the branches do not remain in the relative stabilizer of a chosen normalization branch. The identical computation for a non-Galois cover is Bertin–Romagny Lemma 6.23 and Proposition 6.24. Consequently the normalized morphism is, *étale* locally on the target, a finite disjoint union of finite *étale* gerbes. Equivalently it is such a gerbe over a representable finite *étale* cover of the target root stack. No representability assertion is made for the Hurwitz discriminant map itself; the nonrepresentability warning in Proposition 6.14 is essential here.

Because every e_i is prime to p, the target roots so obtained are cofinal among the prime-to-p roots of the Hurwitz boundary. The source map also acts on these tame groupoids: it extends to the stable compactifications, and every source node parameter pulls back to a monomial in Hurwitz node parameters times a unit. The pullback of a prime-to-p Kummer cover is again such a cover. All inverse images of target marks have been retained; this is what makes the intermediate quotient curves stable and avoids the unmarked quotient problem noted in the source.

A finite *étale* gerbe with fibre group D has the fundamental-group sequence

$$D \longrightarrow \pi_1(\mathcal{H}, u) \longrightarrow \pi_1(\mathcal{Q}, q) \longrightarrow 1.$$

The image of D is normal and finite; this is Noohi Proposition 10.4 and Corollary 10.5 (Noohi, 2004). The same assertion over a stack base is proved by *étale* descent: neutralize locally, check it for BD, and observe that finite covers with trivial relative-inertia action descend uniquely to $\mathcal{Q}$. In general the left map need not be injective. Pass to the cofinal root system first. The limit sequence remains exact: surjectivity follows from the finite-intersection property for compact lift sets; an element in the limit kernel comes from a single element of the finite group D, again by the finite-intersection property. The source map on this limit sends D to the canonical subgroup $D \subset \mathrm{Aut}(V) \subset \pi_1^{\mathrm{bt}}(\mathcal{M}, V)$, which is injective by E.10. Hence the left map is injective in this application. This supplies the missing check that a nontrivial gerbe transgression cannot kill deck transformations.

Thus a target tame path and a starting cover determine a unique endpoint isomorphism class of covers and admit a lift in the Hurwitz tame groupoid. Two lifts to the same endpoint differ by a deck element. Their source paths therefore agree modulo the action of that deck group.

Define an arrow by a pair (u, α), where $u : V \to Y$ is one of the permitted covers and $\alpha : X \rightsquigarrow V$ is a boundary-tame marking. Identify (u, α) with $(u', h\alpha)$ when $h : V \xrightarrow{\sim} V'$ is an isomorphism over Y. For $(u, \alpha) : X \to Y$ and $(v, \beta) : Y \to Z$, write $v : W \to Z$, transport u

along β to $u' : V' \to W$, and let $\gamma : V \rightsquigarrow V'$ be its source marking. Set

$$(v, \beta)(u, \alpha) = (vu', \gamma\alpha).$$

The deck-kernel assertion proves independence of the lift and of the chosen representatives. For associativity use the Hurwitz stack of a chain of covers. A composite has Galois closure contained in the corresponding wreath product, whose order is still prime to p. Its common Galois closure and the intermediate subgroups transport the chain simultaneously. Their marked quotient models are stable by the result already checked. Both bracketings give the same composite-cover map and source path on that stack. Identities are $(\mathrm{id}, \mathrm{id})$.

Actual covers map to (u, id). If two such maps became equal, an isomorphism of their sources would have trivial canonical marking. E.10 makes that isomorphism identity, so the inclusion is faithful. Factorization and cover transport have been built into the construction.

Tame Riemann–Hurwitz gives $c(V) = d\,c(Y)$. The maximal prime-to-p curve group is topologically finitely generated, so it has only finitely many subgroups of a given index. This proves the stated finite indexing of the Hom formula. A path torsor is profinite and D_u is finite; its quotient is compact Hausdorff. On each component, endpoint transport depends on a finite quotient, and after the endpoint is fixed composition is induced by continuous maps of Hurwitz path torsors and group multiplication. Composition is therefore continuous. A degree-one cover is an isomorphism, giving the asserted automorphism group.

Under $k \subset K$, E.3 identifies fixed-target covers, their source types, and their deck groups. E.10 identifies the marking torsors. The Hom formula proves full faithfulness. The moduli stack of each hyperbolic type is geometrically connected, and any K-curve has a type represented over k. A marking connects it to such a base-changed curve. This proves essential surjectivity, without asserting that the individual curve descends as a scheme. □

E.12 (Proved). *Fix p, and let $\mathcal{T}^{\mathrm{cov}(p')}$ be the characteristic-zero covering category with its full profinite moduli markings, retaining precisely the covering arrows whose Galois closures have order prime to p. After choosing good reduction base models for the types and paths between type base points, there is a specialization functor*

$$\mathcal{T}^{\mathrm{cov}(p')} \longrightarrow \mathcal{T}_k^{\mathrm{bt}}.$$

It preserves type and is surjective on each Hom space between the chosen type base objects. On automorphism groups it is

$$\widehat{\mathrm{Mod}}_{g,[n]} \twoheadrightarrow \pi_1^{\mathrm{bt}}(\mathcal{M}_{g,[n],k}).$$

On Hom spaces it preserves the finite cover-component index set and is the disjoint union of the surjections

$$D_u \backslash \mathrm{Path}_{\mathrm{et}}(X, V_u) \longrightarrow D_{u_s} \backslash \mathrm{Path}_{\mathrm{bt}}(X_s, V_{u_s}).$$

The deck groups correspond under specialization. With compatible path identifications and a group map $q : \Gamma \twoheadrightarrow \Gamma_s$, the fibre above the coset of $q(\gamma)$ consists of the left-D_u classes in $D_u \ker(q)\gamma$.

Proof. Take $R = W(k)$. For every type, smoothness of the pointed moduli stack lifts a chosen special curve to a smooth marked model over R; use a smooth chart through its point and completeness, or formal existence for the proper curve. Fix one such model per type. The full subcategories on their geometric generic fibres and their special fibres are equivalent to the respective full categories, because markings connect all objects of one type. Characteristic-zero base-extension invariance identifies the generic category with the stated one. One may use a common algebraically closed overfield; no embedding of an arbitrarily large field into $\mathbb{C}$ is required.

Let $u : V \to Y$ be a permitted cover of a chosen generic target. By E.3 it corresponds to a special cover with the same degree, source type and deck group. It has a smooth marked model over the chosen target model after a finite extension of R: lift the corresponding special cover, identify its geometric generic fibre with u, and descend that identification to a finite extension. Uniqueness of the marked stable model makes this model and its deck action compatible with further extensions.

Represent an arrow by (u, α), with α now a full characteristic-zero moduli marking. Both endpoints of α have models after the preceding finite extension. E.10 specializes the marking through the proper root stacks. Define

$$(u, \alpha) \longmapsto (u_s, \mathrm{sp}(\alpha)).$$

The proper finite-*étale* comparisons are compatible with extension of the DVR. The separated stable moduli stack gives the unique extensions of generic model isomorphisms. These two facts give independence from extensions, representatives and model identifications. The normalized root Hurwitz source and target maps commute with specialization. Applying that compatibility to the common-closure chain used in E.11 proves compatibility with composition; hence this is a functor.

Degree and the marked source type are preserved by E.3. For automorphisms, E.10 gives the surjection from the full generic moduli group; the characteristic-zero comparison identifies it with the full profinite mapping class group. More explicitly the map first passes to the inverse limit of generic root groups with indices prime to p, and then specializes. It retains the target's finite interior p-monodromy; it is not the maximal prime-to-p quotient of the moduli group.

Finally apply the normal forms of E.11 and the fixed-target cover equivalence E.3. Every cover component corresponds to exactly one special component, with its same deck group. Specialization of path torsors is surjective by E.10. It therefore induces the displayed surjection on each component. After identifying the torsors with their groups, two classes have equal images exactly when their representatives differ by a deck element and an element of $\ker(q)$, giving the stated fibres. No injectivity of q or faithfulness of this specialization functor is asserted. E.2 proves that its restriction on covering arrows cannot simply be removed. □

The former obligation E.U1 is superseded by E.10–E.12. They prove the repaired construction with boundary-tame paths retaining interior p-monodromy and the restricted-source specialization. The literal full-path and unrestricted-source requirements remain subject to E.1–E.2.

5.6 The exact remaining arithmetic-image obligation

E.U2 (Unresolved). *We do not identify the image of $G_{\mathbb{Q}}$ as compatible outer automorphisms of all $\widehat{M}_{g,n}$ with all covering and puncture-forgetting operations. E.4 supplies the published genus-zero arithmetic subgroup characterization. Extending that characterization to the specified entire category, with all compatibilities and with the conjugacy ambiguity accounted for, is the unproved step here. In particular we do not replace it by the stronger and different equality $\widehat{\mathrm{GT}} = G_{\mathbb{Q}}$.*

E.8 resolves the earlier comparison obligation for the explicitly Hausdorff algebraic presentation. It does not assert that the unseparated abstract quotient of finite words in profinite paths already has Hausdorff Hom spaces. That additional assertion is unneeded for the completion theorem requested in E3.

6 Part F. Reconstruction of schemes and the inertia comparison

6.1 Two different conditions that the prompt calls pinned

The manuscript locus is cote 149, manuscript p. 25v, scan 52. The transcription says "1-fidèle" and then asks about full faithfulness after universal homeomorphisms have been inverted. The actual scan is not in the supplied image selection; we make no palaeographic assertion about it. We distinguish a *relative* geometric morphism, which includes a specified commutation isomorphism over BG_k, from a *closed-point-preserving* relative morphism. Carlson–Haine–Wolf call the latter *pinned*, in Definition 2.19 of the supplied preprint (Carlson et al., 2024). This is an additional condition. LM5-F.5 states it correctly; the definition in R27 of the present prompt omits it. We use $\mathrm{Hom}^{\mathrm{cp}}_{BG_k}$ for that restricted Hom-groupoid below, so that no change of definition is hidden.

F.1 (Refuted). *With the literal definition of "pinned" in R27, F1 is false. More strongly, there is no characteristic-zero field over which all relative geometric morphisms between the small* étale *topoi of finite-type schemes are induced by scheme morphisms, even after absolute weak normalization. This failure already occurs for source* $\operatorname{Spec} k$ *and target* $\mathbb{P}^1_k$.

Proof. Choose separable closures compatibly with $k(t) \subset k((t))$. In equal characteristic zero the decomposition group D_0 of the t-adic valuation in $G_{k(t)}$ has an exact sequence

$$1 \longrightarrow \widehat{\mathbb{Z}}(1) \longrightarrow D_0 \longrightarrow G_k \longrightarrow 1.$$

It has a continuous splitting: in a maximal unramified extension act on the coefficients, and extend that action by fixing a compatible system of roots $t^{1/n}$. Every finite extension over the maximal unramified field is tame, and its tame part is obtained by these roots; hence this description defines the action on a separable closure, not only on a finite list of extensions. Restriction to a separable closure of $k(t)$ gives a continuous homomorphism $s : G_k \to G_{k(t)}$ whose composite with $G_{k(t)} \to G_k$ is the identity. Equivalently one can make this construction in the henselization of $k(t)$ at 0.

A continuous homomorphism of profinite groups induces the geometric morphism of their continuous-action topoi with inverse image the restriction functor. The composite

$$BG_k \xrightarrow{Bs} BG_{k(t)} = \operatorname{Spec}(k(t))_{\mathrm{et}} \longrightarrow (\mathbb{P}^1_k)_{\mathrm{et}}$$

is therefore a relative geometric morphism, with its specified commutation isomorphism over BG_k. Its map on the spaces of points sends the sole point to the generic point of $\mathbb{P}^1_k$. Every k-morphism $\operatorname{Spec} k \to \mathbb{P}^1_k$ has closed image. Isomorphic geometric morphisms induce the same map on subterminal objects and hence on the underlying sober spaces, so the displayed morphism is not induced by a scheme morphism. A field is already absolutely weakly normal in characteristic zero; universal homeomorphisms preserve the topological obstruction. This is precisely the example of Carlson et al. (2024, Example 2.30), with the splitting construction made explicit. The example is valid for $k = \mathbb{Q}$, for $\mathbb{Q}^{\mathrm{ab}}$, and for $\mathbb{Q}^{\mathrm{solv}}$. □

F.2 (Proved). *There is, over every characteristic-zero field k, a relative endomorphism of $(\mathbb{G}_{m,k})_{\mathrm{et}}$ which induces the identity on its arithmetic fundamental group and has underlying image the generic point. It is not induced by an algebraic morphism and does not factor through BG_k. It is, however, constant on underlying topological points. Consequently it does not settle the closed-point-preserving, nonconstant test intended in F2.*

Proof. Put $X = \mathbb{G}_{m,k}$ and $E = \pi_1(X)$, retaining the full arithmetic quotient. The map $D_0 \to E$ induced by the generic point is an isomorphism: it is the identity on G_k and the local Kummer inertia maps isomorphically onto $\pi_1(X_{\bar{k}}) = \widehat{\mathbb{Z}}(1)$. Finite *étale* covers of X give the canonical geometric morphism $X_{\mathrm{et}} \to BE$; its inverse image sends a continuous E-set to the corresponding locally constant sheaf. This functor preserves finite limits and all colimits, and has the usual right adjoint. Compose it with

$$BE \simeq BD_0 \longrightarrow BG_{k(t)} \longrightarrow X_{\mathrm{et}}.$$

On π_1 the composite is $E \simeq D_0 \to G_{k(t)} \to E$, the identity. The middle field topos has one underlying point, and the last map takes it to the generic point. An algebraic endomorphism of X is $t \mapsto ct^m$, $c \in k^*$ and $m \in \mathbb{Z}$; none has this point map. Factoring through BG_k would kill the nontrivial geometric kernel on π_1, which the displayed morphism does not do. The last sentence of the claim records the distinction between this fact and nonconstancy of a point function. □

6.2 The repaired cyclotomic statement

We now audit the intended mathematical content of LM5-F.47 and F.50 with the missing closed-point condition restored. The following proof does not use either AI return as a theorem. It identifies the arithmetic inputs and supplies the steps from them to the reconstruction criterion. For an abelian group A write $\widehat{A} = \varprojlim_n A/nA$ and $TA = \varprojlim_n A[n]$, using multiplication transition maps in the second limit. For a field L put $D(L^*) = \bigcap_{n\geq 1}(L^*)^n$.

F.3 (Known, with derivation). *Let k be a subfield of a finitely generated extension M of $\mathbb{Q}^{\mathrm{ab}}$. For every finite extension L/k in a fixed algebraic closure, there is an overfield $E \supset L$ finitely generated over $\mathbb{Q}^{\mathrm{ab}}$ such that*

$$E^*/\mu_\infty \text{ is free abelian}, \qquad D(E^*) = \mu_\infty, \qquad \bigcap_{n\geq 1} nA(L) = 0$$

for every abelian variety A/L. In particular $D(L^*) \subset \mu_\infty$.

Proof. We quote exactly the forms of the three arithmetic inputs used here. Amoroso–Zannier, Theorem 1.1, states that for a number field F, an abelian extension B/F, and a nonzero algebraic number a which is not a root of unity, its logarithmic height has a positive lower bound depending only on F and $[B(a) : B]$ (Amoroso and Zannier, 2000). In particular it is uniformly positive when the latter degree is bounded. Tsujimura, Proposition 4.9 of RIMS-1957, states that if F is a number field and $f \in F$, then the field B obtained by adjoining all roots of f has $B^*/B^{*\infty}$ free abelian (Tsujimura, 2022). With $f = 1$ this gives the needed statement over $F\mathbb{Q}^{\mathrm{ab}}$. Tsujimura, Theorem 3.1, states that for a number field F and an abelian variety A/F the natural Kummer map

$$A(F\mathbb{Q}^{\mathrm{ab}}) \longrightarrow H^1\big(G_{F\mathbb{Q}^{\mathrm{ab}}}, TA\big)$$

is injective (Tsujimura, 2020). Its proof factors it as the natural map to $\varprojlim_n A(F\mathbb{Q}^{\mathrm{ab}})/n$ followed by an injection. Thus its kernel is precisely the intersection of the subgroups $nA(F\mathbb{Q}^{\mathrm{ab}})$.

Take $E = ML$, finite over M, and let E_0 be the algebraic closure of $\mathbb{Q}^{\mathrm{ab}}$ in E. Since E is finitely generated, $E_0/\mathbb{Q}^{\mathrm{ab}}$ is finite, hence $E_0 = F_0\mathbb{Q}^{\mathrm{ab}}$ for a number field F_0. A normal proper geometrically integral E_0-model Z of E has

$$E^*/E_0^* \hookrightarrow \mathrm{Div}(Z).$$

Indeed, a rational function with zero divisor on a proper normal variety is a global unit, and the constant field is E_0. The divisor group is free abelian, as is each of its subgroups. The height theorem shows $D(E_0^*) = \mu_\infty$: arbitrary roots of a nontorsion element would have positive heights tending to zero. The quoted freeness theorem now makes E_0^*/μ_∞ free. The exact sequence with this group as kernel and E^*/E_0^* as quotient splits, because the quotient is free. This proves freeness of E^*/μ_∞. Alternatively, applying the divisor map to an infinitely divisible element and to each of its roots shows they all lie in E_0, proving $D(E^*) = \mu_\infty$ directly.

For the abelian-variety assertion first work over E. Spread A/E and a point $P \in \bigcap_n nA(E)$ to an abelian scheme $\mathcal{A} \to V$ and a section P_V, with V a normal integral variety over $\mathbb{Q}^{\mathrm{ab}}$. A chosen division $Q_n \in A(E)$ is a generic section of the finite *étale* cover $[n]^{-1}(P_V) \to V$. Normality extends this generic section over the same V: its selected component is finite birational and hence isomorphic to V. Therefore every division extends without further shrinking. At each closed point v, the residue field is finite over $\mathbb{Q}^{\mathrm{ab}}$, and the abelian variety and point descend to a number field enlarged by finitely many coefficients. The quoted Kummer theorem gives $P_V(v) = 0$. Closed points are dense and $\mathcal{A}$ is separated, so $P_V = 0$. Finally embed $A(L)$ into $A(E)$ to prove the assertion over L. This is also the content of Tsujimura, Proposition 1.2(ii), whose specialization proof we have just written with the common-domain step explicit. □

F.4 (Proved). *Let L be a finite extension of a field k as in F.3, let $b_1, \dots, b_r \in L$ be distinct, and put $V = \mathbb{A}^1_L \setminus \{b_1, \dots, b_r\}$. Suppose $\xi \in \varprojlim_n H^1(V, \mu_n)$ has an actual Kummer value in L^* at every L-point of V. Then ξ is the Kummer class of a unit in $\mathcal{O}(V)^*$. No discrete valuation of L is assumed.*

Proof. Since $\mathrm{Pic}(V) = 0$ and $\mathcal{O}(V)^* = L^* \times \prod_j (t - b_j)^{\mathbb{Z}}$, write

$$\xi = \gamma + \sum_j \lambda_j [t - b_j], \qquad \gamma \in \widehat{L^*}, \quad \lambda_j \in \widehat{\mathbb{Z}}.$$

We construct $x, y \in V(L)$ such that $r_j = (x - b_j)/(y - b_j)$ are multiplicatively independent modulo roots of unity. Put $F = \mathbb{Q}(b_1, \dots, b_r)$. Choose a smooth integral finite-type model A of F over $\mathbb{Z}[1/N]$, containing the b_j and all inverse differences $(b_i - b_j)^{-1}$. This follows by taking a smooth affine model over $\mathbb{Q}$, adjoining these functions and shrinking, then spreading its finite presentation over $\mathbb{Z}[1/N]$ and shrinking the arithmetic base. Its image contains all but finitely many rational primes. Choose distinct such primes $p_i > r$ and a generic point of a component of the fibre at each p_i. Regularity and reducedness of that fibre make its local ring a discrete valuation ring with uniformizer p_i; denote the valuation by v_i. The residues of the b_j are pairwise distinct. Choose $c_i \in \mathbb{F}_{p_i}$ avoiding them. The Chinese remainder theorem supplies $x, y \in A$ with

$$x \equiv b_i + p_i \pmod{p_i^2 A}, \qquad y \equiv c_i \pmod{p_i A} \quad (1 \le i \le r).$$

Thus $v_i(r_j) = \delta_{ij}$. In particular $x, y \in V(L)$ and the r_j are independent modulo roots of unity. An equality proving dependence in an overfield would be an equality between these same elements of F, and so the independence persists.

Embed L in E as in F.3. The classes of the r_j in the free abelian group E^*/μ_∞ have finite support in a basis and are independent. There are therefore integer-valued characters $h_i : E^*/\mu_\infty \to \mathbb{Z}$ for which the square matrix $N_{ij} = h_i(r_j)$ has nonzero determinant. Let $u_x, u_y \in L^*$ realize the two given evaluations. Applying completed characters to their ratio gives

$$N(\lambda_j)_j = (h_i(u_x/u_y))_i \in \mathbb{Z}^r.$$

The adjugate matrix shows $(\det N)\lambda_j \in \mathbb{Z}$. The diagonal copy of $\mathbb{Q}$ meets $\widehat{\mathbb{Z}}$ in $\mathbb{Z}$, as is checked one prime at a time; hence $\lambda_j \in \mathbb{Z}$. After subtracting these integral unit classes, an evaluation at any L-point identifies γ with an actual constant Kummer class. This gives the required unit. For $r = 0$ the last evaluation alone is the proof. □

F.5 (Proved). *Let C/L be a smooth geometrically connected affine curve, with L as in F.4. Suppose $\xi \in \varprojlim_n H^1(C, \mu_n)$ has an actual multiplicative Kummer value at every point over every finite extension of L. After a finite extension of L, ξ is the class of a regular unit of C.*

Proof. Let $\overline{C}$ be the smooth projective completion and $S = \overline{C} \setminus C$. A finite extension makes S split and supplies a point $x_0 \in C(L)$. Residues give a vector $(\lambda_s)_{s\in S} \in \widehat{\mathbb{Z}}^S$. For a fixed s, choose by Riemann–Roch a finite map $\pi : \overline{C} \to \mathbb{P}^1_L$ with a simple zero at s and nonzero values at all other points of S. Remove its branch values, $0, \infty$ and the images of S, and, after another finite constant extension, split this finite set. This gives a finite *étale* map $W \to V$, with $W \subset C$ and V a punctured affine line. The norm of $\xi|_W$ is a class on V to which F.4 applies: above an L'-point its Kummer value is the product of the norms of the actual values at the finite residual fields in that fibre.

The valuation of a field norm is the sum of residue degrees times the local valuations. Consequently the residue of this norm class at 0 is λ_s: the only boundary point above 0 is the rational point s, and all points inside C have zero residue. This proves $\lambda_s \in \mathbb{Z}$; no ramification multiplier occurs in the divisor-of-norm formula. Set $D = \sum_s \lambda_s[s]$. The localization sequence and the Kummer injection $\operatorname{Pic}(\overline{C})/n \hookrightarrow H^2(\overline{C}, \mu_n)$ imply $[D] \in \bigcap_n n \operatorname{Pic}(\overline{C})$. Its degree is zero, and every one of its division classes has degree zero. The rational point x_0 identifies actual degree-zero line bundles with $J(L)$, so F.3 gives $[D] = 0$. Let g_0 have divisor D.

The class $\xi - [g_0]$ is residue free. At every finite level its extension from C to $\overline{C}$ is unique, since $H^1(\overline{C}, \mu_n) \to H^1(C, \mu_n)$ is injective; uniqueness makes those extensions compatible. Taking limits of the Kummer sequences on $\overline{C}$ gives

$$0 \longrightarrow \widehat{L^*} \longrightarrow \varprojlim_n H^1(\overline{C}, \mu_n) \longrightarrow T \operatorname{Pic}(\overline{C}) \longrightarrow 0.$$

The left system has surjective transition maps, so there is no derived-limit obstruction. Each coordinate of the last Tate module is an infinitely divisible degree-zero Picard class, and F.3 kills it. Thus the remaining class is constant. Evaluation at x_0 makes it the class of an actual element of L^*, completing the proof. □

F.6 (Proved). *Let K be an algebraic closure of k as in F.3 and let $B = \mathbb{Q}^{\mathrm{ab}} \subset K$. Let C/K be a smooth projective connected curve. Suppose a function q on a cofinite subset of $C(K)$ satisfies, for every $b \in K$,*

$$q(x) - b = e_b(x) r_b(x), \qquad e_b(x) \in B^*, \quad r_b \in K(C)^*,$$

outside a finite set depending on b. Then q agrees with one rational function on a cofinite subset. If r_0 is constant, that rational function is constant.

Proof. Every finitely generated field N/B inside K has algebraic constant field $N_0 = N \cap \overline{\mathbb{Q}}$ finite over B. The extension N/N_0 is regular, so for algebraic numbers a, $[N(a) : N] = [N_0(a) : N_0]$. These degrees are unbounded: otherwise $\overline{\mathbb{Q}}/N_0$ would have bounded finite subextension degrees, and hence finite degree, contrary to the height lower bound of F.3 applied to $2^{1/n}$. A fixed finite map of a curve over N to $\mathbb{P}^1_N$ therefore supplies closed points of arbitrarily large degree over N. They can avoid any fixed finite set of K-points; the algebraic points in that finite set have bounded degrees, and the other points cannot equal a closed N-point.

Assume r_0 is nonconstant. Write the representations of q and $1-q$ as $A(x)r(x)$ and $B(x)s(x)$, with coefficients in B^*. Descend C, r, s to a finitely generated field N/B. A nonzero rational function $ar+bs-1$, $a,b \in B^*$, has zero divisor of degree bounded independently of a, b, by the fixed pole divisor. At a point of degree larger than this bound the pointwise equality forces the rational identity $ar+bs=1$. Absorb those two constants into r, s, obtaining $r+s=1$. It follows that

$$(A(x)-B(x))r(x) = 1-B(x),$$

so $q(x)=r(x)$ whenever $r(x) \notin B$, away from finitely many exceptions.

Choose $a \in K \setminus B$ and apply the same argument to $q+a$. Its rational factor cannot be constant: otherwise the cofinite image of the nonconstant r_0 would lie in a vector space of dimension at most two over B, whereas $[K:B]=\infty$. Obtain $q(x)=r_a(x)-a$ where $r_a(x) \notin B$. These two outside-B sets cover the common cofinite domain, since membership of both rational values in B would put both $q(x)$ and $q(x)+a$ in B. Their intersection is infinite: after adjoining all coefficients to N, a point of sufficiently large degree cannot belong to a fibre over a value in B of either of the fixed finite maps r, r_a. On the intersection $r = r_a - a$, hence this is a rational identity. It proves the assertion on the union.

If r_0 is a nonzero constant, rescale q so that its cofinite values lie in B. For $b \notin B$, the factor r_b is constant, because otherwise its cofinite image would lie in $B+Bb$. Two different values of q would then force first $r_b \in B$ and then $b \in B$ by subtracting their two equations. Thus q is constant. This argument verifies the finite-exception and transcendental-coefficient versions explicitly; it does not cite the algebraic-extension version of Tsujimura's quasi-rationality criterion outside its hypotheses. □

F.7 (Proved). *With the closed-point condition restored, LM5-F.50 is valid: if k is any subfield of a finitely generated extension of $\mathbb{Q}^{\mathrm{ab}}$, then for all finite-type k-schemes X, Y the natural map*

$$\mathrm{Hom}_k(X^{\mathrm{awn}}, Y) \longrightarrow \pi_0 \,\mathrm{Hom}^{\mathrm{cp}}_{BG_k}(X_{\mathrm{et}}, Y_{\mathrm{et}})$$

is bijective. In characteristic zero X^{awn} is the seminormalization of X. The statement includes the corrected LM5-F.47 by taking transcendence degree zero.

Proof. We first prove the needed point-function assertion for a smooth geometrically connected affine curve C/L. A closed-point-preserving relative morphism $\phi : C_{\mathrm{et}} \to (\mathbb{G}_m)_{\mathrm{et}}$ gives a G_L-equivariant function $f : C(\overline{L}) \to \overline{L}^*$. An L'-point maps to an L'-point for every finite L'/L: composition gives a relative morphism of closed-point Galois topoi, whose map to $BG_{L'}$ forces the residue field of the selected image to embed in L'. The specified relative isomorphism also identifies the coefficient sheaves μ_n, so pulling back Kummer classes and evaluating at points commute.

F.5 applied to the coordinate class gives, after a finite extension, a unit g_0 such that $f(x)/g_0(x) \in \mu_\infty$ for every geometric point. If the generic image of ϕ is closed, continuity forces its entire point function into that closed point; after a finite extension the function is constant. Otherwise, for each $b \in \overline{L}$, the inverse image of $\mathbb{G}_m \setminus \{b\}$ is a dense open C_b, with finite complement. Applying F.5 to the class of $t-b$ on this open gives a rational function g_b, a unit on C_b, with

$$f(x) - b \in \mu_\infty \, g_b(x) \quad (x \in C_b(\overline{L})).$$

F.6 gives a rational R agreeing with f outside a finite set. For each b, $(R-b)/g_b$ has cofinite image in μ_∞ if it is nonconstant, an impossibility; thus it is a constant root of unity. Fix x and choose $b \neq f(x)$. The identity $R-b = c_b g_b$ first shows that R has no pole at x. For every $b \notin \{f(x), R(x)\}$ it then gives

$$\frac{f(x)-b}{R(x)-b} \in \mu_\infty.$$

If $f(x) \neq R(x)$ the left side, as b varies, fills $\overline{L} \setminus \{0, 1\}$, contradicting the proper subset μ_∞. Hence $f = R$ everywhere. It follows that R is a regular unit. Galois equivariance descends it to L.

If $C(L) \neq \varnothing$, its compatible Kummer class equals the one pulled back by ϕ already over L. Over the finite extension used in F.5, R/g_0 takes only root-of-unity values, so it is constant; evaluation at an L-point makes its class zero. The residual finite-level class over L is geometrically zero, and the low-degree Hochschild–Serre sequence makes it a constant class. Evaluation at the L-point, which splits the constant-class inclusion, makes that residual class zero. This proves equality at all levels without an unwarranted descent of a chosen Kummer root.

We next pass to arbitrary dimension. Let X/k be connected regular affine of finite type, let $x_0 \in X(k)$, and let $\phi : X_{\mathrm{et}} \to (\mathbb{G}_m)_{\mathrm{et}}$ be closed-point-preserving and relative. Regularity in characteristic zero gives smoothness. The algebraic constant field in $k(X)$ embeds in k by evaluation at x_0, so X is geometrically integral. Choose a smooth projective compactification Y with normal-crossings boundary $D_1, \dots, D_s$. The strong resolution and principalization result of Encinas and Hauser (2002, pp. 1–4, "The result" and "Idea of proof") gives this by operations supported off the given smooth open. Its hypotheses are characteristic zero and a finite-type embedded variety, both satisfied here.

Write $h = (h_n)_n$ for the pulled-back coordinate Kummer class. For every boundary prime D_i it has a residue $r_i \in \widehat{\mathbb{Z}}$. After a finite extension choose a rational smooth point q of a geometric component of D_i, away from the other components, and a smooth geometrically integral projective curve through q and x_0, transverse to D_i at q. Here is the precise Bertini input. Stacks, Lemma 33.47.1 says sufficiently high powers of an ample bundle, with prescribed vanishing on a finite subscheme, embed its complement; Lemma 33.47.3 makes their general divisor smooth on that complement; Lemma 37.32.3 makes it geometrically irreducible when the base locus has codimension at least two and the linear system has two independent moving directions (The Stacks Project Authors, 2026a). Surjectivity of the first-jet evaluation for sufficiently high powers allows independent differentials at the prescribed points, including transversality. Intersect these nonempty opens successively, reducing dimension to one. The fields are infinite, so each such open has a rational point. This constructs the specified curve and checks the extra hypotheses, rather than presuming a curve through fixed points.

The curve calculation makes its restricted Kummer class integral; its residue at q is r_i. To check the comparison, in the regular local ring at q the sole boundary has equation $z = 0$. At each finite level, after an *étale* localization, the Kummer class is represented by uz^j: this local ring is regular and factorial, and the unit factors away from z have zero boundary residue. The transverse curve pulls z back to a uniformizer, and u to a unit, so the residues agree modulo every n. Finite separable constant extension has multiplicity one along the divisor. Thus all $r_i \in \mathbb{Z}$.

The integral divisor $D = \sum_i r_i D_i$ has zero Kummer Chern class modulo every n, hence $[D] \in \bigcap_n n \operatorname{Pic}(Y)$. This intersection vanishes. Indeed the quotient of $\operatorname{Pic}(Y)$ by its algebraically trivial subgroup is a subgroup of the finitely generated geometric Néron–Severi group, by Cheng et al. (2021, Theorem 7.3). That theorem assumes a smooth projective scheme over an algebraically closed field, and is applied to $Y_{\bar{k}}$. Kill the finite torsion of the quotient by an integer e. If x is divisible by all integers in $\operatorname{Pic}(Y)$, its quotient is zero, and divisions by em yield divisions of x by m in the algebraically trivial subgroup. This subgroup injects into the k-points of the Picard abelian variety; F.3 therefore gives $x = 0$. The same argument kills $T \operatorname{Pic}(Y)$, since each coordinate of a compatible torsion system is infinitely divisible. The possible Brauer obstruction only restricts the subgroup of Picard-scheme points represented by actual line bundles; it does not spoil this injection, which follows from Hilbert 90.

Choose $g \in k(Y)^*$ with divisor D. The class $h - [g]$ is unramified along every prime of Y. A finite-level Kummer torsor unramified in codimension one on a regular scheme extends *étale*

over it by purity. The exact form used is Stacks, Lemma 58.31.2: a finite *étale* cover of a dense open, unramified along the missing prime divisors, extends when the missing local rings are regular (The Stacks Project Authors, 2026b). Its extension is unique by normality and generic injectivity. The limit Kummer sequence and $T\operatorname{Pic}(Y) = 0$ make the remaining class constant. Evaluation at x_0 supplies an actual $c \in k^*$, so $h = [cg]$ and $cg(x_0) = \phi(x_0)$.

For any $a \in X(\bar{k})$ choose a finite extension making it rational and a curve through a and x_0 by the same Bertini argument. The curve's actual unit f_C and $cg|_C$ have equal Kummer classes. Their ratio is divisible by every integer in the unit group. Its divisor on the projective completion is therefore zero, so it is constant. Its value at x_0 is one. It follows that $cg(a) = \phi(a)$ for every a. The case $\dim X = 1$ is the curve argument, and $\dim X = 0$ is immediate.

Finally Carlson–Haine–Wolf, Theorem 4.18, for an arbitrary field, states the equivalence between full reconstruction and its condition (4.18.5): for every affine regular connected finite-type source with a rational point, every pinned map to $(\mathbb{G}_m)_{\mathrm{et}}$ has the geometric-point function of a morphism to $\mathbb{G}_m$ (Carlson et al., 2024). We have proved exactly that condition, and for all finite extensions used in its descent. Applying the theorem gives the bijection stated above. This final use is essential: equality of point functions for one map alone would not identify the entire inverse-image functor. The theorem's proof uses the same assertion on sliced *étale* objects, reconstructs affine coordinate maps, and applies h-descent in the source. Thus neither a fundamental-group isomorphism nor a point-function calculation is substituted for the full topos claim. □

F.8 (Known). *For a characteristic-zero field k, the following is an exact necessary and sufficient criterion for* closed-point-preserving étale *reconstruction: for every connected smooth affine X/k with $X(k) \neq \varnothing$, every closed-point-preserving relative morphism $X_{\mathrm{et}} \to (\mathbb{G}_m)_{\mathrm{et}}$ has on $\bar{k}$-points the values of one regular unit of X.*

Proof. This is the equivalence (4.18.1)⇔(4.18.5) of the opened Carlson–Haine–Wolf theorem, with "regular" replaced by "smooth" in characteristic zero and $X^{\mathrm{awn}} = X$ for regular X. The forward implication is restriction to the stated source and target; the reverse implication is the reconstructed affine-coordinate and h-descent argument specified at the end of F.7. This is a topos-theoretic criterion, not an intrinsic classification in terms of the field's arithmetic. □

Literature comparison. The September 2026 version of Carlson–Stix has been opened as well. Its Theorem 4.1 proves reconstruction over sub-p-adic fields; Theorem 6.12 adds a relative result for componentwise universally open schemes over geometrically unibranch noetherian Nagata bases with sub-p-adic function fields (Carlson and Stix, 2026). Proposition 3.17 assumes Kummer-faithfulness for its point-theoretic open-image dichotomy. These hypotheses do not cover $\mathbb{Q}^{\mathrm{ab}}$ or $\mathbb{Q}^{\mathrm{solv}}$. Indeed, the algebraic constant field in a finitely generated extension of $\mathbb{Q}_p$ is finite over $\mathbb{Q}_p$, whereas either of these fields contains roots of unity of unbounded p-power order. This also corrects R27's attribution: LM5-F.38 actually cites Carlson–Stix 2026, not Carlson–Haine–Wolf 2024. No opened source or targeted search located the full statement of LM5-F.47 or F.50. This records the search outcome and makes no assertion of priority or of absence from the literature.

F.9 (Unresolved). *An intrinsic necessary-and-sufficient arithmetic classification of all characteristic-zero fields satisfying closed-point-preserving reconstruction for every finite-type scheme remains unresolved here. The corrected $\mathbb{G}_{m,\mathbb{Q}^{\mathrm{solv}}}$ endomorphism test is proved in F.23 below, and F.31 proves all-scheme reconstruction over every finite extension of $\mathbb{Q}^{\mathrm{solv}}$. Neither result supplies an intrinsic classification of all fields.*

The earlier F.3–F.7 proof does not extend directly to $\mathbb{Q}^{\mathrm{solv}}$. Its difficulty is not lack of a discrete valuation: F.4 already removes that assumption. That proof uses the freeness

of an ambient multiplicative group modulo μ_∞ and the absence of divisible abelian-variety points. Over $\mathbb{Q}^{\mathrm{solv}}$, the multiplicative group is divisible. Indeed, for $a \in (\mathbb{Q}^{\mathrm{solv}})^*$ and $n > 0$, choose a finite solvable Galois extension $L/\mathbb{Q}$ containing a and μ_n. Adjoining n-th roots of all the conjugates of a gives an abelian extension of L and a Galois solvable extension of $\mathbb{Q}$. It therefore lies in $\mathbb{Q}^{\mathrm{solv}}$ and supplies the desired root. In particular the multiplicative Kummer class of 2 is zero although 2 is not a root of unity. Thus the coefficient-character argument and the residual-root-of-unity conclusion cannot be applied. An elliptic curve over $\mathbb{Q}$ also has its entire 2-primary torsion rational over $\mathbb{Q}^{\mathrm{solv}}$: the groups $\mathrm{GL}_2(\mathbb{Z}/2^n)$ are solvable, since their reductions to $\mathrm{GL}_2(\mathbb{F}_2) \simeq S_3$ have 2-group kernels. These points are divisible by every integer in that field and invalidate F.3's abelian-variety input. Neither failure constructs a nonalgebraic pinned morphism of the full *étale* topos. A system of finite-cover maps must still account for all open subtopoi and their partly ramified charts. F.20–F.23 supply that comparison for the specific rational endomorphism test. F.24–F.31 then supply a different all-scheme proof by finite quotient detection, arithmetic equivariance over a number field, and the semiabelian Tate-Hom theorem; they do not use the failed abelian-variety input. The generic-valued map F.2 has constant underlying image and is not a counterexample to that corrected test. Under R27's literal definition the universal-field criterion has the empty solution set by F.1, and therefore cannot simultaneously contain the incorrectly phrased F.50.

F.16 and F.18–F.19 below give further finite-extension constraints: the induced profinite exponent is an integer, and the complete point function is an algebraic monomial. F.20–F.23 complete the comparison on full topoi by finite-quotient detection and normalization of charts. The higher-genus and higher-dimensional extensions are proved separately in F.24–F.31. F.17 refutes the literal relative-only nonconstant assertion.

6.3 What cyclotomic normalization does and does not prove

F.10 (Known, with explicit derivation). *Let K have characteristic zero, let X/K be smooth of finite type, let D be a normal-crossings divisor, and put $U = X \setminus D$. At a geometric point $\bar{x}$ with branch set J, the strict local covering-germ group is T^J, $T = \widehat{\mathbb{Z}}(1)$. A chosen path gives a continuous homomorphism $j_x : T^J \to \pi_1(U_{\bar{K}})$, well defined up to geometric conjugacy. The local decomposition group acts on T^J by the cyclotomic character times the permutation of branches. A compatible monomial map*

$$t'_j = u_j \prod_i t_i^{m_{ji}}$$

induces the integer contact matrix (m_{ji}) on the two geometric local inertia groups. Units change tangential splittings by Kummer cocycles but do not change this geometric inertia map.

For a family of smooth hyperbolic curves over a smooth finite-type base, a horizontal puncture group injects into the geometric fibre and total groups. If U is a hyperbolic polycurve and X is any smooth proper normal-crossings compactification, the diagonal $T \to T^J \xrightarrow{j_x} \pi_1(U_{\bar{K}})$ has nonzero image. At a rational boundary point every section factoring through the punctured henselian neighbourhood therefore has nontrivial cyclotomic centralizer. Global injectivity of the full *group T^J is an additional assertion; it does not follow from these facts.*

Proof. SGA 1, XIII, Proposition 5.2 assumes a regular local scheme and a normal-crossings divisor given by part of a regular parameter system; it says a finite cover tamely ramified on the divisor becomes *étale* after adjoining roots of those parameters, with orders prime to the residue characteristic. Corollary 5.3 then identifies the tame fundamental group of the punctured strict localization with $\prod_{\ell \neq p} \mathbb{Z}_\ell(1)^J$ (Grothendieck and et al., 2003). Here $p = 0$, so every prime occurs. On the cover $z_i^n = t_i$, an inertia tuple acts by $z_i \mapsto \zeta_i z_i$. Conjugating

by an element of the local arithmetic group sends ζ_i to its cyclotomic power and permutes the parameter branches. Units have roots in the strict henselian ring, which proves both the contact-matrix formula and its independence of those roots on geometric inertia. Paths account exactly for the geometric conjugacy ambiguity.

For a punctured surface of type (g, r) with $2g - 2 + r > 0$, $r > 0$, its fundamental group is free on $a_1, b_1, \ldots, a_g, b_g, c_1, \ldots, c_{r-1}$; the last puncture is the inverse of $\prod_i [a_i, b_i] \prod_j c_j$. Every puncture word is nontrivial. A cyclic subgroup generated by a nontrivial word in a free group has its full profinite topology: represent a cyclically reduced word by an immersed circle in a rose, complete that finite labelled graph to a finite covering without identifying its vertices, and use its primitive homology class to map the resulting finite-index subgroup to $\mathbb{Z}/n$. Taking the core gives finite quotients in which the word has order divisible by every prescribed n. Thus its completed cyclic subgroup is $\widehat{\mathbb{Z}}$ and injects. SGA 1, XII, Theorem 5.1 identifies finite algebraic and analytic *étale* covers of a locally finite-type complex scheme; Corollary 5.2 identifies its algebraic fundamental group with the finite-index completion of its topological group. Descent of the finite-type pair to a finitely generated field and characteristic-zero geometric base change transport this argument to $\bar{K}$. This proves the horizontal statement, including a single puncture, whose inertia can vanish in abelianization while remaining injected in the full group. Injection from the geometric fibre to the total group is Hoshi, Proposition 2.4(i), valid for a hyperbolic curve over a connected noetherian separated normal characteristic-zero base (Hoshi, 2014). The present smooth base satisfies these hypotheses.

For diagonal nonvanishing, the further input is Hoshi, Lemma 2.8: a point of a relative hyperbolic curve over the fraction field of a strictly henselian trait extends over the trait exactly when its map to the total fundamental group is trivial. Its base has the same noetherian, separated, normal, characteristic-zero hypotheses. Induction gives this criterion for a polycurve: first extend to the preceding stage, then apply the relative-curve result. In completed branch coordinates at $\bar{x}$, send every boundary parameter to t and all other parameters to zero. This defines an arc $\operatorname{Spec} \bar{K}[[t]] \to X_{\bar{K}}$ with generic point in U and closed point $\bar{x}$. The Kummer covers identify its map on fundamental groups with j_x composed with the diagonal. If that map were zero, the criterion would extend the generic point to U. Separatedness of X identifies this extension with the original arc, contradicting its boundary closed point. Finally branch permutations fix the diagonal and the cyclotomic kernel acts trivially on it. Equivariance gives the centralizer assertion for every locally factoring section, including nonsplit branches. This is precisely the extra nonvanishing statement of CK-C8; it is weaker than full local injection. □

F.11 (Proved). *The following precise group-theoretic formulation supplies the injection step of LM5-F.39. Suppose* $1 \to F_r \to G \to Q \to 1$, $r \geq 2$, *is an extension of discrete groups. Then* $\widehat{F_r} \to \widehat{G}$ *is injective and*

$$1 \longrightarrow \widehat{F_r} \longrightarrow \widehat{G} \longrightarrow \widehat{Q} \longrightarrow 1$$

is exact. Suppose in addition free abelian local lattices A, A_B *map compatibly to* G, Q, *with contact map* $M : A \to A_B$, *the map* $\widehat{A}_B \to \widehat{Q}$ *injective, and* $\ker M$ *either zero or infinite cyclic whose generator maps to a nontrivial element of* F_r. *Then* $\widehat{A} \to \widehat{G}$ *is injective. This applies at each stage of a recursively compatible stable pointed curve tower with affine hyperbolic fibres and injected base meridian lattices: the node kernel is generated by the essential vanishing cycle, and the marked kernel by an essential puncture cycle.*

Proof. First $Z(\widehat{F_r}) = 1$. For a finite quotient $F_r \to A_0$ with kernel L, choose a prime $\ell \nmid |A_0|$. The cellular chains of the finite regular rose covering give

$$0 \to H_1(L, \mathbb{F}_\ell) \to \mathbb{F}_\ell[A_0]^r \to \mathbb{F}_\ell[A_0] \to \mathbb{F}_\ell \to 0.$$

Semisimplicity gives $H_1(L,\mathbb{F}_\ell) \simeq \mathbb{F}_\ell \oplus \mathbb{F}_\ell[A_0]^{r-1}$, a faithful A_0-module. In the finite quotient $F_r/[L,L]L^\ell$, the image in A_0 of a central element of $\widehat{F_r}$ must act trivially on this module, so must be one. This for every A_0 proves centrelessness.

The automorphism group of a finitely generated profinite group is profinite: intersections of its finitely many open subgroups of bounded index give a characteristic open basis. Therefore conjugation $G \to \mathrm{Aut}(\widehat{F_r})$ extends to $\widehat{G}$. Its composite with $\widehat{F_r} \to \widehat{G}$ is the inner-automorphism map, which is injective. The quotient of $\widehat{G}$ by the closure of F_r classifies exactly the finite quotients of Q, proving the asserted exactness.

Smith normal form shows $\ker \widehat{M} = \widehat{\ker M}$: the image of M is free, the sequence onto that image splits, and its inclusion in A_B stays injective after completion. A kernel element of $\widehat{A} \to \widehat{G}$ first maps to zero in $\widehat{A}_B$, then lies in $\widehat{\ker M}$. In the cyclic case F.10's full-topology argument and the first paragraph of this claim make the latter map injective, so the element is zero.

In a stable curve chart $xy = t_1^{a_1} \cdots t_s^{a_s}$ the normal lattice has, relative to the base, one kernel direction represented by the two opposite branch meridians. On a smooth fibre it is the vanishing cycle. A vanishing cycle bounding an unmarked disc would cut off an unmarked genus-zero tree attached at one node; summing special points over that tree contradicts stability. Thus it is essential. At a marking the corresponding kernel is a puncture cycle. Compatible root changes restrict these lattices to finite index; smooth toroidal subdivisions insert essential annular directions and preserve the same kernel calculation. The hypotheses that these are the local lattice maps and that the base lattices already inject are exactly what is required for the induction; an arbitrary blowup of a base boundary is not included.

For the geometric application, the required full exact sequence at a smooth tower stage, including its left injection, is also given directly by Hoshi, Proposition 2.4(i). The base is smooth, connected, separated, noetherian and normal. Apply the identical kernel chase to this profinite sequence. Thus the application does not presume that a merely right-exact homotopy sequence is left exact, or that arbitrary profinite completion preserves kernels.

We record CK-B5's logically separate group assertion as well. Every open subgroup V of $\widehat{F_r}$ is the completion of the corresponding finite-index discrete free subgroup: cores of finite-index subgroups show that the induced topology is full. The graph-covering formula gives its rank $1 + [\widehat{F_r} : V](r-1) \geq 2$, so the centre calculation above applies to V. The group $\widehat{F_r}$ is torsion free: if a nonidentity finite-order z survives modulo an open normal N, put $V = N\langle z\rangle$. Its nonzero cyclic quotient V/N factors through the torsion-free abelianization of V, where the finite-order element z must vanish, a contradiction. For an open normal V, its centralizer injects into the finite group $\widehat{F_r}/V$ and is therefore trivial. Passing to a normal subgroup inside an arbitrary open H proves $C_{\widehat{F_r}}(H) = 1$. This property passes to extensions: project a centralizer element to the slim quotient, then centralize the open intersection with the slim kernel. It passes to surjective inverse limits because the image of an open subgroup under each quotient projection is open, and all its coordinate centralizers vanish. Rank one is excluded, since $\widehat{\mathbb{Z}}$ centralizes itself. This proves the CK-B5 comparison without converting slimness into an inertia recognition theorem. □

F.12 (Refuted). *Cyclotomic normalization alone does not imply global injection of a full normal-crossings inertia group, and finite Kummer root changes cannot repair a nonzero kernel. A finite root stack also changes the group and can enlarge the local inertia kernel.*

Proof. Let $U = (\mathbb{P}^1_{\mathbb{Q}} \setminus \{0,1,\infty\})^2$ and blow up the point $(0,2)$ of its product compactification. Write local coordinates $x = uv$, $y - 2 = v$ on one blowup chart. The two boundary components at $u = v = 0$ are the strict transform and the exceptional divisor. Their meridians both map to the old $x = 0$ meridian; the loop at $y = 2$ is null-homotopic in the second factor. Thus,

identifying the injected old cusp group with T, the local map is

$$T^2 \longrightarrow T, \qquad (a,b) \longmapsto a+b,$$

with kernel $\{(a,-a) : a \in T\}$. Both factors have the stated cyclotomic action. The diagonal survives by multiplication by two. The discrete matrix and its reductions are checked in `code/F_checks.py`; the displayed calculation proves the whole profinite statement.

If $A \neq 0$ is a closed subgroup of $\widehat{\mathbb{Z}}^r$ and L is a finite-index lattice, choose n with $n\widehat{\mathbb{Z}}^r \subset L$. Multiplication by n is injective, so $0 \neq nA \subset A \cap L$. A finite root *cover* only passes to such a lattice and an open global group, so the kernel persists. For a finite root *stack* with orders n_i, the global group is the quotient by the closed normal group generated by the n_i-th powers of boundary inertia. This follows directly from the local charts $[\operatorname{Spec} A[z_i]/(z_i^{n_i} - t_i)/\prod_i \mu_{n_i}]$: its *étale* covers are exactly the tame covers whose inertia orders divide n_i. For $(\mathbb{A}^1, 0)$ the original inertia map is the identity of T, whereas its map to the n-th root stack has kernel nT. Taking all divisible orders recovers the original kernel because every finite quotient has finite inertia orders. These are the distinct cover and stack statements in LM5-F.22, and neither contradicts F.11. □

F.13 (Proved). *Trait-inertia conditions satisfied by a homomorphism do not, by themselves, make it the homomorphism of a rational point. This holds both for trivial inertia and for the positive-multiplicity, injective, finite-index cusp conditions of F149-C3. Likewise the injection of actual tangent vectors into the completed Kummer torsor in F149-C8 is not surjective, already over* $\mathbb{Q}$.

Proof. Set $R = \mathbb{Q}[[t]]$, $K = \mathbb{Q}((t))$, and let $U/R = \mathbb{P}^1_R \setminus \{0, 1, \infty\}$ be constant. Write $E = \pi_1(U_{\mathbb{Q}})$ for the group of the total good model. The coefficient splitting identifies $G_K = T \rtimes G_{\mathbb{Q}}$, and the tangent vector 1 at the rational cusp 0 identifies its normalizer $D_0 \subset E$ with $T \rtimes G_{\mathbb{Q}}$, with cyclotomic action in both products.

For the zero-inertia example compose $G_K \to G_{\mathbb{Q}}$ with this cuspidal splitting. It kills inertia. If it came from a K-point, the local Kummer formula would force its proper extension to avoid the boundary; its reduction would be a point of $U(\mathbb{Q})$. The resulting section would then be both a point section and a cuspidal section over $\mathbb{Q}$, contrary to Stix, Theorem 14. That theorem assumes injectivity for proper hyperbolic curves over every finite extension; here it follows from the abelian-variety Kummer injectivity established in F.3. Thus all its hypotheses are met.

For the positive-inertia example choose $\lambda \in \widehat{\mathbb{Z}}$ with $\lambda_2 = 1$ and $\lambda_\ell = 0$ for all odd primes. It is not an ordinary integer: an integer divisible by every odd prime is zero, incompatible with its 2-adic coordinate. Let $a = \lambda\kappa_{\mathbb{Q}}(2) \in H^1(G_{\mathbb{Q}}, T)$ and choose a continuous cocycle representative. For an integer $\mu > 0$ define

$$h : G_K \longrightarrow D_0 \subset E, \qquad (z,g) \longmapsto (\mu z + a(g), g).$$

The cocycle identity proves that h is a continuous homomorphism. It is injective, its inertia is μT, and its image has index μ in D_0. Via the good-reduction pullback $\pi_1(U_K) = E \times_{G_{\mathbb{Q}}} G_K$, it also determines a section of the generic arithmetic extension.

Suppose the map to E were, up to geometric conjugacy, induced by a point $x \in U(K)$. Its nontrivial inertia forces its proper extension to meet a cusp, and the intersection theorem for cusp groups (Stix, Proposition 11 and Corollary 12) makes that cusp 0. Write $x = t^\mu u$, $u \in R^*$. Pulling back all covers $z^{1/n}$ shows that its residual cocycle is $\kappa_{\mathbb{Q}}(\bar{u})$: the group $1 + tR$ is uniquely divisible in characteristic zero. A conjugation preserving the nonzero open subgroup μT of the cusp inertia belongs to D_0, and its geometric part belongs to T; it changes only the cocycle representative. Applying the completed 2-adic-place valuation $\widehat{\mathbb{Q}^*} \to \widehat{\mathbb{Z}}$ to the residual

class gives the ordinary integer $v_2(\bar{u})$ on the point side and λ on the constructed side. They cannot agree. This is a whole-profinite-group obstruction, not a finite-level experimental claim.

The same class a gives a splitting of $D_0 \to G_{\mathbb{Q}}$ not induced by any actual nonzero tangent vector. Actual vectors form a $\mathbb{Q}^*$-torsor; after choosing 1, their images are precisely $\kappa_{\mathbb{Q}}(\mathbb{Q}^*)$. Their 2-valuations are integers, excluding a. The finite CRT residues and the semidirect-product law used above are checked in `code/F_checks.py`. □

F.14 (Known, with derivation). *For a hyperbolic curve over a characteristic-zero field k which distinguishes tori from abelian varieties, a nontrivial closed subgroup $A \subset \pi_1(U_{\bar{k}})$ is contained in a cusp inertia group if and only if it is procyclic and essentially cyclotomically normalized. A continuous section is cuspidal if and only if its image cyclotomically normalizes a nontrivial procyclic geometric subgroup. This supplies a recognition theorem which the local normalization/injection theorem F.10–F.11 does not supply. The hypothesis holds for every k as in F.3, in particular every field finitely generated over $\mathbb{Q}$.*

Proof. Stix, Theorem 20, states the equivalence of containment in inertia, procyclic cyclotomic normalization, and procyclic essential cyclotomic normalization, under exactly the displayed field and hyperbolicity hypotheses. Corollary 21 identifies the maximal such subgroups, and Corollary 22 gives the section statement (Stix, 2008b). To spell out the last implication, Theorem 20 places A in I_c. If an element of the section image normalizes A, the two conjugate cusp groups meet nontrivially. Proposition 11 makes them equal, and Corollary 12 identifies their arithmetic normalizer with the cusp decomposition group. The section image therefore lies in that group. The reverse direction is the local Kummer action. The section surjects onto G_k, so the cusp is k-rational.

To verify the field hypothesis, split a given torus after a finite extension k'/k. A nonzero equivariant map $TS \to TA$ would give, at some prime ℓ, a nonzero map $\mathbb{Z}_\ell(1) \to T_\ell A$. Restriction to $k'(\mu_\infty)$ gives nonzero invariant compatible ℓ-power torsion. Every coordinate is divisible by all positive integers in $A(k'(\mu_\infty))$, since multiplication by integers prime to ℓ is invertible on the compatible ℓ-primary tower. This field is again a subfield of a finitely generated extension of $\mathbb{Q}^{\mathrm{ab}}$, so F.3 excludes it. Hence $\mathrm{Hom}_{G_k}(TS, TA) = 0$.

No part of this argument identifies every noncuspidal section with a rational point. That would require the section-surjectivity assertion separately retained in Part D. □

F.15 (Proved: exact scope of the eight-row comparison). *None of F149-C1–C6, C8, C9 asserts that an arbitrary section comes from a rational point solely because of its inertia. C2, C3, and C5 do contain converses within the class of homomorphisms already known to come from rational geometric sections. Those conditional converses are valid and are stronger than the bare injection statement of LM5-F.39. They do not turn into an arbitrary-section recognition theorem. The same distinction applies to C9's tangential specialization data.*

Proof. Here is the comparison, keeping the quantifiers of the original rows. The manuscript references are those of the opened row statements.

Row	Actual assertion and locus	Converse and its disposition
C1	Hartogs extension of a rational section of a smooth pair or tower; 149, pp. 32, 34–35, 41; scans 65, 70–72, 83.	No inertia-recognition assertion. Its input is an actual section on a dense open.

Row	Actual assertion and locus	Converse and its disposition
C2	An actual trait point extends iff its map to the total good model kills inertia; pp. 36–38, scans 74–78.	A genuine conditional converse. Proper extension and the nonzero Kummer multiplicity prove it. The arbitrary-section extension is false by F.13's zero-inertia example.
C3	For an actual point meeting a cusp, inertia is multiplication by its positive contact order, with the stated index; pp. 37–39, 44, scans 75–79, 90.	Equivalent conditions for that already geometric point. The positive-inertia converse for an arbitrary homomorphism is false by F.13's $\lambda\kappa(2)$ example.
C4	Exact homotopy and reconstruction of a centre-free extension from its outer action; p. 39, scan 79, also 64, 73.	No claim about which sections are geometric. Neither exactness nor centrelessness implies section surjectivity.
C5	An actual rational section extends over a smooth base iff it kills divisorial inertia; pp. 32, 34–36, 41–42; scans 65, 70–74, 83–84.	A genuine conditional converse using C2 and C1. An arbitrary unramified section can be cuspidal and need not be an interior point section; F.13 refutes that extension.
C6	Normalizer of a specified cusp group and its Chern-class extension; pp. 39–41, scans 79–82.	The subgroup is already specified as cusp inertia. A section lying in its normalizer is cuspidal by definition. Recognition from cyclotomic normalization is the extra theorem F.14.
C8	Actual normal vectors inject into the completed Kummer torsor of splittings; p. 42, scan 85.	No surjectivity is asserted. Such surjectivity is false over $\mathbb{Q}$, by F.13. Replacing normal vectors by conormal frames changes the Kummer sign.
C9	An actual rational section fails to extend on a divisorial locus and has its leading-term tangential specialization; pp. 46–47, scans 93–95.	The rational section is part of the input. An arbitrary contact-order datum need not be geometric, again by F.13.

For completeness the geometric implications underlying this table can be checked directly. Over a trait a point extends uniquely to the proper curve. If it meets the boundary with equation $z = t^{\mu}u$, the pullback of $z^{1/n}$ gives inertia $\tau \mapsto \mu\tau$ in the injected cusp group. If it avoids the boundary the map factors through the trait. These are exactly the two directions of C2 and the positive-multiplicity equivalences in C3. For a higher-dimensional regular base, killing divisorial inertia extends the point at every height-one local ring. Its domain then has complement of codimension at least two. The extension step C1 is geometric: after resolving the rational map to a proper positive-genus curve, exceptional projective-space fibres map constantly, since no nonconstant map from $\mathbb{P}^1$ to that curve exists. Descend each blowup using a finite birational graph over the normal base. A boundary intersection is an effective Cartier divisor, so cannot be supported in codimension two. For genus zero with at least two removed points, trivialize one boundary section *étale*-locally and apply Hartogs to the resulting regular function; the other boundary intersections are again Cartier divisors. The extensions glue by separatedness. Induction on a tower uses its actual Cartesian pullbacks. This proves the

conditional converse in C5 and explains the divisorial statement in C9. Its leading coefficient transforms under $t \mapsto at$ as $u \mapsto a^{-\mu}u$, giving the stated Kummer change $-\mu\kappa(a)$.

The sign in C6 is also consistent with F.10. If N is the normal line, choose frames $e_j = g_{ij}e_i$ and scalar coordinates $w_i = g_{ij}w_j$. Roots $r_i^n = w_i$ transform by $r_i = h_{ij}r_j$, $h_{ij}^n = g_{ij}$. Their triple defect $h_{ij}h_{jk}h_{ki} \in \mu_n$ is $+c_1(N) \bmod n$ with the positive Kummer convention. The conormal line is $N^\vee$ and has the opposite sign. Thus this is extra extension-class information, not a recognition of geometric point sections.

LM5-F.21 concerns the local tube and its possible Chern transgression; F.22 concerns root stacks and covers; F.23 changes a cuspidal model to a stable model; F.35 is the abstract closed-subgroup criterion for extension of a covering germ as a sub-germ; F.39 is the injection and normalization induction of F.10–F.11. These are compatible with the eight rows just listed. CK-B5 is a slimness statement, not an inertia-recognition statement: the centralizer of an open subgroup vanishes in nonabelian free profinite groups, and the property passes through extensions and surjective inverse limits. Rank one is excluded. CK-C8 asserts nonzero diagonal boundary inertia and a boundary-local centralizer implication; it explicitly allows the blowup kernel of F.12. Neither it nor LM5-F.48 asserts its arbitrary-section converse. The distinction between local injection, cyclotomic recognition in F.14, and point section surjectivity in Part D is therefore essential in all three returns. □

6.4 A finite-extension constraint for the unresolved solvable-field test

The following additional test uses finite extensions of $\mathbb{Q}^{\mathrm{solv}}$. It rules out a proposed source of exotic endomorphisms without asserting reconstruction of the full topos.

F.16 (Proved). *Put $k = \mathbb{Q}^{\mathrm{solv}}$. For every finite extension L/k,*

$$D(L^*) = k^*.$$

Consequently, if $\phi : X_{\mathrm{et}} \to X_{\mathrm{et}}$ is a closed-point-preserving relative geometric morphism for $X = \mathbb{G}_{m,k}$, then its pullback of the canonical compatible Kummer class is

$$\phi^*[t] = m[t] \quad \text{for one } m \in \mathbb{Z}.$$

For every $\alpha \in \overline{\mathbb{Q}}^*$ its induced point value satisfies

$$\phi(\alpha) = c_\alpha \alpha^m, \qquad c_\alpha \in k^*.$$

In particular a nonintegral profinite exponent cannot give the endomorphism requested in F2. This claim alone does not prove that c_α is independent of α or that $m \neq 0$ for a nonconstant point map; F.19 below settles those two point-function issues. This Kummer calculation alone does not identify the full geometric morphism; F.20–F.23 provide the additional comparison.

Proof. We first establish a descent calculation. Enlarge a finite extension of k to $L = Mk$, where $M/\mathbb{Q}$ is finite Galois, and put $B = M \cap k$. Both M/B and k/B are Galois and their intersection is B, so restriction identifies

$$\mathrm{Gal}(L/B) = \mathrm{Gal}(M/B) \times \mathrm{Gal}(k/B).$$

Write $G = \mathrm{Gal}(L/k)$ for the first factor. Given a finite list of elements of L, choose a number field $S \subset k$ containing B such that the list lies in the number field $F = MS$. Suppose $z^n = a \in F^*$ with $z \in L^*$. For $\sigma \in G$ and $\tau \in \mathrm{Gal}(L/F)$ we have $\tau z/z \in \mu_n \subset k$. The two factors commute, and σ fixes these roots of unity. Hence

$$\tau\left(\frac{\sigma z}{z}\right) = \frac{\sigma(\tau z/z)}{\tau z/z}\frac{\sigma z}{z} = \frac{\sigma z}{z}.$$

It follows that $\sigma z/z \in F$ and

$$\frac{\sigma a}{a} \in (F^*)^n. \tag{1}$$

The important point is that F can be fixed before the root z and the integer n are chosen.

A number field satisfies $D(F^*) = \{1\}$. Indeed the valuations of an element in this intersection all vanish, and Dirichlet's unit theorem then reduces the assertion to a finitely generated abelian group, whose intersection of all power subgroups is trivial. If $a \in D(L^*)$, choose F containing a and apply (1) for every n. Thus $\sigma a/a = 1$ for every $\sigma \in G$, so $a \in k^*$. Conversely k^* is divisible by the radical argument in F.9. This proves $D(L^*) = k^*$ for the enlarged Galois extension, and then for every intermediate finite extension as well. In particular a proper finite extension of k is not radically closed: any of its elements outside k fails some power-divisibility test.

The finite Kummer sequence and $\mathrm{Pic}(X) = 0$ give

$$H^1(X, \mu_n) = \mathcal{O}(X)^*/\mathcal{O}(X)^{*n} = \mathbb{Z}/n,$$

because $\mathcal{O}(X)^* = k^* \times t^{\mathbb{Z}}$ and k^* is divisible. The relative rigidification identifies the pullback of μ_n with μ_n, compatibly in n. We may therefore write $\phi^*[t] = \lambda[t]$ with $\lambda \in \widehat{\mathbb{Z}}$. For a finite L/k and $\alpha \in L^*$, the composite $\mathrm{Spec}(L)_{\mathrm{et}} \to X_{\mathrm{et}} \xrightarrow{\phi} X_{\mathrm{et}}$ is pinned. The elementary field-source reconstruction statement of Carlson et al. (2024, Section 3, proposition on reconstruction for fields) identifies it with a unique $\beta \in L^*$: it factors through its closed image point, whose residue field is finite separable over k, and the relative map of field topoi gives its embedding in L. Functoriality of Kummer theory gives

$$\kappa_L(\beta) = \lambda \kappa_L(\alpha). \tag{2}$$

There exist a finite Galois $M/\mathbb{Q}$, an element $\sigma \in \mathrm{Gal}(Mk/k)$, and $\alpha \in M^*$ such that $\sigma\alpha/\alpha$ is not a root of unity. Here is a concrete existence argument requiring no assumption on finite extensions of k. The polynomial $T^5 - 6T + 3$ is Eisenstein at 3. Its derivative has two real zeros $\pm(6/5)^{1/4}$, with positive and negative polynomial values, respectively. Thus it has exactly three real roots, and complex conjugation is a transposition in its transitive degree-five Galois group. A transitive group of prime degree is primitive; the graph of conjugates of that transposition is connected, so those transpositions generate S_5. Its splitting field M therefore has group S_5, and $\mathrm{Gal}(Mk/k) = A_5$. Choose $\sigma \in A_5$ moving a root θ. Since M contains only finitely many roots of unity, all but finitely many integers N have $\alpha = \theta + N \neq 0$ and $\sigma\alpha/\alpha$ not a root of unity: for each root of unity $\zeta \neq 1$, the equation $\sigma\theta + N = \zeta(\theta + N)$ excludes at most one N, and $\zeta = 1$ excludes none.

For this $L = Mk$ and α, let $\beta = \phi(\alpha)$, and choose the fixed number field $F = MS$ containing α and β. If $a_n \in \mathbb{Z}$ represents λ modulo n, (2) says $\beta\alpha^{-a_n} = z_n^n$ for some $z_n \in L^*$. Applying the descent calculation gives

$$\frac{\sigma\beta}{\beta}\left(\frac{\sigma\alpha}{\alpha}\right)^{-a_n} \in (F^*)^n \quad \text{for every } n.$$

The group $F^*/\mu(F)$ is free abelian: its unit subgroup modulo torsion is free of finite rank, and its quotient is a subgroup of the free group of fractional ideals, so that exact sequence splits. Choose a homomorphism $h : F^*/\mu(F) \to \mathbb{Z}$ which is nonzero on $\sigma\alpha/\alpha$. The displayed relations imply

$$h(\sigma\beta/\beta) = \lambda h(\sigma\alpha/\alpha) \quad \text{in } \widehat{\mathbb{Z}}.$$

Since both character values are integers and the second is nonzero, $\lambda \in \mathbb{Q} \cap \widehat{\mathbb{Z}} = \mathbb{Z}$. Write it as m. For an arbitrary finite L/k and $\alpha \in L^*$, (2) now gives $\beta\alpha^{-m} \in D(L^*) = k^*$, which proves the stated point formula. The remaining distinctions in the last sentence of the claim are essential: a scalar calculation in the finite-*étale* subtopos supplies neither the maps on all open subtopoi nor their compatibility. □

F.17 (Refuted). *With R27's literal relative-only definition of pinned, the nonconstant endomorphism assertion in F2 is false over every characteristic-zero field. There is a relative geometric endomorphism of* $(\mathbb{G}_{m,k})_{\mathrm{et}}$ *whose point map sends* 1 *to* 1 *and every other point to the generic point. It is nonconstant on points and is not algebraic. It is not closed-point preserving, so this does not refute F2 with CHW's actual definition restored.*

Proof. Let $X = \mathbb{G}_{m,k}$, $Z = \{1\}$, $U = X \setminus Z$, and

$$T = \operatorname{Spec} \mathcal{O}^h_{X,1}, \qquad K_h = \operatorname{Frac}(\mathcal{O}^h_{X,1}), \qquad D = G_{K_h}.$$

Write $j : U \hookrightarrow X$ and $i : Z \hookrightarrow X$. If $I \subset D$ is inertia, then $I = \widehat{\mathbb{Z}}(1)$ and $D/I = G_k$. The coordinate $u = t - 1$ defines $\operatorname{Spec} K_h \to \mathbb{G}_{m,u}$ and an isomorphism

$$D \xrightarrow{\sim} \pi_1(\mathbb{G}_{m,u}).$$

Indeed it induces the identity on G_k and the Kummer isomorphism on the geometric kernels; all ramification of the strict henselian characteristic-zero trait is tame. Composing the map induced by $u : U \to \mathbb{G}_{m,u}$ with this identification gives

$$q : U_{\mathrm{et}} \longrightarrow BD$$

over BG_k. Its restriction to the local decomposition group at 1 is the identity $D \to D$.

For a continuous discrete D-set B, the nearby-stalk calculation therefore gives a natural G_k-equivariant isomorphism

$$i^* j_* q^* B \simeq B^I. \tag{3}$$

The geometric stalk on the left is obtained by taking sections over the punctured strict henselization at 1, hence by taking invariants under its local inertia. The preceding decomposition map identifies both that action and the residual G_k-action with the ones on the right. This calculation applies to every continuous D-set: such a set is a disjoint union of finite D-orbits, and the punctured strict henselization is connected. Thus q^*B is a union of finite locally constant sheaves; no common *étale* trivialization of an arbitrary infinite union is assumed.

Artin gluing identifies X_{et} with the category of triples

$$(A, C, v), \qquad A \in BG_k, \quad C \in U_{\mathrm{et}}, \quad v : A \longrightarrow i^* j_* C.$$

For T_{et} the corresponding triples are $(A, B, v : A \to B^I)$, with $B \in BD$. These are statements about all sheaves of sets; the exact gluing theorem and its henselian-trait form are recalled in Cornut (2024, pp. 1–2), citing SGA 4, IV, 9.5. Define a functor on these full triple categories by

$$\Phi^*(A, B, v) = (A, q^*B,\ A \xrightarrow{v} B^I \xrightarrow{\sim} i^* j_* q^* B).$$

Finite limits and all colimits of the triple categories are computed on their two components, with the induced connecting map; the gluing functors preserve finite limits. Consequently Φ^* preserves finite limits and colimits. For completeness its right adjoint can be given explicitly. For (A', C, v') put $B = q_* C$, use the natural map

$$B^I \simeq i^* j_* q^* q_* C \longrightarrow i^* j_* C,$$

and take $A = A' \times_{i^* j_* C} B^I$, with the projection $A \to B^I$. The adjunction follows directly from $q^* \dashv q_*$ and the fibre-product condition. Thus Φ^* is the inverse image of a geometric morphism $\Phi : X_{\mathrm{et}} \to T_{\mathrm{et}}$. On both components it commutes with the structural inverse image from BG_k;

(3) respects that commutation. Hence Φ is a relative geometric morphism with the required isomorphism.

Compose Φ with the morphism $T_{\mathrm{et}} \to X_{\mathrm{et}}$ induced by the henselization. The generic open of T pulls back to U, and its closed stratum pulls back to Z, with identity on BG_k. The point map of the composite is therefore exactly the one in the claim. Its two values show nonconstancy. A morphism of finite-type k-schemes sends closed points to closed points, whereas the displayed morphism sends the rational point 2 to the generic point. It is consequently nonalgebraic. Finally X is smooth, so absolute weak normalization does not alter this example. □

F.18 (Proved). *Put $k = \mathbb{Q}^{\mathrm{solv}}$. Finite extensions of k have unbounded degrees, even when required to contain a prescribed finite extension L/k. Let $U \subset \mathbb{G}_{m,L}$ be a nonempty open, and let $\xi \in \varprojlim_n H^1(U, \mu_n)$. Suppose that every evaluation of ξ at every point over every finite extension E/L is the compatible Kummer class of an actual element $b_x \in E^*$. Then ξ is the compatible Kummer class of an actual unit $h \in \mathcal{O}(U)^*$, and*

$$b_x/h(x) \in k^*$$

for every such point. The statement applies to arbitrary finite L/k, although these fields are not in general radically closed.

Proof. There is a proper finite extension of k, by the S_5 polynomial in F.16. If E/k is any proper finite extension, choose $a \in E \setminus k$. Since $D(E^*) = k^*$, some n-th root of a does not belong to E. Adjoining it gives a proper finite extension of E. Iterating gives degrees at least doubling. The construction can start with any prescribed proper finite L/k; when $L = k$, first use the displayed proper extension. The primitive element theorem then supplies points of arbitrarily large degree over L. This argument uses F.16's divisibility calculation and not an unproved assertion about faithful Kummer lines.

Write $P_0 = t$ and let $P_1, \dots, P_r \in L[t]$ be the distinct monic irreducible polynomials for the other removed closed points. Since this coordinate ring is a localization of a principal ideal domain, its Picard group is zero and

$$\mathcal{O}(U)^* = L^* \times \prod_{j=0}^{r} P_j(t)^{\mathbb{Z}}.$$

Choose $x_0 \in U(L)$, and an actual $b_0 \in L^*$ representing the evaluation there. The Kummer sequence expresses ξ as

$$\xi = \kappa_L(b_0) + \sum_{j=0}^{r} \lambda_j\big([P_j(t)] - [P_j(x_0)]\big), \qquad \lambda_j \in \widehat{\mathbb{Z}}. \tag{4}$$

Thus a possibly nontrivial arithmetic constant class over L has been removed by one actual evaluation, not assumed to vanish.

Take a proper finite extension L'/L as above. Choose a finite Galois number field $M/\mathbb{Q}$ containing generators of L and L' over k, all coefficients and roots of the P_j, and x_0, b_0. Then $L' \subset Mk$, and we can choose a nonidentity $\sigma \in \mathrm{Gal}(Mk/L)$. Its restriction to M is nonidentity; it fixes the coefficients of every P_j, but need not fix their roots.

There are infinitely many rational primes splitting completely in M. One elementary justification suffices here. For a primitive integral element of the Galois field M, infinitely many primes divide nonzero values of its minimal polynomial. Indeed, if the possible primes were $p_1, \dots, p_s$, choose $f(a) \neq 0$ and apply this claim to the nonconstant integer polynomial

$$\frac{f(a + f(a)p_1 \cdots p_s T)}{f(a)},$$

which is congruent to 1 modulo every p_i: an integer value of absolute value greater than 1 has a new prime divisor. Outside the finitely many index and discriminant primes, a linear factor of the minimal polynomial gives a residue-degree-one prime in M, which forces complete splitting because $M/\mathbb{Q}$ is Galois. Choose distinct completely split primes p_i, $0 \le i \le r$, avoiding coefficient denominators, discriminants and pairwise resultants of the P_j, and with $p_i > \sum_j \deg P_j$.

Choose a place v_i of M over p_i and a root $r_i \in M$ of P_i. Complete splitting and $\sigma \neq 1$ imply $v_i \neq \sigma^{-1}v_i$. All these places for different i lie over distinct rational primes. In the ring of integers of M with the finitely many required denominators inverted, the Chinese remainder theorem provides α satisfying

$$\alpha \equiv r_i + p_i \pmod{\mathfrak{p}_{v_i}^2}, \qquad \alpha \equiv c_i \pmod{\mathfrak{p}_{\sigma^{-1}v_i}},$$

where $c_i \in \mathbb{F}_{p_i}$ avoids all roots of all the P_j. The residue-size bound ensures such c_i exist. Taylor expansion at the simple root and the fixed-coefficient identity give

$$v_i(P_j(\alpha)) = \delta_{ij}, \qquad v_i(P_j(\sigma\alpha)) = v_{\sigma^{-1}v_i}(P_j(\alpha)) = 0.$$

In particular $\alpha \in U(Mk)$, and the ratios

$$R_j = P_j(\sigma\alpha)/P_j(\alpha) \quad \text{satisfy} \quad v_i(R_j) = -\delta_{ij}.$$

The second congruence is imposed at $\sigma^{-1}v_i$ precisely because σ may permute the roots of P_j.

Let $b_\alpha \in Mk$ realize the evaluation of ξ. Choose one fixed number field $F = MS \subset Mk$ containing b_α, where $S \subset k$ is finite and contains $M \cap k$. For integer representatives $a_{j,n}$ of λ_j modulo n, (4) gives $z_n \in Mk$ with

$$\frac{b_\alpha}{b_0} \prod_j \left(\frac{P_j(\alpha)}{P_j(x_0)} \right)^{-a_{j,n}} = z_n^n.$$

Apply F.16's fixed-number-field descent to σ. Since σ fixes b_0 and every $P_j(x_0)$, it yields

$$\frac{\sigma b_\alpha}{b_\alpha} \prod_j R_j^{-a_{j,n}} \in (F^*)^n.$$

The integer-valued characters $h_i = v_i \circ N_{F/M} : F^* \to \mathbb{Z}$ satisfy $h_i(R_j) = -[F : M]\delta_{ij}$. Consequently

$$[F : M]\lambda_i = -h_i(\sigma b_\alpha / b_\alpha) \in \mathbb{Z}.$$

As $\mathbb{Q} \cap \widehat{\mathbb{Z}} = \mathbb{Z}$, every λ_i is an integer. The normalized expression for ξ is now the Kummer class of

$$h(t) = b_0 \prod_j \left(\frac{P_j(t)}{P_j(x_0)} \right)^{\lambda_j} \in \mathcal{O}(U)^*.$$

At any finite-extension point, the quotient $b_x/h(x)$ has zero compatible Kummer class. F.16 identifies that kernel with k^*, proving the last assertion. □

F.19 (Proved). *Let $k = \mathbb{Q}^{\mathrm{solv}}$. Every closed-point-preserving relative geometric endomorphism ϕ of $(\mathbb{G}_{m,k})_{\mathrm{et}}$ has, on all geometric closed points, the form*

$$\phi(\alpha) = c\alpha^m, \qquad c \in k^*, \quad m \in \mathbb{Z}.$$

If $m = 0$, the entire geometric morphism is the algebraic constant map. Thus a nonconstant pinned endomorphism has $m \neq 0$ and agrees with an algebraic monomial on all geometric closed points. These point-function calculations alone do not identify the nonconstant full-topos morphism*; F.20–F.23 below complete that additional step.*

Proof. Let f denote the induced $\overline{\mathbb{Q}}$-point function. F.16 already gives an integer m and $f(\alpha) = c_\alpha \alpha^m$ with $c_\alpha \in k^*$. Suppose first that $m = 0$. Choose $\beta \notin k$ and put $L = k(\beta)$. Every geometric closed-point value of ϕ_L lies in k, so none is β. The inverse image of the closed point β is closed in $\mathbb{G}_{m,L}$. Any nonempty closed subset of this Jacobson scheme contains a closed point. Thus this inverse image is empty; the generic point cannot be an isolated exception. The morphism ϕ_L therefore factors through the open complement of β. Pull back its unit $t - \beta$ and apply F.18 on all of $\mathbb{G}_{m,L}$. The resulting unit has the form at^r, $a \in L^*$, $r \in \mathbb{Z}$, and

$$f(\alpha) - \beta = d_\alpha a \alpha^r, \qquad d_\alpha \in k^*.$$

The left-hand side lies in L. If $r \neq 0$, a point of degree $[L(\alpha) : L] > |r|$ supplied by F.18 would therefore satisfy $\alpha^r \in L$, a contradiction. Hence $r = 0$. Comparing with $\alpha = 1$ gives

$$\frac{f(\alpha) - \beta}{f(1) - \beta} \in k^*.$$

Since both $f(\alpha)$ and $f(1)$ belong to k while β does not, this equation forces $f(\alpha) = f(1)$. The inverse image of that closed rational point is closed and contains all closed points of the source, hence is the whole source. By the closed-subtopos factorization, ϕ factors through that $\mathrm{Spec}(k)_{\mathrm{et}}$. Relativity over BG_k identifies this factorization with the ordinary algebraic constant morphism.

Now assume $m \neq 0$. The point map is nonconstant: otherwise ϕ would factor through a closed point and its geometric Kummer exponent would be zero. For each $b \in k^*$ the inverse image of b is a proper closed subset of $\mathbb{G}_m$, hence finite. In particular its complement U_b is a nonempty open. Pull back $t - b$ on the target complement of b. F.18 provides $h_b \in \mathcal{O}(U_b)^*$ with

$$f(\alpha) - b = d_{\alpha,b} h_b(\alpha), \qquad d_{\alpha,b} \in k^*, \quad \alpha \in U_b(\overline{\mathbb{Q}}).$$

For $b = 1$ this gives

$$c_\alpha \alpha^m - d_{\alpha,1} h_1(\alpha) - 1 = 0.$$

After clearing the fixed denominators of t^m and $h_1(t)$, the numerator degree is bounded independently of the two scalar values. Choose α outside the finite exceptional fibre with degree over k greater than this bound. A nonzero polynomial of that bounded degree cannot vanish at α. We therefore obtain an identity of rational functions, and hence

$$h_1(t) = At^m + B, \qquad A, B \in k^*.$$

If $\alpha \notin k$, then $\alpha^m \notin k$, since k is radically closed and contains every root of unity. Such an α cannot satisfy $f(\alpha) = 1$ either. Comparing the coefficients of 1 and α^m in the preceding evaluation equation gives

$$c_\alpha = -A/B =: c, \qquad f(\alpha) = c\alpha^m \quad (\alpha \notin k).$$

For any $b \in k^*$, apply the same bounded-degree argument to h_b at a point of sufficiently large degree, where this monomial formula is already known. It gives

$$h_b(t) = e_b(ct^m - b), \qquad e_b \in k^*.$$

Every zero of $ct^m - b$ in $\mathbb{G}_m$ must lie in the removed fibre $f^{-1}(b)$, because h_b is a unit on U_b. For any $x \in k^*$ choose $b = cx^m$. The zero at x then forces $f(x) = b$. This proves the monomial formula at every geometric closed point, including the rational ones.

The last distinction in the claim is substantive. The proposition on reconstruction from points in Section 4 of Carlson et al. (2024) requires algebraic realizations after pullback to

every separated quasicompact *étale* target chart. Such charts and their pullbacks include covers of punctured $\mathbb{G}_m$ with arbitrary genus. F.18 uses a rational source with zero Picard group; it supplies no vanishing theorem for the Picard Tate modules of these other curves. Thus this point-function argument alone does not establish equality of full-topos morphisms. F.20–F.23 use finite-quotient detection and normalization of charts to complete that comparison without asserting vanishing of those Picard Tate modules. □

6.5 Completion of the corrected solvable-field endomorphism test

F.20 (Proved). *Put $k = \mathbb{Q}^{\mathrm{solv}}$, and let $U \subset \mathbb{A}^1_k$ be a nonempty open. For every finite quotient $\pi_1(U) \twoheadrightarrow G$ and every $n \geq 5$, there is a closed point $x \in U$ of degree n over k whose arithmetic decomposition group maps onto all of G:*

$$G_{k(x)} \longrightarrow \pi_1(U) \longrightarrow G \quad \textit{is surjective.}$$

This detects arithmetic finite quotients, including their constant quotients. It does not assert that k is Hilbertian or that k-rational points suffice.

Proof. Let $Y \to U$ be the connected finite *étale* G-torsor corresponding to the quotient. Descend the equations, the entire G-action, and the torsor isomorphism to a finite Galois solvable number field $F/\mathbb{Q}$ contained in k. The descended cover is connected, since its base change to k is connected. Retain the notation $Y \to U$ for it, and let M be the algebraic closure of F in $F(Y)$. There is an exact sequence

$$1 \longrightarrow N \longrightarrow G \longrightarrow P := \mathrm{Gal}(M/F) \longrightarrow 1,$$

where N is the geometric group. Connectedness after base change implies $M \cap k = F$; both extensions of F are Galois.

We have $F^{\mathrm{solv}} = k$. Indeed, the normal closure over $\mathbb{Q}$ of a finite solvable Galois extension of F is solvable: its group over F embeds into a product of the solvable groups of its conjugates, and its quotient over $\mathbb{Q}$ is $\mathrm{Gal}(F/\mathbb{Q})$. Conversely every element of k lies in a finite solvable Galois extension of F. Thus every solvable quotient of P is trivial, since its fixed subfield would lie in $M \cap k = F$. In particular P is perfect.

Form the ordered n-fold fibre product of Y over its common constant field M, and remove the loci where two of its images in U coincide. Call the resulting scheme T. The factors are geometrically integral over M, so T is geometrically integral over M and connected over F. Over the unordered configuration space

$$B = \mathrm{Sym}^n(U)_{\mathrm{dist}}$$

it is a connected finite *étale* Galois cover with group

$$H = \big(G \times_P \cdots \times_P G\big) \rtimes S_n. \tag{5}$$

The coordinate automorphisms must induce the same automorphism of M; this is the fibre-product condition. Permutation of the ordered factors supplies the literal subgroup S_n. Together these actions are free and transitive on the geometric fibres over an unordered configuration. The source T has *not* been quotiented by S_n: that quotient would have smaller degree and would not in general be H-Galois.

Since $U \subset \mathbb{A}^1_F$, the coefficients of the monic polynomial of a configuration identify B with a nonempty open of $\mathbb{A}^n_F$. Hilbert irreducibility over the number field F gives a connected specialization with Galois group H, say Ω/F. Precisely the needed statements are Serre (2007, Proposition 3.3.1 and Theorem 3.4.1, pp. 23, 25). The specialization proposition assumes an

irreducible Galois cover; it does not require its total space to be geometrically irreducible over F. Thus the nontrivial constant field M causes no extra hypothesis here.

Let H^∞ be the stable term of the derived series of H. Since $k = F^{\mathrm{solv}}$,

$$\Omega \cap k = \Omega^{H^\infty}, \qquad \mathrm{Gal}(\Omega k/k) = H^\infty.$$

We verify that a coordinate stabilizer in H^∞ projects onto the whole corresponding copy of G. First, the literal A_n is perfect and hence is contained in H^∞. The image of H^∞ in the permutation group is exactly A_n. Second, $H^\infty \to P$ is onto, because the image of every derived term is the corresponding derived term of the perfect group P. Multiplication by the inverse literal permutation therefore shows that

$$B_0 = H^\infty \cap \ker(H \to S_n)$$

still surjects onto P. Finally $[N^n, A_n] \subset B_0$, and its projection to any selected coordinate contains all of N: put $a \in N$ in that coordinate and commute it with an even permutation moving the coordinate elsewhere. The chosen coordinate of the resulting balanced tuple is a or a^{-1}. It follows that the projection $B_0 \to G$ contains N and surjects onto $G/N = P$, hence is surjective.

The permutation image A_n is transitive. The specialized configuration consequently gives over k one closed point x of degree n. Its decomposition group inside $\Omega k/k$ is the stabilizer of one coordinate. The preceding calculation shows that this stabilizer surjects onto the Galois group of the corresponding G-torsor fibre. Equivalently the original $G_{k(x)} \to G$ is onto, as required. □

F.21 (Proved). *Let $k = \mathbb{Q}^{\mathrm{solv}}$, let $X = Y = \mathbb{G}_{m,k}$, and let $f(t) = ct^m$ with $c \in k^*$ and $m \neq 0$. If a pinned relative geometric morphism $\phi : X_{\mathrm{et}} \to Y_{\mathrm{et}}$ agrees with f on all geometric closed points, then their generic finite* étale *pullback functors are naturally isomorphic over BG_k.*

Proof. The generic point maps to the generic point: otherwise its image would be closed, and continuity would make the whole point map constant. Hence for every nonempty open $V \subset Y$ the inverse-image open is

$$\phi^{-1}(V) = f^{-1}(V) =: U,$$

again an open rational curve. A finite separable algebra over $k(Y)$ extends to a finite *étale* cover over some V. Pull it back along $\phi|_U$ and take its generic fibre. Shrinking V does not change that fibre, so this defines the generic pullback functor. It preserves finite limits, finite coproducts and covers, and hence defines a continuous generic Galois homomorphism. This construction does not presuppose an unproved factorization theorem for arbitrary generic-valued topos morphisms.

Choose generic fibre functors compatibly with the specified identification on the base field. The two maps are continuous homomorphisms over G_k,

$$h_\phi, h_f : G_{k(X)} \longrightarrow G_{k(Y)}.$$

Let $q : G_{k(Y)} \twoheadrightarrow Q$ be any finite quotient. Choose V over which its cover is finite *étale*; both qh_ϕ and qh_f factor through $\pi_1(U)$. Form their *joint* image

$$J = \mathrm{im}\big((qh_\phi, qh_f) : \pi_1(U) \to Q \times Q\big).$$

F.20 supplies a closed point $x \in U$ whose decomposition group surjects onto J.

The two point-source morphisms from $\mathrm{Spec}\, k(x)_{\mathrm{et}}$ to Y_{et} are isomorphic as relative geometric morphisms. Indeed each factors through its closed image. The residue field of that image is

finite separable over k, and its topos is the corresponding slice of BG_k. Relative morphisms into that slice correspond to actual k-embeddings of the residue field into $k(x)$; equality of geometric point values identifies these embeddings. This is also the field-source reconstruction statement in Section 3 of Carlson et al. (2024).

It follows that the restrictions of the two homomorphisms to the entire decomposition group are conjugate by one element. They are not merely elementwise conjugate. Choosing the paths to x over the same geometric point of $\operatorname{Spec} k$ makes the conjugator geometric. Let a_Q be its image in Q. Surjectivity of the decomposition group onto the joint image J then gives

$$qh_\phi = \operatorname{Int}(a_Q)\, qh_f$$

on the whole generic source group.

Put $\Gamma^{\mathrm{geom}} = G_{\bar{k}(Y)}$ and define

$$C_Q = \{a \in \Gamma^{\mathrm{geom}} : qh_\phi = \operatorname{Int}(q(a))\, qh_f\}.$$

These are nonempty closed subsets of a compact profinite group. The quotient by the intersection of finitely many kernels refines all the corresponding quotients; applying the same joint-image argument to it gives their finite-intersection property. Consequently there is one $a \in \bigcap_Q C_Q$. It conjugates the two generic homomorphisms on all finite quotients. Since a is geometric, the induced natural isomorphism of generic pullback functors respects the identification over BG_k. □

F.22 (Proved). *Let X, Y be smooth connected separated finite-type curves over a characteristic-zero field, and let $f : X \to Y$ be finite* étale*. Suppose a geometric morphism $\psi : X_{\mathrm{et}} \to Y_{\mathrm{et}}$ takes the generic point to the generic point and is equipped with a natural isomorphism of its generic finite* étale *pullback functor with that of f. Then $\psi \simeq f_{\mathrm{et}}$ on the full small* étale *topoi. If the given generic comparison is relative to an identified base-field topos, the full comparison is relative as well.*

Proof. We give the step from generic covers to all charts explicitly. By Carlson et al. (2024, Theorem A.8 and Corollary A.9), the inverse image under ψ of a separated quasicompact *étale* chart is represented by a separated quasicompact *étale* scheme. The source is topologically noetherian and the target is qcqs, exactly as required there. Its proof uses the complemented diagonal to obtain a separated *étale* algebraic space, which is a scheme by quasi-finiteness; no additional representability assumption is being made.

A separated *étale* chart over a normal connected curve is canonically an open in the normalization in its finite generic algebra (Grothendieck and et al., 2003, Exposé I, Proposition 10.1). Quasicompactness leaves finitely many generic components, and normalization is finite because the curves are excellent. Let A be a finite separable $k(Y)$-algebra, let $S \to Y$ be its finite normalization, and put $Z = S^{\mathrm{et}}$. The *étale* base change f^*S is normal, so it is the normalization of X in $k(X) \otimes_{k(Y)} A$. Moreover

$$(f^*S)^{\mathrm{et}} = f^*Z, \tag{6}$$

because being *étale* is local for the *étale* topology on the base. The prescribed generic comparison therefore identifies ψ^*Z with an open in f^*S. Since ψ^*Z is *étale* over X, it lies in the locus in (6), giving a canonical inclusion

$$\psi^*Z \hookrightarrow f^*Z. \tag{7}$$

First this inclusion forces equality of the underlying point maps. Suppose a closed $x \in X$ has $\psi(x) = y' \neq y = f(x)$. If y' is closed, choose $h \in k(Y)^*$ with $v_y(h) = 1$ and $v_{y'}(h) = 0$; if it

is generic, only the first condition is needed. Approximation for these discrete valuations gives such h. Explicitly, one can use a common affine neighbourhood and the Chinese remainder theorem, or prescribe a first jet at y and a nonzero value at y' by Riemann–Roch on a smooth proper completion. The quadratic extension $k(Y)(\sqrt{h})$ is totally ramified at y, and is unramified at y' when the latter is closed, since h is then a unit and 2 is invertible. Its maximal *étale* locus Z has empty geometric fibre at y and nonempty fibre at y' (including the generic case). Inclusion (7) makes the fibre of ψ^*Z at x empty, whereas the inverse-image stalk formula makes it nonempty. This contradiction proves $\psi(x) = f(x)$. The generic image is already prescribed, so the point maps agree. In particular a closed-point-preservation assumption is not needed for this rigidity lemma.

Now take any finite normalization S and its maximal *étale* locus Z. At every geometric closed point $\bar{x}$, equality of the underlying image points gives

$$|(\psi^*Z)_{\bar{x}}| = |Z_{\psi(\bar{x})}| = |Z_{f(\bar{x})}| = |(f^*Z)_{\bar{x}}|.$$

The fibres are finite, and the inclusion in (7) is therefore onto on them. Generic fibres already agree. Thus that inclusion is an isomorphism, canonically extending the prescribed generic one. Use it to form the sliced geometric morphism

$$\psi_Z : (f^*Z)_{\mathrm{et}} \longrightarrow Z_{\mathrm{et}}.$$

Its generic functor is identified with that of the finite *étale* algebraic map $f_Z : f^*Z \to Z$. Apply the ramification argument just proved to these curves. It shows that the underlying point maps of ψ_Z and f_Z agree. If the curves are disconnected, their finitely many clopen components are assigned according to their generic supports, and the argument applies on each relevant connected component. This uses only the already proved point-map step, not the desired full comparison recursively.

Every open subobject $V \subset Z$ consequently has the same inverse-image open under ψ_Z and f_Z. Every separated quasicompact *étale* chart $V \to Y$ is such an open in the maximal *étale* locus of its finite normalization. We have therefore obtained $\psi^*V \simeq f^*V$ on every such chart, extending the fixed generic comparison. This includes partly ramified completions: ramified and unramified points may lie over the same base point, and the maximal *étale* locus followed by its open subobjects selects exactly the latter points and any desired subset of them. We have not assumed that finite covers of Zariski opens alone generate every chart.

For a morphism between two charts, the two maps induced by these identifications agree on generic fibres. Their source is reduced with dense generic points and their target is separated over X. The equalizer is therefore the whole source, so the identifications are natural. Separated quasicompact *étale* charts form a basis for the small *étale* topology (Carlson et al., 2024, Lemma A.4); inverse-image functors preserve colimits, so this natural isomorphism extends to all sheaves. Finally it agrees with the specified base-field comparison on finite constant objects by generic uniqueness. Every continuous discrete G_k-set is a coproduct of finite orbits. Thus the full isomorphism is relative when the generic one is. □

F.23 (Proved). *With the actual closed-point-preserving meaning of pinned, the endomorphism test in F2 has an affirmative answer:*

$$\mathrm{End}_k(\mathbb{G}_m) \xrightarrow{\sim} \mathrm{Hom}^{\mathrm{cp}}_{BG_k}((\mathbb{G}_m)_{\mathrm{et}}, (\mathbb{G}_m)_{\mathrm{et}}), \qquad k = \mathbb{Q}^{\mathrm{solv}},$$

where the right-hand side denotes isomorphism classes with their specified relative structures. Every such endomorphism is induced by exactly one $t \mapsto ct^m$, $c \in k^*$, $m \in \mathbb{Z}$, and it is nonconstant exactly when $m \neq 0$. This endomorphism theorem alone does not imply all-scheme reconstruction; that property is proved separately in F.31.

Proof. F.19 proves the entire constant case and identifies the point function in the nonconstant case with $f(t) = ct^m$, $m \neq 0$. This algebraic map is finite *étale* in characteristic zero. F.21 gives the relative generic comparison, and F.22 extends it to a relative isomorphism of the full small *étale*-topos morphisms. Uniqueness follows from the point functions, equivalently from the ordinary description of the units of $k[t, t^{-1}]$. Both the rational-source hypothesis in F.18–F.19 and the rational configuration space in F.20 have been used; a general source has not been substituted without proof. The literal relative-only statement remains refuted by F.17. Thus the two definitions have different answers even for this specific endomorphism test. □

6.6 Detection beyond rational sources

F.24 (Proved). *Let $k = \mathbb{Q}^{\mathrm{solv}}$, let L/k be finite, and let U/L be a smooth separated geometrically connected curve. Every finite quotient $\pi_1(U) \twoheadrightarrow G$ is detected by a closed point: there exists $x \in U$ for which*

$$G_{L(x)} \longrightarrow \pi_1(U) \longrightarrow G$$

is surjective. Such detecting points have unbounded degrees over L. In particular the rational-source restriction in F.20 is not necessary for finite-quotient detection. No Hilbertianity assertion about L is made.

Proof. Let $Y \to U$ be the connected finite *étale* G-torsor, and let C be the smooth proper completion of U. Descend C, U, Y, the entire G-action, and the torsor isomorphism to a number field $F \subset L$. Enlarge F inside L until $L = Fk$. The extension L/F is Galois and prosolvable, since it is a base change of $k/\mathbb{Q}$. The model C/F is geometrically connected. The descended Y is connected because Y_L is connected. If M is the algebraic closure of F in $F(Y)$, there is an exact sequence

$$1 \longrightarrow N \longrightarrow G \longrightarrow P = \mathrm{Gal}(M/F) \longrightarrow 1,$$

with N the geometric group. The curve Y is geometrically connected over M, and connectedness of Y_L gives $M \cap L = F$. Here P need not be perfect. Put $B = C \setminus U$. Normalization extends the cover to a finite map $\bar{Y} \to C$ of smooth proper curves, ramified only over B.

Choose a line bundle A/F on C of degree $n \geq \max(5, 2g(C) + 3)$. A suitable pair of F-rational sections defines a degree-n morphism

$$r : C \longrightarrow \mathbb{P}^1_F$$

with only simple ramification, distinct branch values, no ramification at B, distinct images of the geometric points of B, and $r(B)$ disjoint from all branch values. For completeness, these generic-pencil assertions follow from jet counts. For every length-at-most-four subscheme $Z \subset C$ we have $H^1(C, A(-Z)) = 0$, so restriction of sections to those jets is onto. A base point imposes two equations with one varying point parameter. Ramification order at least three likewise imposes two equations on the local ratio jet. Two distinct ramification points with the same value impose three equations, with two point parameters; their diagonal limit is covered by the preceding higher-ramification condition. Ramification at a fixed boundary point, or equality of two fixed boundary images, imposes one equation. A ramification point sharing the image of a fixed boundary point imposes two equations with one varying point parameter. The diagonal limit is excluded by unramifiedness at that boundary point. Thus every bad incidence has smaller dimension than $H^0(C, A)^2$. The complement of their closures contains a nonempty open defined over F, which has F-rational points because F is infinite. Base-point-freeness gives $r^*\mathcal{O}(1) = A$ and hence degree n. Tensor powers of an ample line bundle supply unbounded such degrees; a curve's index may restrict the available integers.

Let $V \subset \mathbb{P}^1_F$ be the complement of the branch values and $r(B)$. Let H be the arithmetic Galois group of the Galois closure of $F(\bar{Y})/F(\mathbb{P}^1)$. Label the n blocks over a geometric point of V, and choose a point in each G-torsor fibre inducing the same embedding of M. Monodromy then embeds

$$H \subseteq (G \times_P \cdots \times_P G) \rtimes S_n. \tag{8}$$

Every coordinate induces the same automorphism of M. Restriction $H \to P$ is surjective. The full constant field of the Galois closure may exceed M; no equality is needed. The geometric monodromy H^{geom} lies in $N^n \rtimes S_n$ and acts transitively on the $n|N|$ points of one geometric M-component. That action is faithful: monodromy commutes with the global deck action of G, whose translates exhaust the other components.

Embed $\bar{F}$ into $\mathbb{C}$ and apply SGA 1, XII, Theorem 5.1 and Corollary 5.2 (Grothendieck and et al., 2003). Small loops around the removed values generate the geometric monodromy. Their actions have exactly two possible forms. At a simple branch value of r, the action is a decorated transposition $r_{ij}(a)$: it swaps blocks i, j with mutually inverse labels $a, a^{-1} \in N$ and fixes the others. It has order two, since $\bar{Y} \to C$ is unramified over that value. At the image of a boundary point the action is a single-coordinate element $e_i(b)$, $b \in N$. The pencil conditions ensure that all other blocks are unramified there. Conjugating local loops to the chosen base point preserves these two forms.

The transposition graph on the n blocks is connected, since the projection is the transitive monodromy of the geometrically connected curve C. Choose a spanning tree. Conjugating by one element of N^n, chosen successively along that tree, makes all its edge labels one. Its edge transpositions now generate the literal $S_n \subset H^{\mathrm{geom}}$. Let $N_0 \subset N$ be the subgroup generated by all remaining reflection labels and single-coordinate labels. Every generator has coordinates in N_0, so the orbit of $(1, i)$ lies in $N_0 \times \{1, \dots, n\}$. Transitivity on the chosen component forces $N_0 = N$. Multiplying a remaining reflection by its literal transposition gives

$$d_{ij}(a) = e_i(a)e_j(a^{-1}) \in H^{\mathrm{geom}} \cap N^n.$$

Together with the single-coordinate generators, these provide base elements of support at most two whose coordinate labels generate N. We do not assert that the unrestricted group N^n lies in geometric monodromy; an Abel–Jacobi constraint can prevent that stronger assertion.

Let H^∞ be the stable derived term of H. The literal A_n is perfect for $n \geq 5$, hence lies in H^∞. For one of the preceding base elements u and a selected coordinate i in its support, choose $\tau \in A_n$ with $\tau^{-1}(i)$ outside that support. Then

$$[u, \tau] \in H^\infty \cap N^n, \qquad [u, \tau]_i = u_i u^{-1}_{\tau^{-1}(i)} = u_i.$$

Normality of H^∞ and conjugation by the literal S_n move these labels to every coordinate. Consequently $H^\infty \cap N^n$ projects onto N at every coordinate. This is the balanced-label assertion needed below, even in the presence of the Abel–Jacobi constraint.

Hilbert irreducibility over F gives a specialization Ω/F with full group H, by Serre (2007, Proposition 3.3.1 and Theorem 3.4.1). As in F.20, irreducibility over F suffices; geometric irreducibility is not required. Place Ω in $\bar{\mathbb{Q}}$ with its constant copy of M equal to the chosen M, and set

$$J = \mathrm{Gal}(\Omega L/L) = \mathrm{Gal}(\Omega/(\Omega \cap L)) \subset H.$$

Since L/F is prosolvable, H/J is solvable, so $H^\infty \subset J$. Since $M \cap L = F$, restriction $J \to P$ is onto.

Fix a block i and put $J_i = J \cap \mathrm{Stab}(i)$. Its coordinate projection to G is a homomorphism and its image contains N by the commutator calculation. It also surjects onto P: for $h \in J$ with a prescribed P-image, choose a literal $\tau \in A_n$ moving $h(i)$ back to i. Then $\tau h \in J_i$ has

the same P-image. Thus $J_i \to G$ is surjective. This uses only transitivity of A_n; it does not assume P perfect or require an odd permutation in J.

Finally $A_n \subset J$ makes the specialized degree-n fibre of $C \to \mathbb{P}^1$ over L transitive, hence one closed point x of degree n. It lies in U because the specialization avoids $r(B)$. The image of $G_{L(x)}$ in J is precisely J_i, and its action on Y_x is the coordinate projection $J_i \to G$. The latter is onto, proving the claim. □

F.25 (Proved). *Let $L/\mathbb{Q}^{\mathrm{solv}}$ be finite, and let $X, Y/L$ be smooth separated geometrically connected curves. Suppose a pinned relative geometric morphism $\phi : X_{\mathrm{et}} \to Y_{\mathrm{et}}$ and a nonconstant algebraic map $f : X \to Y$ have the same geometric closed-point values. Their generic finite* étale *pullback functors are naturally isomorphic over BG_L. If f is finite* étale*, then $\phi \simeq f_{\mathrm{et}}$ on the full small topoi.*

Proof. Continuity forces the generic image to be generic. For every nonempty target open V, the inverse-image open is the same $U = f^{-1}(V)$ for both morphisms. The generic functors are therefore defined as in F.21. For a finite generic target quotient, extend its cover over V and take the joint image of the two maps from $\pi_1(U)$. F.24 detects that entire finite arithmetic joint image at one closed point. Field-source reconstruction supplies a single relative conjugator on its whole decomposition group, hence on the joint image. The same compactness argument in the geometric kernel as in F.21 gives one relative generic comparison. If f is finite *étale*, F.22 extends it over all charts and sheaves. □

The remaining obstruction in this $\mathbb{Q}^{\mathrm{solv}}$ approach is consequently not a failure of closed-point detection on higher-genus curves. F.24 detects all finite quotients, and F.25 compares morphisms once an algebraic point function is available. The next step, supplied in F.26–F.28 below, is to construct that point function for an arbitrary pinned morphism $C_{\mathrm{et}} \to (\mathbb{G}_m)_{\mathrm{et}}$. The rational-source Kummer calculation F.18 used $\mathrm{Pic}(C) = 0$; a general curve can have a nonzero Picard Tate module. Finite-quotient detection does not, by itself, turn pointwise actual Kummer values into one rational unit. F.26–F.28 add arithmetic equivariance, the semiabelian Tate-Hom theorem, and norm-residue integrality. No vanishing of the whole Picard Tate module or intrinsic field classification is asserted.

6.7 Point-theoretic classes on curves: removal of the Picard obstruction

F.26 (Proved). *Let $k = \mathbb{Q}^{\mathrm{solv}}$, let L/k be finite, and let U/L be a smooth separated geometrically connected curve with $x_0 \in U(L)$. Suppose $\xi \in \varprojlim_n H^1(U, \mu_n)$ has actual multiplicative Kummer values at every point over every finite extension of L. After a finite constant extension splitting the boundary, choose a basis $u_1, \dots, u_s$ of the regular units modulo constants, normalized by $u_j(x_0) = 1$. Then*

$$\xi = \kappa(b_0) + \sum_{j=1}^{s} \lambda_j \kappa(u_j), \qquad b_0 \in L^*, \quad \lambda_j \in \widehat{\mathbb{Z}},$$

where b_0 is the given value at x_0. Thus the Picard Tate component of this particular point-theoretic class vanishes. For a proper curve the class is constant. No vanishing of the whole Picard Tate module is asserted.

Proof. Write C for the smooth proper completion and $D = C \setminus U$. After the indicated finite extension, descend C, D, U, x_0 to a number field $F \subset L$ with $L = Fk$, with all boundary points

rational over F. Let $A = J_D$ be the reduced-modulus generalized Jacobian and $j : U \to A$ its Abel–Jacobi map, normalized at x_0. Its universal property identifies

$$\mathrm{Hom}(A, \mathbb{G}_m) \simeq \{u \in \mathcal{O}(U)^* : u(x_0) = 1\}.$$

Its Tate module is the geometric abelian fundamental group. These facts are the curve case of Spieß and Szamuely (2003, Section 3 and Propositions 4.1–4.3). For clarity, the general source has a finite kernel in its abelian fundamental-group comparison; that kernel is killed by homomorphisms to $\mathbb{Z}_\ell(1)$. For curves the Néron–Severi quotient in that statement has no torsion: it is $\mathbb{Z}$ in the proper case and zero after a geometric boundary point is present. Evaluation along j is the usual point Kummer cocycle, as is seen directly by pulling back the multiplication-by-ℓ^r covers of A.

Fix a prime ℓ, put $H = G_L$, and set

$$V = V_\ell A, qquad W = V_H = V/\langle (h-1)v : h \in H,\ v \in V\rangle_{\mathbb{Q}_\ell}.$$

The subspace is closed because V is finite dimensional. Since H is normal in G_F, W is a G_F-representation on which H acts trivially. The image W_0 of $T_\ell A$ is a G_F-stable full $\mathbb{Z}_\ell$-lattice. For a point $x \in U(E)$, E/L finite, project the Kummer cocycle of $j(x)$ to obtain

$$c_x : G_E \longrightarrow W_0.$$

It is a homomorphism independent of division-point choices, since $G_E \subset H$ acts trivially on W. Its $\mathbb{Q}_\ell$-span is unchanged by a finite extension of E: restricting a homomorphism to a finite-index subgroup changes its image by finite index.

The images of all c_x span W. Otherwise an annihilating nonzero functional, rescaled to be integral on W_0, defines a nonzero homomorphism $\pi_1(U) \to \mathbb{Z}_\ell$, zero on the base-point copy of G_L and on every closed-point decomposition group. Here the rational section splits the arithmetic extension, and $\mu_\infty \subset L$ makes the coefficient action trivial. A nonzero finite quotient of this homomorphism contradicts F.24. Therefore finitely many points $x_1, \dots, x_r$ suffice to span W.

Subtract $\kappa(b_0)$ from ξ. The normalized class has geometric functional $\lambda : T_\ell A \to \mathbb{Z}_\ell(1)$ and is zero at x_0. It factors through W rationally. If b_i is the actual value at x_i, evaluation says

$$\lambda(c_{x_i}(h)) = \kappa_\ell(b_i/b_0)(h).$$

Choose one number field $M \supset F$ containing the coordinates of every x_i, every b_i, and b_0, and put $L' = Mk$. Restriction to $G_{L'}$ preserves the spanning property. Both W and $\mathbb{Q}_\ell(1)$ have trivial $G_{L'}$-action. For $g \in G_M$ and $h \in G_{L'}$, the cocycle identity gives

$$c_{x_i}(ghg^{-1}) = g c_{x_i}(h), \qquad \kappa_\ell(b_i/b_0)(ghg^{-1}) = \chi_\ell(g)\kappa_\ell(b_i/b_0)(h).$$

The points and actual values are M-rational, and $G_{L'}$ is normal in G_M, which are the hypotheses needed here. Combining these formulas proves $\lambda(gv) = \chi_\ell(g)\lambda(v)$ on each spanning image, hence on all of W. Thus the original integral functional $T_\ell A \to \mathbb{Z}_\ell(1)$ is G_M-equivariant.

Now apply the integral Tate-Hom theorem for semiabelian varieties over the *number field* M:

$$\mathrm{Hom}_M(A, \mathbb{G}_m) \otimes \mathbb{Z}_\ell \xrightarrow{\sim} \mathrm{Hom}_{G_M}(T_\ell A, \mathbb{Z}_\ell(1)).$$

This is Jossen (2009, Lemma 6.3.2, pp. 81–82), whose proof and hypotheses have been opened. It is not being applied directly over the infinite field L. In the proper case the Hom-group is zero; equivalently, at a good prime away from ℓ, Frobenius eigenvalues on $T_\ell J$ have absolute value $q^{1/2}$ and those on $\mathbb{Q}_\ell(1)$ have absolute value q, as in that source's Proposition 6.3.1.

The normalized unit lattice is independent of M and ℓ. Indeed every geometric unit has divisor supported on the F-rational boundary. Each Galois conjugate has the same divisor; after normalization at x_0 it also has the same value there, so the quotient is the constant one. Thus every normalized geometric unit is already defined over F. This lattice is a finitely generated free subgroup of the boundary-divisor lattice. The integral Tate-Hom theorem places the normalized class in its tensor product with $\mathbb{Z}_\ell$ for every ℓ. Equality of arithmetic classes follows from their equal geometric functionals and zero values at x_0. One fixed basis therefore gives the displayed coefficients in $\widehat{\mathbb{Z}}$.

There is no assumption that the lattice of principal boundary divisors is primitive. The integral theorem gives the assertion directly. Alternatively, after the rational Tate-Hom theorem, the Kummer exact sequence gives saturation: the quotient of integral H^1 by the completed unit group is $T_\ell \operatorname{Pic}(U)$, which is torsion free. An integral class in the rational unit span consequently lies in the integral completed unit group. This explains why torsion boundary classes do not create an unexamined denominator. □

F.27 (Proved). *Let $L/\mathbb{Q}^{\mathrm{solv}}$ be finite and let U/L be a smooth separated geometrically connected curve. Every compatible multiplicative Kummer class with actual values at all points over finite extensions is, after a finite constant extension, the class of a regular unit h. At a point over any further finite field E, its actual value divided by $h(x)$ belongs to $(\mathbb{Q}^{\mathrm{solv}})^*$.*

Proof. Extend L finitely to split the boundary and provide x_0. Use the unit-lattice expression of F.26. For each boundary point d, Riemann–Roch supplies a finite map $\pi : C \to \mathbb{P}^1_L$ with a simple zero at d and nonzero values at the other boundary points. Delete the branch values, $0, \infty$, and all boundary images. We obtain a finite *étale* map $W \to V$ with $W \subset U$ and V a punctured rational line. The norm of the restricted class has actual values: at a rational point over any finite extension, its value is the product of the field norms of the given values at the residual points of the fibre. F.18 makes this norm class the class of an actual unit on V.

The residue at 0 is the original residue λ_d. The only boundary point over 0 is the rational point d; all interior points have residue zero. As in F.5, norms weight residues by residue degree, not ramification index. Therefore every λ_d is an ordinary integer. The divisor matrix of the basis units in F.26 is an injective integer matrix. An invertible square minor and its adjugate show that each profinite unit exponent is rational. Since $\mathbb{Q} \cap \widehat{\mathbb{Z}} = \mathbb{Z}$, all exponents are ordinary integers. The resulting unit, multiplied by the actual constant value at x_0, represents the whole class. F.16 identifies the kernel of its evaluations with $D(E^*) = (\mathbb{Q}^{\mathrm{solv}})^*$, proving the last assertion. □

F.28 (Proved). *Let $L/\mathbb{Q}^{\mathrm{solv}}$ be finite and let U/L be a smooth separated geometrically connected curve. The geometric point function of every pinned relative morphism $U_{\mathrm{et}} \to (\mathbb{G}_{m,L})_{\mathrm{et}}$ is the point function of one regular unit of U. If that function is constant, the entire topos morphism is the algebraic constant map.*

Proof. Put $k = \mathbb{Q}^{\mathrm{solv}}$ and write q for the geometric point function. Field-source reconstruction supplies actual values. After a finite Galois constant extension, F.27 gives $h \in \mathcal{O}(U)^*$ and

$$q(x) = c_x h(x), \qquad c_x \in k^*.$$

We first work over this enlarged L.

Suppose $h = h_0$ is constant. Choose a nonzero algebraic β with $\beta/h_0 \notin k$ and extend constants to contain it. The inverse image of β is closed and has no closed point, so it is empty. Pull back $t - \beta$ on all of U and use F.27 to obtain a unit g with

$$q(x) - \beta = d_x g(x), \qquad d_x \in k^*.$$

All necessary further constant extensions are finite. The left side belongs to the current constant field for every geometric x. If g were nonconstant, a closed point of degree larger than $\deg(g : C \to \mathbb{P}^1)$, available by F.24, would have $g(x)$ outside that field, a contradiction. Thus g is constant. Comparing with one fixed x_0 gives

$$\frac{h_0 c_x - \beta}{h_0 c_{x_0} - \beta} \in k,$$

which forces $c_x = c_{x_0}$ because $\beta/h_0 \notin k$. The point map is constant. Its factorization through that closed subtopos and its relative structure give the full algebraic constant map.

Now suppose h is nonconstant. The point map cannot be constant, by the same high-degree argument. For $b \in k^*$, let U_b be the nonempty open complement of $q^{-1}(b)$. Apply F.27 to the pullback of $t - b$, obtaining, after a finite constant extension if necessary,

$$q(x) - b = d_{x,b} h_b(x), \qquad d_{x,b} \in k^*, \quad h_b \in \mathcal{O}(U_b)^*.$$

Each removed fibre is finite, since it is a proper closed subset of a curve. For $b = 1$, the rational functions $c_x h - d_{x,1} h_1 - 1$ have one fixed bound on the degree of their pole divisors. F.24 gives a closed point, outside the finite exceptional set and over the current finite constant field, whose degree exceeds this bound. Vanishing at that point forces a rational identity, hence

$$h_1 = Ah + B, \qquad A, B \in k^*.$$

For any x with $h(x) \notin k$, the relation

$$(c_x - d_{x,1} A) h(x) = 1 + d_{x,1} B$$

forces $c_x = -A/B =: c$. Such x cannot lie in the removed fibre. Thus $q(x) = ch(x)$ whenever $h(x) \notin k$.

For arbitrary $b \in k^*$, choose a high-degree point also of degree greater than $\deg h$, so its h-value is outside the current constant field and hence outside k. The same bounded pole-divisor argument yields

$$h_b = e_b(ch - b), \qquad e_b \in k^*.$$

Because h_b is a unit on U_b, every zero of $ch - b$ inside U belongs to $q^{-1}(b)$. At any remaining point with $h(x) \in k^*$, choose $b = ch(x)$. This forces $q(x) = ch(x)$ there as well. The identities have coefficients in k even when a particular U_b required further constants.

The reconstructed unit is unique because it agrees with the point function on all geometric points. Galois equivariance of the original point function therefore makes it invariant under the initial finite Galois constant extension, so it descends to L. The reconstructed unit also has the pulled-back Kummer class: their difference vanishes at every finite-extension point, so F.24 applied to each finite Kummer quotient makes it zero. This proves the assertion without replacing pointwise agreement by an unsupported full-topos comparison. □

6.8 All-scheme reconstruction over the maximal solvable extension

F.29 (Proved). *Let $L/\mathbb{Q}^{\mathrm{solv}}$ be finite and let X/L be a smooth geometrically connected quasi-projective variety. Every finite quotient of $\pi_1(X)$ is detected surjectively by a closed-point decomposition group.*

Proof. The zero-dimensional case is $X = \operatorname{Spec} L$, and the curve case is F.24. Let $Y \to X$ be the connected finite *étale* Galois cover of a prescribed quotient, and suppose $\dim X = d \geq 2$. Embed X locally closed in $\mathbb{P}^N_L$. Each geometric component Y_i of Y is geometrically irreducible, since it is smooth and connected, and its composite to $\mathbb{P}^N$ is quasi-finite with image $X_{\bar{L}}$. Its nonempty fibres therefore all have dimension zero. By Poonen and Slavov (2020, Theorem 1.5), the constructible locus of hyperplanes whose pullback to Y_i is not geometrically irreducible has dimension at most

$$\operatorname{codim}(X_{\bar{L}}, \mathbb{P}^N) + 1 = N - d + 1 < N.$$

Remove the closures of these finitely many bad loci and the bad smoothness locus for X. The remaining nonempty open is Galois invariant and defined over L. Since L is infinite, it has an L-rational hyperplane. The section of X is smooth and geometrically connected, and every geometric component of its pulled-back cover is still irreducible. The Galois group continues to permute these components transitively, so the pulled-back cover is connected over L. Thus arithmetic constants, not only geometric monodromy, are retained. Iterate to a smooth geometrically connected curve $C \subset X$ with Y_C connected. F.24 gives a closed point of C whose decomposition group surjects onto the chosen Galois group. As a point of X it has the same residue field and supplies the required detection. □

F.30 (Proved). *Let $L/\mathbb{Q}^{\mathrm{solv}}$ be finite and let X/L be a smooth geometrically connected affine variety with $x_0 \in X(L)$. Every compatible multiplicative Kummer class having actual values at all finite-extension points is, after finite constants, the class of an actual regular unit. Consequently the point function of every pinned relative morphism $X_{\mathrm{et}} \to (\mathbb{G}_{m,L})_{\mathrm{et}}$ is one regular unit of X.*

Proof. The generalized Albanese A_X pointed by x_0 has the universal property

$$\operatorname{Hom}(A_X, \mathbb{G}_m) \simeq \{u \in \mathcal{O}(X)^* : u(x_0) = 1\}$$

(Spieß and Szamuely, 2003, Section 3). The exact cohomological comparison needed here is integral. Over an algebraic closure, that source's Proposition 4.1 gives

$$0 \longrightarrow \operatorname{Hom}(A_X[\ell^r], \mu_{\ell^r}) \longrightarrow H^1(X_{\bar{L}}, \mu_{\ell^r}) \longrightarrow \operatorname{Tor}(S, \mathbb{Z}/\ell^r) \longrightarrow 0,$$

where S is the finite torsion subgroup of the Néron–Severi quotient by boundary classes. The transition on its torsion term is multiplication by ℓ; its inverse limit is zero. The finite left terms satisfy the Mittag–Leffler condition. Passing to the limit therefore gives

$$H^1(X_{\bar{L}}, \mathbb{Z}_\ell(1)) \simeq \operatorname{Hom}(T_\ell A_X, \mathbb{Z}_\ell(1)). \tag{9}$$

Corollary 4.2 states this inverse-limit consequence explicitly. Equivalently, Proposition 4.3 gives a finite kernel in $\pi_1(X_{\bar{L}})^{\mathrm{ab}} \to TA_X$, killed by homomorphisms to torsion-free $\mathbb{Z}_\ell(1)$. All hypotheses hold: characteristic zero supplies a smooth projective compactification by resolution, and every prime is prime to the characteristic. The comparison is the one induced by the pointed Albanese map, so evaluations are the corresponding point Kummer cocycles.

Choose a smooth projective normal-crossings compactification $\bar{X}$ preserving X, as in F.7. After a finite constant extension its geometric boundary components are individually defined. Descend $X, \bar{X}, x_0, A_X$ to a number field $F \subset L$ with $L = F\mathbb{Q}^{\mathrm{solv}}$. For each ℓ, use the quotient $(V_\ell A_X)_{G_L}$ and its integral image lattice. The spanning, finite-number-field, and conjugation arguments in F.26 apply verbatim, with F.29 replacing F.24 and (9) replacing the curve comparison. More explicitly, a missing span would give a nonzero normalized finite Kummer quotient invisible at all closed points, contradicting F.29. Finitely many spanning points and their actual values lie in one number field M. Restriction to $G_{M\mathbb{Q}^{\mathrm{solv}}}$ retains their

spans, and conjugation by G_M makes the geometric functional arithmetic. Jossen's integral semiabelian Tate-Hom theorem then expresses it in the normalized regular-unit lattice (Jossen, 2009, Lemma 6.3.2).

This is one fixed finitely generated free lattice: a normalized geometric unit has divisor on the split boundary, and its Galois conjugates have the same divisor and the same value one at x_0. They are therefore equal. This supplies, for one basis $u_1, \dots, u_s$, an expression

$$\xi = \kappa(b_0) + \sum_j \lambda_j \kappa(u_j), \qquad \lambda_j \in \widehat{\mathbb{Z}},$$

with the actual value b_0 at x_0. This conclusion concerns the class ξ, not vanishing of the whole Picard Tate module.

For a boundary component D_i, choose a smooth geometric point on it away from the other components. After finite constants, sufficiently ample complete intersections with specified first jets give a smooth geometrically connected curve in $\bar{X}$ through that point and transverse to D_i. The finite-jet Bertini predicates are precisely those checked in F.7, using The Stacks Project Authors (2026a, Lemmas 33.47.1, 33.47.3 and 37.32.3). Its open part in X carries a point-theoretic restriction of ξ. F.27 says this restriction is an actual unit class. Its residue at the chosen boundary point is exactly the global residue on D_i, because the intersection multiplicity is one. At finite Kummer level this is the local calculation with a parameter z and, after *étale* localization, a representative uz^a; a transverse parameter pulls z to a unit times a parameter. Thus every global residue is an ordinary integer. The injective integer divisor matrix of the u_j and $\mathbb{Q} \cap \widehat{\mathbb{Z}} = \mathbb{Z}$ force every λ_j to be an ordinary integer. We obtain an actual unit h representing ξ, normalized by $h(x_0) = b_0$.

Now let ξ be the pullback of the coordinate class under the pinned morphism in the statement, and let q be its geometric point function. Work first after the finite constant extension just used. For any geometric point x, extend constants finitely to make it rational. If $\dim X \geq 2$, sufficiently ample complete intersections through x_0 and x, with independent first jets there, give a smooth geometrically connected curve $C \subset X$. Smoothness and irreducibility off the finite base locus and smoothness at the prescribed points are the same finite-jet Bertini assertions used in F.7. For $\dim X = 1$ take $C = X$; dimension zero is the field-source case. F.28 realizes the restricted point function by a regular unit v_C and gives

$$\kappa(v_C) = \xi|_C = \kappa(h|_C).$$

The unit $v_C/h|_C$ has zero compatible Kummer class. Its ordinary divisor is divisible by every integer and hence zero, so it is constant; F.16 puts that constant in $(\mathbb{Q}^{\mathrm{solv}})^*$. Its value at x_0 is one, so $q(x) = h(x)$. This also works when $h|_C$ is constant. Thus the entire point function equals h. Finally uniqueness on geometric points and Galois equivariance descend h through the initial finite constant extension. □

F.31 (Proved). *Every finite extension $L/\mathbb{Q}^{\mathrm{solv}}$ has closed-point-preserving* étale *reconstruction. Precisely, for all finite-type L-schemes X, Y, the natural map*

$$\mathrm{Hom}_L(X^{\mathrm{awn}}, Y) \xrightarrow{\sim} \pi_0 \,\mathrm{Hom}^{\mathrm{pin}}_{BG_L}(X_{\mathrm{et}}, Y_{\mathrm{et}})$$

is bijective, with pinned understood in its actual closed-point-preserving sense. This supplies the all-scheme property for $\mathbb{Q}^{\mathrm{solv}}$ itself. It does not classify all characteristic-zero reconstruction fields intrinsically.

Proof. Apply Carlson et al. (2024, Theorem 4.18, equivalence of (1) and (5)). Its condition (5) asks that, for every connected regular affine finite-type source over L having an L-point,

every pinned relative morphism to $\mathbb{G}_m$ have the point function of one regular unit. In characteristic zero, regular finite-type sources over a field are smooth. Connectedness together with an L-point makes such a smooth source geometrically connected: its geometric connected components form a transitive Galois set, and the component containing that rational point is fixed. F.30 supplies exactly the required unit on exactly these sources. The theorem then reconstructs the full small-topos Hom-groupoids using its affine-coordinate, *étale*-chart, and h-descent arguments; no full-topos conclusion is inferred merely from pointwise agreement. Characteristic-zero absolute weak normalization is seminormalization, as specified in F.7. The relative-only interpretation of R27 remains refuted by F.1 and F.17. The new theorem proves its corrected version over this additional field class while leaving F.9's intrinsic classification request unresolved. □

7 The additional lead: local covering germs

We give "sub-germ" a precise geometric meaning: a connected finite cover of the punctured strict henselian neighbourhood is a connected component of the pullback of a finite global cover. A weaker meaning, that the local cover is dominated by such a component, gives the same criterion. The covers throughout this section are finite étale; a ramified germ crossing the divisor is understood by its restriction to the complement.

G.1 (Proved). *Let $i : H \to G$ be a continuous homomorphism of profinite groups. The following assertions are equivalent:*

(i) i is injective.

(ii) Every transitive finite continuous H-set is isomorphic to an H-orbit in the restriction of a finite continuous G-set.

(iii) Every transitive finite continuous H-set is a quotient of an H-orbit in the restriction of a finite continuous G-set.

Thus the germ criterion in the prompt is true for the geometric local tame group $H = \widehat{\mathbb{Z}}(1)^{I(x)}$ in characteristic zero, with $G = \pi_1(U_{\overline{K}})$, provided that the local point is geometric and the normal crossings divisor has its branches separated by strict henselization. It is a criterion, not an assertion that local inertia is always injective.

Proof. Assume (i). A continuous injection from a compact space into a Hausdorff space is a homeomorphism onto its closed image. We identify H with that image. A transitive finite continuous H-set has the form H/V for an open subgroup V of H. The subspace topology supplies an open normal subgroup N of G with $N \cap H \subseteq V$. Then VN is a subgroup of G, because N is normal, and it is open, because it contains N. Moreover

$$H \cap VN = V.$$

Indeed $h = vn$ with $h, v \in H$ implies $n \in H \cap N \subseteq V$. Consequently the orbit of VN in G/VN is exactly H/V. This proves (ii), including its stronger exact-component conclusion. Assertion (ii) implies (iii). If (iii) holds, every element of $\ker i$ acts trivially on every finite transitive H-set. In particular it acts trivially on H/V for each open normal subgroup V of H, so belongs to $\bigcap_V V = 1$. This proves (i).

For the geometric application choose a geometric generic base point of the punctured strict henselian neighbourhood mapping to a geometric base point of U. Pullback of finite étale covers is the restriction functor for the induced continuous map i. The equivalence between finite covers and finite continuous fundamental-group sets identifies connected covers

with transitive sets and connected components with orbits. In a smooth normal crossings chart in characteristic zero, Kummer covers of all orders in the branch parameters are cofinal: Abhyankar's lemma removes the ramification, and a finite unramified cover of a strictly henselian local ring is trivial. The local group is therefore the stated product of Tate modules. Changing paths conjugates i in G, leaving every assertion invariant. The precise local inputs are SGA 1 XIII, Propositions 5.1–5.2; the abstract finite-cover criterion is also SGA 1 V, Corollary 6.8 (Grothendieck et al., 2003). These are statements about all finite covers and the whole profinite group, not merely a chosen collection of cyclic quotients. □

G.2 (Proved; explicit boundary cases). *The criterion holds for $U = \mathbb{G}_m^r \times \mathbb{A}^{d-r}$ at the intersection of the coordinate boundary divisors in $\mathbb{A}^d$. It fails for $\mathbb{A}^1 \subset \mathbb{P}^1$ at infinity. At $[0:0:1] \in \mathbb{P}^2$ with divisor $xy = 0$, the local-to-global map is*

$$\widehat{\mathbb{Z}}(1)^2 \longrightarrow \widehat{\mathbb{Z}}(1), (a, b) \longmapsto a - b,$$

and has a diagonal kernel.

Proof. Over an algebraically closed characteristic-zero field the Kummer covers of the coordinate torus give the first isomorphism of groups; the affine-space factor has trivial fundamental group. Over $\mathbb{C}$ these statements also follow by deformation retraction and the finite-cover comparison, and descend to characteristic zero. The affine line has no nontrivial connected finite étale cover: Riemann–Hurwitz applied to a hypothetical cover of $\mathbb{P}^1$ branched only at infinity gives $2g - 2 = -2d + \sum(e_i - 1) \leq -d - 1$, impossible for $d > 1$. Its local group at infinity is nevertheless $\widehat{\mathbb{Z}}(1)$. For the last example use coordinates $u = x/y$ and $v = z/y$ on the complement; it is $\mathbb{G}_m \times \mathbb{A}^1$. In the local chart $z = 1$, the global coordinate is $u = x/y$, so its Kummer character has valuations $(1, -1)$ on the two branches. The formula and its kernel follow. The finite reductions of these maps, and the explicit failed extension of a local cyclic cover, are checked in `code/root_checks.py`. □

Numbered statements and manuscript loci

This register assigns the manuscript locus to every numbered statement. The CSV supplies the dependencies and the exact qualifications about doubtful readings. Repeated loci identify the question being answered; they do not attribute our proofs to the manuscript.

Statement	Status	Cote; manuscript page; scan
A.1	Proved	cote 149, section 1, manuscript p.17; scans 35-36
A.2	Proved	cote 149, section 1, manuscript p.17; scans 35-36
A.3	Proved	cote 149, section 1, manuscript p.17; scans 35-36
A.4	Proved	cote 149, section 1, manuscript p.17; scans 35-36
A.5	Proved	cote 149, section 1, manuscript p.17; scans 35-36
A.6	Proved	cote 149, section 1, manuscript p.17; scans 35-36
A.7	Proved	cote 149, section 1, manuscript p.17; scans 35-36
A.8	Proved	cote 149, section 1, manuscript p.17; scans 35-36
A.9	Proved	cote 149, section 1, manuscript p.17; scans 35-36
A.10	Proved	cote 149, section 1, manuscript p.17; scans 35-36
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A.17	Proved	cote 149, section 1, manuscript p.17; scans 35-36
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A.25	Proved	cote 149, section 1, manuscript p.17; scans 35-36
A.26	Proved	cote 149, section 1, manuscript p.17; scans 35-36
A.27	Proved	cote 149, section 1, manuscript p.17; scans 35-36
A.28	Refuted	cote 149, section 1, manuscript p.17; scans 35-36; an attempted sufficient route to A1, not an assertion of the manuscript
A.29	Proved	cote 149, section 1, manuscript p.17; scans 35-36
A.29a	Proved	cote 149, section 1, manuscript p.17; scans 35-36; a concrete obstruction to one attempted extension, not a manuscript assertion
A.29b	Proved	cote 149, section 1, manuscript p.17; scans 35-36; a concrete method obstruction, not a manuscript assertion
A.29c	Proved	cote 149, section 1, manuscript p.17; scans 35-36
A.30	Unresolved	cote 149, section 1, manuscript p.17; scans 35-36
A.C2.1	Proved	149; manuscript pp.17–17v; scans35–36: main point-separation question A1 and covering condition A2. This is a modern result or audit criterion, not a claimed manuscript assertion.
A.C2.2	Proved	149; manuscript pp.17–17v; scans35–36: main point-separation question A1 and covering condition A2. This is a modern result or audit criterion, not a claimed manuscript assertion.
A.C2.3	Proved	149; manuscript pp.17–17v; scans35–36: main point-separation question A1 and covering condition A2. This is a modern result or audit criterion, not a claimed manuscript assertion.
A.C2.4	Proved	149; manuscript pp.17–17v; scans35–36: main point-separation question A1 and covering condition A2. This is a modern result or audit criterion, not a claimed manuscript assertion.
A.C2.5	Proved	149; manuscript pp.17–17v; scans35–36: main point-separation question A1 and covering condition A2. This is a modern result or audit criterion, not a claimed manuscript assertion.
A.C2.6	Unresolved	149; manuscript pp.17–17v; scans35–36: main point-separation question A1 and covering condition A2. This is a modern result or audit criterion, not a claimed manuscript assertion.
A.C2.7	Proved	149; manuscript pp.17–17v; scans35–36: main point-separation question A1 and covering condition A2. This is a modern result or audit criterion, not a claimed manuscript assertion.
A.C2.8	Proved	149; manuscript pp.17–17v; scans35–36: main point-separation question A1 and covering condition A2. This is a modern result or audit criterion, not a claimed manuscript assertion.
B.1	Proved	149; manuscript pp.49–51; scans99–104
B.2	Proved	149; manuscript pp.49–51; scans99–104
B.3	Known	149; manuscript pp.49–51; scans99–104
B.4	Proved	149; manuscript pp.49–51; scans99–104
B.5	Known	149; manuscript pp.49–51; scans99–104
B.6	Proved	149; manuscript pp.49–51; scans99–104
B.7	Proved	149; manuscript pp.49–51; scans99–104
B.8	Proved	149; manuscript pp.49–51; scans99–104
B.8u	Proved	149; manuscript pp.49–51; scans99–104
B.9	Refuted	149; manuscript pp.49–51; scans99–104
B.10	Proved	149; manuscript pp.49–51; scans99–104
B.11	Known	149; manuscript p.50; scan102 NB ell=p

Statement	Status	Cote; manuscript page; scan
B.11a	Proved	149; manuscript p.50; scan102 NB ell=p
B.11b	Proved	149; manuscript pp.49–50; scans99–102
B.11c	Known	149; manuscript p.50; scan102 NB ell=p
B.11f	Proved	149; manuscript p.50v; scan102, residue-prime NB; modern criterion, not attributed to the manuscript
B.12	Proved	149; manuscript p.50; scan102 NB ell=p
B.13	Proved	149; manuscript pp.37–39,45; scans74–78,90
B.14	Proved	149; manuscript pp.37–39,45,51; scans74–78,90,103–104
B.15	Proved	149; manuscript pp.49–51; scans100–104; modern B10 extension
B.15a	Proved	149; manuscript pp.27,30,49–51; scans55–56,61,99–104; modern arithmetic lemma, not manuscript attribution
B.15u	Proved	149; manuscript pp.49–51; scans99–104; modern B10 extension, not manuscript attribution
B.16	Known	149; manuscript pp.49–51; scans99–104
B.16u	Proved	149; manuscript pp.49–51; scans99–104
B.17	Proved	149; manuscript p.49; scan99
B.18	Proved	149; manuscript pp.49–51; scans99–104
B.19	Proved	149; manuscript pp.49–51; scans99–104
C.1	Known	149; manuscript pp.49–51; scans99–104; finite-residue extension of question
C.2	Known	149; manuscript pp.49–51; scans99–104; finite-residue extension of question
C.3	Proved	149; manuscript p.50; scans101–102; residue-prime correction
C.4	Proved	149; manuscript pp.49–51; scans99–104; L9 extension
C.5	Proved	149; manuscript pp.49–51; scans99–104; finite-residue extension of question
C.6	Proved	149; manuscript pp.49–51; scans99–104; modern C4 extension
C.7	Proved	149; manuscript pp.49–51; scans99–104; modern C4 test
C.7u	Unresolved	149; manuscript pp.49–51; scans99–104; modern C4 extension
C.8	Proved	Modern lead L10 from CONNES_CONSANI_MAP note L11; no direct cote/page/scan attribution.
B.20	Proved	149; manuscript pp.49–51; scans99–104; modern positive-characteristic extension, not manuscript attribution
B.21	Proved	149; manuscript pp.49–51; scans99–104; modern positive-characteristic extension, not manuscript attribution
B.22	Proved	149; manuscript pp.49–51; scans99–104; modern positive-characteristic extension, not manuscript attribution
B.23	Proved	149; manuscript pp.49–51; scans99–104; modern positive-characteristic extension, not manuscript attribution
B.24	Proved	149; manuscript pp.49–51; scans99–104; modern characteristic-two edge case, not manuscript attribution
B.11d	Proved	149; manuscript p.50; scan101; residue-prime NB; modern criterion, not manuscript attribution
B.11e	Proved	149; manuscript p.50; scan101; residue-prime NB; modern criterion, not manuscript attribution
D.1	Known	149; manuscript pp.22-31, 62v-63; scans45,48,56,59-63,126-127
D.2	Proved	149; manuscript pp.22-31, 62v-63; scans45,48,56,59-63,126-127
D.3	Proved	149; manuscript pp.22-31, 62v-63; scans45,48,56,59-63,126-127
D.4	Refuted	149; manuscript pp.22-31, 62v-63; scans45,48,56,59-63,126-127
D.5	Known	149; manuscript pp.22-31, 62v-63; scans45,48,56,59-63,126-127
D.6	Proved	149; manuscript pp.22-31, 62v-63; scans45,48,56,59-63,126-127
D.7	Proved	149; manuscript pp.22-31, 62v-63; scans45,48,56,59-63,126-127
D.8	Proved	149; manuscript pp.22-31, 62v-63; scans45,48,56,59-63,126-127
D.9	Proved	149; manuscript pp.22-31, 62v-63; scans45,48,56,59-63,126-127
D.U1	Unresolved	149; manuscript pp.22-31, 62v-63; scans45,48,56,59-63,126-127
D.U2	Unresolved	149; manuscript pp.22-31, 62v-63; scans45,48,56,59-63,126-127

Statement	Status	Cote; manuscript page; scan
D.U3	Unresolved	149; manuscript pp.22-31, 62v-63; scans45,48,56,59-63,126-127
D.U4	Unresolved	149; manuscript pp.22-31, 62v-63; scans45,48,56,59-63,126-127
E.1	Proved	142; manuscript pp.7-10; scans15-21
E.2	Refuted	142; manuscript pp.7-10; scans15-21
E.3	Known	142; manuscript pp.7-10; scans15-21
E.4	Known	142; manuscript pp.7-10; scans15-21
E.5	Proved	142; manuscript pp.7-10; scans15-21
E.6	Proved	142; manuscript pp.7-10; scans15-21
E.7	Proved	142; manuscript pp.7-10; scans15-21
E.U2	Unresolved	142; manuscript pp.7-10; scans15-21
G.1	Proved	Additional lead of 13 November 2026; comparison with 149, manuscript pp.36v-38v,44v; scans74-78,90
G.2	Proved	Additional lead of 13 November 2026; comparison with 149, manuscript pp.36v-38v,44v; scans74-78,90
D.10	Proved	149; manuscript pp.29-31; scans59-63
D.11	Proved	149; manuscript pp.29-31; scans59-63
D.12	Proved	cote 149, pp.62v–63, scans126–127; D5 abstract semilinear action
D.13	Proved	D5 local obstruction to a cyclotomic fixed completed pro-point
D.14	Refuted	cote149 pp.62v–63 scans126–127; universal first clause of D5
E.8	Proved	142; manuscript pp.8v-10; scans18-21
E.9	Proved	142; manuscript pp.8v-10; scans18-21
E.10	Proved	cote 142, (Z.2), characteristic-p repair in E1
E.11	Proved	cote 142, (Z.2), requested tame category in E1
E.12	Proved	cote 142, (Z.2), requested specialization in E1
F.1	Refuted	MPL149, section 2, manuscript p.25v, scan52
F.2	Proved	MPL149, section 2, manuscript p.25v, scan52
F.3	Known	MPL149, section 2, manuscript p.25v, scan52; field extension issue of LM5-F.47,F.50
F.4	Proved	MPL149, manuscript p.25v, scan52; integrality step in LM5-F.47,F.50
F.5	Proved	MPL149, manuscript p.25v, scan52; curve step in LM5-F.50
F.6	Proved	MPL149, manuscript p.25v, scan52; quasi-rationality step in LM5-F.47,F.50
F.7	Proved	MPL149, section 2, manuscript p.25v, scan52
F.8	Known	MPL149, manuscript p.25v, scan52; prompt F2 field criterion
F.9	Unresolved	MPL149, manuscript p.25v, scan52; LM5-F.44 and prompt F2
F.10	Known	MPL149, manuscript pp.63-67, scans128-135; LM5-F.21-F.23,F.39,F.48; CK-C8; F149-C2,C3,C6,C8,C9
F.11	Proved	MPL149, manuscript pp.6-7 and66-67, scans13-15,133-135; LM5-F.39,F.48; CK-B5
F.12	Refuted	MPL149, manuscript pp.63-67, scans128-135; especially withdrawn injectivity at scan130; LM5-F.22,F.39,F.48; CK-C8
F.13	Proved	MPL149, manuscript pp.36-39,42,44,46-47; scans74-79,85,90,93-95; F149-C2,C3,C5,C8,C9
F.14	Known	MPL149, manuscript pp.39-41, scans79-82; F149-C6; comparison with LM5-F.39 and prompt F3
F.15	Proved	C1: pp32,34-35,41/scans65,70-72,83; C2: pp36-38/scans74-78; C3: pp37-39,44/scans75-79,90; C4: p39/scan79; C5: pp32,34-36,41-42/scans65,70-74,83-84; C6: pp39-41/scans79-82; C8: p42/scan85; C9: pp46-47/scans93-95
F.16	Proved	MPL149, manuscript p.25v, scan52; prompt F2 and LM5-F.44
F.17	Refuted	MPL149, manuscript p.25v, scan52; literal R27 and prompt F2
F.18	Proved	MPL149, manuscript p.25v, scan52; prompt F2
F.19	Proved	MPL149, manuscript p.25v, scan52; intended closed-point-preserving version of F2

Statement	Status	Cote; manuscript page; scan
F.20	Proved	MPL149, manuscript p.25v, scan52; corrected closed-point-preserving test in prompt F2
F.21	Proved	MPL149, manuscript p.25v, scan52; corrected closed-point-preserving test in prompt F2
F.22	Proved	MPL149, manuscript p.25v, scan52; full-topos obligation in prompt F2
F.23	Proved	MPL149, manuscript p.25v, scan52; corrected closed-point-preserving form of prompt F2
F.24	Proved	MPL149, manuscript p.25v, scan52; extension of the finite-quotient detection route toward prompt F2
F.25	Proved	MPL149, manuscript p.25v, scan52; higher-genus comparison route toward prompt F2
F.26	Proved	MPL149, manuscript p.25v, scan52; higher-genus point-function step toward prompt F2
F.27	Proved	MPL149, manuscript p.25v, scan52; general-curve unit reconstruction in prompt F2
F.28	Proved	MPL149, manuscript p.25v, scan52; arbitrary-curve point-function step in prompt F2
F.29	Proved	MPL149, manuscript p.25v, scan52; higher-dimensional detection step for prompt F2
F.30	Proved	MPL149, manuscript p.25v, scan52; regular affine source criterion in prompt F2
F.31	Proved	MPL149, manuscript p.25v, scan52; corrected reconstruction-property question in prompt F2

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